GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog

GeoVac Project

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Abstract

The GeoVac framework decomposes into two structurally separate layers. **Layer 1** is a bare combinatorial graph: quantum-number labels $(n, \ell, m, \ldots)$, integer eigenvalues, rational matrix elements, no physics. **Layer 2** is a finite family of named projections from the bare graph onto physical settings. Every physical variable, every dimensional unit, and every transcendental constant that appears in any GeoVac calculation enters through exactly one projection, and is fully traceable to it.

We catalogue twenty-eight projection mechanisms and tag each one along three independent axes: (a) which physical variables it introduces (charge Z, coupling α, energy scale $\hbar\omega$, internuclear distance R, gauge coupling β, cutoff Λ, ...); (b) which physical dimension or unit it attaches to graph data (energy, length, dimensionless, solid angle); and (c) its transcendental signature in the sense of Paper 18 (rational, π^{2k}, half-integer Hurwitz, odd ζ, Dirichlet L, ...). The projection axis is a structural sibling of Paper 18's transcendental axis: each transcendental tier corresponds to a specific projection type, and reading the dictionary in either direction recovers the other.

We then assemble the empirical matches catalogue: a tagged table of GeoVac quantities that match standard known physics values to machine precision (or stated uncertainty). Each row is annotated by projection(s), variable(s), dimension, transcendental class, and match precision. Reading the catalogue surfaces a falsifiable structural prediction:

Prediction 1 (Layer-2-presence bound.)**.** *The residual error of a catalogue row is bounded above by the larger of (i) the framework's basis-quality residual at zero Layer-2 inputs, and (ii) the largest single Layer-2 input consumed by the chain. Equivalently:* $|\epsilon(\textit{observable})| \leq \max\bigl(\epsilon_{basis}(\textit{framework}), \max_i |L_2^{(i)}|\bigr)$*. The genuine axis ordering catalogue precision is observable-class (zero-, one-, multi-Layer-2-input) crossed with the framework's basis-quality limit; chained projection depth is a downstream proxy that fails in both directions.*

This replaces an earlier naive form that scaled residual linearly with chain depth; the May 2026 synthesis pass falsified the depth form on the catalogue itself (depth-3 chains exist at residual 0 and at -211 ppm; depth-4 chains exist at $+2$ ppm and at -0.534%). The Lamb shift example (depth-4 chain, -0.534% residual at converged basis) is consistent with the framework's one-loop QED basis-quality limit; the H 1S polarizability example (depth-3 chain, residual 0) is consistent with the zero-Layer-2-input class for rational-analytic-answer observables.

Apart from Remarks 1 and 2, which report new measurements with dedicated backing tests, this paper introduces no new computation. It is a structural consolidation of the framework's self-presentation: language for discussing where physics enters, what carries dimensions, and which transcendentals are forced by which projection.

1 Introduction: The Two-Layer Thesis

The recurring difficulty in discussing GeoVac is that the framework sits between two languages. In one language, GeoVac is a sparse graph Hamiltonian framework for quantum chemistry: nodes labeled by quantum numbers, edges from selection rules, eigenvalues that predict atomic and molecular spectra (Paper 14, Paper 17, Paper 20). In the other language, GeoVac is a discrete spectral triple in the Marcolli–van Suijlekom lineage of graph-spectral-triple constructions (Paper 32), with a heat-kernel expansion that reproduces the QED β-function at one loop (Paper 28) and recovers seven of eight QED selection rules with a single Hopf-base calibration constant $1/(4\pi)$ per loop (Paper 33).

These two languages are not in conflict, and we have not been able to choose between them, because they are descriptions of two different layers of the same construction. This paper names the layers and the bridge between them.

Definition 1 (Layer 1: the bare graph)**.** *The Layer 1 object is the dimensionless combinatorial graph. Nodes are quantum-number labels $(n, \ell, m, \kappa, m_j, \ldots)$. Its spectral data are integers or half-integers: the Casimir label $n^2 - 1$ on the Coulomb S^3 graph and $n + 3/2$ on Dirac S^3 (in both cases the* continuum *operator spectra that the graph encodes as labels — the bare graph Laplacian itself is a distinct, positive-semidefinite object), and the total quanta N on the Bargmann–Segal S^5 graph. Matrix elements of bare graph operators (Casimir, ladder, adjacency) are rational, integer, or algebraic over $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ (from Wigner $3j$ algebra); π does not appear in any Layer 1 quantity (Paper 24's π-free certificate; the same applies on the Coulomb side). Layer 1 carries no physical units, no coupling constants, no observable energies.*

Definition 2 (Layer 2: the projection family)**.** *The Layer 2 object is a finite family of named maps from the bare graph to physical settings. Each projection is a structural construction with a name in the literature (Fock 1935 conformal projection; Hopf bundle; Bargmann–Segal transform; Connes–Chamseddine spectral action; Sturmian basis at exponent $\lambda = Z/n$; Camporesi–Higuchi spinor lift; Wigner $3j$ angular coupling; Wigner D-matrix rotation between molecular centers; Wilson plaquette; molecule-frame hyperspherical separation; vector-photon promotion; Mellin/heat-kernel regularization). Each projection adds (i) specific physical variables, (ii) specific physical dimensions, and (iii) specific transcendental content to the bare graph data on which it acts.*

Observation 1 (Single-projection origin of physical content)**.** *Every coupling constant, energy scale, length scale, and transcendental constant that appears in any GeoVac computation can be traced to exactly one projection in Layer 2. The bare graph does not contain α, Z, $\hbar\omega$, R, Λ, π, or $\zeta(2k)$. These quantities are introduced by, and only by, the projection that the calculation invokes.*

Remark 1 (The minimal presentation: two primitives, and the geometry–dynamics split)**.** *Observation 1 admits a sharper packaging. The framework's verified content is generated by* two *primitive kinds of data: (P1) the Layer-1 skeleton (Definition 1); and (P2) the projection family (Definition 2), each member carrying its scale parameter and its tagged transcendental. Operators that the framework's presentation treats as separate objects keep collapsing into these primitives. Four verified instances: (i) the chemistry Dirac operator equals the one-body Hamiltonian bit-exactly under the Marcolli–van Suijlekom gauge-network assembly* — measured but not yet promoted to a paper*, so it is recorded here as supporting rather than as one of the paper-backed instances; (ii) the N-electron isoenergetic Sturmian secular equation is a pure-number matrix plus a single scale p_κ, with energies as eigenvalues — an* atomic (single-centre) *property; two centres make T^0 non-diagonal and p_κ-dependent (Paper 60); (iii) the multi-center configuration operator $F = \sum_i P_i$ is similar to the union-basis Gram matrix* given intra-centre orthonormality *— a load-bearing* *hypothesis, not a technicality (Paper 32, `rem:config_operator_is_metric`); (iv) the atomic one-electron Hamiltonian in the shared-k Coulomb–Sturmian basis is* $h_1 = k^2\big(\mathbb{1} - S/2\big) - Zk\,\mathrm{diag}(1/n)$ *— a textbook Sturmian identity (the Sturmian equation plus potential-weighted orthonormality* $\langle\chi_m|r^{-1}|\chi_n\rangle = (k/n)\,\delta_{mn}$*), re-read here as an information statement and verified against an independent quadrature route at relative deviation* 4×10^{-9} *at the backing test's* $N_g = 500$ *grid* (6×10^{-10} *at* $N_g = 800$*; the ratio* $6.7 \approx 1.6^4$ *over* $500 \to 800$ *pins the truncation order at* $O(N_g^{-4})$*) (`tests/test_paper34_minimal_presentation.py`).*

Instance (iv) exhibits the split cleanly: all geometric content sits in the integer labels and the metric S; dynamics enters only through the single scale $k = p_0$*, which is the Fock focal length* ($p_0^2 = -2E$) *— the energy scale* is *the projection parameter. The same split holds in instances (i)–(iii).*

The boundary of the reduction is equally sharp, and is measured in the same test file: the two-electron tensor is not *metric-generated (best fit by Kronecker words in* S *leaves a* 39% *residual, against* $\sim 10^{-12}$ *for a genuinely metric-generated control), and the multi-center composition algebra and the Class-1 calibration inputs are established skeleton-external elsewhere in the corpus. The derived/independent boundary for the atomic sector sits exactly at the electron–electron channel.*

Tier: *instances (ii)–(iv) are individually proven or measured* at the citations given*; instance (i) is measured but carries no paper citation and is therefore not load-bearing here. The claim that the two-primitive set is* minimal *is an Observation. Its falsifier: a verified operator within the one-center, one-electron sector that carries information beyond (labels, metric, scale).*

Remark 2 (The electron–electron boundary admits an exact partial split)**.** *The boundary stated in Remark 1 — the two-electron tensor is not metric-generated — is sharper than a residual: the tensor splits* exactly*, and the irreducible part is a single low-rank object. Each multipole kernel obeys the elementary min/max identity*

$$\frac{r_<^L}{r_>^{L+1}} = \underbrace{\tfrac{1}{2}\big[\varphi(r_1)\psi(r_2) + \varphi(r_2)\psi(r_1)\big]}_{\textit{separable, smooth}} - \underbrace{\tfrac{1}{2}\big|\varphi(r_1)\psi(r_2) - \varphi(r_2)\psi(r_1)\big|}_{W_L \ge 0} \qquad (\varphi = r^L,\ \psi = r^{-(L+1)}),$$

so the channel-L tensor is a rank-two *head* $\tfrac{1}{2}(A\otimes B + B\otimes A)$*, with* $A = \langle r^L\rangle$ *(bandwidth* $L+1$*, measured) and* $B = \langle r^{-(L+1)}\rangle$*, minus a remainder built from* W_L*. At* $L = 0$ *the head is pure labels* $\times$ *metric (*$A = S$*,* $B = (k/n)\,\delta_{mn}$*), and the full dynamical dependence is one power of the focal length,* $g(k) = k\,g(1)$ *with* $g(1)$ *rational (*$\langle 1s^2|g|1s^2\rangle = 5k/8$ *exactly); at* $L \ge 1$ *the head is still rank-two one-body* $\times$ *one-body but* B *is dense —* a product*, hence genuinely two-body, so the fold-into-*h_1 *below is available only in the* $L = 0$ *channel. Verified properties of the split (`tests/test_paper34_ee_split.py`): (i) in the* $L = 0$ *channel the separable head supports* no correlation *— its full-CI equals the dressed one-body problem* $h_1 + \tfrac{1}{2}(N-1)V$ *bit-exactly for* $N = 2, 3$*, so all correlation lives in* W *(at* $L \ge 1$ *the head is a product with dense* B *and is genuinely two-body, per the preceding sentence, so the fold does not apply); (ii)* W *carries the kernel's entire non-smoothness at electron coalescence; (iii)* W *is not small (*$\langle 1s^2|W\rangle/\langle 1s^2|g\rangle = 3/5$ *exactly) but is* compressible at basis-independent rank*: full-CI is reproduced to* 1 *mHa at* W*-rank* 3–4 *and to* 0.01 *mHa at rank* 5–7 *across* $n_s = 3$–6 *(while* n_s^2 *grows* 9–36*), the compressibility improving with* L *(Frobenius tail after four terms:* 1.6%, 0.7%, 0.1%, 0.03% *for* $L = 0$–3*). The payoff survives angular coupling: in a Gaunt-coupled s+p full-CI with the* $L = 0, 1, 2$ *channels all active, truncating every* W_L *at rank* 3–4 *reproduces the full energy to under* 1 *mHa, flat as the radial-pair space grows from* 10 *to* 28. Resource consequence ($L = 0$; measured and bounded): *in the* $L = 0$ *channel the head is exactly one-body (its kernel* $\tfrac{1}{2}(r_1^{-1} + r_2^{-1})$ *sums over pairs to* $\tfrac{1}{2}(N-1)\sum_i r_i^{-1}$*), so it folds into* h_1 *with no approximation, lowering the Jordan–Wigner LCU* 1*-norm by* 15–17% *at bit-identical* *energy, and beating a plain density-fitting control (eigen-truncation of g itself at matched rank) by* 13–17% *at matched accuracy. The advantage is a* constant factor that degrades with basis size — $\lambda_{split}/\lambda_{full} = 0.83, 0.85, 0.89, 0.91, 0.94$ *at* $n_s = 4, 6, 8, 10, 12$, *monotone toward unity — so it changes no exponent and is not claimed as a scaling result.*

Lineage: *numerical low-rank structure of electron-repulsion integrals is classical — it is the basis of the Cholesky-decomposition approach of Beebe–Linderberg [61] and of density fitting — and the compressibility per se is not claimed as new. What this remark adds is the placement in the minimal presentation: an* exact *rank-two labels-×-metric head, the bit-exact no-correlation property of that head, the localization of the coalescence non-smoothness in W, and the measured flatness of the remainder's rank in basis size.* Tier: *the split identity and the rank-two head are exact (machine-verified); the no-correlation property is exact for fixed N in the $L = 0$ channel (there, and only there, a one-body operator in disguise); the rank-flatness is MEASURED over $n_s \le 6$ one-center s-only, and separately in a Gaunt-coupled s+p full-CI with $L = 0, 1, 2$ active, where rank* 3–4 *holds to under* 1 *mHa as the radial-pair space grows from* 10 *to* 28.

Two sharpenings (`tests/test_paper34_ee_split.py`, second block). (a) The $2n-1$ rank law. *The s-wave pair densities $R_m R_n r^2 e^{-2kr}$ are polynomials of degree $\le 2n-2$ (on the r^2 floor) times one fixed exponential, a function space of dimension exactly $2n-1$. Hence the matricization of the two-electron tensor has exact rank $\le 2n-1$ —* linear *in n, for* every *radial kernel — with explicit integer dependencies among the densities ($-\rho_{11} + 2\rho_{12} - 3\rho_{13} + 2\rho_{22} \equiv 0$ at $n = 3$, verified symbolically) and total nullity $(n-1)^2$. Measured: g, W, and a kinked control kernel all saturate the ceiling at every $n_s \le 6$, while very smooth kernels may sit below it; the separable head stays rank* 2. *The flat accuracy-rank* 3–7 *of the previous paragraph sits under this exact linear ceiling.* (b) The closed-form question, resolved negative. *At $k = 1$ every entry is exactly rational ($\langle 1s^2|W|1s^2\rangle = 3/8$), but the active characteristic polynomials are* irreducible *over $\mathbb{Q}$ (degree $2n-1$; at $n = 2$ the cubic $131072\lambda^3 - 50688\lambda^2 - 25200\lambda - 675$, pinned in the backing test), so the finite eigenpairs are algebraic numbers of full degree with no low-degree closed form; and the Löwdin-compressed W spectrum* grows *with n_s (the kernel operator is unbounded), so the finite eigenvectors do not converge to fixed continuum eigenfunctions. The skeleton-form characterization that survives is differential: in reciprocal radius $u = 1/r$ the Coulomb kernel is $\min(u_1, u_2)$ — the Green's function of $-d^2/du^2$ with Dirichlet boundary at $u = 0$, equivalently the manifestly positive resolution $1/r_> = \int_0^\infty \mathbf{1}[r_1<R]\,\mathbf{1}[r_2<R]\,dR/R^2$ — and W's kernel $\frac{1}{2}|u_1 - u_2|$ obeys $d^2K/du^2 = \delta(u - u')$, so eigenfunctions of any weighted compression satisfy the second-order ODE $\lambda f''(u) = w(u)\,f(u)$ (grid-verified at 10^{-5}).* Tier: *the rank law, rationality, integer dependencies, and irreducibility are exact (machine-verified over $\mathbb{Q}$); the spectral growth is MEASURED.*

The remainder of this paper is the dictionary.

2 Layer 1: The Bare Graph

2.1 Quantum-number labelling

Nodes of the bare GeoVac graph are tuples of quantum numbers. The specific tuple depends on which natural geometry is in play:

- Atomic Coulomb S^3: (n, ℓ, m_ℓ) for the scalar sector, (n, κ, m_j) for the Dirac sector.
- Bargmann–Segal S^5 (3D HO): (N, ℓ, m_ℓ) in SU(3) $(N, 0)$ irreps.
- Composed molecular: per-block tuples of the above, with inter-block edges labeled by the multipole order L of the cross-center expansion.

No physical quantities are required to define these labels. They are pure representation-theoretic data of compact Lie groups.

2.2 π-free certificate

On both the Coulomb S^3 Fock graph and the Bargmann–Segal S^5 graph, every adjacency matrix entry, every diagonal eigenvalue, and every CG-coupling matrix element is exact rational or algebraic in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15}, \ldots]$ (Paper 24 [12], Paper 28's graph-native QED certificate). No π appears in Layer 1 by exact computation.

2.3 Layer-1 observables

Some observables are computable entirely within Layer 1. These are the "zero-projection" matches in the catalogue (Section 5). Examples:

- Pauli term count $N_{\text{Pauli}} = 11.10 \cdot Q$ (across 35 composed molecules; Paper 14).
- Magic numbers $2, 8, 20, 28, 50, 82, 126$ (HO + spin-orbit graph eigenvalue counting; Paper 23).
- Angular sparsity density 1.44% at $\ell_{\max} = 3$ (Paper 22, potential-independent).
- All Gaunt selection rules (CG sparsity).

These exist on Layer 1 alone. They have no π, no α, no energy scale, no transcendentals of any kind, and they match experiment exactly because there is nothing to project—they are consequences of compact-Lie-group representation theory acting on discrete quantum-number labels.

3 Layer 2: The Projection Family

We enumerate twenty-eight projection mechanisms. For each: source and target spaces, physical variables introduced, dimensions introduced, transcendental signature, and existing GeoVac papers that exercise the projection. Projections 14 (rest-mass) and 15 (observation / temporal-window) were added in sprint KG[1] / Paper 35 (May 2026); projection 16 (two-body Dirac / Breit retardation) was added in the precision-catalogue sprint of 2026-05-08, motivated by the positronium 1S–2S framework-native residual that surfaced as the sixth (and structurally cleanest) instance of the multi-focal-composition wall named in CLAUDE.md § 2. Projections 17–19 (nuclear charge-density / Foldy–Friar, magnetization-density / Zemach, and tensor multipole) were added in the intra-particle hunting sprint of 2026-05-09, after the audit memo `debug/paper34_v_base_unit_audit_memo.md` flagged magnetization-density as a single 17th-projection candidate and the follow-up hunt established that the three mechanisms are categorically distinct (different Hamiltonian operators, different selection rules; documented in `debug/intra_particle_L_hunting_memo.md`). A boundary subsection (§ 3.29) names the internal-graph-derived vs. Layer-2-input distinction, with the external-field hunting memo (`debug/external_field_hunting_memo.md`, same date) verifying that uniform external classical fields (Zeeman, Stark) are Layer-2 inputs rather than § 3 projections.

3.1 Fock conformal projection: $\mathbb{R}^3 \to S^3$

Source: continuum Coulomb Hamiltonian $H = -\frac{1}{2}\nabla^2 - Z/r$ on $\mathbb{R}^3$.
Target: unit S^3 via stereographic-like map at focal length $p_0^2 = -2E$.
Variables introduced: Z (nuclear charge), E (energy eigenvalue).

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

Dimension introduced: energy (in atomic units, Hartree). The dimensional content is carried by the choice $m_e = \hbar = e = 1$.
Transcendental signature: rational prefactor $\kappa = -1/16$ (the Rydberg-to-graph-eigenvalue conversion). π enters through $\mathrm{Vol}(S^3) = 2\pi^2$ when subsequent spectral integrals are performed.
Used in: Paper 7 (S^3 proof), Papers 13/15/17 (hyperspherical hierarchy), Paper 18 § III (Peter–Weyl realization), all atomic energy computations.

3.2 Hopf bundle: $S^3 \to S^2 \times S^1$ (locally)

Source: $S^3 \cong \mathrm{SU}(2)$.
Target: S^2 base $\times S^1$ fibre.
Variables introduced: α (the fine-structure constant treated as a candidate observable in Paper 2's Observation).
Dimension introduced: dimensionless.
Transcendental signature: $\pi = \mathrm{Vol}(S^2)/4$ (Hopf measure factor; Sprint A α-PI identification).
Used in: Paper 2 (fine-structure Observation), Paper 25 (Wilson–Hodge gauge structure), Paper 30 (SU(2) Wilson sibling).

3.3 Bargmann–Segal transform: $\mathbb{R}^3$ HO $\to S^5$ Hardy space

Source: 3D isotropic harmonic oscillator $H_{\mathrm{HO}} = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 r^2$.
Target: holomorphic Hardy-space sector of S^5, restricted to SU(3) $(N, 0)$ symmetric irreps.
Variables introduced: $\hbar\omega$ (HO frequency).
Dimension introduced: energy.
Transcendental signature: *none.* The Bargmann–Segal graph is bit-exactly π-free in rational arithmetic at every finite $N_{\max}$ (Paper 24). π appears only as $\mathrm{Vol}(S^5) = \pi^3$ in continuum integration measures, never in lattice data.
Used in: Paper 24 (HO discretization), Paper 31 (universal vs. Coulomb partition).

3.4 Stereographic / conformal coordinate change

Source: compact sphere (S^2, S^3, S^5).
Target: flat-space coordinates ($\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^5$).
Variables introduced: length scale (radial coordinate r).
Dimension introduced: length.
Transcendental signature: conformal factor producing the $1/r$ Coulomb potential as coordinate distortion (Paper 7).
Used in: Paper 7 (Coulomb-as-coordinate), all calculations that quote results in physical bohr/Hartree.

3.5 Sturmian reparameterization at $\lambda = Z/n$

Source: bare Coulomb S^3 graph.
Target: same graph relabelled as a Coulomb Sturmian basis spanning bound states plus a discretized continuum.
Variables introduced: Z, n (re-used; no new variables).
Dimension introduced: preserves existing.
Transcendental signature: preserves rationality of Layer 1. No new π or other transcendentals enter at this step; calibration-tier transcendentals (e.g. Bethe logarithm) enter through subsequent

spectral mode sums.
Used in: Papers 8–9, Paper 19, sprint LS-2 (Bethe logarithm from Sturmian sum at 2–3%).

3.6 Connes–Chamseddine spectral action / heat kernel

Source: Dirac operator D on bare S^3 graph (or extension).
Target: heat-kernel expansion $\operatorname{Tr} f(D^2/\Lambda^2)$ with cutoff function f and scale Λ^2.
Variables introduced: Λ (UV cutoff), α (when gauge sector included).
Dimension introduced: energy via Λ.
Transcendental signature: $\sqrt{\pi} \times \mathbb{Q}$ Seeley–DeWitt coefficients on unit S^3 ($a_0 = a_1 = \sqrt{\pi}$, $a_2 = \sqrt{\pi}/8$); resulting observable coefficients are $\pi^{2k} \times \mathbb{Q}$ at one loop (T9 theorem, Paper 28).
Used in: Paper 28 (QED on S^3), Paper 32 (spectral triple synthesis), sprint LS-1 (Uehling shift -27.13 MHz from $\Pi = 1/(48\pi^2)$).
Structural reading (master Mellin engine framing). This projection is the $k = 2$ sub-case of the master Mellin engine of Paper 32 § VIII and Paper 18 § III.7: the proper-time integration $\int_0^\infty dt\, f(t) \operatorname{Tr} e^{-tD^2}$ that converts D-spectral data to physical observables. The same Mellin technology applied at $k = 0$ to $\operatorname{Tr} e^{-tD^2}$ recovers the M1 Hopf-base measure content (§ 3.2 and § 3.11), and applied at $k = 1$ recovers the M3 vertex-parity Hurwitz content (composed with § 3.7 and vertex restriction; Paper 28 § IV). The Mellin transform itself is the *evaluator* (a meta-mechanism organizing the spectral-side projection family), not a § III peer projection; consequently the Schwinger proper-time formalism and the Connes–Chamseddine spectral action are the same operation under two names, both absorbed into the present entry (Sprint 3 Track 12 scoping verdict, 2026-05-15; `debug/sprint3_drafts/track_12_heat_kernel_regularization_scoping_memo.md`). The α-power index n in iterated applications of this projection counts loop order; per-iteration transcendental content is the $\sqrt{\pi} \cdot \mathbb{Q}$ ring at the heat-kernel level (T9 theorem, Paper 28) and the $1/(4\pi)$ vector-photon Hopf-base calibration per loop (§ 3.11 in Coulomb gauge, § 3.26). The finite parts of α^n observables are framework-native at every n; UV-divergent multi-loop integrands are reproduced structurally (LS-8a verdict, Paper 36 § VII) but their counterterm-renormalized finite extraction requires external $Z_2/\delta m$ input not autonomously generated by the bare spectral action (Paper 35 § VII.3 Refined Prediction 1; Sprint 3 Track 11 absorption verdict). The open question about fractional-s behaviour of the Mellin engine (§ 9 (a)) is structurally separate from the present entry and remains as catalogued.

3.7 Camporesi–Higuchi spinor lift

Source: scalar bare S^3 graph.
Target: Dirac-on-S^3 graph with eigenvalues $|\lambda_n| = n + 3/2$, degeneracies $g_n = 2(n{+}1)(n{+}2)$.
Variables introduced: α enters through $(Z\alpha)^2$ relativistic content.
Dimension introduced: dimensionless (the spinor lift itself is unit-preserving).
Transcendental signature: half-integer Hurwitz zetas $\zeta(s, n + 3/2)$, producing odd-ζ content $\zeta(3), \zeta(5), \ldots$ at integer s (Paper 28 § IV); Catalan $G = \beta(2)$ and Dirichlet $\beta(4)$ at the vertex level via quarter-integer shifts $\zeta(s, 3/4), \zeta(s, 5/4)$. Lives in the spinor-intrinsic algebraic ring $R_{\mathrm{sp}} = \mathbb{Q}(\alpha^2)[\gamma]/(\gamma^2 + (Z\alpha)^2 - 1)$ (Paper 18 § IV).
Used in: Paper 28 § IV, Paper 33, Tier-2 spin-ful composed qubit encoding (Paper 14 § V).

3.8 Wigner $3j$ angular coupling

Source: pair of angular-momentum bare-graph nodes.
Target: coupled angular-momentum eigenstate.

Variables introduced: none.
Dimension introduced: dimensionless.
Transcendental signature: pure rational ($\mathbb{Q}[\sqrt{2k+1}]_k$).
Used in: all Gaunt integral evaluations, Paper 22 angular sparsity, every Slater F^k computation.

3.9 Wigner D-matrix rotation between molecular centers

Source: composed orbital block at center A.
Target: same block in molecular coordinate frame at center B.
Variables introduced: nuclear position vector $\vec{R}_{AB}$.
Dimension introduced: length.
Transcendental signature: preserves rationality with $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$ algebraic content from the Wigner d-matrix at non-collinear angles.
Used in: multi-center molecules in Paper 17 and Paper 14 (LiF, CO, N_2, NaCl, CH_2O, C_2H_2, C_2H_6, F_2).

3.10 Wilson plaquette: edge → 2-cell

Source: Hopf graph 1-cochains.
Target: 2-cells (plaquettes) carrying SU(2) link variables.
Variables introduced: $\beta = 1/g^2$ (gauge coupling).
Dimension introduced: dimensionless.
Transcendental signature: non-abelian SU(2) Haar content; in the maximal-torus reduction yields the abelian $U(1)$ content of Paper 25.
Used in: Paper 30 (SU(2) Wilson lattice gauge), Paper 25 (Hodge-1 Laplacian as gauge propagator), transverse photon construction (Paper 28 § transverse_photon).

3.11 Vector-photon promotion

Source: scalar 1-cochain photon on bare graph.
Target: vector-harmonic photon with explicit (q, m_q) quantum numbers and Wigner $3j$ vertex coupling.
Variables introduced: none (combinatorial labels).
Dimension introduced: dimensionless; specifically a solid-angle measure on the Hopf S^2 base.
Transcendental signature: $1/(4\pi)$ per loop, identified as the S^2 Weyl exchange constant of the Hopf base (Paper 33).
Used in: Paper 33 (recovers six angular-momentum selection rules), sprint VP-1 (vertex correction calibration).

3.12 Molecule-frame hyperspherical separation

Source: composed atomic blocks.
Target: molecule-frame hyperspherical coordinates $(\rho, \alpha, \theta_{12}, \ldots)$ with internuclear distance R appearing in the charge function.
Variables introduced: R (internuclear distance); for polyatomics, additional bond angles.
Dimension introduced: length.
Transcendental signature: piecewise-smooth in R at the adiabatic potential level (Track S, Paper 15); Gaunt integrals preserve rationality at the angular level.
Used in: Paper 15 (Level 4), Paper 17 (composed geometry), all H_2/HeH^+/H_2O calculations.

3.13 Drake–Swainson asymptotic subtraction

Source: divergent finite-basis spectral sum $J_v(nl)/I_v(nl)$ for $\ell > 0$ Bethe-logarithm states, where the velocity-form closure $I_v(nl) = (Z^4/n^3)\,\delta_{\ell,0}$ vanishes exactly for $\ell > 0$ on the bare Sturmian-projected graph.
Target: K-independent regularized value $\ln k_0(n,\ell) = J_v(n,\ell)/D_{\mathrm{drake}}(n,\ell)$, with the structural denominator

$$D_{\mathrm{drake}}(n,\ell) = \frac{2(2\ell+1)\,Z^4}{n^3}.$$

This recovers $D_{\mathrm{drake}}(nS) = I_v(nS) = 2Z^4/n^3$ for $\ell = 0$ (matching the closure value), and provides a non-vanishing structural normalization for $\ell > 0$.
Variables introduced: subtraction scale K (transient; the final result is K-independent over a plateau spanning ~ 3.6 orders of magnitude in K, as verified at sprint LS-4).
Dimension introduced: energy via K (transient, cancelled in $\beta_{\mathrm{low}}(K) + \beta_{\mathrm{high}}(K) = J_v$); the resulting $\ln k_0$ is dimensionless.
Transcendental signature: flow tier (Paper 18 § IV). D_{drake} itself is rational ($Z^4 \cdot \mathbb{Q}$); the transcendental content sits in J_v, where the spectral-sum logarithms $\ln|2\Delta E_m/\mathrm{Ry}|$ are inherited from the Sturmian projection and carry calibration α-dependent pieces through the matching to the standard Bethe formula. The structural denominator $D_{\mathrm{drake}}(n,\ell) = (\mathrm{spin})\cdot(\text{angular degeneracy})\cdot Z^4/n^3$ is a purely combinatorial quantity (spin $\cdot\, 2\ell{+}1 \cdot$ hydrogenic density), making the projection's structural content combinatorial-rational while its operational content is calibration.
Used in: sprint LS-4 (Bethe logs for 2P at +2.4%, 3P at +6.3%, 3D at −0.24% versus Drake-Swainson 1990 tabulated values). The combined hydrogen Lamb shift via this projection chained with Sturmian and spectral-action gives −0.39% residual at the sweet-spot $N = 40$ (transient cancellation; converges to LS-1 baseline −3.10% at $N \to \infty$ — the pre-LS-6a convention, superseded by the LS-6a fix at −0.534%; and per Sprint LS-7 the residual is not primarily an α^5 multi-loop ceiling). The projection is structurally a sibling of the Schwartz (1961) integral form, simplified to act on the discrete Sturmian pseudostate basis directly.

3.14 Rest-mass projection

Source: bare relativistic spectrum on $S^3 \times \mathbb{R}$ (or any natural-geometry spectrum), in the massless limit.
Target: massive spectrum $\omega_n^2 = \omega_n^{(0)\,2} + m^2$, where $\omega_n^{(0)}$ is the massless eigenvalue from the ambient natural geometry.
Variables introduced: m (rest mass; equivalently rest-energy mc^2 at $c = 1$).
Dimension introduced: mass = energy at $c = 1$.
Transcendental signature: *trivial.* The bare-graph algebraic-extension ring is preserved when $m^2 \in \mathbb{Q}$ (or any ring extension of $\mathbb{Q}$ already present in the ambient spectrum). Verified explicitly in sprint KG-1 for the Klein-Gordon spectrum on $S^3 \times \mathbb{R}$: $\omega_n = c_n\sqrt{d_n}$ with $c_n \in \mathbb{Q}$ and d_n a positive square-free integer for all $n \in [1, 50]$ across the rational m^2 panel $\{0, 1, 1/4, 2\}$ (Paper 35). The mass projection is parameter-only; it does not introduce π or any other transcendental.
Used in: muonic / electronic / positronium reduced-mass corrections; Paper 23 nuclear-electronic composed Hamiltonian ($I \cdot S$ hyperfine coupling); any future Klein-Gordon or Dirac-on-S^3 relativistic mass calculation; Paper 35 Proposition 1.A.

3.15 Observation / temporal-window projection

Source: bare relativistic spectrum on a spatial natural geometry crossed with open temporal direction $\mathbb{R}$.
Target: same spectrum on the spatial geometry crossed with periodic Euclidean time S^1_β of period β (equivalently: finite-time observation window of duration β in real time; finite-temperature partition function at temperature $T = 1/\beta$). For bosons, temporal Matsubara modes $\omega^t_k = 2\pi k/\beta$, $k \in \mathbb{Z}$; for fermions with antiperiodic time, $\omega^t_k = (2k+1)\pi/\beta$.
Variables introduced: β (or $T = 1/\beta$).
Dimension introduced: time = inverse energy.
Transcendental signature: $2\pi \cdot \mathbb{Q}$ per Matsubara mode in the spectrum directly; in spectral integrals over the compactified direction, $\pi^{2k} \cdot \mathbb{Q}$ from $\zeta_R(2k)$ values, including the Stefan-Boltzmann constant $\pi^2/90$ in the high-T limit of $S^3 \times S^1_\beta$. The first π-bearing eigenvalue of the compactified Klein-Gordon spectrum is the $(n{=}0, k{=}1)$ Matsubara mode at $\omega^2 = 4\pi^2/\beta^2$, with no earlier step injecting π (verified in sprint KG-2; Paper 35).
Used in: all finite-temperature observables (Casimir at $T \neq 0$, free-energy density, Stefan-Boltzmann constants); all QED loop integrals that involve continuous time integration (Schwinger proper-time, Mellin-Barnes, etc.); spectral-action evaluation Tr $f(D^2/\Lambda^2)$ when the trace runs over a compact-time direction; Paper 35 Proposition 1.B. This projection is structurally what an observer does when integrating an instantaneous spectrum over a finite measurement window.[2]

3.16 Two-body Dirac / Breit retardation

Source: framework-native single-particle Eides § 3.2 self-energy / Lamb-shift bracket on the Sturmian-reparameterized Fock graph, valid in the $\lambda_l \ll \lambda_n$ hierarchy where the $1/M_{\text{nucleus}}$ recoil expansion is regular.
Target: full two-body Breit / Bethe–Salpeter Hamiltonian for the bound state at order α^4, retaining all retardation and crossed-photon terms that the $1/M$ expansion drops as subleading.
Variables introduced: the rest-mass ratio m_l/m_n (and not the absolute mass m already introduced by the rest-mass projection of § 3.14). This ratio does *not* vanish in the equal-mass leptonic

[2] Bisognano–Wichmann reading. Under the standard Wick-rotation chain (Hartle–Hawking 1976 [38]; Sewell 1982 [36]; Bisognano–Wichmann 1976 [35]), the temporal-window projection's β IS the period of the modular-flow / Lorentz-boost orbit on a wedge or static-exterior algebra: Bisognano–Wichmann establishes that the modular automorphism of the algebra of observables localized in a Rindler wedge, with respect to the vacuum state, IS the unitary representation of the Lorentz boost subgroup that maps the wedge to itself; Sewell 1982 generalizes this to globally hyperbolic Lorentzian manifolds with bifurcate Killing horizons, including Schwarzschild at $\beta_{\text{cigar}} = 8\pi M$. Sprint TD Track 4 (`debug/sprint_td_track4_memo.md`, May 2026) reproduces this $\beta = 8\pi M$ from the framework's M1 Hopf-base measure / $\mathrm{Vol}(S^1_\tau)$ mechanism alone. Track D (`debug/bisognano_wichmann_track_d_memo.md`, May 2026) documents the structural correspondence between the framework-side 2π and the Lorentzian-side modular-flow / boost-orbit period. The algebraic structure underwriting this reading is the Tomita–Takesaki modular flow on the wedge / static-exterior algebra (modular operator $\Delta = e^{-K}$, modular conjugation J, $\sigma_t(A) = \Delta^{it} A \Delta^{-it}$ closing with $\sigma_{i\cdot 2\pi}$ = identity at the BW wedge); Sprint 3 Track 10 (2026-05-15; `debug/sprint3_drafts/track_10_tomita_takesaki_scoping_memo.md`) absorbed Tomita–Takesaki into the present footnote and § 3.27, with the operator-system-level framework-internal realization (modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$) named as the open extension in § 9. The Hawking temperature $T_H = 1/(8\pi M)$ enters the catalogue as an off-precision row in § 5.1 (error class C, M external Einstein-equations input, machinery-witness only); the Unruh pendant (`debug/unruh_pendant_memo.md`, May 2026, a sprint-scale follow-up to Track D) closes the Track D §5.4 flag with $T_U = a/(2\pi)$ as the third Wick-rotation witness via the same M1 Hopf-base measure / $\mathrm{Vol}(S^1)$ mechanism (Hawking–Sewell–BW–Unruh four-witness chain). See Paper 32 §VIII Remark on the Bisognano–Wichmann reading and Paper 35 §VIII for the framework-level structural correspondence. This is a structural correspondence, not a literal identification: the operator-level statement ($\sigma_{i\cdot 2\pi}$ = identity on the truncated metric spectral triple) is the principal falsifier flagged in Track D §3.1, a follow-up beyond sprint scale.

limit $m_l = m_n$ where the $1/M$ expansion is invalid. Distinct from § 3.14: rest-mass projection rescales single-particle observables under $m_e \to m_{\rm red}$; this projection adds the genuine two-body Breit kinematic content that has no single-particle analog.
Dimension introduced: energy.
Transcendental signature: $\alpha^4 \cdot \mathbb{Q}$. Pure rational coefficient $\times\alpha^4$, no transcendental injected beyond α already carried by the spectral-action projection. Structurally the cleanest tier: same calibration class as the leading magnetic dipole-dipole structure, ring-preserving over $\mathbb{Q}(\alpha)$. Distinct from the magnetization-density / Zemach projection class (Paper 34 catalogue Layer-2 inputs), which couples to nuclear spatial structure; this projection is a relativistic two-body kinematic effect at the leading α^4 order.
Used in: positronium 1S–2S two-photon transition (sprint precision-catalogue 2026-05-08, equal-mass leptonic limit $\lambda_a = \lambda_b = 0.5$); positronium $1^3\mathrm{S}_1$ HFS (annihilation channel separated as a distinct Layer-2 mechanism); future positronium fine-structure tests; future tests of the equal-mass μ^+e^- limit if measured.
Empirical anchor: Ps 1S-2S framework-native at +64.75 ppm vs Fee 1993 1,233,607,216.4(3.2) MHz; the Layer-2 α^4 Breit input (the standard coefficient $-(11/48)$; Ps theory per Karshenboim 2005 § 9 — the literal fraction is the framework's own reduction, verified numerically at 0.57%, its textbook source an open verification item; complete Ps theory at $m\alpha^6$ from Pachucki–Karshenboim 1998, PRL 80, 2101) at the ~ 80 GHz scale closes the residual to sub-MHz.
Honest scope. This projection has one empirical anchor as of 2026-05-08 (Ps 1S-2S). Its definition is structural — the construction (single-particle Eides bracket $\to$ full two-body Breit Hamiltonian) is established physics (Bethe–Salpeter 1957, Karplus–Klein 1952, Pachucki–Karshenboim 1998, Czarnecki–Melnikov–Yelkhovsky 1999, Karshenboim 2005 § 9) — but the framework's catalogue verification rests on this single observable until additional equal-mass or near-equal-mass two-body precision data are tested.
Operator-level verification (2026-05-09, Sprint Calc-Ps-1S2S Track 4 autopsy, §5.2.13): PARTIALLY VERIFIED. The retardation kernel $r_<^k/r_>^{k+3}$ is implemented as production module `geovac.breit_integrals` (26 tests; exact Fraction/sympy arithmetic). Closed-form matrix elements at $Z = 1$, $k = 0$ for the $(1s, 1s)$, $(1s, 2s)$, $(2s, 2s)$ orbital pairs that participate in the Ps 1S–2S level structure live in $\mathbb{Q}[\log 2, \log 3]$ (Paper 18 embedding-log content) for the diagonal orbital pairs and in $\mathbb{Q}$ for the $(1s, 2s; 1s, 2s)$ exchange, *e.g.* $R^0_{\rm BP}(1s, 1s; 1s, 1s) = -5 + 8\log 2$, $R^0_{\rm BP}(1s, 2s; 1s, 2s) = 4/81$ (production-verified by `tests/test_breit_integrals.py` and `tests/test_paper34_projection_spot_checks_batch1.py`; the $(1s, 2s; 1s, 2s)$ value is a pure rational, no log content). Roothaan multipole termination at $L_{\max} = 2l_{\max}$ is preserved at the equal-mass limit $\lambda_a = \lambda_b = 0.5$ because it is angular content (Wigner-3j triangle inequality, § 3.8), not a small-parameter expansion. *Named structural extension:* state-specific bound-state matrix element evaluation $\langle \mathrm{Ps}, 1S|H_B|\mathrm{Ps}, 1S\rangle - \langle \mathrm{Ps}, 2S|H_B|\mathrm{Ps}, 2S\rangle$ in equal-mass two-body Dirac normalization requires the Bethe–Salpeter quasi-energy expansion machinery, structurally outside the framework's single-particle Eides § 3.2 SE bracket; this is the W2a-class infrastructure target. The $\sim 0.57\%$ match between the calibration-by-construction Layer-2 input ($-79{,}861.9$ MHz $= -0.2279 \times m_e c^2\alpha^4/h$) and the canonical Karshenboim 2005 § 9 closed-form $-(11/48) = -0.2292$ is consistency evidence at the magnitude level.
Structural reading. This is the *structurally cleanest known instance* of the multi-focal-composition wall named in CLAUDE.md § 2. The wall's other five instances (Sprint H1 Yukawa, LS-8a $Z_2/\delta m$, HF-3 recoil, HF-4 Zemach, HF-5 multi-loop a_e) all surface in the multi-loop / renormalization sector at α^5 or higher, where two-body-projection failure is entangled with renormalization failure. The Breit retardation content surfaces at α^4 — one full order *before* the LS-8a renormalization wall — so the two-body-projection failure is isolated and visible without renormalization confounds. This

is also what places the projection in the Layer-2 input class (structurally a sibling of magnetization-density / Zemach in § 3.18): both take the framework's single-particle output and add a structural correction the bare spectral action does not autonomously generate.

3.17 Nuclear charge-density correction (Foldy/Friar)

Source: bare Coulomb $V_{\rm ne}(\vec{r}) = -Ze^2/|\vec{r}|$ on the Fock-projected S^3 graph (point-source nucleus).
Target: convolved Coulomb $V_{\rm ne}(\vec{r}) \to \int d^3r'\, \rho_E^{\rm nuc}(\vec{r}')\,(-Ze^2/|\vec{r}-\vec{r}'|)$, with $\rho_E^{\rm nuc}$ the nuclear electric charge distribution; at leading order in the small-r expansion the correction reduces to the Foldy / Friar contact term $\Delta V = +\frac{2\pi}{3} Z\alpha\, \langle r^2\rangle_E\, \delta^3(\vec{r})$.
Variables introduced: $\langle r^2\rangle_E$ (or equivalently the charge radius r_E for a given species).
Dimension introduced: length (one new $[L]$ scale per nuclear species: proton r_p, deuteron r_d, etc.; structurally distinct from atomic / internuclear length scales).
Transcendental signature: ring-preserving over $\mathbb{Q}(\alpha)$. The Foldy / Friar correction is rational in the charge form-factor moments at every order of the small-r expansion; the only transcendental content is the foundational α already present in the spectral-action chain.
Used in: hydrogen Lamb shift r_p contribution (currently consumed externally via Eides Tab. 7.3 / Friar 1979); μH Lamb shift Friar moment leading-order (Sprint MH Track A; 4.3% at leading); future framework-native operator-level Foldy module (sibling of `geovac/magnetization_density.py`, not yet implemented).
Empirical anchor: proton charge radius $r_p = 0.84075(64)$ fm (CODATA 2022); deuteron charge radius $r_d = 2.1415(45)$ fm (Pohl et al. 2017). Hydrogen $2S$ Lamb shift contribution from r_p at the $\sim +0.138$ MHz level (corrected 2026-08-28 from +1.18, which is the 1S value; Sprint LS-7 reframing; Eides Tab. 7.3).
Honest scope. The framework currently consumes r_p, r_d as external scalars (`R_PROTON_BOHR` and analogs) at the point-source level. Operator-level construction of $V_{\rm ne} \to V_{\rm ne} \star \rho_E$ is the natural next step but has not been implemented; the catalogue rows that depend on this projection treat r_p as a Layer-2 input.
Structural reading. Categorically distinct from § 3.18: the Hamiltonian operator modified is $V_{\rm ne}$ (Coulomb), not $A_{\rm hf}^{\rm contact}$ (Fermi contact). Empirical separation: hydrogen Lamb shift depends on r_p but only trivially on r_Z, while 21 cm hyperfine depends on r_Z but only trivially on r_p.

3.18 Nuclear magnetization-density correction (Zemach)

Source: bare Fermi-contact hyperfine operator $A_{\rm hf}^{\rm contact} = (8\pi/3)\,\mu_e\cdot\mu_N\,\delta^3(\vec{r})$ on the Fock-projected S^3 graph (point-source nuclear magnetic moment).
Target: extended-magnetization Fermi-contact operator $A_{\rm hf}^{\rm contact}(1 - 2Z\alpha m_e r_Z + O(r_Z^2))$, where $r_Z = \int d^3r\, d^3r'\, \rho_E^{\rm nuc}(\vec{r}')\, \rho_M^{\rm nuc}(|\vec{r}-\vec{r}'|)\,|\vec{r}|$ is the Zemach radius defined as the convolution moment of the nuclear charge and magnetization profiles.
Variables introduced: r_Z per nuclear species (sub-features: nuclear magnetic radius r_M and Friar moment $\langle r^3\rangle_{(2)}$ enter as profile-shape parameters).
Dimension introduced: length (a new $[L]$ scale, structurally distinct from § 3.17).
Transcendental signature: ring-preserving over $\mathbb{Q}(\alpha)$ at leading order; profile-dependence at sub-leading order admits Gaussian, exponential, and dipole forms with structurally identical leading $-2Z\alpha m_e r_Z$ behaviour (profile independence at leading order verified in Sprint HF Track 4 / Sprint MH Track C).
Used in: hydrogen 21 cm hyperfine HF-4 row (Sprint HF; $r_Z = 1.045$ fm yields -39.5 ppm framework-native shift); μH 1S Zemach mass-enhancement (Sprint MH Track B/C; factor 185.94×

ep enhancement at leading order); deuteron 1S hyperfine (Sprint precision-catalogue; $r_Z(D) = 2.593$ fm yields -98.0 ppm framework-native shift).

Empirical anchors: proton Zemach radius $r_Z = 1.045(20)$ fm (Eides–Grotch–Shelyuto 2007 atomic compilation) or $r_Z = 1.013(10)(12)$ fm (lattice QCD 2024); deuteron Zemach radius $r_Z(D) = 2.593(16)$ fm (Friar–Payne 2005).

Honest scope. Operator-level construction is mature: `geovac/magnetization_density.py` ($\sim$ 600 lines, 57 tests; Sprint MH Track C + W1b recoil-mixing extension, May 2026) implements the convolved Fermi-contact operator with the `lepton_mass` parameter eliminating manual mass-scaling for muonic systems and an opt-in `include_recoil_mixing` flag extending the kernel to next-to-leading order with the recoil-mixing prefactor $m_l/(m_l + m_n)$ (per arXiv:2604.06930 eq. 95) and the Friar moment correction $(1/2)(Z\alpha m_{\rm red})^2\langle r^2\rangle_{(2)}$ (per Friar 1979 / Eides §6.2). At leading order the kernel matches Eides analytical to floating-point precision; with recoil-mixing on, the framework muH 1S HFS prediction matches Krauth 2017's full-theory Zemach line to $\sim 1\%$ (vs $\sim 3\%$ overshoot at LO-only). r_Z values themselves are Layer-2 inputs (cannot be predicted from framework structure; consumed from QCD lattice or atomic spectroscopy). The W1b recoil-mixing extension sprint (May 2026, see `debug/w1b_recoil_mixing_extension_memo.md`) showed that this kernel improvement does NOT close the ~ 0.22 fm muH-alone vs Eides offset diagnosed by Sprint Calc-rZG-extended; that offset is now attributable to Layer-2 itemization-convention mismatch between Krauth 2017, Eides–Grotch–Shelyuto 2007 [48], and Antognini-Krauth-Pohl 2015, not to kernel approximation. Operator-level verified at I=1 in §5.2.7 (D 1S HFS autopsy, May 9 2026): substituting $r_Z(D) = 2.593$ fm reproduces the Eides analytic LO scalar at $\sim 1.45 \times 10^{-14}\%$ of the LO shift (machine precision); profile (Gaussian vs exponential) independence preserved at I=1; the operator does not depend on nuclear spin (it operates on the spatial register only, with r_Z as a scalar moment). Pauli encoding remains 4 terms (II, Z_e, Z_p, Z_eZ_p) at I=1, identical minimal sparse encoding as I=1/2.

Structural reading. Sibling of § 3.17 (different Hamiltonian operator) and of § 3.16 (Breit retardation: both take the framework's single-particle output and add a structural correction the bare spectral action does not autonomously generate). Categorical distinction from charge-density is empirical: 21 cm hyperfine depends on r_Z but trivially on r_p, electronic Lamb shift depends on r_p but trivially on r_Z.

3.19 Nuclear tensor multipole coupling

Source: multipole expansion of the nuclear charge potential $V_{\rm ne}(\vec{r}) = -Ze^2/|\vec{r}|$, truncated at the monopole on the bare Fock-projected S^3 graph (spherical-symmetric nucleus).

Target: same potential with $L \geq 2$ multipole corrections retained. At leading rank-2: quadrupole coupling $H_Q = -e\, Q_N\, T^{(2)}_{ij}\, (\partial_i E_j)/6$, with Q_N the nuclear electric quadrupole moment and $T^{(2)}_{ij}$ a rank-2 spin tensor on the nuclear-spin Hilbert space. Higher ranks (octupole, hexadecapole) follow the same pattern at higher angular momentum.

Variables introduced: Q_N per nuclear species (rank-2 leading); higher-rank moments enter at higher L.

Dimension introduced: length-squared (the rank-2 quadrupole has dimension $[L]^2$; rank-L multipoles have dimension $[L]^L$). This is the only projection in § 3 whose dimensional injection is not strictly $[L]^1$ or $[E]$; the squared-length structure reflects the tensor character of the operator.

Transcendental signature: ring-preserving over $\mathbb{Q}$ at the angular level (multipole expansion uses Wigner $3j$ at every step). Sub-leading corrections involve the same algebraic ring as § 3.17.

Used in: deuteron 1S hyperfine s-state contribution (sub-ppm; *not* load-bearing in current catalogue); future molecular HD / D_2 rotational hyperfine where electronic gradients at the nucleus are

nonzero and Q_d couples at the percent level (Komasa–Pachucki 2020 sub-percent precision regime); future heavy-nucleus hyperfine spectra where $L \geq 1$ valence orbitals couple to Q_N at leading order.
Empirical anchors: deuteron quadrupole moment $Q_d = 0.285699(15)(18)$ fm^2 (Komasa–Pachucki 2020); ^{17}O and ^{14}N quadrupole moments for molecular spectroscopy benchmarks; sub-percent precision spectroscopy in HD / D_2 rotational HFS as the load-bearing test.
Honest scope. Forward-looking projection slot. No catalogue row currently uses this projection at leading order (deuteron 1S HFS s-state sees only sub-ppm contribution; the load-bearing tests are in molecular rather than atomic spectroscopy). Operator-level construction has not been implemented; this projection occupies a structural slot, with the empirical anchor in molecular HD / D_2 HFS not currently catalogued.
Operator-level test status. Track 4 of the multi-track Roothaan autopsy sprint (2026-05-18) produced an order-of-magnitude scoping autopsy on HD molecule $J = 1$ deuteron quadrupole coupling $\chi_d \approx 224.54$ kHz (Code-Ramsey 1971; Komasa-Pachucki 2020 ab initio 224.51(5) kHz). Using only quantities the framework currently exposes analytically (multipole expansion of the partner nucleus's bare Coulomb potential at $R_{\text{eq}} = 1.4011$ bohr, plus a free-atom 1s electron screening estimate, plus the $J = 1, m_J = 0$ rigid-rotor Wigner-3j angular reduction $\langle V_{zz} \rangle = -(2/5)\, V_{zz}^{\text{mol}}$), the framework-side OoM estimate gives $\chi_d^{\text{OoM}} = 135$ kHz vs experiment 224.54 kHz, residual -89.5 kHz $= -39.9\%$. The bare-nuclear-only contribution alone reaches 195 kHz, 87% of experiment; the 40% gap is the absence of molecular bond-charge redistribution from the free-atom picture. The architecture is confirmed structurally consistent (the framework's existing multipole machinery captures the dominant LO contribution), but sub-percent reproduction requires a position-space electronic-density operator at the deuteron coordinate. This is the named follow-on sprint `HD-EFG-position-space` (beyond sprint scale; analog of the W1a-D Roothaan recoil and §3.18 operator-level extensions). Source: `debug/HD_rotational_HFS_scoping_track4_memo.md`.
Structural reading. Categorically distinct from § 3.17 and § 3.18: tensor coupling to a gradient is structurally different from scalar coupling to a contact density. Different selection rules: Q_N requires $l \geq 1$ on the electronic side (does not contribute to s-states at leading order), while charge-density and magnetization-density Foldy / Friar / Zemach contributions are pure $l = 0$ contact corrections. The Wigner–Eckart theorem ensures the tensor and scalar projections are mutually orthogonal: they cannot be unified by a single profile-broadening prescription.

3.20 Phillips–Kleinman / core–valence orthogonality

Source: a valence-only Hamiltonian on the Fock-projected S^3 graph in which the core orbitals $\{\phi_c\}$ of a heavy atom are *frozen out* of the active Hilbert space, leaving the valence electrons without any structural memory of the Pauli-exclusion constraint they must satisfy with respect to the core.
Target: same valence-only Hamiltonian with the Phillips–Kleinman projection added:

$$H_{\text{val}} \;\longrightarrow\; H_{\text{val}} \;+\; \sum_{c \in \text{core}} (E_v - E_c)\, |\phi_c\rangle\langle\phi_c|\,,$$

or equivalently at the matrix-element level $\Delta H_{pq}^{\text{PK}} = \sum_c (E_v - E_c)\, S_{pc}\, S_{cq}$ for valence indices p, q and cross-overlap $S_{pc} = \langle \phi_p \,|\, \phi_c \rangle$, with $\{\phi_c, E_c\}$ the (external) frozen-core orbitals and binding energies and E_v a valence reference energy (typically $E_v = 0$ for the standard absolute-PK barrier, which is purely repulsive whenever $E_c < 0$).
Variables introduced: the indexed set $\{\phi_c, E_c\}_{c \in \text{core}}$ of core orbital wavefunctions and binding energies. Structurally one bookkeeping integer (the core orbital count) plus an external Layer-2 input for each E_c and each radial profile ϕ_c. These are not generated by the framework's bare graph or projection chain; they are consumed from Clementi–Roetti HF tabulations or from the

framework's own core-shell solution (Paper 17 § IV's `CoreScreening` object, which solves the same-center core-electron problem independently and hands its eigenvectors to the valence solver).
Dimension introduced: none. Orthogonality is dimensionless; the projector $\sum_c |\phi_c\rangle\langle\phi_c|$ takes the same dimension as its argument. The valence reference E_v and the core eigenvalues E_c carry energy, but they were already part of the spectrum-of-states ledger before the projection was applied. In this sense Phillips–Kleinman is in the same dimensionless class as Drake–Swainson asymptotic subtraction (§3.13) — both are bookkeeping projections that re-organize an already- dimensioned quantity rather than introduce a new physical scale.
Transcendental signature: ring-preserving. No transcendental is injected beyond what the source spectrum already carries. In the finite hydrogenic / Slater basis used by the composed and balanced architectures, the cross-overlaps S_{pc} are evaluated in closed form (radial integrals via the hypergeometric Slater machinery `geovac/hypergeometric_slater.py`) and live in $\mathbb{Q}$ at the angular level (Wigner $3j$, §3.8) and in the same algebraic-extension ring as the underlying orbitals at the radial level. The PK weights $(E_v - E_c)$ are external Layer-2 numerical inputs whose transcendental content is whatever the source compilation delivers. No new π, no new ζ value, no new Catalan-class transcendental enters from the projection itself.
Used in: the same-center PK barrier in the composed architecture (Paper 17 § IV, module `geovac/ab_initio_pk.py`) for LiH and BeH^+ at $l_{\max} \leq 2$; the l-dependent variant (Paper 17 § IV, channel-blind $\to l = 0$ projected) which reduces LiH R_{eq} error from 6.9% to 5.3%; the cross-center PK barrier (Paper 19 § 6, module `geovac/phillips_kleinman_cross_center.py`) for second-row chemistry where the heavy-atom core is frozen out via a `FrozenCore` potential rather than carried as an explicit electron block.
Empirical anchor: LiH balanced-coupled R_{eq} at 5.3% error with the channel-blind l-dependent PK at $l_{\max} = 2$ (Paper 17 § IV, Track CD baseline; the PK contribution is necessary — removing it produces an unbound PES, see Track CB record in CLAUDE.md § 3); core-valence orthogonality satisfied numerically in the explicit-core architectures (LiH balanced-coupled at $n_{\max} = 2$ has the Li-core block diagonalized simultaneously with the LiH-bond block; Pauli is enforced by Slater-determinant antisymmetry in the FCI sector and the PK projector is redundant for the explicit-core slice but load-bearing for the frozen-core slice).
Honest scope. Three layers of scope-honesty are required here, all flagged for CLAUDE.md § 1.5 rhetoric compliance.

First, in the composed / balanced architecture as implemented, the PK projection is *essential* — it is not redundant in second quantization. The Track CB sprint (CLAUDE.md § 3 record, April 2026) tested whether explicit cross-block ERIs could substitute for the PK barrier by carrying core-valence repulsion explicitly; the result was that adding cross-block V_{ee} without simultaneously balancing cross-center V_{ne} produces a 29% FCI energy error at LiH $n_{\max} = 2$ (worst of three tested configurations, CLAUDE.md § 3 record). The composed orbital basis uses different effective charges Z on different blocks, and PK is the infrastructure that handles the corresponding orthogonality without requiring two-center one-body integrals in a single-center hydrogenic basis. PK is *not* a calibration choice; it is the projection that makes the composed architecture's Hilbert-space factorization consistent with Pauli exclusion.

Second, the same-center PK in Paper 17 § IV [6] closes the core-valence-orthogonality problem cleanly at the $l_{\max} = 2$ operating point. Six prior PK modification attempts (channel-blind, projector, spectral, self-consistent, R-dependent, l-dependent on the 2D solver; see the six-row PK modifications digest in the CLAUDE.md § 3 failed-approaches table) all carry the same lesson: PK provides coordinate-space exclusion that angular projectors cannot replicate. The $l_{\max}$ divergence (Paper 17 § V and outlook section) at higher $l_{\max}$ is a structural feature of the composed architecture's interaction between PK and angular basis expansion, not a deficiency of the PK projection itself; the 2D variational solver reproduces the same drift, confirming that PK is not the bottleneck (Paper 17 Track BQ record).

Third, the cross-center PK barrier (Paper 19 § 6, May 2026) extends the same projection to the heavy-atom-frozen-core regime ($Z \geq 11$, [Ne] / [Ar] / [Kr] / [Xe] cores) and was tested as the named engineering closure for the W1c-residual orthogonality wall identified in the Phase C chemistry sprint. The honest verdict was NEGATIVE on binding outcome but POSITIVE on architectural completeness: PK cross-center reduces the W1c-residual NaH overattraction by 14.6% (0.357 Ha $\rightarrow$ 0.305 Ha at standard absolute-PK $E_v = 0$) but does *not* close the wall. The NaH PES remains monotonically descending with no internal equilibrium minimum. The remaining residual ($\sim$ 290 mHa, 4$\times$ experimental $D_e \approx 75$ mHa) lives in mechanisms beyond W1c + PK orthogonality: the hydrogenic $Z_{\text{orb}} = 1$ basis on the Na valence side is qualitatively wrong (it is not the actual Na 3s orbital but a generic $Z_{\text{eff}} = 1$ hydrogenic function), the cross-block ERI between H 1s (proper) and Na "valence" (wrong shape) inherits this error, and a strong-PK probe at unphysical $E_v = 100$ Ha shows a minimum-depth signature (0.148 Ha descent) that represents parameter tuning rather than derivation. The CLAUDE.md § 3 failed-approaches table carries the explicit row "Phillips-Kleinman cross-center barrier for second-row chemistry binding (post-Track-2 sprint, 2026-05-08)" documenting this honest negative. The PK projection remains a clean architectural addition; the wall is structurally deeper than PK orthogonality alone.

Structural reading. Phillips–Kleinman sits in the Layer-2 $\rightarrow$ Layer-2 input class, structurally a sibling of the three nuclear-structure projections (§3.17, §3.18, §3.19): each takes the framework's bare single-particle output and adds a structural correction whose load-bearing data is external to the framework's graph and projection chain. For the nuclear-structure trio the external data is a QCD-internal scalar ($\langle r^2 \rangle_E$, r_Z, Q_N); for Phillips–Kleinman the external data is the set of frozen-core orbitals $\{\phi_c, E_c\}$ derivable either from an independent atomic Hartree–Fock solve (Clementi–Roetti) or from the framework's own core-shell Level 3 solution. In both cases the projection mechanism is structurally well-defined and ring-preserving; in both cases the input scalars / orbitals are calibration content the framework does not autonomously derive.

Two further structural distinctions deserve naming. First, Phillips–Kleinman differs categorically from the three nuclear-structure projections in its targeted Hilbert-space sector: the nuclear-structure projections modify the electron-nucleus or hyperfine operators by convolving with a nuclear density profile, while Phillips–Kleinman modifies the kinetic-energy *plus* potential-energy effective single-particle Hamiltonian on the valence electrons via an orthogonality projector. Different mechanism, same input class.

Second, the same-center and cross-center variants of PK are structurally one projection in two architectures. In the same-center (explicit core block) architecture, the core orbitals live inside the FCI sector and Pauli is enforced by Slater-determinant antisymmetry; the PK barrier in Paper 17 § IV is the residual content that handles the kinetic-energy cost of orthogonalizing the angular component of the valence basis against the radial core. In the cross-center (frozen-core) architecture, the core orbitals are external to the FCI Hilbert space (they live in a one-body local $Z_{\text{eff}}(r)$ potential), and the PK barrier (Paper 19 § 6) is the non-local one-body operator that re-injects core-valence orthogonality the frozen-core approximation removed. Same operator-form projection, two architectures, two load-bearing roles, the same load-bearing limitation: PK gets the projection structure right but cannot independently fix a wrong valence basis.

3.21 Multipole expansion / Gaunt termination

Source: bare two-body Coulomb kernel $1/|\vec{r}_1 - \vec{r}_2|$ (for V_{ee}) or $1/|\vec{r} - \vec{R}_B|$ (for cross-center V_{ne}) acting on a product of angular-momentum eigenstates on the Fock-projected S^3 graph.

Target: the same kernel re-expressed as a finite sum over multipoles

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} \;=\; \sum_{L=0}^{L_{\max}} \frac{r_<^L}{r_>^{L+1}}\, P_L(\cos\theta_{12}),$$

with the angular content $P_L(\cos\theta_{12})$ contracted against spherical-harmonic vertices via Gaunt integrals $G(l_1, L, l_2) = 2\begin{pmatrix} l_1 & L & l_2 \\ 0 & 0 & 0 \end{pmatrix}^2$, i.e. Wigner-$3j$ angular coupling (§ 3.8) at each vertex.

Variables introduced: the multipole order L (integer, runs $0, 1, \ldots, L_{\max} = 2l_{\max}$). No physical parameter is introduced; L is an internal summation index whose upper bound is fixed by the angular content of the basis.

Dimension introduced: dimension-preserving. Each multipole term $r_<^L/r_>^{L+1}$ carries the same $[E]$ scaling as the parent Coulomb matrix element; the decomposition reorganizes the angular content of an existing operator without injecting a new $[L]$, $[E]$, $[M]$, or $[T]$ scale.

Transcendental signature: ring-preserving over the Layer 1 algebraic ring. The Gaunt $3j$ coefficients live in $\mathbb{Q}[\sqrt{2k{+}1}]_k$ (the Wigner-$3j$ ring of § 3.8); the radial R^L matrix elements are algebraic in the hydrogenic / Z_{eff}-rational basis (Slater-type integrals evaluate to rationals at integer screening and to incomplete-gamma combinations at fractional screening, per § 3.8 sibling derivations and Paper 19 § III.C). No transcendental is introduced at any step *when both densities sit on the same centre.*

Two-centre scope (added 2026-08-13). The ring-preserving statement above is a one-centre statement, and it is worth saying exactly where it stops. When the two densities sit on *different* centres the same projection is carried out in prolate spheroidal coordinates with the Neumann ordered-$\xi_</\xi_>$ split (§ 3.5; Paper 12, Paper 58), and there the radial factor is no longer a Slater integral. A native closed-form evaluation of all four two-centre classes shows what enters, and it is a short list: the Level-2 Stieltjes seed $e^a E_1(a)$ (Paper 18 § Level 2), the logarithm — inherited from the Neumann kernel $Q_0(\xi) = \frac{1}{2}\ln\big[(\xi{+}1)/(\xi{-}1)\big]$, so it is the *same* embedding object seen from the kernel side rather than the moment side — and the Euler–Mascheroni constant γ_E, which arrives inseparably with E_1 through $E_1(z) = -\gamma_E - \ln z + \mathrm{Ein}(z)$. All three sit in the **embedding** tier of Paper 18's taxonomy, the same tier as the $1/r_{12}$ cusp content: they are what the continuum two-centre geometry costs, not what the graph contains. The seed set grows with class difficulty — none for $(AA|BB)$, $\{E_1, \ln\}$ for the hybrid classes, $\{E_1, \ln, \gamma_E\}$ for exchange — but the *weight* does not: every class closes at transcendence weight one.

Two negatives carry more structural content than the list itself. First, **no π survives**: it enters through the spherical normalisations and cancels identically against the 4π of the Coulomb kernel, so the calibration tier is untouched and the master-Mellin sub-mechanisms $M_1/M_2/M_3$ (§ 3.2, § 3.6) are simply not in this chain. Second, **no weight-two object appears** — no Li_2, no $\zeta(2)$ — even though the ordered simplex integral is formally a period of exactly the type that produces dilogarithms. The two together are what make the $(AA|BB)$ entries *decidable*: the closed form reduces to $A_0(R) + \sum_j A_j(R)e^{-\lambda_j R}$ with A_j rational and λ_j rational, so Lindemann–Weierstrass decides vanishing exactly rather than by threshold (Paper 58, decided census: 195/195 permitted entries nonzero, zero accidental zeros). Had π or a dilogarithm survived, that decision procedure would not exist. The chain is Fock conformal (§ 3.1) $\to$ Sturmian (§ 3.5) $\to$ the present multipole projection evaluated across two centres; three-axis tag $(L, \text{dimension-preserving}, \{E_1, \ln, \gamma_E\}$ at weight one).

Three-centre continuation (added 2026-08-16). Carried to a *third* centre—the two-body three-centre integral $(XY|XZ)$, two two-centre densities sharing one centre on different axes—the same Fock $\to$ Sturmian $\to$ multipole chain, evaluated in momentum space, leaves the two densities on

Fock spheres of *different* radii, and the two-centre negatives do not survive. The radial kernel lives on the elliptic curve $y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1)$, whose $D = 0$ period is a complete elliptic integral K (modulus $m = 1 - c_{\min}/c_{\max}$ over the Feynman family, degenerate only at $c_1 = c_2$). The transcendental axis is thus genus-one elliptic—an elliptic polylogarithm, a new ring within the embedding tier—so the embedding tier is genus-graded: genus zero at two centres, genus one at three, the third centre leaving the polylogarithm/MZV tower rather than climbing it. Three-axis tag (L, dimension-preserving, {elliptic K / elliptic polylog} at genus one); the elliptic period is verified numerically to 31 digits, the closed form not derived (Paper 18 § Level 2, three-centre genus grading).

Used in: essentially every multi-electron matrix element in the codebase. Specific instances: Level 4 mol-frame hyperspherical V_{ee} and V_{ne} coupling (Paper 15 § V, split-region Legendre expansion); cross-center V_{ne} in the balanced coupled architecture (Paper 19 § III.C; `geovac/shibuya_wulfman.py`); single-center Slater F^k and G^k integrals at every level (Paper 7 § VI.B, Paper 12); the Roothaan cross-register V_{eN} kernel at multi-focal $\lambda_a \neq \lambda_b$ (`geovac/cross_register_vne.py`); the multipole structure underlying § 3.19 (nuclear rank-L couplings).

Empirical anchor: the multipole expansion underwrites the angular sparsity of the Pauli-string Hamiltonian across the catalogue (Paper 22 angular sparsity theorem; ERI density 1.44% at $l_{\max} = 3$ depends on the exact $L \leq l_1 + l_2$ truncation). Direct numerical verification in Sprint CD (April 2026, Paper 19 § III.C, Track CD v2.0.39): cross-center V_{ne} for LiH at $n_{\max} = 2$ contributes exactly 33 nonzero one-body matrix elements (multipole sum runs $L = 0, 1, 2$, terminates at $L_{\max} = 2$); at $n_{\max} = 3$, 168 nonzero elements ($L_{\max} = 4$). Machine-precision evaluation via the regularized incomplete gamma function (`scipy.special.gammainc` / `gammaincc`) replaces grid-based trapezoid quadrature with zero truncation error at the angular level.

Honest scope. The termination $L \leq L_{\max} = 2l_{\max}$ is *exact*, not asymptotic. It is forced by the Wigner-$3j$ triangle inequality $|l_1 - l_2| \leq L \leq l_1 + l_2$ applied to the angular Gaunt integral at each vertex (§ 3.8): for $L > l_1 + l_2$ the $3j$ symbol vanishes identically, so the multipole sum has no nonzero contribution past $L_{\max}$. There is no truncation parameter to tune, no UV cutoff to remove, no convergence series to evaluate. This is categorically different from the slowly convergent Shibuya–Wulfman *basis* expansion (Paper 19 § III.B), which expands a complete orbital basis on one center in the Sturmian basis of another center and does *not* terminate; the multipole expansion of the $1/|\vec{r}_1 - \vec{r}_2|$ *kernel* terminates trivially while the Shibuya–Wulfman expansion of the *basis* does not. Computational note: where the radial integrand at fractional screening is non-polynomial (the smooth $[Z_A - Z_{\text{eff}}(r)]/r$ correction in composed geometries), the algebraic expansion does not apply and Gauss–Legendre quadrature is used for the radial integral only; the angular content remains exactly truncated regardless (Paper 15 § V, closing paragraph).

Structural reading. Layer 1 $\to$ Layer 1. This projection reorganizes the angular content of the Coulomb kernel *prior to* any focal-length, mass, or observation projection; it does not move the chain across the Layer-1/Layer-2 boundary of § 3.29. Categorically, it is the coordinate-side counterpart of § 3.8: the multipole expansion of $1/|\vec{r}_1 - \vec{r}_2|$ produces the angular vertices at which Wigner-$3j$ coupling is then applied, so the two projections are inverse images of each other across the kernel-versus-vertex divide ($P_L(\cos\theta_{12}) \leftrightarrow G(l_1, L, l_2)$). Operationally they almost always compose: in the framework's matrix-element pipeline the multipole expansion is the kernel-side step that produces the $3j$-symbol calls at each vertex. Structurally distinct from § 3.19 (nuclear tensor multipole coupling), which uses the same Wigner-$3j$ angular machinery but applies it to nuclear-structure tensor moments (one new $[L]^L$ length scale per rank); the present projection introduces no physical scale at all and sits cleanly inside Layer 1. This is the workhorse projection that underwrites the entire angular-sparsity story of Paper 22 and the exactly Q-linear composed Pauli count ($N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis) of Paper 14: every claim about ERI density, selection-rule survival, or composed-architecture sparsity ultimately rests on the exact $L \leq 2l_{\max}$ termination of this projection.

3.22 Bipolar harmonic / Drake combining

Source: pair of single-electron angular tensor operators acting on distinct electrons of a multi-electron state on the Fock-projected S^3 graph, in the LS coupling scheme.
Target: a bipolar tensor harmonic $\{Y^{(k_1)}(\hat{r}_1) \otimes Y^{(k_2)}(\hat{r}_2)\}^{(K)}_M$ coupling the two angular factors into a single rank-K object on the joint two-electron angular space, with K-dependent matrix elements in the coupled $|LSJ\rangle$ basis given by $(-1)^{L+S+J}\, 6j\{L\,S\,J;\ S\,L\,k\}$ (rank-k Racah J-pattern).
Variables introduced: the bipolar coupling indices (k_1, k_2) and the total angular rank K (integer triple with triangle constraint $|k_1 - k_2| \leq K \leq k_1 + k_2$). These are internal-graph-derived combinatorial labels in the sense of § 3.29; they introduce no new dimensioned scale and no continuous parameter, only an enumeration of Gaunt-allowed angular channels.
Dimension introduced: preserved (the projection is a purely angular reorganization on the multi-electron Hilbert space; no $[L]$, $[E]$, or $[T]$ scale enters).
Transcendental signature: pure rational. The bipolar harmonic expansion of a two-electron tensor operator (Breit retardation, spin-spin, spin-other-orbit, orbit-orbit) decomposes on the angular side into Wigner $3j$ Gaunt coefficients (one per single-electron factor) times rank-k $6j$ symbols (for the J-pattern in $|LSJ\rangle$) times rank-(k_1, k_2, K) $9j$ symbols (for the bipolar recoupling to the coupled two-electron space). Each ingredient is exactly rational over $\mathbb{Q}[\sqrt{2k+1}]_k$ — the same Wigner ring as § 3.8. The Drake (1971) combining coefficients $(\frac{3}{50}, -\frac{2}{5}, \frac{3}{2}, -1)$ for spin-spin and spin-other-orbit on the helium triplet are pure rationals (Paper 18 intrinsic tier). No transcendental injection at any step of the angular reduction.
Used in: two-body Breit-Pauli spin-spin (rank-K=2 spatial, rank-2 coupled spin tensor $[s_1 \otimes s_2]^{(2)}$) and spin-other-orbit (rank-K=1 spatial, rank-1 sum tensor $\vec{s}_1 + 2\vec{s}_2$) interactions on the relativistic composed qubit encoding (Paper 14 § V); helium-class fine-structure predictions through the He 2^3P Roothaan autopsy (§ 5.2.6); structurally compatible with all two-body retardation kernels in `geovac/breit_integrals.py`; angular machinery underlying Drake's $M^K_{\mathrm{dir}}/M^K_{\mathrm{exch}}$ direct/exchange decomposition of the helium triplet, with Sprint 5 DV characterizing the bipolar-to-Drake basis change for the He $(1s)(2p)\,{}^3P$ configuration as a convention-dependent piecewise recombination of Region-I and Region-II radial pieces.
Empirical anchor: He 2^3P fine structure dominant intervals at NIST precision. Framework-native operator-level decomposition (Sprint multi-track Roothaan autopsy 2026-05-09, § 5.2.6) reproduces $E(^3P_0) - E(^3P_1) = +29{,}612.91$ MHz vs NIST $+29{,}616.951(6)$ MHz at -0.014% and $E(^3P_0) - E(^3P_2) = +31{,}844.07$ MHz vs NIST $+31{,}908.131(6)$ MHz at -0.20%, with the angular content (J-pattern, spin reduced matrix elements, bipolar Gaunt selection) verified *sympy-exact* at machine precision via the rank-k $6j$ and $9j$ identities $(-1)^{L+S+J}\, 6j\{1, 1, J; 1, 1, k\}$ at $k = 1$ (SOO) and $k = 2$ (SS), normalized so $f(J{=}1) = 1$. Cross-reference: Paper 14 § V and the sympy-verification tests `tests/test_breit_integrals.py::test_drake_f_SS_pattern_from_6j` and `test_drake_f_SOO_pattern_from_6j`. At the operator level, the bipolar (k_1, k_2) channels for the He $(1s)(2p)$ configuration reduce uniquely (Sprint 5 DV) to $(k_1{=}0, k_2{=}2)$ for the SS direct path and $(k_1{=}1, k_2{=}1)$ for the SS and SOO exchange paths; the SOO direct path is *Gaunt-forbidden* at the bipolar level (parity selection forces $k_1 = 0$, then the $(0, k_2, 1)$ triangle inequality requires k_2 odd, contradicting the (1)(1) Gaunt parity constraint).
Honest scope. The small interval $E(^3P_1) - E(^3P_2)$ framework-native at $+2{,}231.16$ MHz vs NIST $+2{,}291.176(15)$ MHz shows a -2.62% relative residual. This is *partial-cancellation amplification* of the same $\sim$60 MHz absolute residual that sits inside the LS-8a one-loop QED budget on the

dominant intervals: SO ($-58{,}396$ MHz) + SS ($-9{,}499$ MHz) + SOO ($+70{,}126$ MHz) combine to a net $+2{,}231$ MHz, a structural cancellation factor $(|\mathrm{SO}|+|\mathrm{SS}|+|\mathrm{SOO}|)/|\mathrm{total}| \approx 61.9\times$. The framework's absolute precision is uniform across the three intervals; the small interval's fractional residual is amplified by the cancellation. This is a feature of how the SO + SS + SOO operators combine on the $(1s)(2p)\,^3P$ configuration, not a framework limitation. Higher-order corrections (multi-loop QED at $\alpha^4 \cdot \alpha/\pi$, finite-mass recoil, hyperfine averaging) are external Layer-2 inputs (sibling status: Drake 1971 § IV). The direct/exchange mixing coefficients $(\frac{3}{50}, -\frac{2}{5}, \frac{3}{2}, -1)$ in A_{SS} and A_{SOO} are pure rationals at the angular level but *not closed from* $9j$ *alone* on the He triplet: Sprint 5 DV showed they are convention-dependent identities specific to Drake's piecewise $M^K_{\mathrm{dir,I}} + M^K_{\mathrm{dir,II}}$ basis rather than a direct Racah-algebra output (a fully first-principles derivation remains open; see `debug/dd_drake_derivation.md` and `debug/dv_drake_bipolar_memo.md`). The bipolar harmonic projection *itself* is closed; the open question is the basis change to Drake's specific M^K partitioning.

Structural reading. Sibling of § 3.8 (Wigner $3j$ angular coupling): both are angular-content-only projections living in the pure-rational ring $\mathbb{Q}[\sqrt{2k+1}]_k$ with no dimensional or transcendental injection. The structural distinction is multiplicity: Wigner $3j$ couples a *single* pair of angular nodes (one electron's orbital and one external rank-k multipole), whereas the bipolar harmonic / Drake combining projection couples *two single-electron angular tensors* into a joint rank-K object on the two-electron Hilbert space via the rank-(k_1, k_2, K) $9j$ recoupling. In dictionary terms, § 3.8 is the single-particle special case of this projection at $(k_1, k_2) = (0, k)$ on the two-electron space (one electron inert). The two projections compose: every bipolar reduction decomposes into a product of two single-particle Wigner-$3j$ Gaunt factors at the Slater integral level, together with a $9j$ for the two-electron recoupling and a $6j$ for the J-pattern in $|LSJ\rangle$. *Internal multi-focal angular-only finding*: on configurations like He $(1s)(2p)$ where the two electrons sit at structurally distinct focal lengths ($Z_{\mathrm{eff}}(1s)/Z_{\mathrm{eff}}(2p) = 2$), the bipolar Gaunt selection (k_1, k_2) channels are determined by angular content alone — the integer parity and triangle constraints — and are *independent of the focal-length ratio*. Roothaan multipole termination at $L_{\max} = 2l_{\max}$ (here $K \le 2$ for $l_{\max} = 1$) holds verbatim irrespective of the relative magnitudes of the two electrons' Slater exponents; this is the *internal* analog of the cross-register Roothaan termination identified at the multi-focal-composition wall (§ 3.16 structural reading). The operator-level realization of internal multi-focal architecture is therefore not a new projection: it is bipolar harmonic / Drake combining acting at $\lambda_a \neq \lambda_b$ on two electrons in the same atomic block. Different focal-length ratio, same angular projection, same rational ring.

3.23 Symmetry projection / Young tableau

Source: the unrestricted N-particle tensor product $\mathcal{H}^{\otimes N}$ on the single-particle Fock-projected S^3 graph (or on any single-particle natural-geometry sector; § 3.1, § 3.3, § 3.12). At this stage indistinguishability of identical fermions has not yet been enforced; the Hilbert space contains symmetric, antisymmetric, and mixed-symmetry components without distinction.

Target: the λ-irrep subspace $\mathcal{H}_\lambda \subset \mathcal{H}^{\otimes N}$, obtained via the character-based orthogonal projector

$$P_\lambda \;=\; \frac{d_\lambda}{|S_N|} \sum_{g \in S_N} \chi_\lambda(g)\, \rho(g),$$

where λ is a Young tableau (partition of N), d_λ is the dimension of the corresponding irreducible representation of S_N, $\chi_\lambda(g)$ is its character at the permutation g, and $\rho(g)$ is the natural action of g on $\mathcal{H}^{\otimes N}$ by index permutation. For fully indistinguishable spin-$\frac{1}{2}$ fermions in the unrestricted formulation, $\lambda = [1^N]$ (antisymmetric); in the Slater-determinant / spatial-spin coupled formulation,

λ is the conjugate of the spin Young diagram (Paper 16 convention), so that S_N antisymmetry is implemented *jointly* on spatial and spin indices.

Variables introduced: the irrep tableau λ itself (a discrete combinatorial label; equivalently a partition of N). For $N = 2$: $\lambda \in \{[2], [1,1]\}$ (triplet vs singlet under spatial-spin coupling). For $N = 4$ closed shell: $\lambda = [2,2]$ is the spin-singlet sector (Paper 17 §VIII, Track AJ). For general N with total spin S: $\lambda_1 = N/2 + S$ and $\lambda_2 = N/2 - S$ (Paper 16 convention).

Dimension introduced: *none.* P_λ is an orthogonal projection on the N-particle Hilbert space; it does not introduce a new physical scale, focal length, or coordinate. The projector reduces the linear dimension of the working space from $(\dim \mathcal{H})^N$ to a λ-irrep-counted subspace, but no new $[L]$, $[E]$, or $[T]$ enters the framework.

Transcendental signature: *none.* The character table of S_N is integer-valued (Frobenius character formula); the projector P_λ has rational matrix entries (denominators dividing $|S_N| = N!$, numerators integer); the dimension d_λ is integer (hook-length formula). No π, no ζ, no Hurwitz content enters at this step. The projection is the unique § 3 entry whose entire algebraic content lives in $\mathbb{Z}/N!$, making it the cleanest example of a *combinatorial* Layer 1 $\to$ Layer 1 projection.

Used in: all $N \geq 2$ computations in the framework, at multiple levels of explicitness:

- **Level 3 (He, $N = 2$).** Singlet $[1,1]$ vs triplet $[2]$ spatial-spin coupling is the projection that splits the He 1^1S ground state from the 2^3S first excited state. Spatial-spin coupling for two electrons with total spin $S \in \{0,1\}$ yields $\lambda_{\text{spatial}} = [1,1]$ for $S = 0$ (antisymmetric spatial, symmetric spin: singlet) and $\lambda_{\text{spatial}} = [2]$ for $S = 1$ (symmetric spatial, antisymmetric spin: triplet).

- **Level 4N (LiH, $N = 4$).** The S_4 $[2,2]$ Young diagram projection is the character-based projector onto the closed-shell singlet sector of the 4-electron mol-frame hyperspherical Hilbert space on S^{11} (Paper 17 §VIII; Track AJ, `geovac/n_electron_solver.py`). The projection reduces the raw SO(12) angular channel count by a tableau-dependent factor and makes the 4-electron PES computationally tractable; it is empty at $l_{\max} = 0$ (no antisymmetric 4-electron state can be built from s-waves), gives 2 channels at $l_{\max} = 1$, 12 channels at $l_{\max} = 2$, and 40 channels at $l_{\max} = 3$.

- **Graph-native CI (all N).** The Slater-determinant basis used in `geovac/casimir_ci.py` is itself the *Slater-determinant realization* of the $[1^N]$ antisymmetric projector: each determinant $\det[\phi_{i_1}(\vec{r}_1) \cdots \phi_{i_N}(\vec{r}_N)]$ is the image under $P_{[1^N]}$ of the single Hartree product $\phi_{i_1}(\vec{r}_1) \otimes \cdots \otimes \phi_{i_N}(\vec{r}_N)$, up to a normalization factor.

- **Composed atomic classifier (Paper 16).** The atomic structure-type assignment (A/B/C/D/E) for $Z = 1$–30 is itself read off the S_N irrep shape: structure-type letters correspond to specific tableau patterns under the periodic-table indexing of Paper 16. The classifier is the boundary at which the S_N projection becomes computationally explicit in the production composed-Hamiltonian pipeline.

- **Hyperfine and cross-register couplings (Track NI, Paper 23 §VI).** In the composed nuclear–electronic deuterium Hamiltonian, the Clebsch–Gordan multiplicity $(2I+1)/2$ relating $F = I + J$ to $F = I - J$ hyperfine levels (D HFS autopsy, §5.2.7) is itself a small-N tableau projection applied to a coupled spin Hilbert space. See § 3.8 for the angular-momentum coupling content; the present projection is its S_N analog on the indistinguishable-particle side.

Empirical anchor: He 2^1S–2^3S exchange splitting at 6421.46 cm^{-1} (NIST experimental) is the cleanest finite-energy witness of the singlet-vs-triplet ($[1,1]$ vs $[2]$ spatial) projection in the catalogue. The splitting is not a small Layer-2 correction; it is the energy difference between two distinct λ-irrep sectors of the same single-particle basis, generated entirely by the exchange-integral content of the two-electron V_{ee} matrix elements under spatial antisymmetrization vs symmetrization. Sprint Calc-He-2S-2S-splitting and Sprint Hylleraas-r_{12} (2026-05-09) returned this entry at +11.86% (graph-native CI, $n_{\max} \leq 11$) and +209% (single-α Hylleraas-3p, $\omega = 5$) respectively, with the diagnosis that the residuals trace to single-exponent basis limitations for the excited λ-irrep state rather than to the projection itself (§ 5.1 He singlet-triplet row). At the projection level, the splitting is structurally clean: the projector $P_{[1,1]}$ vs $P_{[2]}$ acts identically on either side; only the V_{ee} matrix elements respond differently to the two sectors.

Honest scope. At modest N ($N \leq 4$) the framework implements the character-based projector P_λ directly: explicit construction of the projector matrix in the Slater / configuration-state-function basis, or via the equivalent Young symmetrizer in the unrestricted tensor product. This is exact and combinatorial. At large N (heavy atoms, $N \gg 10$), the framework falls back on *compositional approximations* to the tableau projection: (i) single-Slater-determinant Hartree–Fock reductions project onto a single representative of the $[1^N]$ irrep rather than spanning it, (ii) frozen-core constructions (`geovac/neon_core.py` for [Ne]-core, [Kr]-core, [Xe]-core species) project the antisymmetric structure onto a product of a fixed core determinant with a smaller-N valence tableau projection, (iii) the composed-block architecture for multi-center systems (§ 3.12) treats each block's antisymmetric subspace independently, neglecting inter-block antisymmetry except through cross-block ERI / cross-center V_{ne} corrections. Each of these is itself a structured projection onto a coarser tableau pattern. The boundary between exact character-projection and compositional approximation is empirical, set by the largest N at which $|S_N| = N!$ direct construction remains tractable. As of v2.40.0 this boundary sits at $N \approx 6$ for the atomic classifier's explicit construction (extended via lookup tables to higher N where the periodic-table structure-type pattern repeats), and at $N = 4$ for the Level 4N solver (`geovac/n_electron_solver.py`).

Structural reading. This is the projection at which **Pauli antisymmetry enters the framework explicitly** rather than implicitly. In many quantum-chemistry frameworks antisymmetrization is folded into the coordinate or operator choice (e.g. second-quantized anticommutators built into the Hamiltonian's algebraic structure); in GeoVac the dimensionless graph is symmetry-agnostic by construction, and Pauli antisymmetry is recovered as a separate, named, character-based projection at the multi-particle level. This is the structural sibling of § 3.7 (which introduces the spin-$\frac{1}{2}$ Hilbert space at the single-particle level): the spinor projection commits to particle statistics (fermionic half-integer spin), while the present projection commits to the many-body consequence of those statistics (N-particle antisymmetry). Composition $P_\lambda \circ \mathrm{spinor\_lift}$ realizes the standard spatial-spin coupling scheme; composition $P_\lambda \circ \mathrm{Fock}$ realizes the N-fermion graph Hilbert space on which graph-native CI operates. Categorically distinct from § 3.8 (angular-momentum coupling, $SU(2)$ Clebsch–Gordan, continuous-symmetry character content) and § 3.9 (rotation between rotated coordinate frames, $SO(3)$ representation theory). The shared abstraction across these three is that all three are character-based projections onto irreducible representations of finite or compact groups acting on the Hilbert space; the structural distinction is the group: S_N for the present projection, $SU(2)$ for § 3.8, $SO(3)$ for § 3.9. The S_N character table has integer entries while the $SU(2)$ and $SO(3)$ projector matrix elements introduce algebraic-extension and trigonometric content respectively (Table 1); the absence of transcendental content in the symmetry projection is therefore not coincidental but reflects the discrete finite-group structure underlying Pauli exclusion.

3.24 Adiabatic / Born–Oppenheimer projection

Source: a single combined Hamiltonian H acting on two coupled degrees of freedom whose dynamical time-scales differ by a parametrically large ratio (the small parameter $m_e/M_n \ll 1$ for electron–nucleus systems, m_e/M_{cluster} for inner electrons versus an outer cluster, etc.). In the foundational setting this is the molecular Hamiltonian $H = T_n(\vec{R}) + T_e(\vec{r}) + V_{nn}(\vec{R}) + V_{ne}(\vec{r}, \vec{R}) + V_{ee}(\vec{r})$ on a tensor product of nuclear and electronic Hilbert spaces; in the Level 3 hyperspherical setting it is the combined kinetic and potential operator on the joint (R, Ω) space of hyperradius and hyperangle.
Target: a *two-stage* decomposition in which the fast degree of freedom (electronic coordinates $\vec{r}$, hyperangle Ω, valence electrons) is solved at every fixed value of the slow degree of freedom (nuclear coordinate $\vec{R}$, hyperradius R, core configuration), producing a parameter family of fast-sector eigenproblems

$$H_{\text{fast}}(\vec{R})\,\Phi_\nu(\vec{r};\vec{R}) = E_\nu(\vec{R})\,\Phi_\nu(\vec{r};\vec{R}),$$

parameterized by the slow variable. The slow degree of freedom then sees the fast-sector eigenvalues $E_\nu(\vec{R})$ as an effective potential, producing the slow-sector Schrödinger equation $[T_n + E_\nu(\vec{R})]\chi(\vec{R}) = E\,\chi(\vec{R})$ on the slow-only Hilbert space. The full wavefunction is reconstructed as $\Psi(\vec{r},\vec{R}) \approx \Phi_\nu(\vec{r};\vec{R})\chi(\vec{R})$ (single-channel adiabatic) or as a sum $\sum_\nu \Phi_\nu(\vec{r};\vec{R})\chi_\nu(\vec{R})$ (coupled-channel; see § 3.25, the sibling projection that solves the parameterized fast-sector problem and supplies the coupling between channels).
Variables introduced: the slow continuous parameter $\vec{R}$ itself, which enters the previously fast-sector-only graph computation as a *continuous* index over a family of Hamiltonians. The small parameter controlling the projection's validity is the mass ratio m_e/M_n (and, more generally, the ratio of fast and slow time-scales). Both quantities are content the bare graph does not carry: the bare graph computation is formulated at fixed nuclear configuration with no notion of a parametric "slow" variable, and the mass ratio enters only when the slow-sector kinetic operator $T_n = -\nabla_R^2/(2M_n)$ is restored.[3]
Dimension introduced: preserved at the projection step. The fast-sector effective potential $V_{\text{eff}}(\vec{R}) \equiv E_\nu(\vec{R})$ has the same energy dimension $[E]$ as the original Hamiltonian, and the new slow-sector variable $\vec{R}$ carries length $[L]$ (already present in the projection chain via the stereographic or molecule-frame hyperspherical projections at § 3.4 or § 3.12). This projection does not add a new dimension to the Layer-2 dictionary; it *reorganizes* the existing dimensional content into a slow $\vec{R}$ + fast $\vec{r}$ split.
Transcendental signature: *none at the projection step itself.* The Born–Oppenheimer scale-separation is parameter-only: it indexes a family of Hamiltonians by a continuous variable without injecting π or any other transcendental. Transcendentals appear *downstream* of this projection, in the solution of the parameterized fast-sector eigenproblem (§ 3.25): Level 3 hyperspherical $\mu(R)$ is algebraic over $\mathbb{Q}(\pi, \sqrt{2})$ (CLAUDE.md § 12, Track P1, v2.0.12); Level 4 molecule-frame hyperspherical $\mu(\rho)$ is piecewise-smooth in ρ, not algebraic over any closed ring (Track S, v2.0.13); Level 5 composed-geometry effective potentials inherit the full transcendental content of every prior projection in the chain. The clean separation is: the BO projection introduces the continuous parameter $\vec{R}$ with no transcendental cost; solving the parameterized fast-sector eigenproblem then introduces

[3] In mass-independent (Bohr-natural) atomic units the projection still operates: it is the structural operation of treating the nuclear configuration as a parameter rather than as a dynamical variable, even when the mass ratio is not numerically manifest in the units chosen. See Paper 4 (archived) reduced-mass convention discussion — the adiabatic / Born–Oppenheimer projection is what makes the electronic problem mass-independent (at leading order) by separating the slow nuclear sector from the fast electronic sector; the residual mass-dependence enters at the diagonal Born–Oppenheimer correction (DBOC) and at non-adiabatic couplings $P_{\nu\mu}$, both subleading in m_e/M_n.

transcendentals according to the natural geometry of the fast sector and the operator-order content of $H_{\text{fast}}(\vec{R})$.

Used in: every Level 4 molecular calculation (Paper 15; H_2, HeH^+, every two-electron diatomic); every Level 3 hyperspherical calculation that uses the adiabatic effective potential (Paper 13; He, Li, Be, all atomic species with one slow hyperradius); every composed-geometry molecular calculation (Paper 17; LiH, BeH_2, H_2O, all polyatomics, second-row diatomics); the *double*-adiabatic separation of Paper 15 § VII.C where the core–valence fiber bundle introduces a third time-scale (slowest nuclear / slow valence-hyperradius / fast valence-angular, plus a separate slow core-hyperradius / fast core-angular on the core fiber); all the ab initio molecular spectroscopy results of Paper 13 § IX (rovibrational H_2); all D_e benchmarks reported across Papers 13, 15, 17; every cross-register / cross-block calculation in the multi-focal-composition arc (May 2026) that treats nuclear coordinates as classical parameters in `cross_register_vne.py`. This is the most heavily-used projection in the framework's molecular pipeline; very few molecular results do not pass through it.

Empirical anchor: H_2 at 96.0% D_e (Level 4 mol-frame hyperspherical + Schwartz cusp correction, Paper 15); LiH R_{eq} at 5.3% error (Level 5 composed, Paper 17); HeH^+ at 93.1% D_e (Paper 15 § VI.E); diagonal Born–Oppenheimer correction $\Delta E_{\text{DBOC}} \approx +0.0015$ Ha for He (Paper 13 § VI; explains the apparent variational violation $E = -2.9052$ Ha vs. exact -2.9037 Ha at the single-channel adiabatic level by exactly the magnitude expected from the kinetic energy cost of angular wavefunction variation with R); BeH^+ as a transferability test of the ab initio Phillips–Kleinman + adiabatic composition (Paper 17). Each of these is a quantitative test of the BO projection in combination with the natural-geometry sector that follows it.

Honest scope. The strict-adiabatic single-channel approximation is the leading order in the small parameter m_e/M_n; it breaks down where two electronic surfaces become near-degenerate (conical intersections, autoionization channels, non-adiabatic transitions) and where the kinetic energy of the slow variable is comparable to electronic-level spacings. The framework's response to this breakdown is structural, not perturbative: the 2D variational solver (Paper 13 § XIII, Paper 15 § VI.D) treats the slow and fast variables on equal footing, eliminating the adiabatic approximation entirely. Track AR (CLAUDE.md § 2, v2.0.23) showed that the 4-electron LiH adiabatic solver's apparent D_e violation was an artifact of the single-channel approximation, not of the natural geometry: the 2D variational solver gives $E_{\min} = -7.79$ Ha (variational bound respected, D_e unbound) versus the adiabatic solver's $D_e = +0.49$ Ha (5.3× exact). Track BQ (CLAUDE.md § 2, v2.0.32) further established that the R_{eq} drift +0.15–0.22 bohr/$l_{\max}$ in LiH composed geometry is *not* from the adiabatic approximation: both the adiabatic and 2D variational solvers produce identical drift, isolating the residual to the composed-geometry PK architecture rather than the BO projection itself. The named extension when adiabatic breaks down is therefore the coupled-channel / 2D-variational sibling at § 3.25.

Structural reading. Categorically distinct from § 3.14. The rest-mass projection is a ring-preserving Bohr-like rescaling of a *single* particle's spectrum under $m_e \to m_{\text{red}}$; it modifies the foundational energy scale (Ry) without introducing a new continuous parameter, and the spectrum on the projected manifold is the same up to a rational rescaling factor. The Born–Oppenheimer projection is a *scale-separation between two coupled degrees of freedom*: it takes a single Hilbert space $\mathcal{H}_n \otimes \mathcal{H}_e$ and produces a parameter family $\{\mathcal{H}_e \mid \vec{R} \in \mathcal{M}_{\text{nuc}}\}$ together with a slow-sector Schrödinger equation on the parameter manifold $\mathcal{M}_{\text{nuc}}$. The small parameter m_e/M_n controls the projection's accuracy but is not itself the new variable; the new variable is the slow coordinate $\vec{R}$, which becomes a continuous index on a previously discrete graph computation. The two projections answer different structural questions: rest-mass asks "what happens to the spectrum when the single-particle mass changes," while Born–Oppenheimer asks "how do two coupled degrees of freedom decouple at parametrically separated time-scales." The two are independent and commute (rest-mass rescaling of the foundational m_e and BO-separation of slow nuclear from fast electronic degrees of freedom can be applied in either order); they co-appear in every reduced-mass molecular calculation, where rest-mass rescales the electronic-sector m_e to m_{red} on the fast side, and Born–Oppenheimer separates the slow nuclear from the fast electronic sector across the full molecular Hamiltonian.

Structurally, this projection is the *parent* of the coupled-channel / adiabatic-curve projection at § 3.25 (sibling, immediately downstream): BO produces the parameterized electronic problem $H_{\mathrm{el}}(\vec{R})$ with its parameter family of eigenproblems, and the coupled-channel projection is the next step that solves the parameterized problem and assembles the adiabatic potential curves $U_\nu(\vec{R}) = E_\nu(\vec{R})$ as effective potentials for the slow-sector dynamics, optionally including the non-adiabatic couplings $P_{\nu\mu}(\vec{R}) = \langle \Phi_\nu | \partial/\partial\vec{R} | \Phi_\mu \rangle$ that the strict-adiabatic single-channel approximation drops. The two projections are conceptually inseparable in the molecular pipeline — one does not invoke Born–Oppenheimer in practice without then solving the parameterized problem — but they are taxonomically distinct: BO is the *scale-separation operation* (parameter-only, no transcendental content, Layer-1 $\to$ Layer-2 transition), while coupled-channel / adiabatic-curve is the *solution operation* (introduces the transcendental content specific to the fast-sector geometry, and introduces the Berry-connection-like non-adiabatic coupling content if the single-channel approximation is relaxed).

The double-adiabatic structure of Paper 15 § VII.C is a structural *composition* of this projection with itself at two distinct time-scale ratios. In LiH composed geometry the outer BO separates nuclear (slow) from electronic (fast); within the electronic sector, an inner BO separates core (slow on the electronic time-scale, *very* slow on the molecular time-scale) from valence (fast), producing the core–valence fiber bundle of Paper 17 § II. Iterated composition of the BO projection is the structural mechanism by which the framework reaches polyatomic molecules with multiple distinct dynamical time-scales (Paper 17 LiH/BeH_2/H_2O composed geometries; second- and third-row frozen-core diatomics, Sprint 7 v2.19.4). The projection's own structural content does not change under iteration; only the count of independent parameter manifolds $\{\mathcal{M}_{\mathrm{nuc}}, \mathcal{M}_{\mathrm{core}}, \mathcal{M}_{\mathrm{val}}, \ldots\}$ grows.

3.25 Coupled-channel / adiabatic curve

Source: parameterized electronic Hamiltonian $H_{\mathrm{el}}(R)$ inherited from the Born-Oppenheimer / adiabatic projection (§ 3.24): a family of angular operators on the Fock-projected S^3 graph, indexed by a continuous radial coordinate (ρ at Level 3 for the He hyperspherical lattice, internuclear distance R at Level 4 for the molecule-frame hyperspherical lattice; same continuous coordinate at different geometric levels per § 3.12).
Target: the discrete set of adiabatic potential curves $\{\mu_\nu(R)\}_{\nu=1,2,\ldots}$ obtained by solving the angular eigenvalue problem at each R, together with the non-adiabatic coupling matrix $P_{\mu\nu}(R) = \langle \Phi_\mu(R) | \partial_R | \Phi_\nu(R) \rangle$ that couples adjacent channels when the parameterized angular wavefunctions $\Phi_\nu(R; \hat{\Omega})$ vary with R. At the matrix-pencil level

$$H_{\mathrm{ang}}(R) \;=\; H_0 + R \cdot V^{\mathrm{coupling}},$$

with H_0 carrying the free-SO(6) (or SO(3N) at higher N) Casimir eigenvalues and V^{coupling} encoding the nuclear and electron–electron angular couplings (Paper 13 Eqs. 31–32, §XII). The eigenvalues $\mu_\nu(R)$ are the adiabatic potential curves; the coupled-channel radial solver then propagates each radial amplitude $F_\nu(R)$ in the matrix-valued effective potential built from $\{\mu_\nu(R), P_{\mu\nu}(R), Q_{\mu\nu}(R)\}$.
Variables introduced: the radial coordinate R (or ρ) itself, as the continuous-parameter axis of the channel family. The adiabatic curve index ν is a discrete combinatorial label inherited from the SO(6) (or SO(3N)) representation theory of the parent angular problem and is not a new physical

variable.
Dimension introduced: length (one new $[L]$ scale per geometric level; Level 3 introduces the hyperradius ρ, Level 4 the internuclear distance R). Energy dimension is preserved: $\mu_\nu(R)$ carries the same $[E]$ scaling as the parent angular eigenvalues from which it is built, and the curve family is therefore a one-parameter $[E]$-valued function family on the new $[L]$ axis. This is the same $[L]$ injection logged at § 3.12 for the Level 4 case; at Level 3 the radial coordinate is the hyperradius ρ of the SO(6) hyperspherical lattice (Paper 13 §II), structurally analogous but distinct from internuclear R.
Transcendental signature: algebraic-implicit at Level 3, piecewise-smooth at Level 4 — and the distinction is structural, not numerical. At Level 3 (He hyperspherical, single-center, two-electron) the angular Hamiltonian $H_{\text{ang}}(R)$ is a *linear matrix pencil* $H_0 + R \cdot V^{\text{coupling}}$, so its eigenvalues satisfy the global characteristic polynomial

$$P(R, \mu) \;=\; \det\!\big(H_0 + R \cdot V^{\text{coupling}} - \mu I\big) \;=\; 0$$

with coefficients in $\mathbb{Q}(\pi, \sqrt{2})[R, \mu]$ (Paper 13 §XII, Track P1). The π content is inherited from the upstream calibration of § 3.1 (normalisation on $[0, \pi/2]$); the $\sqrt{2}$ content is inherited from the V_{ee} Gaunt integrals (§ 3.8); the radial parameter R contributes no new transcendental content of its own. The function $\mu_\nu(R)$ is therefore *algebraic over* $\mathbb{Q}(\pi, \sqrt{2})$ (an algebraic-implicit function in the Paper 18 §IV.4 sense), of degree $l_{\max}+1$ in both variables at angular truncation $l_{\max}$. Branch points of $P(R, \mu) = 0$ occur at zeros of the discriminant $\Delta(R) = \text{disc}_\mu(P)$, where eigenvalue sheets meet with square-root singularities — pointwise numerical diagonalisation at each R is computational convenience, not mathematical necessity (Paper 13 §XII.B, Track P1, v2.0.12).

At Level 4 (mol-frame hyperspherical, two-center, two-electron) the matrix-pencil form survives but the entries of V^{coupling} themselves depend on ρ piecewise via split-region Legendre factors $\min(s, \rho)/\max(s, \rho)^{k+1}$ from the nuclear charge function expansion (Paper 15 §V). These factors are C^0 but not C^1 at the region boundaries $s = \rho$, equivalently $\alpha = \arccos \rho$ and $\alpha = \arcsin \rho$. The pencil is therefore *piecewise-linear in* ρ rather than linear, and Track S (Paper 15 §VI.7, v2.0.13) established the irreducibility result: no global $P(\rho, \mu) = 0$ exists across the region boundary, and no algebraic-blow-up, Lie-algebra elevation, or $S^3 \times S^3$ hybrid restores one. The function $\mu_\nu(\rho)$ is *piecewise-algebraic within each region* (sub-eigenvalues on each side of $\rho = 1/2$ satisfy local $P_{\text{left}}(\rho, \mu) = 0$ and $P_{\text{right}}(\rho, \mu) = 0$) but is not algebraic globally. This is the *piecewise* tier of Paper 18 Table II (Paper 18 § IV.4), structurally distinct from the algebraic-implicit tier of Level 3.

The two transcendental regimes share the same operator-level shape (linear matrix pencil) but differ in how the R-dependence enters the pencil coefficients. Level 3 admits a single global polynomial because V^{coupling} is R-independent; Level 4 does not because $V^{\text{coupling}}(\rho)$ carries piecewise ρ-dependence inherited from the two-center nuclear charge function.
Used in: Paper 13 (He hyperspherical, $l_{\max} = 2$–7 coupled-channel convergence to 0.022% raw at $l_{\max} = 7$, 0.004% with Schwartz cusp correction at $l_{\max} = 4$ via the 2D variational solver, Track DI v2.6.0); Paper 15 (H_2 Level 4, 96.0% D_e at $l_{\max} = 6$ $\sigma + \pi$ with 61 coupled channels); Paper 17 (composed-geometry diatomics and polyatomics, where the Level 3 and Level 4 coupled-channel solvers run on each electron group's natural geometry block); Paper 18 § IV.4 (this projection is where the algebraic-implicit and piecewise tiers live at operator level); production solver modules `geovac/hyperspherical_adiabatic.py` (Level 3) and `geovac/level4_multichannel.py` (Level 4).
Empirical anchor: He hyperspherical at Level 3 reaches 0.019% error (Track DI v2.6.0, Sprint 1; 2D variational solver on S^5 with $l_{\max} = 7$, $n_R = 35$, self-consistent cusp correction; the underlying

coupled-channel pipeline supplies the adiabatic curves $\mu_\nu(R)$ that the variational ansatz interpolates across). H_2 at Level 4 reaches 96.0% D_e at $l_{\max} = 6$ $\sigma + \pi$ with 61 coupled channels and Schwartz cusp correction (Paper 15, Track X v2.0.14). Graph-native CI on the He pair-state lattice at $n_{\max} = 9$ (3,927 configurations, exact algebraic float integrals) reaches 0.20% (Paper 13, Track DI Sprint 3C) — this is the SAME observable computed by a different projection chain (direct full-CI on the discrete graph, bypassing § 3.25 entirely) and serves as an independent transcendental-content cross-check: the residual matches the small-Z graph-validity-boundary artifact, not the Schwartz cusp tail, demonstrating that the algebraic-implicit transcendental content of $\mu_\nu(R)$ is captured cleanly by both the coupled-channel and the graph-CI projection chains.

Honest scope. This is non-adiabatic correction territory. The bare adiabatic curve set $\{\mu_\nu(R)\}$ ignores the ∂_R acting on the parameterized angular wavefunctions $\Phi_\nu(R;\hat{\Omega})$; capturing that derivative requires the Hellmann–Feynman P-matrix $P_{\mu\nu}(R) = \langle\Phi_\mu(R)|V^{\text{coupling}}|\Phi_\nu(R)\rangle/(\mu_\mu(R)-\mu_\nu(R))$ and the closure-approximated $Q_{\mu\nu}(R) \approx \sum_\kappa P_{\mu\kappa}P_{\kappa\nu}$ in the coupled radial equations (Paper 13 §XIII.A, Eqs. 70–71). Both quantities are exact at every R in the algebraic spectral basis (no finite-difference noise) but the closure sum runs over an in-principle-infinite set of intermediate channels κ; in practice it is truncated together with the included channel set.

The framework's 2D variational solver (`geovac/level3_variational.py` for He, the `level4_method='variatio` path for Level 4; Paper 15 §VI.4) bypasses adiabatic factorisation entirely by treating R and the angular coordinate simultaneously in a single tensor-product basis. This is structurally a different projection altogether: it removes the Born-Oppenheimer § 3.24 step, so the adiabatic curve family of the present projection never appears. It comes at the cost of giving up the channel-by-channel decomposition that makes the adiabatic picture chemically interpretable. Both routes are legitimate; the choice between them is a Layer-2 control of which projections the chain commits to.

Algebraic registry status (CLAUDE.md §12): hyperradial overlap and kinetic matrices S, K are fully algebraic at Levels 3 and 4 via the three-term Laguerre recurrence (Track H, v2.0.10; pentadiagonal M_2 moment matrix for S, tridiagonal derivative kernel for K, machine-precision agreement with quadrature). The effective radial potential matrix elements $V^{\text{eff}}(R)$ are algebraic at $l_{\max} = 0$ (single Stieltjes moment seed $e^a \cdot E_1(a)$; Track P2, v2.0.13) and numerical-required at $l_{\max} \geq 1$ because of a radical obstruction — the discriminant $\sqrt{\Delta(R)}$ enters at $l_{\max} = 1$ and Cardano-type irrationals at $l_{\max} = 2$. Pointwise diagonalisation at each R to extract $\{\mu_\nu(R)\}$ is the residual numerical step that survives all the upstream algebraic optimisation, and it is irreducible in the sense that the algebraic curve $P(R, \mu) = 0$ describes the eigenvalues implicitly without admitting a closed-form root extraction. Spectral Laguerre at Level 2 is fully algebraic (one transcendental seed $e^a \cdot E_1(a)$, Track J v2.0.10) and provides the cleanest reference case — the moment the radial coordinate becomes the *family index* of an eigenvalue problem rather than the argument of a single function, the framework crosses from purely algebraic into algebraic-implicit or piecewise-smooth content.

Structural reading. Categorically distinct from its parent § 3.24: the Born-Oppenheimer projection produces the parameterised Hamiltonian $H_{\text{el}}(R)$ and commits to the slow/fast separation as a structural assumption, but it does not itself solve the eigenvalue problem at each R. The present projection makes that commitment: it converts a family of operators into a family of channel functions plus a non-adiabatic coupling matrix.

This is the projection that names the *transcendental boundary* of the framework's algebraic structure. Levels 1 and 2 (the latter in σ states, where the algebraic Laguerre matrix elements give a single transcendental seed and no others) are fully algebraic in the strong sense. Higher levels are not, and the failure mode is this projection: the moment one indexes an eigenvalue problem by a continuous radial coordinate, the eigenvalue itself becomes an algebraic-implicit function (Level 3) or a piecewise-smooth function (Level 4) of that coordinate. The Level 3 case is mild — the implicit function admits a global polynomial description and shares its transcendental content ring $\mathbb{Q}(\pi, \sqrt{2})$ with the upstream calibration and embedding constants, introducing no new transcendentals at the R-dependence step. The Level 4 case is stronger: the piecewise structure is irreducible (Track S) and the function does not admit a global polynomial description, although it remains piecewise-algebraic within each region.

The distinction matters for the broader projection taxonomy because it splits what Paper 34 § 3.12 calls "piecewise-smooth in R at the adiabatic potential level" into two different transcendental classes. Level 3 is algebraic-implicit and satisfies $P(R, \mu) = 0$ globally; Level 4 is piecewise-smooth and does not. Paper 18 §IV.4 already separates these tiers in its exchange-constant taxonomy (the "alg. (implicit)" row for Level 3 hyperspherical, the "piecewise" row for Level 4 mol-frame); the present projection is where that taxonomic split lives at the projection-chain level. The Table 1 entry for this projection (algebraic-implicit at Level 3, piecewise-smooth at Level 4) makes the dual notation explicit; the operator-level distinction in this subsection is unambiguous regardless of how the table is summarised.

Sibling to § 3.6 in the structural sense that both rely on a continuous parameter (Λ for the spectral action, R for the adiabatic curve) to label a one-parameter family of finite-rank spectral problems; categorically distinct from § 3.14 (which introduces $m^2 \in \mathbb{Q}$ as a ring-preserving scalar shift on D^2, not as a family index of an eigenvalue problem) and from § 3.15 (which introduces $\beta = 1/T$ as a compactification radius rather than as a Hamiltonian parameter). The genuine peer projection is § 3.12, which provides the radial coordinate that the present projection then converts into the family index of the adiabatic curve set.

3.26 Gauge choice (Coulomb / Lorenz / Feynman)

Source: the framework's $U(1)$ connection on the Fock-projected S^3 Hopf graph (with SU(2) and SU(3) siblings on the same graph and on S^5 respectively, per Paper 30 and Sprint ST-SU3); equivalently, the photon 1-cochain in the Hodge decomposition of edge data, prior to any specific gauge representative being selected for propagator computations.
Target: the same connection with a specific gauge representative fixed. Three canonical choices are in play:

- *Coulomb gauge* ($\vec{\nabla} \cdot \vec{A} = 0$, equivalently $d^\dagger A = 0$ on the graph): the photon decomposes into an instantaneous longitudinal Coulomb part and a transverse radiative part; the framework's vector-photon construction (Paper 28 §vector_photon_qed, Paper 33 § VI) operates in this gauge by treating the Hopf-base 1-form sector as the propagating photon and computing $G_\gamma = L_1^+$ as the transverse propagator (see also § 3.10, where the weak-coupling expansion $L_1 = B^\top B$ is a Coulomb-class gauge fixing of the SU(2) Wilson construction).

- *Lorenz gauge* ($\partial_\mu A^\mu = 0$, equivalently the covariant retarded condition on the spacetime 4-potential): the photon propagator carries explicit retardation in time and the transverse and longitudinal pieces remain coupled.

- *Feynman gauge* (the propagator $-ig_{\mu\nu}/k^2$): the simplest covariant choice, used in Paper 28 § 6 for the one-loop electron self-energy.

The projection step is the act of fixing one of these representatives *before* intermediate Green's-function, propagator, or matrix-element computations begin.
Variables introduced: *none*. A gauge choice is a selection among equivalent classes of representatives, not the introduction of a continuous parameter. No physical scale, focal-length, mass, or coordinate is injected by the gauge fixing itself. This places gauge choice alongside § 3.11 in the

(no-variable) row of Table 1.
Dimension introduced: dimension-preserving. The gauge choice operates on the existing connection without injecting a new $[L]$, $[E]$, $[M]$, or $[T]$ scale. All three gauges produce the same physical observable with the same physical dimension; what differs is the intermediate-step organisation of the propagator content.
Transcendental signature: *gauge-dependent at intermediate steps; gauge-invariant on physical observables.* This is the distinguishing structural property of the gauge-choice projection. At the intermediate-propagator level, the three gauges produce categorically different transcendental content:

- In Coulomb gauge — the framework's implicit choice throughout Papers 28, 30, and 33 — the vector-photon construction's per-loop factor is $1/(4\pi) = \mathrm{Vol}(S^2)/(4 \cdot 4\pi^2)$, identified in Paper 33 § VI as the S^2 Weyl exchange constant of the Hopf base (the same Hopf-base measure $\pi = \mathrm{Vol}(S^2)/4$ that appears in the M1 master-Mellin sub-mechanism, per Paper 32 § VIII and Paper 18 § III.7). The intermediate transcendental content sits cleanly in the calibration π-tier of Paper 18.
- In Lorenz gauge, the retarded propagator structure would produce different intermediate angular content, with π entering through the retardation kernel rather than through the static Hopf-base solid angle. This route has not been implemented in the framework.
- In Feynman gauge, the explicit $-ig_{\mu\nu}/k^2$ propagator gives the textbook covariant intermediate transcendental content, with π appearing through the standard Feynman-parameter denominators rather than through a Hopf-base normalisation. Paper 28 § 6 names this gauge for the one-loop self-energy formula but the production machinery operates in Coulomb gauge throughout.

On physical observables (cross-sections, transition rates, energy shifts), all three gauges produce identical results — this is the content of the Ward identity (Rule 6 of Paper 33's eight QED selection rules; see Paper 33 § II for the framework's status), and it is gauge-invariance in its standard QED sense.
Used in: the framework's one-loop QED machinery (Paper 28) implicitly assumes Coulomb gauge throughout the 1-cochain Hodge construction; Paper 33's 1+6+1 selection-rule partition is stated in Coulomb gauge; Paper 30's SU(2) Wilson lattice gauge theory identifies $L_1 = B^\top B$ as the weak-coupling kinetic term via a Coulomb-class gauge fixing of the link variables. Feynman gauge is named at one point in Paper 28 § 6 (the textbook formula for the one-loop electron self-energy) but is not used in production calculations. No production computation has been performed in Lorenz gauge.
Empirical anchor: the Ward identity itself, i.e. gauge-invariance of physical observables. Standard QED requires $k_\mu \mathcal{M}^\mu = 0$ for the photon-emission amplitude $\mathcal{M}^\mu$, and equivalently the equality of physical predictions across gauge choices. This is item R6 in Paper 33's selection-rule census, where the framework's status is recorded honestly: "gauge invariance under the $U(1)$ phase of the photon, which the 1-cochain photon has only in temporal gauge and only approximately" (Paper 33 § II, R6). The Ward identity is therefore the standard external anchor for the gauge-choice projection; gauge-invariance of physical observables is the precise condition under which the projection itself is structurally well-posed.
Honest scope. This is the single most under-tested entry in the projection dictionary, and the entry must be tagged as such.

The framework has *not* been tested across gauges. Every production calculation in Papers 28, 30, 33, and the Sprint HF / Sprint LS-1..LS-7 / Lamb-shift sub-percent closure of Paper 36 runs implicitly in Coulomb gauge: the photon is represented as a Hodge 1-cochain on the GeoVac edge complex, the propagator is $G_\gamma = L_1^+$ (the transverse pseudoinverse of the edge Laplacian), and the per-loop calibration constant $1/(4\pi)$ is identified through the Hopf-base S^2 solid-angle normalisation. This is a clean, structurally well-motivated choice — Coulomb gauge respects the framework's natural angular-momentum decomposition on the Fock-projected S^3 graph, and the transverse photon sector sits naturally in the co-exact subspace of the Hodge decomposition of § 3.10 (compare Paper 28 § transverse_photon). But it *is* a choice.

A genuine validation of the gauge-choice projection at framework level would require computing a sample one-loop observable in Lorenz gauge (or Feynman gauge as implemented at full operator level rather than named for one formula), verifying that the physical result agrees with the Coulomb-gauge calculation to the precision of the framework, and confirming that the per-loop calibration constant changes structurally between the two gauges while the physical observable is gauge-invariant. This has not been done. The Ward identity (R6 in Paper 33's census) holds only approximately on the 1-cochain photon even in temporal gauge, and the framework has not closed the gauge-invariance test against a competing gauge.

Sub-leading observation: the choice of Coulomb gauge is not arbitrary on the GeoVac graph. The Hodge decomposition of the edge complex distinguishes exact (d_0-image), co-exact ($d_1^\top$-image), and harmonic sectors, and Coulomb gauge is precisely the projection onto the co-exact (transverse) sector. The framework's natural data structure — L_0 on nodes, $L_1 = B^\top B$ on edges, plaquette Hodge structure from Paper 25 / Paper 30 — comes equipped with this decomposition. Lorenz gauge, by contrast, would require introducing the time direction explicitly (compactified per § 3.15 or extended per the Klein-Gordon construction of Paper 35), and the framework has not been wired for covariant time-dependent gauge fixing. This is a real structural constraint, not a choice that could be flipped without modifying the upstream construction.

Structural reading. Layer 2 → Layer 2. The gauge-choice projection operates on the projected vector-photon Hilbert space (§ 3.11); it does not act on Layer 1 graph data and does not move the chain across the § 3.29 boundary.

This is the *second independence witness* in the (no-variable, no-dimension, transcendental-only) corner of the variable/dimension/transcendental axis grid. Until Sprint 2 the only witness in that corner was § 3.11 (vector-photon promotion, which introduces $1/(4\pi)$ without any new variable or dimension); the gauge-choice projection is its structural sibling. Both witnesses operate on the same object (the framework's $U(1)$ connection) at successive structural steps: vector-photon promotion transforms the scalar 1-cochain into a vector harmonic with (q, m_q) labels, and the present projection fixes a specific gauge representative of the resulting connection before propagators are computed. The fact that the two witnesses are *both gauge-related* sharpens the V/D-correlation observation of § 4 and § 6: the transcendental-only corner of the axis grid is the gauge sector of the framework. It is not an accident that the only two projections preserving both V and D while moving the transcendental axis are gauge-construction steps; the structural content of that corner is "add transcendental content to the gauge sector without adding any physical variable or dimension."

Categorically distinct from § 3.11 despite the shared axis pattern: vector-photon promotion is the structural *construction* of the gauge sector (it builds the vector-harmonic photon from a scalar 1-cochain), while the gauge-choice projection is the *selection* of a representative within the already-built sector. The two compose: the chain Layer 1 graph → § 3.11 → § 3.26 is how the framework reaches a computable QED propagator from bare graph data. Categorically distinct also from § 3.21, which similarly preserves V and D (the multipole index L is an internal integer summation label, not a physical variable; the kernel reorganisation is dimension-preserving) but lives in the kernel-side angular sector rather than the gauge sector and is ring-preserving over $\mathbb{Q}[\sqrt{2k+1}]_k$

rather than gauge-dependent in transcendental content. The multipole / Gaunt projection is an angular-content-only structural projection on the Coulomb kernel; the present projection is a gauge-content-only structural projection on the connection that the Coulomb kernel contracts against. Both are necessary to reach a computable amplitude; both preserve the V/D axes; neither introduces a Layer-2 input.

Sibling in the calibration tier of Paper 18: the per-loop $1/(4\pi)$ that appears in Coulomb gauge is the same calibration π-content identified by Paper 33 § VI as the S^2 Weyl exchange constant. Under the master-Mellin reading of Paper 32 § VIII and Paper 18 § III.7, this calibration π is the M1 sub-mechanism signature ($\mathrm{Vol}(S^2)/4$, the Hopf-base measure). A Lorenz-gauge calculation, if performed, would produce different intermediate π-content (retardation kernels rather than static Hopf-base normalisations) but would necessarily land on the same physical observable. The gauge-choice projection therefore sits at the interface of the framework's gauge construction (§ 3.11 + § 3.10) and the framework's transcendental taxonomy (Paper 18), and it is the structural step at which a Layer-2 chain commits to a specific way of accumulating π at each loop.

Open question (flagged for a future sprint): a serious test of the gauge-choice projection would compute a sample observable (e.g. the one-loop Lamb-shift Bethe-log component of Paper 36, or the hydrogen 21 cm contact term of § 5.2.2) in Feynman gauge as well as the existing Coulomb-gauge implementation, and verify gauge invariance to the precision of the framework ($\sim 10^{-3}$ on Lamb-shift-class observables, $\sim 10^{-4}$ on hyperfine contact terms). If gauge invariance fails at framework precision, the failure localises a specific Layer-2 calibration content that the Coulomb-gauge construction is silently encoding; if it succeeds, the gauge-choice projection is verified as a genuine independence witness in the sense of Observation 5 rather than a tautological relabelling.

3.27 Wick rotation / signature change

Source: the framework's bare spectral-triple machinery on the Fock-projected S^3 graph (or any natural-geometry Euclidean spectral triple, including the $T_{S^3} \otimes T_{S^1_\beta}$ tensor-product construction of Sprint TD Track 1). By construction every operator in the GeoVac inventory — Camporesi–Higuchi Dirac on round S^3, Bargmann–Segal heat kernel on S^5, Wilson plaquette functionals on Hopf graphs, all $\zeta_R(2k)$ values entering through ζ_{D^2} on closed Riemannian manifolds — is computed in Euclidean (Riemannian) signature.
Target: the Lorentzian (Minkowski-signature) physical observable obtained by analytic continuation $\tau \to it$, $\beta \to i\beta$ on the imaginary-time circle (or proper-time circle in the Rindler case). Concrete instances under the Wick-rotation chain [Hartle–Hawking 1976] $\to$ [Sewell 1982] $\to$ [Bisognano–Wichmann 1976] $\to$ [Unruh 1976]: the Hartle–Hawking KMS state at $\beta_{\mathrm{cigar}} = 8\pi M$ on the Euclidean Schwarzschild cigar continues to the Hawking thermal state on the Lorentzian Schwarzschild exterior at $T_H = 1/(8\pi M)$; the periodic Euclidean Rindler coordinate η_E with period 2π continues to the Lorentzian Rindler boost at acceleration a, with the accelerated-observer Unruh temperature $T_U = a/(2\pi)$; the modular automorphism of the algebra of observables localized in a Lorentzian Rindler wedge, with respect to the Minkowski vacuum, is identified by Bisognano–Wichmann with the unitary representation of the Lorentz boost subgroup preserving the wedge [47], and the analytic period of the modular flow is the same $2\pi = \mathrm{Vol}(S^1)$ that appears on the Euclidean side.
Variables introduced: the signature index — a discrete binary choice between Riemannian Euclidean and Lorentzian Minkowskian. Structurally analogous to the gauge-choice projection (§ 3.26): a discrete bookkeeping selection that fixes how the framework's spectral content is read as a physical observable. Continuous rapidity parameters relating two Lorentzian frames at relative velocity v/c are *not* introduced by this projection; a full Lorentz-boost projection at the operator-system level (acting on all particles simultaneously, relating two truncated spectral triples

at different observer rapidities) is a separate structural extension named in § 9.
Dimension introduced: dimension-preserving. Analytic continuation does not introduce a new $[L]$, $[E]$, $[M]$, or $[T]$ scale; it changes the interpretation of an already-dimensioned quantity from a Euclidean (imaginary-time-periodic) to a Lorentzian (real-time-thermal or wedge-modular) reading. In the V/D/P tagging of § 4, the projection sits in the same dimension-preserving class as Sturmian reparameterization (§ 3.5) and the gauge-choice projection (§ 3.26).
Transcendental signature: $2\pi \cdot \mathbb{Q}$ via $\mathrm{Vol}(S^1)$ on the Euclidean time circle (Hopf-base measure / M1 sub-mechanism of the master Mellin engine; see Paper 32 § VIII Remark on the Bisognano–Wichmann reading and Paper 35 § VIII for the framework-level structural correspondence). Critically: this is the *same generator* as the observation / temporal-window projection (§ 3.15), which injects $2\pi \cdot \mathbb{Q}$ per Matsubara mode via compactification of Euclidean time at finite $\beta = 1/T$. The two projections share the M1 transcendental signature but are *structurally distinct* as mechanisms: observation / temporal-window compactifies a single Euclidean time direction at finite period β and reads out finite-temperature observables (Casimir at $T \neq 0$, Stefan–Boltzmann free energy, Matsubara loops), whereas Wick rotation analytically continues between two signatures and reads out Lorentzian observables (Hawking radiation, Unruh thermal bath, Bisognano–Wichmann modular flow) from the same Euclidean data. Both surface the M1 2π on the same S^1 factor; the structural reading of what that 2π *means* differs: a Hopf-bundle measure factor on the Riemannian side; a modular-flow / Lorentz-boost orbit period on the Lorentzian side via the published Wick-rotation prescription.
Used in: the Hawking-temperature off-precision catalogue row in § 5.1 (error class C, M external Einstein-equations input, machinery-witness only); Sprint TD Track 4's reproduction of $T_H = 1/(8\pi M)$ on the Euclidean Schwarzschild cigar from M1 alone; Sprint Unruh-pendant's reproduction of $T_U = a/(2\pi)$ on the Euclidean Rindler wedge from the same M1 mechanism; the structural-correspondence reading of $\beta = 2\pi/(\text{surface gravity})$ that unifies Hawking, Sewell, Bisognano–Wichmann, and Unruh as four faces of a single Wick-rotation theorem. Any future framework-side observable whose physical interpretation requires Lorentzian signature (Wightman-class expectation value of a real-time-ordered operator product, KMS state on a Lorentzian wedge algebra, real-time Schwinger–Keldysh closed time contour) consumes this projection.
Empirical anchor: the four-witness Wick-rotation theorem, each face reproducing a shared $2\pi = \mathrm{Vol}(S^1)$ from the M1 mechanism: (i) Hawking temperature $T_H = 1/(8\pi M) = \kappa_g/(2\pi)$ on the Euclidean Schwarzschild cigar, with surface gravity $\kappa_g = 1/(4M)$ [38]; Sprint TD Track 4 reproduces this from M1 with sympy residual zero on all three identities $\beta = 8\pi M$, $T_H = \kappa_g/(2\pi)$, $\beta = 4M \cdot 2\pi$; (ii) Sewell's identification of the Hartle–Hawking thermal state restricted to the Schwarzschild exterior as a KMS state at $\beta = 8\pi M$ on a globally hyperbolic Lorentzian manifold with a bifurcate Killing horizon [36]; (iii) Bisognano–Wichmann identification of the modular automorphism of the algebra of observables localized in a Rindler wedge, with respect to the vacuum state, with the unitary representation of the Lorentz boost subgroup preserving the wedge [35], with analytic period $\sigma_{i \cdot 2\pi}$ = identity forced by Tomita–Takesaki KMS; (iv) Unruh temperature $T_U = a/(2\pi)$ for a uniformly accelerated observer at proper acceleration a, derived in the same Wick-rotation construction with $\beta_{\text{Rindler}} = 2\pi/a$ on the Euclidean Rindler η_E-circle [39]. Sprint Unruh-pendant reproduces this verbatim with the same M1 mechanism, sympy residual zero, lowest bosonic Matsubara mode = a, lowest fermionic mode = $a/2 = \pi T_U$. All four faces reproduce the shared 2π from the framework's M1 Hopf-base measure on the same S^1 factor; the published Wick-rotation prescription supplies the bridge to the Lorentzian-side modular-flow / boost-orbit vocabulary in each case.
Honest scope. The framework demonstrably reproduces $\beta = 2\pi/\kappa_g$ on all four Wick-rotation faces from M1 alone, and the published Hartle–Hawking / Sewell / Bisognano–Wichmann / Unruh

chain identifies that 2π with the Lorentzian-side modular-flow / Lorentz-boost orbit period.

Sprint L1 closure (2026-05-16): literal identification at operator-system level (Riemannian). The L1-named falsifier ("compute the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state, verify $\sigma_{2\pi}$ = identity at finite $n_{\max}$") is now closed positively at the strongest level. Sprint L1 (`debug/l1_modular_hamiltonian_results_memo.md`, Paper 32 § VIII.F) constructs K on the truncated CH spectral triple via the hemispheric-wedge BW-α geometric realization ($K = J_{\text{polar}}$ with integer spectrum two_m$_j$ on the full-Dirac basis) and verifies the period closure $\sigma_{2\pi}(O) = O$ bit-exactly at $n_{\max} \in \{2, 3, 4, 5\}$ for all four physical witnesses (BW, HH, Sew, Unruh). Maximum periodicity residual: 1.8×10^{-16} at $n_{\max} = 2$ through 1.6×10^{-15} at $n_{\max} = 5$ (machine precision scaling $O(\sqrt{\dim_H} \cdot \varepsilon_{\text{machine}})$). Cross-witness collapse is bit-exact (max consistency residual = 0.0 at every $n_{\max}$): the period closure is independent of κ_g because $e^{i \cdot 2\pi \cdot n} = 1$ for integer spectrum. This lifts the four-witness Wick-rotation theorem from "structural correspondence" to "literal identification at the operator-system level (Riemannian)" at every finite cutoff. The Lorentzian extension to signature $(3, 1)$ via the Bizi–Brouder–Besnard 2018 Krein-space spectral triple remains the named Sprint L2 follow-up (a substantial open program per the Sprint L0 audit).

Sprint L1-tighten (2026-05-16 evening): unified BW-α/BW-γ closure. The L1 closure above is at the geometric BW-α level (K = J_{polar} with integer spectrum). Sprint L1-tighten (`debug/l1_tighten_tomita_results_memo.md`, Paper 32 § VIII.F extension) extends this to the full Tomita-Takesaki BW-γ construction: build $K_{\text{TT}} = -\log \Delta$ from the polar decomposition $S = J_{\text{TT}} \Delta^{1/2}$ on the GNS Hilbert-Schmidt space $H_{\text{GNS}} = M_{\dim_W}(\mathbb{C})$, verify $\sigma_{2\pi}^{\text{TT}}(O) = O$ bit-exactly at the operator-action level for all four witnesses at $n_{\max} \in \{2, 3, 4, 5\}$. Verdict: *UNIFIED_STRONG closure.* Maximum Tomita-level residual: 1.09×10^{-16} ($n_{\max} = 2$) through 4.79×10^{-15} ($n_{\max} = 5$). The BW-α and BW-γ flows are operator-action-level conjugates of each other ($\sigma_t^{\text{TT}} = \sigma_{-t}^{\alpha}$ bit-exact at general t; identical at the period $t = 2\pi$). The state-dependent $J_{\text{TT}}(X) = \rho^{1/2} X^* \rho^{-1/2}$ verifies $J_{\text{TT}}^2 = +I$ at the full Tomita level (not just the canonical $U = I$ sanity case). The honest scope above is therefore tightened: literal identification at the operator-system level (Riemannian) holds via *both* the geometric BW-α *and* the Tomita-Takesaki BW-γ constructions, not just one. The L1 "BW-α-only" caveat is removed.

Two further scope distinctions are required. First, *Wick rotation is not the full Lorentzian extension.* This projection lifts the framework's metric-functional Riemannian output (a Mellin trace, a heat-kernel coefficient, a Wilson expectation value) to its Lorentzian analytic continuation in cases where the bridge is a single Wick rotation of a temporal circle. It does *not* construct a Lorentzian (Minkowski-signature) spectral triple with Krein-space Hilbert space and fundamental symmetry $\mathcal{J}^2 = I$; that extension would carry the framework's signature index from $(m, n) = (3, 0)$ in the Bizi–Brouder–Besnard 2018 [41] classification to $(3, 1)$. A clean Lorentz-boost projection at the metric / curvature level, acting on *all* particles in the system simultaneously and relating two truncated metric spectral triples at different observer rapidities, requires that full Lorentzian-signature spectral triple plus a Lorentzian analog of the Latrémolière propinquity used in Paper 38 [63] for the WH1 PROVEN result. The propinquity literature is purely Riemannian as of May 2026 (Latrémolière, Toyota, Hekkelman–McDonald, Leimbach–van Suijlekom); the parallel synthetic Lorentzian-geometry program (optimal transport in Lorentzian synthetic spaces, timelike Ricci curvature lower bounds [62]) provides a metric-geometric analog on the commutative side but does not yet supply an operator- algebraic propinquity. Constructing a Lorentzian propinquity is original NCG-mathematics work, a substantial open program beyond sprint scale. That open extension is flagged in § 9 and is the structural deepening of this § 3.27 entry, not a separate candidate.

Second, *distinct from existing γ-extensions in the catalogue.* The Dirac-sector $\gamma = \sqrt{1 - (Z\alpha)^2}$ in the Camporesi–Higuchi spinor lift (§ 3.7) is a bound-state radial Lorentz factor for a single-particle hydrogenic wavefunction in a central potential; it is not a frame transformation between two observers, and the Wick rotation discussed here does not introduce or modify it. The rest-mass projection (§ 3.14) rescales a single particle's mass under $m_e \to m_{\mathrm{red}}$ within Euclidean signature; the Breit retardation projection (§ 3.16) introduces the rest-mass ratio m_l/m_n as a two-body kinematic control parameter. None of these is a signature change. Wick rotation sits structurally above all three: those projections operate within a fixed Euclidean spectral triple; this projection toggles between the Euclidean and Lorentzian readings of the *same* computed quantity, with the toggle bridged by the published Wick-rotation chain.

Structural reading. Wick rotation sits in the Layer-2 $\to$ Layer-2 class (cf. § 3.29): it operates on already-projected Euclidean spectral content, not on the bare graph or its initial projection chain. Structurally it is a *metric-functional-level* projection: the framework computes a Mellin trace, heat-kernel coefficient, or other Euclidean spectral-action functional (Paper 32 § VIII case-exhaustion theorem ensures the transcendental content is M1, M2, or M3); Wick rotation then reads that functional as a Lorentzian physical observable via the published Hartle–Hawking / Sewell / Bisognano–Wichmann / Unruh chain.

The structural-correspondence-not-literal-identification verdict crystallizes the framework's position on Lorentzian physics in general: at the metric functional level (specifically the M1 $2\pi = \mathrm{Vol}(S^1)$ signature on a Wick-rotated time circle), the framework's Euclidean output IS the Lorentzian modular-flow / boost-orbit period, with no extension required. At the operator-system level (Tomita–Takesaki modular automorphism acting on the truncated algebra at finite $n_{\max}$, Krein-space representation, frame-level Lorentz-boost between truncated metric spectral triples at different rapidities), the framework currently sees the same 2π but cannot internally close the period identity; that closure is the named open extension.

Three observations distinguish this projection from § 3.15 despite shared M1 generator and shared 2π transcendental content. First, mechanism: observation / temporal-window compactifies a single Euclidean time direction at finite period $\beta = 1/T$ and sums Matsubara modes $\omega_k^t = 2\pi k/\beta$; Wick rotation analytically continues a Euclidean computation to Lorentzian signature without changing the periodicity of the underlying time circle. Second, target physics: observation / temporal-window targets finite-temperature observables (Casimir at $T \neq 0$, partition functions, Stefan–Boltzmann); Wick rotation targets Lorentzian-signature observables read from Euclidean computations (Hawking radiation, Unruh thermal bath, BW modular flow). Third, falsifier sharing: the operator-level falsifier for § 3.15's Bisognano–Wichmann footnote (closing $\sigma_{i\cdot 2\pi}$ on $\mathcal{T}_{n_{\max}}$) is identical to the falsifier for the four faces of this projection, because the bridge in every case runs through the same KMS–Tomita–Takesaki algebra; one operator-system-level proof lifts all four faces simultaneously. Same structural correspondence, same M1 generator, same falsifier, but distinct mechanisms and distinct target classes of physical observable. These are not the same projection.

Cross-references: Paper 32 § VIII Remark on the Bisognano–Wichmann reading documents the spectral-triple-side statement and the case-exhaustion-theorem context; Paper 35 § VIII subsection on the Bisognano–Wichmann reading of the temporal-window projection develops the cigar identification at Hartle–Hawking $\beta = 8\pi M$; Paper 35 § VIII subsection on the Unruh pendant develops the Rindler-wedge identification at $\beta = 2\pi/a$; § 3.15 carries the original Bisognano–Wichmann reading footnote on the temporal-window projection. The structural ground for the present § 3.27 entry is the Sprint TD Track 4 + Sprint Unruh-pendant + Track D documentation chain (May 2026), which promotes the metric-functional-level content of the Wick-rotation reading from a footnote (§ 3.15) and an open-question candidate (§ 9 entry on the Lorentz boost; see § 9) to a named projection slot in § 3. The full operator-system-level Lorentzian extension remains open as a major NCG-mathematics structural extension and is documented as a deepening of this entry, not as a separate candidate.

Sprint L2-E (2026-05-17): Krein-space period closure, signature-blind at finite cutoff. A Lorentzian-side analog of the Sprint L1 period closure was constructed (the strong "literal identification" reading recorded below was subsequently *withdrawn* — see the 2026-06 status update at the end of this entry). Sprint L2 (sub-sprints L2-A scoping, L2-B Krein space, L2-C Lorentzian Dirac, L2-D Connes axiom audit, L2-E Krein-level modular Hamiltonian; Paper 32 § VIII.E.E) builds the Lorentzian Krein space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ via van den Dungen 2016 Proposition 4.1 (arXiv:1505.01939) and the Bizi–Brouder–Besnard 2018 $(m,n) = (4,6)$ classification of Lorentzian signature $(s,t) = (3,1)$ [41], then constructs the hemispheric wedge $W_L = P_W^{\mathrm{spatial}} \otimes P_{t\geq 0}$ and the wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ under the BW choice $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$. The Krein-space period-closure identities close bit-exactly at every tested $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$: BW-$\alpha$ period closure $\sigma_{2\pi}^{L,\alpha}(O) = O$, BW-$\gamma$ Tomita period closure $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$, flow conjugacy $\sigma_t^{L,\mathrm{TT}} = \sigma_{-t}^{L,\alpha}$ at general t, and six-witness collapse all at residual $\leq 4 \times 10^{-16}$ (machine precision). At $N_t = 1$ the Lorentzian-side construction reduces bit-identically to Paper 42's Riemannian construction (Frobenius residual exactly 0.0); the $N_t > 1$ content is temporal-derivative structure layered on the same compact KMS $\beta = 2\pi$ circle (see the 2026-06 status update below). Substantive structural finding: Paper 42 § 7.2 open question O3 ("$H_{\mathrm{local}} \neq D_W$") is **signature-INDEPENDENT at the Riemannian limit** and refined upward at $N_t > 1$ by temporal-derivative content (the gap between the spectral-action object and the modular-Hamiltonian generator widens with temporal cutoff). **Net status (as computed, May 2026):** the period-closure identities $\sigma_{2\pi}^L(O) = O$ hold bit-exactly at every tested $(n_{\max}, N_t)$, and the $N_t = 1$ slice reduces to Paper 42's Riemannian construction.

Status update (2026-06 — the "literal identification" reading is WITHDRAWN). The subsequent Paper 45 K^+-annihilation theorem (2026-06-09) and the compact-boost closure (2026-06-19) show that these period-closure identities do *not* constitute a Lorentzian metric identification at finite cutoff. The truncated Bisognano–Wichmann boost generator $K = \mathrm{diag}(2m_j)$ has *integer* spectrum, so $\mathrm{e}^{2\pi \mathrm{i} K} = I$ bit-exactly at every finite cutoff: the modular flow is the compact KMS $\beta = 2\pi$ *circle*, not a non-compact boost, and a compact circle carries only a Euclidean (forward-triangle) metric. The Krein K^+ compression is signature-blind (the restricted Lipschitz seminorm is unchanged under a temporal Wick involution), so the Lorentzian *signature* is metrically invisible to any finite-cutoff seminorm; the Lorentzian reverse-triangle requires the non-compact continuum boost (continuous spectrum), reached only as $n_{\max} \to \infty$. The finite-cutoff statement is therefore Euclidean/convention, and the earlier "literal identification at the Krein operator-system level" and "genuine Lorentzian extension" framing is withdrawn (Paper 45 descope; the group1 synthesis honest-scope note). What survives is the norm-resolvent / period-closure computation, read as the compact-side KMS statement. No published finite-cutoff Lorentzian propinquity construction exists. Cross-manifold W2b ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$) is structurally distinct from the $(3,1)$ extension closed here and remains open per Paper 32 § VIII.D "frontier-of-field framing." Files: `geovac/modular_hamiltonian_lorentzian.py`, `tests/test_modular_hamiltonian_lorentzian.py` (74 fast + 3 slow, all pass), `debug/l2_e_modular_hamiltonian_lorentzian_memo.md`. Paper 32 § VIII.E.E and Paper 42 § 11 carry the synthesis writeup; Paper 43 (Lorentzian-extension math.OA standalone, outlined in `papers/group1_operator_algebras/paper_43_lorentzian_extension_outline.md`) is the natural Zenodo deposit for the standalone math.OA form.

3.28 Apparatus identity / state-side reduction

Source: the framework's spectral-side output — a Hamiltonian H on the Hilbert space $\mathcal{H}$ of a Fock-projected S^3 graph (or its Bargmann–Segal / molecule-frame analogue), with eigenvalues $\{\lambda_n\}$ and eigenstates $\{|\psi_n\rangle\}$ supplied by the upstream projection chain. All twenty-seven preceding entries of § 3 (§ 3.1 through the Wick-rotation entry of Sprint 2) operate on this spectral side: they introduce variables, dimensions, and transcendentals at the operator level of H, D, or $\mathcal{H}$ itself.

Target: the state-side observables associated with a density matrix ρ on $\mathcal{H}$ — specifically, the von Neumann entropy $S(\rho) = -\mathrm{Tr}\,\rho \log \rho$, the reduced density matrices $\rho_A = \mathrm{Tr}_B \rho_{AB}$ for any bipartite split $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, the mutual information $I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$, and the classical–quantum apparatus identity

$$S_{\mathrm{thermo}}(T) \;=\; k_B \cdot S_{\mathrm{microstate}}(\rho_\beta), \qquad \rho_\beta = e^{-\beta H}/Z,\ \beta = 1/T,$$

which says that the macroscopic thermodynamic entropy of a system in contact with a heat bath at temperature T equals (in units of k_B) the von Neumann entropy of the corresponding Gibbs density matrix. This projection is the rule that converts spectral-side data into state-side observables.

Variables introduced: the density matrix ρ on $\mathcal{H}$, or, equivalently for thermal applications, the inverse-temperature $\beta = 1/T$ that indexes the Gibbs family $\rho_\beta = e^{-\beta H}/Z$. The framework's spectral-side machinery never references ρ; the apparatus identity is what introduces it. (When this projection follows § 3.15, β is already present as the Matsubara compactification radius; the apparatus identity then names the state-side use of that β. When it stands alone on a non-thermal density matrix — e.g. the ground-state reduced density matrix $\rho_A = \mathrm{Tr}_B |\psi_0\rangle\langle\psi_0|$ of Paper 27 — ρ is the new variable.)

Dimension introduced: preserves. Von Neumann entropy $S(\rho) = -\mathrm{Tr}\,\rho \log \rho$ is dimensionless by construction (the eigenvalues of ρ are pure probabilities and the logarithm has no unit). Temperature carries energy dimension via $k_B T$, but the dimensional content of T is consumed externally through Boltzmann's constant rather than generated by the projection itself; the projection's structural role is the dimensionless operator-to-number map $\rho \mapsto S(\rho)$.

Transcendental signature: von Neumann class. The functional $S(\rho) = -\mathrm{Tr}\,\rho \log \rho$ involves the logarithm of an operator and is generically transcendental at the value level for any non-trivial ρ. Structurally, however, this transcendental class is *categorically distinct from the master Mellin engine* $M_1 \cup M_2 \cup M_3$ that organises every spectral-side transcendental documented in Paper 34 §§ 3.1–3.27 and Paper 32 § VIII. This is not a framing claim — it is a verified PSLQ negative. Sprint TD Track 5 (CLAUDE.md §2, 2026-05-08) computed the framework's atomic correlation entropies $S_{\mathrm{full}}(\mathrm{He}) = 0.040811051\ldots$, $S_{\mathrm{full}}(\mathrm{Li}^+) = 0.011211717\ldots$, and $S_{\mathrm{full}}(\mathrm{He}, n_{\max}{=}4) = 0.041879430\ldots$ to 150 dps, then ran PSLQ at 100 dps against a 12,312-form mechanical basis spanning the M1 (Hopf-base measure), M2 (Seeley–DeWitt $\sqrt{\pi}$), and M3 (vertex-parity Hurwitz) sub-rings plus algebraic and rational control classes. Result: zero hits across all three systems. Stress tests at coefficient ceiling 10^6 against the ten simplest ring elements $(1, \pi, \pi^2, 1/\pi, \zeta(2), \zeta(3), G, \log 2, \sqrt{2}, \varphi)$ returned zero hits at 10^{-50} tolerance. Cross-system ratios $S(\mathrm{He})/S(\mathrm{Li}^+) = 3.6400\ldots$ and the $n_{\max}$-ratio $0.9744\ldots$ also null. Six orders of magnitude in coefficient ceiling do not change the verdict. GeoVac correlation entropy is genuinely a state-side von Neumann quantity, structurally disjoint from the spectral-side Mellin ring.

Used in: every place the framework computes an entropy or a reduced density matrix. Paper 27 (entropy as projection artifact; $S_{\mathrm{kin}}/S_{\mathrm{full}} \sim 10^{-14}$ for He at $n_{\max} = 2, 3$; V_{ee}-only generation of correlation entropy; the universal $S_B = A(\tilde{w}_B/\delta_B)^\gamma$ scaling with $\gamma_\infty \approx 1.96$ in the $Z, n_{\max} \to \infty$ limit) exercises this projection throughout; the apparatus identity is the structural mechanism that makes "entropy" a well-defined Layer-2 observable in the first place. Sprint TD Track 2 (apparatus identity operational verification: $S_{\mathrm{thermo}} = k_B \cdot S_{\mathrm{microstate}}$ to machine precision, max residual 4.8×10^{-14}, across H/He/Li^+ at 30 log-spaced temperatures each). Future bipartite-entropy / mutual-information analyses of atomic correlation (Paper 26 § VI Z-scaling extrapolation). Any future quantum-information-theoretic comparison between the framework's qubit Hamiltonians (Paper 14) and competing encodings, where fidelity and trace distance between candidate ground states

sit on the same state-side as $S(\rho)$.
Empirical anchors. Two independent verifications, both at machine precision.

Operational identity (Sprint TD Track 2, 2026-05-08). The classical-quantum apparatus identity $S_{\mathrm{thermo}}(T) = k_B \cdot S_{\mathrm{microstate}}(\rho_\beta)$ is verified to floating-point machine precision across the framework's hydrogenic-class systems. Maximum residual across H/He/Li^+ at 30 log-spaced temperatures each: 4.8×10^{-14}. The $T \to 0$ limit reproduces Paper 27's ground-state correlation entropy $S_{\mathrm{full}}(\mathrm{GS})$ bit-identically: He 0.040811 (residual 3.1×10^{-10}), Li^+ 0.011212 (residual 3.2×10^{-10}), H residual identically zero by single-determinant ground-state rigidity. The $T \to \infty$ limit approaches $\log(\dim \mathcal{H})$ (e.g. $\log 14 \approx 2.639$ for hydrogen at $n_{\max} = 3$), the Boltzmann maximum mixing. The Lindblad concavity bound holds at every β. These verifications confirm that the apparatus identity is operative as a state-side reduction rule on the framework's hydrogenic systems without anomaly.

Master-Mellin-engine disjointness (Sprint TD Track 5, 2026-05-08). PSLQ at 100 dps against a 12,312-form mechanical basis (frozen before testing, W3-protocol style) returned zero hits for $S_{\mathrm{full}}(\mathrm{He})$, $S_{\mathrm{full}}(\mathrm{Li}^+)$, and $S_{\mathrm{full}}(\mathrm{He}, n_{\max}{=}4)$. Twenty random comparable targets at coefficient ceiling 100 also returned zero hits (control null distribution genuinely zero). Stress tests at coefficient ceiling up to 10^6 against the simplest ring elements, and against cross-system ratios, also null. The verdict is clean and sharp: the framework's correlation entropies are not algebraic combinations of the spectral-side master-Mellin transcendentals.
Honest scope. State-side enumeration is currently *incomplete* in Paper 34's dictionary. This entry is the first state-side projection; the natural continuations (mutual information, conditional entropy, fidelity, trace distance, relative entropy $S(\rho\|\sigma)$, Wasserstein–Kantorovich distance on the state space $P(\mathcal{H})$ of Paper 38's propinquity machinery) are not enumerated here. The structural status of those candidate projections is named in the open-question section (§ 9, candidate state-side enumeration); operationally, they have appeared in the framework only as scaffolding inside specific calculations (Paper 26 bipartite entanglement, Paper 38 Wasserstein–Kantorovich identification of the propinquity limit) and have not yet been promoted to dictionary entries with their own focal-length variable / dimension / transcendental tagging. The state-side dictionary should be read as deliberately opened in this sprint, not as fully populated.

A second honest scope note: the apparatus identity is operationally verified on the framework's hydrogen-class systems (H, He, Li^+), where the spectral data is dense and the Gibbs ensemble is small enough to enumerate. Extension to molecular spectral triples (Papers 15, 17, 19) and to the nuclear–electronic composed triple (Paper 23 § VI) is mechanical but has not been exercised; in those settings the relevant ρ is potentially high-dimensional and the von Neumann entropy may require sampling or low-rank approximation methods rather than direct diagonalisation. This is an engineering scope rather than a structural one: the projection is well-defined on every spectral triple the framework produces; the operational cost of computing $S(\rho)$ scales with the dimension of $\mathcal{H}$.

A third scope note concerns the relationship between this projection and the foundational fact that pure-state von Neumann entropy of an ungrouped system is zero. Apparatus identity / state-side reduction is non-trivial only when ρ is mixed (Gibbs ensemble) or when a bipartite trace $\rho_A = \mathrm{Tr}_B \rho_{AB}$ is taken on a pure state. This is consistent with Paper 27's framing: entropy is a projection artifact in the precise sense that it requires choosing either a thermal-bath partition (apparatus identity in the Gibbs form) or a Hilbert-space partition (reduced density matrix). The state-side reduction is the structural operation that names that choice and supplies the resulting dimensionless number.
Structural reading. Categorically distinct from *every* prior entry of § 3, in the precise sense that

those entries operate on the spectral data $(\mathcal{A}, \mathcal{H}, D)$ of the spectral triple, while this entry operates on density matrices ρ on $\mathcal{H}$. The spectral / state divide is the deepest taxonomic split in the dictionary so far: it is sharper than the variable / dimension / transcendental axes (Theorem T3 in § 6), sharper than the master-Mellin sub-mechanism partition (M1 / M2 / M3), and sharper than the Layer 1 / Layer 2 partition itself (which lives entirely on the spectral side: Layer 1 is the bare graph D-data, Layer 2 is the projected D-data). The apparatus identity is structurally *the* sibling of § 3.6 (Connes–Chamseddine spectral action): both project the framework's finite-rank spectral data into a single dimensionless number, but one does so via $\operatorname{Tr} f(D/\Lambda)$ on the spectral side while the other does so via $\operatorname{Tr} \rho \log \rho$ on the state side. Sprint TD Track 5's PSLQ negative is the quantitative statement that these two procedures generate disjoint transcendental classes: spectral-side traces of operator functions of D live in the master Mellin engine $M_1 \cup M_2 \cup M_3$ (Paper 32 § VIII case-exhaustion theorem; Sprint TS-E3, May 2026), while state-side traces of $\rho \log \rho$ do not.

The structural complement to § 3.24 is also worth naming explicitly. The Born–Oppenheimer projection creates a parameterised family of Hamiltonians $H(\vec{R})$, and a thermal ensemble of nuclear configurations weighted by Boltzmann factors $e^{-\beta E(\vec{R})}$ is precisely the density matrix $\rho_\beta(\vec{R}) = e^{-\beta H(\vec{R})}/Z$ on which the present projection acts. In the language of the multi-focal-composition arc (May 2026, CLAUDE.md § 2), adiabatic / BO supplies the Layer-1.5 parameter family; the apparatus identity is the Layer-2 reduction that converts that family into a single thermodynamic number. The two projections compose naturally and the composition is exercised operationally in Sprint TD Track 2.

Paper 38's Wasserstein–Kantorovich state space $P(S^3)$ (Proposition "Limit identification") is the natural metric container for the state-side complement of the dictionary. In the $n_{\max} \to \infty$ propinquity limit, the truncated spectral triple $\mathcal{T}_{n_{\max}}$ converges to $\mathcal{T}_{S^3}$ with the round-S^3 Wasserstein–Kantorovich distance as the metric on its state space. This identifies a clean future-direction target: state-side projections beyond apparatus identity should sit naturally in the Wasserstein–Kantorovich complex — mutual information, fidelity, and trace distance all have known metric structures on state spaces of finite-rank operator algebras, and Paper 38's machinery suggests these are reachable in the GH-convergent limit without requiring additional NCG framework extension. The state-side dictionary is therefore named here as deliberately incomplete but *not* blocked: extension is structurally available and is the natural next direction for the dictionary's axis-completeness.

Born rule paragraph (added 2026-05-26 after sprint `ahha_born_rule_attempt`). The apparatus identity construction produces Born-rule probabilities directly via the standard GNS construction: for a Layer 1 pure state $\psi \in \mathcal{H}_{\mathrm{GV}}$ and apparatus subalgebra $\mathcal{A}_W$ with apparatus observable $\hat{A} = \sum_a a\, P_a$, the state-side reduction ω_ψ assigns probability $\omega_\psi(P_a) = \langle\psi|P_a|\psi\rangle = \|P_a\psi\|^2 = |\langle a|\psi\rangle|^2$ to outcome a when $P_a = |a\rangle\langle a|$. The squared-modulus structure is inherited from the complex inner product on $\mathcal{H}_{\mathrm{GV}}$, which is itself inherited from the standard Peter–Weyl construction on $L^2(S^3, \Sigma)$ (Paper 32 § III, where $\mathcal{H}_{\mathrm{GV}}$ is taken as input data of the spectral triple). No GeoVac-specific operation reduces the Born rule to more primitive structures: in this framework, as in standard algebraic quantum mechanics, the Born rule follows from the Hilbert-space + GNS combination via Gleason's theorem (1957) and the standard projection-measure formalism. The attempt of 2026-05-26 to derive a non-Gleason / non-standard Born expression from §III.28 plus Pythagorean orthogonality (Paper 43 § 10.2) plus the master Mellin engine (Paper 32 § VIII) returned the same Born expression by all three routes, indicating that the framework is structurally consistent with Gleason but does not reach further than Gleason at the substrate level. The honest framework-internal classification of the Born rule is therefore that it sits at the *calibration-data tier* in the sense of Paper 18 § IV.6 (inner-factor input data) and CLAUDE.md § 1.7 WH8 (the Born measure as the observation projection's exchange constant, external calibration input): alongside the Yukawa couplings, the fine-structure constant α, the gauge-group lower-bound calibration, and the multi-loop QED counterterms (LS-8a renormalization gap), the probability-weighting structure of measurement is external to the substrate, not derivable from the packing axiom + Fock projection + spectral-triple data. The Pythagorean orthogonality theorem of Paper 43 § 10.2 contributes a decoherence-flavored constraint on *which observables* the apparatus's preferred basis selects (the centralizer of H_{local} in $\mathcal{A}_W$) but does not modify the Born probability assignment. This is a closed structural position, not an open question; the framework is in the Wigner / calibration-tier camp on Born rule rather than the Deutsch–Wallace / Zurek / Bub–Pitowsky derivation camps. See § 9 "Born rule as calibration data" for the corresponding §VIII closure note.

A final structural reading: the apparatus identity / state-side reduction is the projection that names "how an observer extracts a number from the system," as a structural complement to the spectral-side projections that name "how the framework constructs the system in the first place." In the master-Mellin-engine partition of Paper 32 § VIII and Paper 18 § III.7, the spectral side has three sub-mechanisms (M1, M2, M3); the state-side — as documented by Sprint TD Track 5's PSLQ negative — is a fourth mechanism class structurally outside the engine. Whether the state-side admits its own internal sub-mechanism partition (analogous to the M1 / M2 / M3 split on the spectral side) is the open structural question this entry leaves for future work. Current evidence (one verified entropy class, machine-precision apparatus identity, PSLQ-disjoint from spectral side) is consistent with either: a single state-side sub-mechanism that produces all state-side observables uniformly, or multiple sub-mechanisms whose distinctions are not yet operationally visible. Either outcome would sharpen the structural-skeleton-scope statement of CLAUDE.md § 2 beyond its current spectral-side-only form.

3.29 Boundary of §3: internal graph-derived variables

The twenty-eight projections enumerated above share a structural property worth naming explicitly: each introduces a variable that is *internal to the framework's graph and projection structure* — a graph-derived integer or rational invariant (such as Z, n, the foundational α, or a Wigner $3j$ coefficient), a continuous parameter that controls a coordinate or focal-length choice (such as ρ, R, Λ, β, m, r_E, r_Z, Q_N), or a function of prior projection variables (compositional projections per Observation 4).

This is structurally distinct from *Layer-2 calibration inputs* that the framework consumes from external sources but does not autonomously generate. Examples: nuclear g-factors g_p, g_d (extracted from precision spectroscopy or QCD lattice), multi-loop QED coefficients (Eides Tab. 7.3; the α^4 Breit coefficient $-(11/48)$ from Karshenboim 2005 § 9, with the complete Ps theory at $m\alpha^6$ from Pachucki–Karshenboim 1998), Yukawa couplings (per CLAUDE.md § 1.7 WH5; Sprint H1 verdict), and external classical-field strengths (Zeeman B, Stark $\vec{\mathcal{E}}$ — verified by the 2026-05-09 external-field hunting sprint to be Layer-2 inputs, not § 3 projections; the bare Landau spectrum is HO-class and π-free, structurally a Bargmann–Segal sibling on the cyclotron plane, with π entering only through the 2D phase-space measure as a calibration π in the M1 Mellin sub-mechanism, not through Matsubara compactification).

The structural test that distinguishes § 3 projection from Layer-2 input is whether the variable's content is internal graph-derived or external classical-field / QCD-lattice-derived. Zemach and charge radii are intrinsic focal-lengths once the framework's spectral content (not external calibration) determines the convolved-potential structure given the radius; nuclear g-factors and

Yukawas are external because no spectral structure of the framework determines their value. This boundary is not absolute: promotion of a Layer-2 input to a projection slot is the structural commitment that the variable acquires a focal-length role beyond bare scalar consumption, as Zemach did when `magnetization_density.py` was implemented at the operator level (§ 3.18). Future Layer-2 inputs may follow if the framework develops operator-level constructions that give them focal-length status.

4 The Three-Axis Dictionary

Table 1 consolidates Section 3 into the three-axis dictionary that this paper proposes as a companion to Paper 18's transcendental classification.

Observation 2 (Two-axis duality)**.** *The transcendental classification of Paper 18 and the projection classification of Table 1 are not independent. Each Paper 18 transcendental tier corresponds to specific projections, and each projection's transcendental signature is a specific Paper 18 tier:*

- *Calibration π^{2k} tier $\leftrightarrow$ spectral action / Hopf bundle / vector-photon promotion projections.*
- *Spinor-intrinsic $(Z\alpha)$ tier $\leftrightarrow$ Camporesi–Higuchi spinor lift.*
- *Odd-ζ tier $\leftrightarrow$ Camporesi–Higuchi spinor lift in combination with Mellin/heat-kernel evaluation at integer s.*
- *Dirichlet L tier $\leftrightarrow$ vertex topology composed with Camporesi–Higuchi at quarter-integer Hurwitz shift.*
- *Embedding tier $\leftrightarrow$ molecule-frame hyperspherical or non-trivial bound-state coupling.*
- *Combinatorial-rational tier $\leftrightarrow$ Wigner $3j$ / D alone (no projection beyond Layer 1); also the structural denominator $D_{drake}(n,\ell) = 2(2\ell+1)Z^4/n^3$ in the Drake–Swainson projection.*
- *Flow tier (Paper 18 § IV) $\leftrightarrow$ Drake–Swainson asymptotic subtraction. This is the projection that takes a divergent finite-basis ratio (where closure makes the denominator vanish) and produces a K-independent regularized value through asymptotic subtraction. The flow content sits in the α-dependent logarithms of the regulated spectral sum, which cancel against the K-cutoff terms.*
- Trivial / ring-preserving tier $\leftrightarrow$ *rest-mass projection. The mass projection introduces a parameter without introducing a transcendental: the bare-graph algebraic ring is closed under the addition of m^2 when $m^2 \in \mathbb{Q}$. This is the only projection in the dictionary that introduces a physical variable while preserving the transcendental signature of the source spectrum. Paper 35 verifies this explicitly for the Klein-Gordon spectrum on $S^3 \times \mathbb{R}$.*
- Calibration $2\pi \cdot \mathbb{Q}$ tier (temporal subtype) $\leftrightarrow$ *observation / temporal-window projection. This is the categorical home of π in the framework: every π that appears in a GeoVac observable can be traced to either (a) a Hopf-base calibration π (Hopf bundle / vector-photon promotion), (b) a spectral-action $\sqrt{\pi} \cdot \mathbb{Q}$ (heat-kernel coefficient), or (c) a temporal Matsubara $2\pi/\beta$ (this projection). The Casimir constant on $S^3 \times S^1_\beta$ is the cleanest illustration: spatial-only Casimir is $1/240$ (rational, no π); adding S^1_β injects π exactly through the Matsubara mechanism.*
- Layer-2 input tier (two-body kinematic subtype) $\leftrightarrow$ *two-body Dirac / Breit retardation projection (§ 3.16). Same ring-preserving character as the rest-mass projection (no transcendental injected beyond the α already present in the spectral-action chain), but distinct mechanism: rest-mass projection rescales single-particle observables by $m_e \to m_{red}$, whereas Breit retardation introduces the rest-mass ratio m_l/m_n as a two-body kinematic control parameter. In*

the equal-mass leptonic limit $m_l/m_n \to 1$ the standard $1/M_{nucleus}$ recoil expansion is invalid, and the projection becomes load-bearing for level energies at α^4. Empirical anchor: Ps 1S–2S framework-native at +64.75 ppm in § 5.1; closure to sub-MHz requires the α^4 Breit input (canonical coefficient $-(11/48)$, Karshenboim 2005 § 9) as Layer-2 content.

Remark 3 (variable–dimension correlation)**.** *The variable axis and dimension axis are not independent: every projection in Section 3 that introduces a physical dimension also introduces a physical variable that carries that dimension. Of the twenty-eight projections, twelve add a fresh dimension and all twelve also add the variable that carries it (perfect correlation on the V/D side); zero projections add a dimension without a variable. The six Sprint 1 additions (§§ 3.20–3.25) preserve this correlation: four (Phillips–Kleinman, multipole expansion, bipolar harmonic, symmetry projection) preserve dimension, and two (adiabatic / Born–Oppenheimer, coupled-channel / adiabatic curve) re-use the existing length variable $\vec{R}$ already injected by stereographic and molecule-frame hyperspherical projections. The three Sprint 2 additions (§§ 3.26–3.28) further strengthen the V/D-correlation observation: gauge choice and Wick rotation both preserve V and D while moving the transcendental axis (joining vector-photon promotion in the (no-variable, no-dimension, transcendental-only) corner — giving* three *independent witnesses there, all gauge-related or signature-related; the structural content of that corner is the gauge-and-signature sector of the framework), and apparatus identity introduces ρ as the first state-side variable while preserving dimension. The transcendental axis remains the only genuinely independent axis; V/D are still perfectly correlated. The three intra-nuclear projections of § 3.17, § 3.18, and § 3.19 preserve this V/D correlation: each adds an intra-nuclear focal length together with its associated dimension ($[L]$, $[L]$, $[L]^2$ respectively). The transcendental axis is the only one that admits an independence witness — vector-photon promotion (§ 3, projection 11) introduces $1/(4\pi)$ without introducing any new variable or dimension. This is the structural content of Section 6 (Theorem 3 composition table): of the three axes, only the transcendental axis is genuinely independent of the other two. The sixteenth projection (two-body Dirac / Breit retardation, § 3.16) preserves the V/D correlation: it adds the variable m_l/m_n and the dimension energy together, and lives in the ring-preserving $\alpha^4 \cdot \mathbb{Q}$ tier.*

4.1 Base-unit decomposition of the projection family

Remark 3 establishes that every dimensioned projection is paired with the variable carrying that dimension, at the level of compound dimensions (energy, length, time, mass). This subsection decomposes the compound dimensions into base-unit constituents and asks: how often is each base unit injected across the twenty-eight projections, and which projections inject which units?

We work in atomic units throughout: $\hbar = m_e = e = 1$, so the mechanical base units are $[L]$, $[M]$, $[T]$, with charge $[Q]$ carried by the integer label Z. Energy is the compound $[E] = [M][L]^2[T]^{-2}$, but in atomic units acts as an effective base for the framework's primary observables; we track $[E]$ alongside the strictly-base units below. We follow the § 3 order of projections for cross-reference with Table 1.

The inverse view — which projections inject each base unit — makes the structure visible:

- $[L]$ (scalar): *six* projections. Fock or Bargmann–Segal anchors the foundational length (a_0 or oscillator length); stereographic re-expresses it as a continuous coordinate r; Wigner D rotation introduces nuclear position $\vec{R}_{AB}$; molecule-frame hyperspherical separation introduces internuclear distance R (Wigner D and mol-frame inject the same physical length scale through different projections); charge-density (§ 3.17) introduces the intra-nuclear charge radius r_E; magnetization-density (§ 3.18) introduces the intra-nuclear Zemach radius r_Z. These last three intra-nuclear length scales are each *categorically distinct* from the atomic / internuclear scales: they modify different Hamiltonian operators (Coulomb V_{ne}, Fermi-contact $A_{\mathrm{hf}}^{\mathrm{contact}}$) with different selection rules, with empirical separation across hydrogen Lamb shift / 21 cm hyperfine / deuteron molecular HFS.

- $[L]^2$ (rank-2 tensor): *one* projection. Tensor multipole (§ 3.19) introduces the nuclear quadrupole moment Q_N at rank-2 leading; higher-rank multipoles ($L \geq 3$) follow at higher $[L]^L$. Categorically distinct from scalar $[L]$ injections: the operator is tensorial and requires $l \geq 1$ on the electronic side.
- $[E]$: *four* projections. Fock or Bargmann–Segal anchors the foundational energy (Ry or $\hbar\omega$); spectral action introduces UV cutoff Λ; Drake–Swainson introduces a transient subtraction scale K (which cancels in the regularized result); two-body Breit retardation introduces $\alpha^4{\cdot}\mathrm{Ry}$ as a correction-energy scale, where α^4 is dimensionless and Ry is foundationally anchored.
- $[M]$: *one* projection (rest-mass). Each particle species in the system requires a separate rest-mass projection; two-particle systems compose two such projections.
- $[T]$: *one* projection (observation / temporal-window). Each measurement window or temperature requires a separate observation projection.
- $[Q]$: integer-counter label Z at the Fock projection only. $[Q]$ does not appear as a continuous scale anywhere in the projection family — charge is always counted, never measured-against-a-scale.

Observation 3 (L–E asymmetry)**.** *Of the mechanical base units (and the charge unit), only $[L]$ and $[E]$ are injected by multiple projections; $[M]$, $[T]$, and $[Q]$ are each instantiated by exactly one projection beyond the foundational atomic-units anchor. After the addition of the three intra-nuclear projections (charge-density, magnetization-density, tensor multipole), the asymmetry is no longer symmetric: $[L]$ admits* six *scalar instances plus one rank-2 tensor instance, while $[E]$ admits four. The framework records more length scales (atomic, internuclear, polyatomic, and three intra-nuclear distribution scales) than energy scales (Rydberg, UV cutoff, multi-loop subtraction, two-body Breit correction). The structural reason: every distinct spatial structure introduces an independent focal length because different intra-nuclear and intra-molecular geometries enter different Hamiltonian operators with different selection rules; spectral content has fewer distinct roles (foundational, regulator, subtraction, correction). The conjugate pair $[L] \leftrightarrow [E]$ remains the only multi-instance axis, but $[L]$ now dominates: physics has more places to put length scales than to put energy scales.*

Observation 4 (Compositional projections)**.** *The two-body Dirac / Breit retardation projection (§ 3.16) is the first identified instance of a structural family of* compositional projections*: projections whose variable is a function (typically a ratio or product) of variables introduced by prior projections. All other projections in Table 2 introduce variables independent of prior content; compositional projections are structurally distinct because they are* gated *by the prior instantiation of two or more distinct values of a foundational variable in the same projection chain.*

We catalogue two confirmed instances of compositional projections:

1. Kinematic composition. *Variable: rest-mass ratio m_l/m_n. Gating: prior instantiation of two distinct $[M]$ scales by separate rest-mass projections (§ 3.14). Manifests at α^4 in two-body bound-state energies, one full order before the α^5 multi-loop renormalization wall. Empirical anchor: positronium 1S–2S framework-native at +64.75 ppm (§ 5; sixth and structurally cleanest instance of the multi-focal-composition wall named in CLAUDE.md § 2).*

2. Angular composition. *Variable: ratio of effective charges Z_a/Z_b within a single atom (e.g., He 1s $Z_{eff} = 2$ vs. 2p $Z_{eff} = 1$). Gating: prior instantiation of two distinct foundational Fock anchors at different effective charges within the same electronic Hilbert space (the "internal multi-focal" pattern of § 5). Manifests at α^2 in fine-structure splittings, where the bipolar harmonic decomposition ($k_1{=}0, k_2{=}2$) direct + ($k_1{=}1, k_2{=}1$) exchange and the Drake combining coefficients ($\frac{3}{50}, -\frac{2}{5}, \frac{3}{2}, -1$) are the structural carrier. Empirical anchor: He 2^3P_0–2^3P_1 at -0.014% and 2^3P_0–2^3P_2 span at -0.20% (§ 5).*

The two instances differ in their gating mechanism (kinematic at the two-body level vs. angular at the two-orbital level) but share the compositional character: each requires the prior projection chain to have instantiated two distinct values of a foundational variable, so that the compositional projection's variable (ratio, product, or function) is well-defined.

The structural prediction the generalization makes: any future GeoVac observable whose computation requires a function of two or more foundationally-anchored variables in the same projection chain belongs to this family, and the order at which it manifests in α-perturbation theory follows the projection at which the function is gated. Candidate compositional projections not yet observed:

- *Ratio of two length scales within a single nucleus (e.g., r_{charge}/r_Z, the Friar/Foldy-versus-Zemach distinction inside one proton; would gate sub-Zemach corrections to hyperfine).*
- *Product of three rest-mass scales (three-body kinematic content beyond Breit retardation; would gate three-body bound-state corrections at α^4 or higher).*
- *Ratio of two temperature scales β_a/β_b in multi-window observation (e.g., simultaneous interrogation at two distinct T).*

The compositional-projection family is open-ended: confirmed instances locate the gating-order pattern (Breit at α^4, internal multi-focal at α^2), and additional empirically anchored instances would refine the structural prediction.

This decomposition sharpens Remark 3: V–D correlation holds at the compound dimension level, but at the base-unit level $[L]$ and $[E]$ each admit four instances while $[M]$, $[T]$ admit one each and $[Q]$ enters only as an integer label. The base-unit decomposition is a finer falsifiable test than the compound dimension alone: an observable that depends on multiple length scales must engage the foundational projection plus the relevant length-injecting projections; an observable that depends on temperature must engage the observation projection; an observable depending on a species mass ratio must engage two rest-mass projections plus Breit retardation. The empirical matches catalogue (§ 5) provides a forward test: each row's projection chain should be traceable to a base-unit signature consistent with the dimensions of the matched observable.

5 The Empirical Matches Catalogue

We list GeoVac quantities that match standard known physics values to machine precision or stated uncertainty. Each row is tagged by projection(s), variables involved, dimension, transcendental class, and match precision. This is a structural index, not an exhaustive benchmark report; the underlying numerical evidence lives in `tests/` and the cited papers.

Match	Projection(s)	Vars	Dim	Trans. class	Match
Zero-projection (Layer 1 only)					
Pauli count $N = 11.10 \cdot Q$	—	none	dimensionless	integer/rational	exact
Magic numbers $2, 8, 20, 28, \ldots$	HO + LS counting	none	dimensionless	integer	exact
Gaunt selection rules	$3j$ algebra	none	dimensionless	rational	exact
π-free certificate (Coulomb)	—	none	dimensionless	rational	exact
π-free certificate (Bargmann)	—	none	dimensionless	rational	exact
Angular sparsity 1.44% at $\ell_{\max} = 3$	$3j$ algebra	none	dimensionless	rational	exact
Discrete $c_1 = 0$ on Hopf-S^3 Fock graph at $n_{\max} = 2, 3$ (Condon–Shortley + Z_2 alternating)	—	none	dimensionless	integer	exact (Fraction)
Single-projection					
Hydrogen $E_n = -Z^2/(2n^2)$ Ha	Fock + κ	Z, E	energy	rational	exact symbolic
$2p$ doublet $\alpha^2/32$ Ha (Z=1)	spinor lift $\circ$ Fock	Z, α	energy	$(Z\alpha)^2 \cdot \mathbb{Q}$	exact symbolic
Dirac fine-structure formula ($n \leq 4$)	spinor lift $\circ$ Fock	Z, α	energy	$(Z\alpha)^2 \cdot \mathbb{Q}$	exact
$\Pi = 1/(48\pi^2)$ VP coefficient	spectral action	Λ, α	dimensionless	$\pi^2 \cdot \mathbb{Q}$	exact symbolic
$\beta(\alpha) = 2\alpha^2/(3\pi)$ one-loop	spectral action	α	dimensionless	$\pi \cdot \mathbb{Q}$	exact symbolic
D^2 heat kernel SD coefficients	spectral action	Λ	energy	$\sqrt{\pi} \cdot \mathbb{Q}$	exact symbolic
$F = \zeta_R(2) = D_{n^2}(d_{\max})$	Fock + Dirichlet at $s = 4$	graph data	dimensionless	$\pi^2 \cdot \mathbb{Q}$	exact symbolic
$\Delta^{-1} = g_3^{\text{Dirac}} = 40$	spinor lift	graph data	dimensionless	integer	exact
$B = 42$ Casimir trace at $m = 3$	—	graph data	dimensionless	integer	exact
$D_{\text{even}}(4)$ closed form	spinor lift (vertex selection)	α	dimensionless	$\pi^2, \pi^4, G, \beta(4)$	PSLQ 80 dps

Match	Projection(s)	Vars	Dim	Trans. class	Match
$E_{\mathrm{Cas}}^{S^3} = 1/240$ (conformal scalar)	Fock + rest-mass projection (conformal value $m^2 = 1$)	none	dimensionless	rational	exact symbolic
$E_{\mathrm{Cas}}^{S^3,\mathrm{Dirac}} = 17/480$ (full Dirac)	Fock + Camporesi–Higuchi spinor lift	none	dimensionless	rational	exact symbolic
$E_{\mathrm{Cas}}^{\mathrm{HO},S^5} = -17/3840$ (3D HO, Bargmann–Segal)	Bargmann–Segal + spectral action (ζ-regularization)	none	dimensionless	rational	exact symbolic
Hydrogen Bohr levels $E_n = -Z^2/(2n^2)$ Ha	Fock + energy-shell rescaling	Z, n	energy	rational	exact symbolic
Static Coulomb-like Green's fn $(L + I)^{-1}$ on Hopf S^3 at $n_{\max} = 2, 3$	Fock (Layer-1 matrix inverse, no continuous integration)	none	dimensionless	rational	exact symbolic at finite $n_{\max}$
Two-projection chain					
Stefan-Boltzmann $\pi^2/90$ on $S^3 \times S^1_\beta$ (high-T)	Fock + observation/temporal-window + spectral action (ζ-regularization)	T or β	energy density	$\pi^2 \cdot \mathbb{Q}$	exact symbolic
Uehling shift -27.13 MHz at $2S$	spectral action $\circ$ Fock	Z, α	energy	$\pi^2{\cdot}\mathbb{Q}$	$< 1\%$
21 cm hyperfine $A_{\mathrm{HF}} = 1418.84$ MHz Bohr-Fermi (HF-1, + external g_p)	spinor lift (Fermi-contact vertex) $\circ$ Fock $\circ$ rest-mass projection (m_p, proton)	$\alpha, g_p, m_e/m_p$	energy	$\alpha^2{\cdot}\mathbb{Q}$	-1102 ppm ($\approx -a_e$)
21 cm hyperfine $A_{\mathrm{HF}} = 1420.488$ MHz + Parker–Toms a_e (HF-2, + external g_p)	spinor lift (Fermi-contact vertex) $\circ$ Fock $\circ$ rest-mass projection (m_p, proton) $\circ$ spectral action (curvature expansion)	$\alpha, g_p, m_e/m_p$	energy	$\alpha^2{\cdot}\mathbb{Q}, c_1 \cdot \mathbb{Q}$	$+58$ ppm

Match	Projection(s)	Vars	Dim	Trans. class	Match
21 cm hyperfine $A_{\mathrm{HF}} = 1420.432$ MHz + external Zemach r_Z (HF-4)	HF-2 chain $\circ$ rest-mass projection $\circ$ magnetization-density (§ 3.18)	r_Z	energy	calibration	+18 ppm (Eides Tab. 7.3 consistent)
μH 1S Bohr–Fermi $\nu_F(\mu p) = 182.4433$ meV (Sprint MH Track B, + external g_p)	spinor lift (Fermi-contact vertex) $\circ$ Fock $\circ$ rest-mass projection (m_μ, muon, lepton swap) $\circ$ rest-mass projection (m_p, proton)	$\alpha, g_p, m_\mu/m_e, m_e/m_p$	energy	$\alpha^2 \cdot \mathbb{Q}$	+2 ppm vs Eides pure-QED $\nu_F = 182.443$ meV
μH 1S Zemach mass-enhancement factor 185.94× ep (Sprint MH Track B; operator-level Track C, May 2026)	rest-mass projection $\circ$ magnetization-density (§ 3.18) $\circ$ Fock; `lepton_mass` parameter on `MagnetizationDensitySpec` eliminates the manual mass-scaling step Track B used in test scripts (Pauli-string sum now carries correct lepton-mass scaling natively for downstream muonic VQE)	r_Z, m_{red}	dimensionless	calibration	0.55% vs Eides muonic target −7300 ppm; bit-identical to Track B manual scaling (intrinsic gap is subleading Eides corrections, not operator construction)
μH $2S$–$2P$ Uehling +205.0074 meV full kernel (Sprint MH Track A)	spectral action $\circ$ Fock $\circ$ rest-mass projection ($m_e \to m_\mu$)	$Z, \alpha, m_\mu/m_e$	energy	$\pi^2 \cdot \mathbb{Q}$	< 1 ppm vs Antognini 2013

Match	Projection(s)	Vars	Dim	Trans. class	Match
μd $2S$–$2P$ Lamb +202.63 meV framework + Layer-2 (multi-track Roothaan sprint Track 1, 2026-05-18; operator-level Roothaan autopsy in §5.2.4)	spectral action $\circ$ Fock $\circ$ rest-mass projection ($m_e \to m_\mu$, nucleus $m_p \to m_d$) + Krauth Layer-2 (multi-loop QED + recoil + Friar HO + deuteron polarizability)	$Z, \alpha, m_\mu/m_e, m_d/m_e, r_d$	energy	$\pi^2 \cdot \mathbb{Q}$, calibration	-0.12% vs CREMA 2016 +202.8785(34) meV (Pohl)
μd $2S$–$2P$ Friar leading -27.85 meV bare Eides at $r_d = 2.1413$ fm (Roothaan Track 1)	Fock $\circ$ charge-density (§3.17) $\circ$ rest-mass projection	$Z, \alpha, m_\mu/m_e, m_d/m_e, r_d$	energy	calibration $+\pi^2 \cdot \mathbb{Q}$ via $(Z\alpha)^4$	$+0.024\%$ vs Krauth 2016 -27.84 meV at r_d (bit-tight; scaling $(m_{\text{red}}/m_e)^3 (r_d/r_p)^2 = 7.577$ reproduced to 4 digits)

Match	Projection(s)	Vars	Dim	Trans. class	Match
$\mu^4\mathrm{He}^+$ $2S$–$2P$ Uehling +1665.77 meV full kernel (multi-track sprint Track B, 2026-05-19; operator-level Roothaan autopsy in §5.2.15)	spectral action ∘ Fock ∘ rest-mass projection ($m_e \to m_\mu$, $Z = 1 \to 2$, nucleus $m_p \to m_{^4\mathrm{He}}$)	$Z, \alpha, m_\mu/m_e, m_{^4\mathrm{He}}/m_e$	energy	$\pi^2{\cdot}\mathbb{Q}$	-0.032% vs Diepold 2018 +1666.305 meV at $\beta = 0.682$ (cleanest single-component muonic VP match at heavy Z; transports kernel verbatim from μH/μd)

Match	Projection(s)	Vars	Dim	Trans. class	Match
$\mu^4\mathrm{He}^+$ $2S$–$2P$ FNS Foldy -295.67 meV bare Eides at $r_\alpha = 1.6755$ fm (Roothaan Track B)	Fock ∘ charge-density (§3.17) ∘ rest-mass projection	$Z, \alpha, m_\mu/m_e, m_{^4\mathrm{He}}/m_e, r_\alpha$	energy	calibration $+\pi^2 \cdot \mathbb{Q}$ via $(Z\alpha)^4$	+2.9% vs Diepold leading, +4.0% vs Diepold full FNS ($Z = 1 \to 2$ amplifies higher-Friar absorption to ~ 11 meV; scaling $(m_{\mathrm{red}}/m_e)^3 (r_\alpha/r_p)^2 \cdot Z^4 = 80.45$ reproduced to 4 digits)
Mu 1S-2S transition 2,455,528,721 MHz (rest-mass rescaling from H experiment, sprint precision-catalogue 2026-05-08)	Fock ∘ rest-mass projection (lepton-unchanged, nucleus $m_p \to m_\mu$)	$\alpha, m_e, m_\mu/m_e$	frequency	$\alpha^2 \cdot \mathbb{Q}$	-0.11 ppm vs Mu-MASS Crivelli 2018
Mu 1S-2S transition 2,455,528,394 MHz (framework Bohr+SE + cross-register recoil)	Fock ∘ spectral action ∘ rest-mass projection ∘ cross-register V_{eN} (Roothaan)	$\alpha, m_e, m_\mu/m_e$	frequency	$\alpha^2 \cdot \mathbb{Q}$	-0.25 ppm vs Mu-MASS

Match	Projection(s)	Vars	Dim	Trans. class	Match
Mu $2S_{1/2}$-$2P_{1/2}$ Lamb shift 1047.63 MHz (framework one-loop QED with full Uehling, sprint precision-catalogue 2026-05-08)	Fock ∘ spectral action ∘ Sturmian ∘ rest-mass projection	$\alpha, m_e, m_\mu/m_e$	frequency	$\alpha^2 \cdot \mathbb{Q}$	+0.013% vs Karshenboim 2005 theory 1047.49(5) MHz
Bethe log $\ln k_0(2S) = 2.786$ (LS-3)	Sturmian (acc.) ∘ Fock	Z	dimensionless	calibration	-0.92%
Bethe log $\ln k_0(1S) = 3.002$ (LS-3)	Sturmian (acc.) ∘ Fock	Z	dimensionless	calibration	+0.60%
D 1S hyperfine $\nu_{\mathrm{HFS}}(D) = 327.2350$ MHz Bohr–Fermi strict (sprint precision-catalogue 2026-05-08, + external g_d)	spinor lift (Fermi-contact vertex) ∘ Fock ∘ rest-mass projection (universal m_e/m_p, g_d); $I = 1$ multiplicity 3/2 and the 1/2 convention divisor on $A_{\mathrm{hf}}\,\hat{I} \cdot \hat{S}$	$\alpha, g_d, m_e/m_p$	frequency	$\alpha^2 \cdot \mathbb{Q}$	-456 ppm vs Wineland–Ramsey 1972 327.384352522(2) MHz
T 1S hyperfine $\nu_{\mathrm{HFS}}(T) = 1515.865$ MHz Bohr–Fermi strict (multi-track sprint Track 2, 2026-05-18, + external g_t; see §5.2.8)	spinor lift (Fermi-contact vertex) ∘ Fock ∘ rest-mass projection (universal m_e/m_p, g_t); $I = 1/2$ multiplicity 1 shared with H 21cm on $A_{\mathrm{hf}}\,\hat{I} \cdot \hat{S}$	$\alpha, g_t, m_e/m_p$	frequency	$\alpha^2 \cdot \mathbb{Q}$	-551 ppm vs Greene 2017 1516.701470773(8) MHz; cleanest mass-vs-spin separation in catalogue (I=1/2 like H, $m_t \approx 3m_p$ like D-extended)

Match	Projection(s)	Vars	Dim	Trans. class	Match
Mu 1S hyperfine $\nu_{\text{HFS}}(\text{Mu}) = 4464.1905$ MHz framework Bohr–Fermi + Schwinger one-loop (sprint precision-catalogue 2026-05-08, muonium triple closure)	spinor lift (Fermi-contact vertex) $\circ$ Fock $\circ$ rest-mass projection (lepton-unchanged, nucleus $m_p \to m_\mu$); leptonic nucleus, no QCD content	$\alpha, m_e, m_\mu/m_e$	frequency	$\alpha^2{\cdot}\mathbb{Q}$	+199 ppm vs Liu 1999 4463.302765(53) MHz; residual cleanly LS-8a multi-loop QED + recoil (no QCD competing budget)
He 2^3P_0–2^3P_1 fine structure +29,612.91 MHz (Sprint 3 BF-D + Sprint 4 DD; Drake combining $(\frac{3}{50}, -\frac{2}{5}, \frac{3}{2}, -1)$, J-pattern from rank-$k$ 6j; sprint precision-catalogue 2026-05-08)	spinor lift $\circ$ Wigner $3j$ ($6j$-coupled) $\circ$ Wigner $3j$ (Gaunt) $\circ$ Fock; internal multi-focal (1s, 2p at distinct Z_{eff}), bipolar $(k_1{=}0, k_2{=}2)$ direct + $(k_1{=}1, k_2{=}1)$ exchange	α, Z	frequency	$\alpha^2{\cdot}\mathbb{Q}$	-0.014% vs NIST +29,616.951(6) MHz; below LS-8a α^3/π budget
He 2^3P_0–2^3P_2 span +31,844.07 MHz (Sprint 3 BF-D + Sprint 4 DD; sprint precision-catalogue 2026-05-08)	same chain	α, Z	frequency	$\alpha^2{\cdot}\mathbb{Q}$	-0.20% vs NIST +31,908.131(6) MHz; within LS-8a one-loop QED budget

Match	Projection(s)	Vars	Dim	Trans. class	Match
Ps 1S-2S transition 1,233,687,096.1 MHz (framework-native: Bohr at $m_{\text{red}}(ee) = 0.5$ + Eides § 3.2 SE Lamb-shift differential; sprint precision-catalogue 2026-05-08, equal-mass leptonic 1S-2S; + external α^4 Breit, Karshenboim 2005 § 9)	Fock ∘ rest-mass projection (equal-mass limit $\lambda_a = \lambda_b = 0.5$) ∘ spectral action (Eides)	α, m_e	frequency	$\alpha^2{\cdot}\mathbb{Q}$	+64.75 ppm vs Fee 1993 1,233,607,216.4(3.2) MHz
Three-projection chain					
Bethe log $\ln k_0(2P) = -0.0307$ (LS-4)	Drake–Swainson ∘ Sturmian ∘ Fock	Z	dimensionless	flow	+2.4% at $N = 40$
Bethe log $\ln k_0(3P) = -0.0406$ (LS-4)	Drake–Swainson ∘ Sturmian ∘ Fock	Z	dimensionless	flow	+6.3% at $N = 40$
Bethe log $\ln k_0(3D) = -0.005236$ (LS-4)	Drake–Swainson ∘ Sturmian ∘ Fock	Z	dimensionless	flow	−0.24% at $N = 40$
Lamb shift 1054 MHz (LS-4 N=40 sweet spot)	Fock ∘ spectral action ∘ Sturmian ∘ Drake–Swainson	Z, α	energy	calibration + flow	−0.39% (T+A cancellation)
Lamb shift 1029 MHz (LS-3)	Fock ∘ spectral action ∘ Sturmian (acc.)	Z, α	energy	calibration	−2.77% (T+A)
μH $2S$–$2P$ Lamb shift 202.17 meV (framework + literature, Sprint MH Track A)	Fock ∘ spectral action ∘ Sturmian ∘ Drake–Swainson ∘ rest-mass projection ($m_e \to m_\mu$)	$Z, \alpha, m_\mu/m_e, r_p$	energy	calibration + flow	−0.10% vs CREMA 202.3706(23) meV
Heisenberg–Euler 1-loop $\alpha^2/(45\pi m_e^4)$	spectral action (Schwinger Mellin) + vector-photon (one loop)	α, m_e	length^4· energy^{-1}	$\alpha^2 \cdot \mathbb{Q}/\pi$	structural (S)

Match	Projection(s)	Vars	Dim	Trans. class	Match
Lyman α emission $A_{21}(2P\to 1S) = 6.2683 \times 10^8$ s^{-1} (Sprint Calc-L, 2026-05-09)	Fock $\circ$ Wigner $3j$ $\circ$ vector-photon promotion	Z, α, ω	1/time	$\alpha^3 \cdot \mathbb{Q}[\sqrt{6}]$	+0.055% vs NIST 6.2649×10^8 s^{-1}; residual is non-relativistic (Schrödinger vs Dirac at $(Z\alpha)^2 \sim 5 \times 10^{-5}$), closes by adding Camporesi–Higuchi spinor lift to the chain (consistent with the Layer-2-presence bound)

Match	Projection(s)	Vars	Dim	Trans. class	Match
H 1S static dipole polarizability $\alpha_p(\mathrm{H}, 1S) = \frac{9}{2}\, a_0^3$ (Sprint Calc-P, 2026-05-09)	Fock $\circ$ Wigner $3j$ $\circ$ Sturmian reparam. at $\lambda = Z$	Z	length^3	rational	exact (Dalgarno–Lewis solution $F\|1s\rangle = -\frac{1}{2}z(r{+}2)\|1s\rangle$ has p-wave radial part lying exactly in $\mathrm{span}\{S_2, S_3\}$ at $N_{\mathrm{basis}} = 2$); Z^{-4} scaling exact at $Z = 1, 2, 3, 4$. Bound-states-only sum on bare Fock graph plateaus at $\sim$ 81.25% of $\frac{9}{2}$ (Bethe–Salpeter 1957 partition); the missing 18.75% continuum content is recovered *exactly* by the Sturmian projection

Match	Projection(s)	Vars	Dim	Trans. class	Match
He $2^1P \to 1^1S$ oscillator strength $f = 0.2761$ (Drake handbook; Hylleraas–Eckart Track 5 closure, 2026-05-18)	Fock $\circ$ Wigner $3j$ $\circ$ vector-photon promotion $\circ$ bipolar harmonic / Schwartz P-state $\circ$ Hylleraas r_{12}	Z, α, ω	dimensionless	$\alpha^2 \cdot \mathbb{Q}[\sqrt{6}, \mathrm{poly}(\beta)]$	$f = 0.2705$ vs Drake 0.2761, residual -2.02% at full Schwartz two-channel $\omega_s = 3$, $\omega_p = 2$; sub-mHa energy match on $E(1^1S)$ and $E(2^1P)$. Closes the 2-electron contact-density cliff (§5.2.10); validates internal multi-focal architecture (§3.22) at the 2^1P transition level.

Observation status

Match	Projection(s)	Vars	Dim	Trans. class	Match
$K = \pi(B+F-\Delta) \approx 1/\alpha$	Fock ∘ Hopf ∘ spinor	α	dimensionless	three-sector mix	4.8×10^{-7} (sum; 8.8×10^{-8} via P2 root)

Table 3: The empirical matches catalogue. Rows are grouped by projection depth. “Trans. class” uses Paper 18’s tier names. “Match” is the precision against standard known values: “exact” or “exact symbolic” means equality at machine precision; numerical percentages are residuals. The bottom row documents Paper 2’s α Observation explicitly as a multi-projection coincidence; the combination rule itself is a numerical observation without first-principles derivation (Paper 18 § III.5; Paper 2 status v2.26.1).

Remark 4 (signature $\neq$ value)**.** *Theorem 1’s signature consistency is a type-level statement: chains that produce the same signature can produce wildly different numerical values. For example, the hydrogen Bohr level $E_n = -Z^2/(2n^2)$ (Fock alone, signature {energy, no π}) and the hydrogen Lamb shift (Fock + spectral action + Sturmian + Drake–Swainson, signature {energy, π}) both have dimension {energy}, yet differ by 13 orders of magnitude. The Layer-2-presence bound of § 7 addresses the value-level question; signature arithmetic addresses only the type-level question.*

5.1 Off-precision matches and error-source classification

The catalogue above records matches at machine precision or stated sub-percent agreement. Equally informative is the catalogue of matches that are *off* precision, classified by the source of residual error. We propose five distinct error sources, each with its own scaling behaviour and remediation pathway:

- **Truncation (T):** finite $n_{\max}$, $\ell_{\max}$, or basis size N. Residual scales as a power law in the cutoff; remediated by extending the cutoff.
- **Basis quality (B):** structural limit of the chosen angular or radial basis at fixed cutoff. E.g. PK $\ell_{\max} = 2$ optimal operating point for composed LiH; higher ℓ degrades. Remediated only by changing basis architecture.
- **Approximation order (A):** leading-order treatment omitting higher-order perturbative terms. Residual is a known fraction of the standard physics (e.g. one-loop QED carries $\sim 5\%$ residual from α^5 multi-loop).
- **Calibration mismatch (C):** chained projections where the intermediate calibration constant for the wrong diagram topology is applied. Confirmed in the Schwinger $\alpha/(2\pi)$ graph-native sprint (VP-1, VP-2): the vacuum-polarization projection constant C_{VP} does not transfer to vertex correction.

- **Structural floor (S):** the framework cannot represent the relevant physics at the chosen layer. E.g. scalar Fock graph cannot represent vector-photon QED selection rules without projection promotion. Remediated by adding the missing projection to Layer 2.

Remark 5 (Error sources are projection-specific)**.** *The error-source classification is not orthogonal to the projection dictionary. Truncation (T) errors are properties of the spectral sums that the Sturmian or spectral-action projections invoke. Calibration mismatch (C) is a property of how multiple projections chain. Structural floors (S) appear where a needed projection is absent from Layer 2 entirely. Approximation order (A) is the "flow" tier of Paper 18: it is the residual after Layer 1 and Layer 2 do their work, and it indicates which higher-order projection or perturbation order should be added next.*

5.2 Roothaan autopsies: multi-component decompositions

The matches catalogue rows in §5 and §5.1 each capture a single observable as a single row. For observables that admit a multi-component literature decomposition — e.g. Eides Tab. 7.3 for the hydrogen Lamb shift, where the standard 1057.85 MHz value decomposes into one-loop SE, one-loop VP, multi-loop QED, finite nuclear size, recoil NLO, and hyperfine averaging — the single-row treatment hides the structure that the projection-chain dictionary makes visible. This subsection hosts *Roothaan autopsies*: decomposition tables that show each measured observable as a sum of focal-length-tagged components, with each component's projection chain made explicit.

The Roothaan autopsy is structurally the same kind of object Roothaan (1951) made for two-electron molecular integrals: a multi-component decomposition where each piece has its own focal length and its own closed-form structure. Applying the same discipline to multi-component *observables* makes the projection-chain dictionary's reach visible at the observable level, not just at the catalogue-row level. The autopsy partitions each component into framework-native (FN — computable from § 3 projections without external numerical constants beyond α) versus Layer-2 (L2 — parameter at operator level, projection chain framework-native), giving an explicit operator-level reading of the structural-skeleton scope statement (CLAUDE.md § 1.7). Several autopsies surface literature convention mismatches (class (a) findings of the multi-observable focal-length decomposition program); these are collected separately in § 5.3.

5.2.1 Hydrogen $2S_{1/2} - 2P_{1/2}$ Lamb shift autopsy

Reference. $\nu(2S_{1/2} - 2P_{1/2}) = 1057.845(9)$ MHz (Lundeen–Pipkin 1981; NIST ASD).

Framework-native subtotal: $+1057.19$ MHz (99.9% of measurement). **Layer-2 net:** $+1.20+0.138-2.40 \approx -1.06$ MHz. **The Layer-2 inputs do *not* cancel in this observable.** Until 2026-08-28 this paragraph reported a net of ≈ -0.02 MHz and called it an "empirical near-cancellation"; that reading was an artifact of the retired $+1.18$ MHz FNS entry and is **withdrawn**. The framework-native components alone still reproduce the measurement at $\approx 6 \times 10^{-4}$ precision — that claim is independent of the FNS row, which is a Layer-2 entry and does not enter the framework-native subtotal.

Multi-focal-composition wall visible. The two non-cancelling Layer-2 components correspond to:

- Component 5 (multi-loop QED) is the LS-8a renormalization-gap wall (CLAUDE.md § 1.7 LS-8a entry). The framework reproduces the two-loop SE structural prefactor $(\alpha/\pi)^2(Z\alpha)^4/n^3$ but cannot autonomously generate the $Z_2/\delta m$ counterterms.

- Component 7 (recoil NLO) is the §3.16 two-body kinematic wall. The framework's Breit retardation projection produces the α^4 kinematic structure but the R_{nl} coefficients enter as Layer-2 input.
- Component 6 (FNS) is *framework-native at the operator level* (the §3.17 promotion); the remaining Layer-2 input is the scalar r_p value, not the operator structure consuming it. (This bullet previously called it "the cleanest beneficiary" of that promotion; at +0.138 MHz rather than the retired +1.18 MHz it is a small contributor, and the showcase claim is withdrawn.)

Structural reading. The autopsy demonstrates Paper 34 §3's reach at finer grain than single-row catalogue entries: eight components, eight different projection chains, every Eides Tab. 7.3 row reproduced. The structural-skeleton scope statement is made concrete — framework-native pieces reproduce the observable to $\approx 6 \times 10^{-4}$, with the Layer-2 inputs (renormalization, Component 5; two-body kinematics, Component 7; FNS, Component 6) netting to -1.06 MHz — they do not cancel, per the withdrawal note above — and FNS now stated in projection-chain language rather than as a black-box external constant.

5.2.2 Hydrogen 21 cm hyperfine autopsy

Reference. $\nu_{\mathrm{HF}} = 1\,420\,405\,751.768$ Hz (Hellwig 1970; CODATA / NIST), the most precisely measured atomic transition.

Operator-level Zemach. Component 4 exercises §3.18 at the operator level via the production module `geovac.magnetization_density`. The bilinear matrix element $\langle \hat{r}_Z \rangle$ on a Sturmian register at the L=0 multipole reduces to $M_1[\rho_M] = r_Z$ at leading order in $r_Z/a_0 \sim 2 \times 10^{-5}$; the operator output -39.495 ppm at $r_Z = 1.045$ fm vs Eides Tab. 7.3 reference -39.500 ppm gives residual $+4.7 \times 10^{-3}$ ppm $= 0.012\%$ of the LO shift — the operator collapses to the Eides scalar at sub-percent precision automatically through the multipole reduction, without any external substitution. Profile independence verified: Gaussian and exponential ρ_M give bit-identical A_{HF} at LO since both are calibrated to first moment $M_1 = r_Z$.

Framework-native subtotal: $A_{\mathrm{HF}}^{\mathrm{native}} = 1420.4318$ MHz (+18.4 ppm above measurement). **Layer-2 net:** $-39.495 + 18.37 \approx -21.1$ ppm of ν_F. The Layer-2 input is the scalar r_Z value; the operator structure consuming it is framework-native via §3.18. This is the cleanest beneficiary of the W1b operator-level extension (Sprint Calc-rZG-extended, 2026-05-09), which promoted Zemach from analytic-substitution to operator-level.

Multi-focal-composition wall visible. The +18.37 ppm residual decomposes per Eides Tab. 7.3 as:

- Multi-loop QED ($\alpha^2(Z\alpha)$, $\sim +6$ ppm) is the LS-8a renormalization gap (CLAUDE.md § 1.7 LS-8a entry).
- Recoil NLO beyond reduced-mass (Bodwin–Yennie, $\sim +6$ ppm) is the W1a-D Roothaan kernel-level recoil-mixing wall.
- Nuclear polarizability Δ_{pol} (+1.4 ppm) and hadronic VP (+0.1 ppm) are W3 inner-factor calibration data (QCD-internal).
- Zemach NLO recoil-mixing $m_e/(m_e + m_p) \approx 5 \times 10^{-4}$ is structural noise in the electronic regime (+0.02 ppm) but the *dominant* systematic in the muonic regime ($f_{\mathrm{recoil}} \approx 0.092$, structural $\sim 10\%$). The W1b NLO opt-in flag covers this cleanly; cf. Sprint MH for the muonic analog.
- Convention drift on r_Z (Eides 1.045 vs Karshenboim 1.054 fm) shifts the Zemach line by 0.34 ppm (≈ 0.5 kHz) — small against the +18.4 ppm residual [corrected 2026-08-29: an

earlier version stated ±5 ppm and ranked this the largest class-(a) sensitivity; at 0.34 ppm it is among the smallest].

(The itemized entries above account for ≈ 13.5 of the +18.4 ppm; the remainder is unitemized higher-order content.)

Structural reading. The autopsy demonstrates Paper 34's multi-observable focal-length decomposition program (CLAUDE.md § 1.8) at the operator level on the most precisely measured atomic transition: four components, four different projection chains, +18 ppm residual attributable in large part (≈ 73% itemized) to LS-8a (multi-loop QED) + W1a-D (Roothaan recoil) + W3 (nuclear polarizability), as classified in CLAUDE.md § 1.7. The structural-skeleton scope statement is verified: the framework reproduces the observable to $\sim 2 \times 10^{-5}$ precision via four named projections, and the itemized ≈ 73% of the residual budget decomposes into already-named walls without surfacing any new class (a) literature convention mismatch beyond the known r_Z-value drift.

5.2.3 Muonic hydrogen $2S_{1/2}$–$2P_{1/2}$ Lamb shift autopsy

Reference. $\Delta E = 202.3706(23)$ meV (CREMA 2010, Pohl et al.; muonic convention $E(2P_{1/2}) - E(2S_{1/2})$).

The CREMA 2010 muonic hydrogen Lamb shift decomposes under five named projection-chain components plus three companion Layer-2 inputs.

Headline framework computation. The full Uehling integration matches the Antognini 2013 / Pachucki 1996 reference at −0.10 ppm — the cleanest single-component match in the muonic catalogue. The $\beta = 2/(m_{\mathrm{red},\mu}\,\alpha) = 1.475$ muonic regime, where the muonic Bohr scale and the e^+e^- Compton wavelength are comparable, is handled correctly by the full kernel where the contact-form Uehling overshoots by $\sim 3.6\times$. This is the muonic-regime analog of the electronic Uehling row in §5.2.1 (Component 2 of §5), where contact form is sufficient because $\beta_{\mathrm{ep}} \approx 274$.

Convention-mismatch surface. Antognini 2013 and Krauth 2017 itemizations agree on the bottom-line total but differ at $\sim$ 100 ppm in per-component split (VP one-loop and SE-recoil aggregation). Specifically, Krauth absorbs $\sim$ 0.021 meV of higher-order VP into the one-loop line (+205.0282 meV), while Antognini itemizes it separately as "VP-VP mixed" (+0.150 meV), with one-loop reported at +205.0074 meV. The framework's −0.10 ppm match against Antognini is 1000× tighter than the 102 ppm Antognini-vs-Krauth gap. At this precision level the choice of Layer-2 reference compilation matters; this autopsy anchors throughout to Antognini 2013. See § 5.3 for catalogue context (entry § 5.3.2).

Operator-level Friar moment. Computed via the §3.18 module `geovac.magnetization_density` with `lepton_mass` $= m_{\mathrm{red},\mu}$. The operator-level result −4.33 meV vs closed-form −3.68 meV reflects the Gaussian profile's $\langle r^2\rangle/\langle r\rangle^2 = 3\pi/8 \approx 1.178$ factor (4.33/3.68 = 1.1766, matching to 0.1%; an earlier version quoted the two-dimensional constant $4/\pi \approx 1.27$, which fails by 8%) — a profile-convention issue between calibrating to the first moment $\langle r\rangle$ versus the RMS radius $\sqrt{\langle r^2\rangle}$. Eides Eq. 2.35 closed-form derivation takes $\langle r^2\rangle^{1/2} = r_p$ (RMS radius); the operator-level realization via §3.18 takes $\langle r\rangle = r_p$ (first moment). The discrepancy is exactly the Gaussian profile factor. The flagged extension is a sibling `MagnetizationDensitySpec` calibrated to RMS radius for the Lamb-shift FNS channel, distinct from the existing first-moment calibration for the HFS Zemach channel. See § 5.3 for catalogue context (entry § 5.3.4).

Structural reading. The autopsy realizes the structural-skeleton-scope statement at the operator level for a cross-register multi-focal observable in the muonic regime. Three framework-native components reproduce +200.50 meV (the five-projection-chain skeleton); three Layer-2 inputs close the remaining +1.67 meV gap, attributable cleanly to LS-8a multi-loop renormalization (Component 4) and W3 inner-factor (Component 5). The same architecture that closes the H 1S Lamb shift at §5.2.1 closes μH $2S$–$2P$ here, with Uehling-kernel kinematics (contact-form $\to$ full kernel) and Friar-profile calibration (first moment $\to$ RMS) being the only two ingredients that change between regimes.

5.2.4 Muonic deuterium $2S_{1/2}$–$2P_{1/2}$ Lamb shift autopsy

Reference. $\Delta E = 202.8785(34)$ meV (CREMA 2016, Pohl et al., *Science* **353**, 669; muonic convention $E(2P_{1/2}) - E(2S_{1/2})$).

The CREMA 2016 muonic deuterium Lamb shift decomposes under the same five-component skeleton as the muonic hydrogen autopsy (§5.2.3), with three architectural substitutions: (a) nuclear charge radius $r_p \to r_d = 2.1413(25)$ fm (factor 2.547 in linear size, 6.487 in r^2); (b) reduced mass $m_{\mathrm{red},\mu p} \to m_{\mathrm{red},\mu d} = 195.74\, m_e$ (factor 1.053, mildly larger via the heavier nucleus); (c) dominant Layer-2 input shifts from Källén–Sabry two-loop VP (at μH) to deuteron polarizability ($\sim 130\times$ larger than proton polarizability, dominant at μd).

Headline framework computation. The Friar moment leading order via the closed-form Eides Eq. 2.35 matches the Krauth 2016 tabulated leading line at the deuteron's measured radius at +0.024% — the cleanest framework-vs-literature match for the nuclear-structure leading line across the muonic catalogue. The scaling

$$\frac{\Delta E_{\mathrm{FNS}}(\mu d)}{\Delta E_{\mathrm{FNS}}(\mu p)} = \left(\frac{m_{\mathrm{red},\mu d}}{m_{\mathrm{red},\mu p}}\right)^3 \left(\frac{r_d}{r_p}\right)^2 = 1.168 \times 6.487 = 7.577 \tag{1}$$

is reproduced by the framework's bare Eides calculation to 4 digits. This confirms the rest-mass projection (§3.14) transports the Friar moment leading-order calculation cleanly across nuclear-isotope variation.

Same sub-leading SE recoil-mixing gap as μH. The framework's self-energy leading-order rest-mass scaling overshoots Krauth's catalogued SE Lamb-shift contribution by +0.25 meV at μd (+41% fractional) vs +0.16 meV at μH (+24% fractional). The *absolute* gap grows by $1.5\times$ between isotopes, while the *fractional* gap grows from +24% to +41% because Krauth's tabulated SE shrinks slightly more. Same physical mechanism (sub-leading $\alpha(Z\alpha)^4 \times (m_{\mathrm{red}}/m_n)$ recoil-mixing pieces, Eides § 4.2, Pachucki 1996, omitted at the leading-order rest-mass projection); the gap is diagnostically robust under nuclear-isotope variation. Phase C-W1a-physics extension at the SE-bracket level would close both observables simultaneously.

Dominant Layer-2 shifts from LS-8a to W3 at μd. In μH, the dominant Layer-2 input is the Källén–Sabry two-loop VP (+1.51 meV, LS-8a renormalization wall). In μd, the dominant Layer-2 input is the deuteron polarizability (+1.69 meV, W3 inner-factor wall) — $\sim 130\times$ larger than the proton polarizability (+0.013 meV in μH) because the deuteron is a weakly bound $n + p$ system whose virtual photonuclear excitations sit at MeV scale. This structural feature (different walls dominate at different nuclei) is direct empirical evidence for the multi-focal-composition wall taxonomy: different observables probe different walls at different relative weights, even within the same projection chain.

Convention-mismatch surface. Carlson–Gorchtein–Vanderhaeghen 2014 catalogues the deuteron polarizability at $+1.690 \pm 0.020$ meV; the older Carlson–Vanderhaeghen 2011 evaluation reported +1.94 meV — a $\sim 13\%$ spread on the *dominant* Layer-2 line at μd. At the framework's -0.12% Lamb-shift residual scale this is the controlling uncertainty. *Flagged as a new V.D-prediction candidate*, class (i) inter-compilation itemization, sibling to §5.3.5 (which addresses sub-component itemization of deuteron HFS polarizability across Friar–Payne 2005 and Pachucki–Yerokhin 2010+).

Structural reading. The autopsy validates the multi-focal architecture under nuclear-isotope substitution at the sub-percent precision level on a precision-frontier observable. Three framework-native components reproduce +198.91 meV; four Layer-2 inputs close the remaining +3.71 meV gap, with the dominant contribution now coming from the W3 inner-factor (deuteron polarizability, QCD-internal NN dynamics) rather than the LS-8a multi-loop QED that dominated at μH. The rest-mass projection (§3.14) transports the entire skeleton (Uehling integrator, SE bracket, Friar leading) verbatim; only the nuclear charge-radius input ($r_p \to r_d$) and the polarizability Layer-2 input change between isotopes. Same architecture, same sub-percent residual (-0.10% at μH, -0.12% at μd), different dominant Layer-2 wall.

5.2.5 He$^+$ $2S_{1/2}$–$2P_{1/2}$ Lamb shift autopsy (Z-scaling test of Paper 36)

Reference. $\nu(2S_{1/2} - 2P_{1/2}) = 14\,041.13(17)$ MHz (van Wijngaarden–Holuj–Drake, *Phys. Rev. A* **63**, 012505 (2000); theoretical reference 14 041.18(13) MHz from Pachucki *Phys. Rev. A* **63**, 042503 (2001) and Drake–Yan *Phys. Rev. A* **46**, 2378 (1992) consistent to ~ 50 kHz). Convention $E(2S_{1/2}) - E(2P_{1/2}) > 0$ matches Paper 36 normal-H.

The He$^+$ Lamb shift is the cleanest Z-scaling test of the Paper 36 one-loop QED architecture (§5.2.1): identical lepton (electron), identical projection chains, only the nuclear charge $Z = 2$ and the nuclear input parameters $m_\alpha = 7294.30\, m_e$, $r_E(^4\mathrm{He}) = 1.6755$ fm change. ^{4}He has $I = 0$, so the entire §3.18 channel is identically zero — no HFS, no Zemach radius, no hyperfine averaging. The Uehling parameter $\beta = 2/(Z\, m_{\mathrm{red}}\, \alpha) = 137.05$ sits in the same large-β regime as H ($\beta = 274$), so contact-form vacuum polarization is valid to ∼ppm and the full Uehling kernel matches contact form at 0.5%.

Z-scaling structural verification (the load-bearing test). Three component ratios He$^+$/H test whether the projection-chain machinery transfers cleanly Z=1 → Z=2:

- Lamb$_{\mathrm{SE}}$ ratio = 13.211 (vs naive $Z^4 = 16$). The deviation from 16 is structurally the $(4/3)\ln(1/(Z\alpha)^2)$ piece of the Eides bracket, which shrinks by $(4/3)\ln(4) \approx +1.85$ at Z=2. Framework predicts $16 \times \mathrm{bracket}(Z = 2)/\mathrm{bracket}(Z = 1) = 16 \times 0.8237 \approx 13.18$, matching observed 13.21 to 0.3%.
- Lamb$_{\mathrm{VP}}$ ratio = 15.71 from the printed rows (-426.2096 full-kernel He$^+$ over -27.13 contact-form H; vs naive $Z^4 = 16$). The 1.8% deviation mixes the β-dependent kernel integral structure with a contact-vs-full-kernel mismatch between the two rows [corrected 2026-08-29: an earlier version quoted 15.878/0.8%, which does not reproduce from the printed rows].
- Lamb$_{\mathrm{FNS}}$ ratio = 63.599 (vs predicted $Z^4 \times (r_E(^4\mathrm{He})/r_p)^2 = 16 \times (1.9925)^2 = 63.52$). Matches to 0.12%, confirming the §3.17 projection Z-scales without architectural modification.

All three structural tests pass. The projection chains are **Z-portable**: substituting $Z = 2$ into the existing pipeline gives He$^+$ with no architectural changes. This is what Paper 34's projection taxonomy was designed to guarantee — projections describe physics, not specific Z values.

Residual attribution. The framework-native subtotal is -200.02 MHz / -1.42% below experiment, which is structurally the same as Paper 36's -5.65 MHz / -0.55% at Z=1 amplified by $Z^4 \times$ ln-amplification. The dominant Layer-2 contribution is the $\alpha(Z\alpha)^6 \ln(Z\alpha)^{-2}$ Drake-Yan 1992 relativistic- Bethe-log correction (+14.7 MHz), which is below precision visibility at Z=1 ($\sim +0.05$ MHz, absorbed into Paper 36's residual) but structurally visible at Z=2. The remaining ~ 180 MHz of the deficit comes from higher-order $\alpha(Z\alpha)^n_{\ln}$ mixed terms that are extensively itemized in Drake-Yan 1992 and Pachucki 2001 but live below the α^3-truncation visibility at H precision.

Cleanest W3 inner-factor contribution in the catalogue. ^{4}He polarizability contributes +0.06 MHz / 4.3×10^{-6} of the 2S Lamb shift [corrected 2026-08-29: an earlier version stated 4×10^{-9}, inconsistent with its own +0.06 MHz row by three orders], the smallest W3 contribution in the catalogue (cf. μd Lamb where deuteron polarizability is +1.69 meV, the dominant Layer-2 line). Mechanism: ^{4}He is tightly bound (binding energy $\sim$ 28.3 MeV total, 7.1 per nucleon, vs the deuteron's 2.2 MeV total), so virtual photonuclear excitations are far above the Lamb-shift scale. He$^+$ Lamb is therefore the cleanest QED-at-Z=2 test with nuclear-structure contamination at the few-ppm level — the regime where the §3.6 LS-8a multi-loop QED becomes the dominant Layer-2 contribution rather than nuclear-structure inputs.

Convention-mismatch surface. Different compilations (Drake-Yan 1992, Pachucki 2001, Eides Tab. 7.6) split the leading SE bracket and the higher-order $\alpha(Z\alpha)^n$ ln corrections differently at $Z = 2$. The framework's α^3 Eides bracket is unambiguous; what depends on compilation is the partition of the Layer-2 residual between "SE α^3 deficit" (in the Pachucki convention) and "$\alpha(Z\alpha)^6$ ln Drake-Yan relativistic-Bethe-log correction" (in the Eides convention). Same total, different per-component itemization. See § 5.3 for catalogue context.

Structural reading. The autopsy demonstrates three things. First, the framework's projection-chain machinery is *Z-portable* at the structural level: all three Z-scaling tests pass cleanly, with the SE bracket's logarithmic Z-dependence reproduced to 0.3% (a matching test, not an exactness claim; the VP ratio is kernel-mixed, see above). Second, the −1.42% native residual is the Z-amplified analog of Paper 36's −0.55% Z=1 residual — same physical content (α^3 leading-order truncation), different size by $Z^4 \times$ ln. Third, He$^+$ joins the multi-focal precision catalogue as the cleanest QED-at-Z=2 test, with nuclear-structure Layer-2 inputs at the few-ppm level (4.3×10^{-6} of the 2S Lamb shift). This is the canonical Z-scaling validation of the Paper 36 one-loop QED closure.

5.2.6 Helium 2^3P fine-structure autopsy (first internal multi-focal entry)

Reference. $\nu(2^3P_0 - 2^3P_1) = +29\,616.951(6)$ MHz, $\nu(2^3P_1 - 2^3P_2) = +2\,291.176(15)$ MHz, $\nu(2^3P_0 - 2^3P_2) = +31\,908.131(6)$ MHz (NIST compilation; Pachucki–Yerokhin 2010 theoretical at sub-kHz vs experiment).

This is the first *internal multi-focal* Roothaan autopsy: two electrons on the same nucleus at structurally distinct effective nuclear charges ($Z_{\text{eff}}(1s) = 2$ full-nuclear, $Z_{\text{eff}}(2p) = 1$ full-shield). Distinct from the cross-register multi-focal autopsies of §5.2.1 (electron + I=1/2 nucleus) and §5.2.3 (electron + muonic nucleus), which couple particles on different registers.

The operator-level Drake (1971) decomposition gives three components:

$$E(^3P_J) = \tfrac{\zeta_{2p}}{2}\, X(J) + A_{SS}\, f_{SS}(J) + A_{SOO}\, f_{SOO}(J) \tag{2}$$

with $X(J) = J(J+1) - L(L+1) - S(S+1)$, $f_{SS}(J) = (-2, +1, -\frac{1}{5})$ from the rank-2 6j symbol $(-1)^{L+S+J} \cdot 6j\{1, 1, J; 1, 1, 2\}$ at $L = S = 1$ (Sprint 4 DD, sympy-exact), $f_{SOO}(J) = (+2, +1, -1)$ from the rank-1 6j, and Drake combining coefficients $A_{SS} = \alpha^2(\frac{3}{50}\, M^2_{\text{dir}} - \frac{2}{5}\, M^2_{\text{exch}})$, $A_{SOO} = \alpha^2(\frac{3}{2}\, M^1_{\text{dir}} - M^1_{\text{exch}})$ (Sprint 3 BF-D, intrinsic-tier rationals).

Bipolar (k_1, k_2) decomposition. For the (1s)(2p) configuration with $L_{\max} = 2l_{\max} = 2$: SS (K=2) admits direct ($\mathrm{k}_1 = 0$, $\mathrm{k}_2 = 2$) and exchange ($\mathrm{k}_1 = 1$, $\mathrm{k}_2 = 1$) channels only; SOO (K=1) admits exchange ($\mathrm{k}_1 = 1$, $\mathrm{k}_2 = 1$) only (the direct path is empty by parity selection). All higher (k_1, k_2) channels are Gaunt-forbidden. This is the INTERNAL analog of the cross-register Roothaan termination (B-W1a-diag finding): the multipole termination at $L_{\max} = 2l_{\max}$ holds *independent of the focal-length ratio* $Z_{\text{eff}}(1s)/Z_{\text{eff}}(2p) = 2$ — the channel structure is angular content, not radial content.

Partial-cancellation mechanism (P_1-P_2 amplification). The small interval is the difference SO (−58.4 GHz) + SS (−9.5 GHz) + SOO (+70.1 GHz) = +2.2 GHz, a structural cancellation by factor 61.9×. The framework's 60 MHz absolute residual on $P_1 - P_2$ is the same order as the 64

MHz residual on $P_0 - P_2$; the -2.62% fractional residual is the 60 MHz absolute divided by the 14×-smaller small interval. Generic feature of multi-component fine-structure splittings (Li 2^2P and Be $2s2p\,^3P$ show analogous partial-cancellation amplification, Sprint 5 CP).

Three-class diagnosis (per CLAUDE.md § 1.8 directive). Class A (literature convention mismatch): NEGATIVE at LO Breit-Pauli — Drake (1971) and Pachucki–Yerokhin (2010) agree on leading-order Breit-Pauli combining coefficients; **but POSITIVE at the $\alpha^3(Z\alpha)^2$ multi-loop level** (V.D-prediction sprint, 2026-05-10) where Pachucki–Yerokhin 2009 reevaluated the relativistic Bethe-log term to resolve a 3σ disagreement on ν_{01} vs Drake's earlier itemization (cross-ref §5.3.6). The convention exposure sits inside the framework's LS-8a $\alpha^3(Z\alpha)^2$ multi-loop wall, NOT at the LO Breit-Pauli sector that the framework computes sympy-exactly. Class B (framework gap, RADIAL): POSITIVE — residual attributes to (i) multi-electron correlation in the (1s)(2p) configuration (largest source, $\sim$ 45 MHz on P_0-P_2 scale, not in framework's single-configuration hydrogenic basis), (ii) $\alpha^3(Z\alpha)^2$ QED on Breit-Pauli (LS-8a vertex sector, $\sim$ 15 MHz), (iii) α^3 recoil (W1a cross-register kinematic wall, $\sim$ 2 MHz), (iv) higher-order $\alpha^4 + \alpha^5 + \alpha^6$ NRQED (LS-8a multi-loop, sub-MHz). Class C (focal-length cataloguing): POSITIVE — partial-cancellation amplification is generic across atomic fine-structure splittings.

Structural reading. The autopsy realizes the structural-skeleton-scope statement at the operator level for an internal multi-focal observable. Five projection chains, three operators, sympy-exact angular content, residuals in the radial sector — the same architecture that closes hydrogen Lamb shift at §5.2.1 closes He 2^3P fine structure here. The internal multi-focal angular-only prediction (Observation 3) holds at machine precision: angular Roothaan termination is independent of focal-length ratio.

5.2.7 Deuterium 1S hyperfine autopsy

Reference. $\nu_{\mathrm{HFS}}(D) = 327\,384\,352.522$ Hz (Wineland–Ramsey 1972), the most precisely measured I=1 atomic HFS transition. Distinguished from §5.2.2 by the nuclear-spin axis: I=1 (deuteron) vs I=1/2 (proton). Distinguished from §5.2.6 by being a cross-register (electron + I=1 nucleus) rather than internal (two electrons on the same I=1/2 nucleus) multi-focal observable.

Operator-level Zemach at I=1. Component 4 exercises §3.18 at the operator level via `geovac.magnetization_density.hydrogen_zemach_eides_leading_order` with $r_Z(D) = 2.593$ fm (Friar–Payne 2005). The bilinear matrix element $\langle \hat{r}_Z \rangle$ on the Sturmian register at L=0 multipole reduces to $M_1[\rho_M] = r_Z$ to machine precision; the operator output -98.001197 ppm vs Eides analytic $-2Zm_e r_Z = -98.001197$ ppm gives reproduction residual $+1.42 \times 10^{-14}$ ppm $= 1.45 \times 10^{-14}\%$ of the LO shift — *machine precision.* No architecture change vs the H 21cm operator-level Zemach in §5.2.2; only the scalar parameter $r_Z(D) = 2.593$ fm replaces $r_Z(p) = 1.045$ fm. Profile (Gaussian vs exponential) independence at machine precision. The Pauli encoding is 4 non-identity terms $(II, Z_e, Z_p, Z_e Z_p)$, the same minimal sparse encoding as for H 21cm — confirming the §3.18 operator does not depend on nuclear spin (it operates on the spatial register only).

Operator-level $\hat{I} \cdot \hat{S}$ Hamiltonian at I=1. The $\hat{I} \cdot \hat{S}$ operator at $I = 1$, $J = 1/2$ has eigenvalues $\{+1/2, -1\}$ on a 6-dim joint nuclear-electronic Hilbert space $(2I+1)(2J+1) = 3 \cdot 2 = 6$. The F=3/2 to F=1/2 splitting is $\frac{1}{2} - (-1) = 3/2$, exactly the Clebsch–Gordan output. At $I = 1/2$ the eigenvalues are $\{+1/4, -3/4\}$ with splitting 1. Multiplicity ratio D/H $= 3/2 = 1.500000000000$ at machine precision. The Pauli encoding extends from $Q_{\mathrm{tot}} = 2$ qubits, 3 terms (I=1/2) to $Q_{\mathrm{tot}} = 3$ qubits, 10 terms (I=1) — one extra qubit for the larger nuclear-spin space, no new structural mechanism. **Operator-level verdict: leading_order_I_independent**, the elevation of the May 8 precision-catalogue verdict from energy level to operator level.

Framework-native subtotal: $\nu_{\mathrm{HFS}}^{\mathrm{native}}(D) = 327.3153$ MHz (99.979% of measurement). **Layer-**

2 net: -210.9 ppm of ν_F (the framework chain *under*shoots). **Correction, 2026-08-28:** this autopsy previously reported a $+285.6$ ppm residual from a Bohr–Fermi baseline of 327.397464 MHz. That baseline was high by 496.5 ppm: the $g^{\text{atomic}} = 2\mu_I/\mu_N$ convention is twice the CODATA g-factor $g_{\text{CODATA}} = \mu_I/(I\mu_N)$, which is what the A-constant requires; the drivers kept the doubled form and additionally used m_e/m_N where the Fermi-contact mass factor is m_e/m_p for every nucleus (the nuclear magneton is defined with the proton mass). The net is $2(m_p/m_N)$, and at $I = 1$ the deuteron's $m_d/m_p = 1.99900750$ nearly cancels the doubling — which is why the error left the baseline looking right to three digits. H 21cm and T 1S HFS were unaffected because their drivers used g_{CODATA} and m_e/m_p directly; the escape is a property of those drivers, not of $I = 1/2$ (He-3 is also $I = 1/2$ and *is* affected, by the factor 0.6682 recorded in §5.2.9). *Correction 2026-08-28:* an earlier version of this note gave the divisor as $2I$ rather than 2; the two agree only at $I = 1$, the case it was derived from. Cross-anchoring the three baselines against each other now reproduces T to 0.02 ppm. **The prior claim that the $+285.6$ ppm residual matched the -286 ppm PY 2010 Layer-2 budget magnitude is withdrawn**; the itemization does not close at the ± 25 ppm level. What replaces it is a sharper structural statement.

Why it does not close: the deuteron's nuclear-structure correction has the opposite sign from the proton's. Dividing each chain by its own Fermi value ν_F (baseline $\times$ reduced mass) and removing the Schwinger a_e factor isolates the net nuclear-structure correction each observable requires:

Observable	Net structure required	Leading Zemach applied	Gap
H 21 cm	-55.9 ppm	-39.5 ppm	-16.4 ppm
T 1S	-63.2 ppm	-66.6 ppm†	$+3.3$ ppm†
D 1S	**+112.9 ppm**	-98.0 ppm	**+210.9 ppm**

† computed at the catalogue's current $r_Z(t) = 1.762$ fm, an input under review (see text). The Gap column is computed from each autopsy's *component* rows (baseline, reduced mass, a_e, Zemach) and equals minus that chain's residual. For H it reads -16.4 ppm against the $+18.4$ ppm printed in §5.2.1's sibling 21 cm table: the printed *final* there (1420.4318) is itself ≈ 2 ppm above what its own components give (1420.4289), a pre-existing inconsistency this table exposes rather than introduces. The sign argument does not depend on it.

For the tightly bound $I = 1/2$ nuclei the required correction is *negative* and the leading Zemach term carries the right sign, capturing 71% of it for hydrogen. (The tritium row is quoted at the $r_Z(t)$ value this catalogue currently uses, 1.762 fm; that input is under review — it coincides with the triton *charge* radius 1.7591 fm to 0.2%, and the proton's $r_Z/r_{\text{charge}} = 1.243$ would put the Zemach radius nearer 2.2 fm, moving the applied shift to ≈ -86 ppm and the gap to $\approx +23$ ppm. The sign of the required correction, which is what the comparison rests on, does not depend on r_Z at all.) For the deuteron the required correction is *positive*: the deuteron is bound by only 2.2 MeV and is correspondingly polarizable, so its polarizability contribution to the hyperfine splitting outweighs the negative Zemach term and reverses the sign of the total. A leading-Zemach-only treatment is therefore not merely incomplete at $I = 1$ — it points the wrong way. The $+210.9$ ppm gap is everything the chain omits — non-Zemach nuclear structure (expected to dominate), multi-loop QED, recoil NLO and finite-size charge — not nuclear structure alone.

This is a boundary of the §3.18 projection, not a bookkeeping error: the projection consumes a single scalar r_Z, which is sufficient where Zemach dominates (H, T) and structurally insufficient where polarizability does (D). Two consequences follow. (i) The $+44$ ppm deuteron-polarizability figure quoted elsewhere in this catalogue is **withdrawn as a unit slip**: the primary literature's total two-photon-exchange (TPE) nuclear-structure correction for electronic D 1S is 44.5(1.1) *kHz*

(Bonilla *et al.*, arXiv:2508.18776, Tab. IV), which on a 327.384 MHz splitting is +135.9 ppm — the numeral 44 was carried across from the kHz total, and it is a *total*, not a polarizability. (ii) Closing the D itemization needs a polarizability channel in the projection dictionary, not a better r_Z — which is the honest open item, and a sharper one than the coincidental magnitude match it replaces.

What the channel has to deliver (2026-08-29, sourced). The primary decomposition is now available and confirms the mechanism this subsection infers from the sign contrast. Bonilla *et al.* split the electronic-D TPE into elastic −41.50(11) kHz, polarizability +110.16(38) kHz, single-proton −33.14(7) kHz and single-neutron +8.95(93) kHz, totalling +44.5(1.1) kHz. In fractions of the splitting: the inelastic (polarizability) channel is +336.5 ppm and *all* elastic-class pieces together are −200.6 ppm, so polarizability outweighs the elastic terms and flips the sign of the total to +135.9 ppm. That is precisely the mechanism argued above, with magnitudes now sourced rather than inferred.

Two quantitative consequences. First, the framework requires +112.9 ppm of total nuclear structure and the literature supplies +135.9 ppm; the +23.0 ppm difference is what this chain omits in *higher-order QED* (it carries Schwinger a_e and reduced mass only). Second — and this is the check that makes the reading non-circular — the H 21cm chain is architecturally identical and, with its own nuclear-structure input consumed, lands at +18.4 ppm (§5.2.2). So a D chain consuming the sourced TPE would sit at +23.0 ppm: the same sign, the same tens-of-ppm class, and the same missing-QED explanation as hydrogen. The $I = 1$ anomaly is then not an anomaly at all — it is the absence of a channel, quantified. *Tier:* MEASURED (the framework side) against a cited Layer-2 evaluation; the +23.0 ppm attribution to higher-order QED is an inference from the H parallel, not an itemization. The Layer-2 inputs are the scalar $r_Z(D)$ value; the PY 2010 budget itemization [**2026-08-28: listed here historically, but the chain does not consume it — the only Layer-2 scalar entering is $r_Z(D)$. This matters for Prediction 2: with $r_Z(D)$ alone as the input, the corrected −210.9 ppm residual is not bounded by $\max(\varepsilon_{\mathbf{basis}}, |L_2|)$ on the ppm-class reading, so this row is an open case for that Prediction rather than a supporting one.**] (−286 ppm convention; alternate Eides Tab 7.3 gives −150 ppm, the source of the well-known ~ 5 fm extraction artifact in the global rZG fit, see §5.3.1). The operator structure consuming the inputs is framework-native via §3.18, identical to the H 21cm autopsy.

Why D Layer-2 is deeper than H Layer-2. The cumulative residual was itemized per Pachucki–Yerokhin 2010 as follows. [**Withdrawn 2026-08-28: these entries sum to $\approx +285$ ppm, the retired residual. The corrected residual is −210.9 ppm and the itemization does not close; the list is retained only as the historical budget it was matched against. The +200 ppm polarizability entry below has since been re-sourced (2026-08-29): its near-coincidence with the +210.9 ppm requirement was accidental, and the primary value is +336.5 ppm with the *total* TPE at +135.9 ppm.**]

- Deuteron polarizability ($\sim +200$ ppm) is the dominant component, $\sim 30\times$ the proton's contribution. [**Re-sourced 2026-08-29: the primary value is +110.16(38) kHz = +336.5 ppm (Bonilla *et al.*, arXiv:2508.18776); the $\sim +200$ figure is neither that nor the +135.9 ppm total. Dominance and sign stand; the magnitude does not.**] This is QCD-internal NN dynamics (W3 inner-factor wall, Paper 18 § IV.6), fundamentally because the deuteron is a weakly bound n+p system with much larger spatial extent ($r_Z(D) = 2.5\, r_Z(p)$).
- Multi-loop QED ($\alpha^2(Z\alpha)$, $\sim +40$ ppm) is the LS-8a renormalization gap, $\sim 7\times$ H scaling because of the larger nuclear extent's $\langle r\rangle$ couplings.
- Recoil NLO beyond reduced-mass ($\sim +30$ ppm) is the W1a-D Roothaan kernel-level recoil-mixing wall.

- Finite-size charge ($\sim +10$ ppm) and higher Friar moments ($\sim +5$ ppm) are Layer-2 inputs via §3.17 and §3.18 NLO opt-in.
- Zemach NLO recoil-mixing $m_e/(m_e + m_d) \approx 2.7 \times 10^{-4}$ is structural noise in the electronic regime (+0.028 ppm).
- Convention drift (± 25 ppm; Eides Tab 7.3 vs Pachucki–Yerokhin 2010) is the largest pre-existing literature-itemization sensitivity, documented in §5.3.1.

Structural reading. The framework-native architecture is identical for H 21cm and D 1S HFS — same operators, same projection chains, same scope-boundary statements. What changes at I=1 is: (a) the Clebsch–Gordan multiplicity (3/2 vs 1, structural angular content), (b) the nuclear-spin Hilbert space dimension (3 vs 2, adding one qubit and 7 Pauli terms to the I·S encoding), (c) the Layer-2 budget magnitude (-286 ppm vs -18 ppm, scaling with the nuclear extent and complexity). The operator-level autopsy verifies that all three changes are structural consequences of nuclear physics, not framework modifications. This is the I=1 verification of the multi-focal architecture: the same framework that reproduces H 21cm at +18 ppm reproduces D 1S HFS at the depth of the deuteron Layer-2 budget, with framework-native components (BF, a_e, recoil, Zemach) at the same operator-level precision. **[2026-08-28: a new convention defect *was* subsequently found — the Bohr–Fermi g/mass-slot error identified in §5.2.9. The sentence below refers only to the itemization axis surveyed here.]** No new convention mismatch surfaced beyond the pre-existing §5.3.1 exposure.

5.2.8 Tritium 1S hyperfine autopsy

Reference. $\nu_{\mathrm{HFS}}(T) = 1\,516\,701\,470.773(8)$ Hz (Greene 2017 hydrogen-maser update of Mathur, Crampton, Kleppner, Ramsey 1967), the most precisely measured I=1/2 atomic HFS transition with a non-proton nucleus. Distinguished from §5.2.2 by the mass-hierarchy axis at fixed nuclear-spin slot ($m_t \approx 3m_p$ at I=1/2; H 21cm has I=1/2 at m_p). Distinguished from §5.2.7 by sharing the I=1/2 nuclear-spin axis with H 21cm rather than the I=1 of D 1S HFS. This is the cleanest separation in the catalogue of nuclear-mass-effect (m_t vs m_p) from nuclear-spin-effect (I=1/2 same as proton).

Operator-level Zemach at $r_Z(t) = 1.762$ fm. Component 4 exercises §3.18 at the operator level via `geovac.magnetization_density.hydrogen_zemach_eides_leading_order` with $r_Z(t) = 1.762$ fm (source under review; the Carlson 2008 attribution is wrong — PRA 78, 022517 is a proton-structure paper). The bilinear matrix element $\langle \hat{r}_Z \rangle$ on the Sturmian register at L=0 multipole reduces to $M_1[\rho_M] = r_Z$ to machine precision; the operator output -66.594 ppm vs Eides analytic $-2Zm_e r_Z = -66.594$ ppm gives bit-identical reproduction at IEEE 754 float64 precision — no architecture change vs the H 21cm operator-level Zemach in §5.2.2 or the D 1S HFS Zemach in §5.2.7; only the scalar parameter $r_Z(t)$ replaces $r_Z(p)$. Profile (Gaussian vs exponential) independence at machine precision. The Pauli encoding is 4 non-identity terms (II, Z_e, Z_p, Z_eZ_p), the same minimal sparse encoding as for H 21cm and D 1S HFS — confirming the §3.18 operator does not depend on nuclear identity (it operates on the spatial register only with r_Z as a scalar parameter; the same operator handles H, D, T at the same 4-Pauli encoding).

Operator-level $\hat{I} \cdot \hat{S}$ Hamiltonian at I=1/2 (shared with H 21cm). The $\hat{I} \cdot \hat{S}$ operator at $I = 1/2$, $J = 1/2$ has eigenvalues $\{+1/4, -3/4\}$ on a 4-dim joint nuclear-electronic Hilbert space $(2I + 1)(2J + 1) = 2 \cdot 2 = 4$. The F=1 to F=0 splitting is $\frac{1}{4} - (-\frac{3}{4}) = 1$, the same multiplicity as hydrogen. Multiplicity ratio T/H = 1 at machine precision (target 1). The Pauli encoding is $Q_{\mathrm{tot}} = 2$ qubits, 3 non-identity terms ($X \otimes X + Y \otimes Y + Z \otimes Z$), the *standard hydrogen hyperfine encoding* — structurally identical between T and H at qubit-encoding level. Distinguished from the D 1S HFS encoding which requires $Q_{\text{tot}} = 3$ qubits and 10 Pauli terms due to the larger I=1 nuclear-spin Hilbert space. **Operator-level verdict: T 1S HFS shares the I=1/2 operator structure with H 21cm exactly**, with only numerical Pauli coefficients differing (proportional to g_t and $r_Z(t)$ rather than g_p and $r_Z(p)$).

Framework-native subtotal: $\nu_{\text{HFS}}^{\text{native}}(T) = 1516.6969$ MHz (99.9997% of measurement). **Layer-2 net:** $-66.6+64.6 \approx -2$ ppm of ν_F, with framework-native -3.0 ppm residual sitting inside the projected Karshenboim 2005 Layer-2 budget of $+12 \pm 5$ ppm. The Layer-2 inputs are the scalar $r_Z(t)$ value (1.762 fm; the Carlson 2008 attribution is wrong — that paper is proton-structure — and the true source is unidentified) and the Karshenboim 2005 itemization for sub-leading multi-loop QED + recoil NLO. The operator structure consuming the inputs is framework-native via §3.18, identical to the H 21cm and D 1S HFS autopsies.

[**Status 2026-08-28: this subsection's -3.0 ppm residual and the "cleanest isolation" reading below both rest on $r_Z(t) = 1.762$ fm, an input flagged as under review in §5.1 — it coincides with the triton *charge* radius, and at the ab-initio Zemach value 2.27(3) fm the residual moves to ≈ -22 ppm, *outside* the quoted $+12 \pm 5$ ppm band. Both claims are provisional pending that re-sourcing.**]

Why T 1S HFS is the cleanest LS-8a isolation in the hadronic catalogue. The cumulative residual decomposes per Karshenboim 2005:

- Multi-loop QED ($\alpha^2(Z\alpha)$, $\sim +6$ ppm) is the LS-8a renormalization gap, identical mechanism to H 21cm (both Z=1).
- Recoil NLO beyond reduced-mass ($\sim +3$ ppm) is the W1a-D Roothaan kernel-level recoil-mixing wall, slightly smaller than H's +5.85 ppm because the tritium recoil-mixing factor is smaller ($m_e/(m_e + m_t) \approx 1.82 \times 10^{-4}$ vs H's 5.45×10^{-4}).
- **Tritium polarizability ($\sim +0.7$ ppm)** is the W3 inner-factor calibration tier — $\sim 300\times$ smaller than deuteron's $\sim +340$ ppm (re-sourced 2026-08-29) because the tritium 2n+1p binding ($B(t) = 8.482$ MeV) is much tighter than the deuteron's n+p binding ($B(d) = 2.225$ MeV), making the tritium nucleus essentially compact and rigid under external fields.
- Hadronic VP $\sim +0.1$ ppm, finite-size charge $\sim +1.5$ ppm, higher Friar moments $\sim +0.5$ ppm are Layer-2 inputs via §3.17 and §3.18 NLO opt-in.
- Zemach NLO recoil-mixing $m_e/(m_e + m_t) \approx 1.82 \times 10^{-4}$ is structural noise in the electronic regime (+0.013 ppm).
- Convention drift (claimed ± 5 ppm; source unidentified — neither Carlson 2008, a proton paper, nor Sick 2014, a ^{3,4}He paper, covers ^{3}H) is the largest literature-itemization sensitivity in this observable.

Cleanest mass-vs-spin separation in the catalogue. Tritium fills the I=1/2 hadronic non-proton cell that was previously empty in the precision catalogue:

	I=1/2	I=1
$m_N = m_p$	H 21cm (+18 ppm)	—
$m_N \neq m_p$ (hadronic)	**T 1S HFS (-3 ppm‡)**	D 1S HFS (-211 ppm)
$m_N \neq m_p$ (leptonic)	Mu 1S HFS (+199 ppm)	—

The framework's structural-skeleton scope statement now applies in tritium with the cleanest isolation: T and H share the I=1/2 nuclear-spin axis exactly (identical Pauli encoding, identical I·S eigenstructure); T and D share the non-proton nucleus axis (both $m_N \neq m_p$); what separates T from BOTH is the combination "I=1/2 like H, mass $\sim 3m_p$ like D-extended". This makes T 1S

HFS the cleanest LS-8a multi-loop QED isolation point‡ for an electron+hadronic-nucleus system after Mu 1S HFS (which has no QCD nucleus at all). No new convention mismatch surfaced beyond the claimed $r_Z(t)$ value drift, whose source is unidentified (neither cited work covers ^{3}H).

‡ Provisional: computed at $r_Z(t) = 1.762$ fm, which coincides with the triton *charge* radius; at the ab-initio Zemach value 2.27(3) fm the residual moves to ≈ -22 ppm, outside the quoted band. See §5.2.8.

Structural reading. The framework-native architecture is identical for H 21cm, D 1S HFS, and T 1S HFS at the operator level — same operators, same projection chains, same scope-boundary statements. What changes between cells of the mass × spin matrix is: (a) the Clebsch–Gordan multiplicity (1 vs 3/2 between I=1/2 and I=1), (b) the nuclear g-factor g_N^{atomic} as numerical Pauli coefficient, (c) the recoil factor $(1 + m_e/m_N)^{-3}$ as numerical multiplicative factor, (d) the Zemach scalar r_Z as input to §3.18, (e) the Layer-2 budget magnitude (small for compact nuclei T/H, large for extended nuclei D/μH). All five changes are structural consequences of nuclear physics, not framework modifications. **The operator-level autopsy verifies that the multi-focal architecture scales across the full mass × spin matrix of light-nucleus HFS observables.**

5.2.9 Helium-3 2^3S_1 hyperfine structure autopsy

Reference. ν_{HFS}(He-3, 2^3S_1) = 6 739 701 177(16) Hz (Schuessler–Fortson–Dehmelt 1969; Prior–Wang 1977; Rosner–Pipkin 1970), the F=3/2 to F=1/2 splitting of the metastable 2^3S_1 state. **First multi-electron HFS autopsy in the catalogue.** The 2^3S_1 state has electronic configuration (1s, 2s) with triplet coupling (S=1, L=0, J=1), and the He-3 nucleus has I=1/2 with *negative* magnetic moment ($\mu(\text{He-3}) = -2.127625\,\mu_N$, $g_{\text{He3}} = -4.25525$).

Distinguished from §5.2.2 / §5.2.7 / §5.2.8 by being a multi-electron HFS at J=1 rather than single-electron HFS at J=1/2. Distinguished from §5.2.6 by being an HFS (cross-electron-nucleus spin-spin) rather than fine-structure (intra-electron spin-orbit + spin-spin); both are multi-electron observables with He nucleus as Layer-2 input but probe categorically different operator structures.

Multi-electron contact density (new architectural content). The 2^3S_1 spatial wavefunction is the antisymmetric Slater determinant $\Psi_T = [\varphi_{1s}(r_1)\varphi_{2s}(r_2) - \varphi_{2s}(r_1)\varphi_{1s}(r_2)]/\sqrt{2}$. The Fermi-contact operator $\sum_i \delta^3(r_i)\,\mathbf{s}_i \cdot \mathbf{I}$ evaluated on the $|J = S = 1, M_S = 1\rangle$ state gives $\langle \sum_i \delta^3(r_i)\, s_{i,z}\rangle_{M_S=1} = (|\varphi_{1s}(0)|^2 + |\varphi_{2s}(0)|^2)/2$. The factor 1/2 is a *structural multi-electron spin projection* that does not appear in single-electron HFS calculations: mechanism is the antisymmetric Slater determinant ($\langle \delta^3(r_i)\rangle = (|\varphi_{1s}|^2 + |\varphi_{2s}|^2)/2$ per electron) weighted by $\langle s_{i,z}\rangle = 1/2$ in the $M_S = 1$ state. This is the §3.22 bipolar-harmonic content for HFS observables (sibling of the He 2^3P spin-spin + spin-other-orbit Drake combining coefficients).

Operator-level §3.18 at He-3. Component 4 exercises the magnetization-density operator at I=1/2, He-3 nucleus, with r_Z(He-3) = 1.965 fm [**input under review, 2026-08-28: this coincides with the He-3 *charge* radius** 1.9661 **fm to three significant figures; Sick 2014 (PRC 90, 064002) reports** $r_Z(^3\text{He}) = 2.528(16)$ **fm; quantum Monte Carlo gives** 2.50(3) **fm (King *et al.*, arXiv:2606.11153, Tab. 2), and a hyperfine determination gives** 2.608(24) **fm, which differs from Sick's value by** 2.8σ **(a documented tension, not a corroboration). All three exceed the value used here by a wide margin. The value used here is therefore low by** $\sim$21–25%**, and this component and the residual built on it are provisional**]. Reproduces Eides analytic leading-order Zemach to machine precision (reproduction residual 0.0 ppm, 0.0% of the LO shift); profile (Gaussian vs exponential) independence preserved at machine precision; 4 non-identity Pauli terms (II, Z_e, Z_p, Z_eZ_p) — same minimal sparse encoding as for H 21cm and D 1S HFS, confirming §3.18 does not depend on multi-electron structure (it acts on the spatial register only).

Operator-level $\hat{I}\cdot\hat{J}$ at I=1/2, J=1. The $\hat{I}\cdot\hat{J}$ operator at $I = 1/2, J = 1$ has eigenvalues

$\{+1/2, -1\}$ on a 6-dim joint nuclear-electronic Hilbert space $(2I+1)(2J+1) = 2\cdot 3 = 6$. The F=3/2 to F=1/2 splitting is $1/2 - (-1) = 3/2$. The D HFS case (I=1, J=1/2) has the *identical* 6-dim Hilbert space with the same $\{+1/2, -1\}$ eigenvalues (Wigner $3j$ I-J swap symmetry): multiplicity ratio He-3/D = 1.000000000000 at machine precision. **Operator-level verdict: $I \leftrightarrow J$ swap symmetric** — the F-multiplet structure depends only on $(2I+1)(2J+1)$ and the total angular momentum content, not on which factor is the nucleus vs the electron-pair.

§3.20 multi-electron screening detectability (THE HEADLINE FINDING). The 2^3S_1 contact density depends sensitively on the screening prescription for the outer 2s electron:

Path	$Z_{\mathrm{eff,1s}}$	$Z_{\mathrm{eff,2s}}$	ν_{HFS} (MHz)	Residual (ppm)
Hydrogenic (unscreened)	2.00	2.00	7311.34	+84 817
Slater (1s $\sigma = 0.85$)	2.00	1.15	6653.41	−12 804
Full screen	2.00	1.00	6600.52	−20 651

Spread across screening prescriptions: 105 468 ppm = 10.5%. This is the *first precision-catalogue entry where the framework's multi-electron screening prescription is the dominant Layer-2 systematic*, well above both the ppm-scale experimental precision (16 Hz / 6.7 GHz = 2.4 ppb) and the LS-8a multi-loop QED budget (∼tens of ppm). §3.20 multi-electron screening is *structurally detectable* at He-3 2^3S_1 HFS precision — the autopsy converts the existing §3.20 dictionary entry from "architectural" (handles core-valence orthogonality in molecular systems) to "load-bearing for HFS precision". The framework-native ab initio path uses `geovac.neon_core.screened_psi_origin_squared` on a no-frozen-core Z=2 system; that is the named follow-on closure target.

Convention finding (new §5.3-class candidate). The Bohr–Fermi formula's mass slot uses m_e/m_p *always*, because the nuclear magneton $\mu_N = e\hbar/(2m_p)$ is referenced to the proton mass regardless of the actual nucleus. Track 5 (D HFS, §5.2.7) used m_d in this slot with a doubled g-factor convention ($g_d^{\mathrm{atomic}} = 2\mu_d/I$). **Corrected 2026-08-28:** this paragraph previously stated that the combination "happens to give the correct splitting by a *factor cancellation* specific to I=1". **The cancellation is not exact.** The convention yields correct $\times\, 2(m_p/m_N)$, and at $I = 1$ that is $2(m_p/m_d) = 1.0004965$ — near unity, but high by 496.5 ppm, which is the error retracted in §5.2.7. The near-miss is why it survived: $m_d/m_p = 1.99900750$ is close enough to 2 that the D baseline looked right to three digits.

The same one-line formula accounts for every known instance: $2(m_p/m_d) = 1.000496$ (D, high by 496.5 ppm), $2(m_p/m_{^3\mathrm{He}}) = 0.6682$ (He-3, low by 3/2 — the discrepancy reported below), and $2(m_p/m_{^7\mathrm{Li}}) = 0.287204$ (Li-7, low by 3.4818× — the error corrected in §5.2.12, where the observed deficit matches this ratio to six digits). Its mass-slot half independently produced the alkali-cliff table's suppression (§5.3.8). Tritium was unaffected: its driver used the standard convention. For I=1/2 nuclei (He-3, H, Mu, T) no cancellation occurs at all; the standard convention ($g_N = \mu/I$, m_p in mass slot) is required throughout. Sanity check: He-3$^+$ 1s (single-electron hydrogen-like ion, no multi-electron factor) matches experiment at +0.05% (framework −8661.25 MHz vs experimental −8665.65 MHz). The Track 5 convention transferred naively to He-3 is off by 3/2; the standard convention generalizes correctly across all I-values. This is not merely a candidate §5.3 class-(i) exposure: it is the **identified root cause of a defect class** that produced one retracted baseline (D), one corrected baseline (Li-7) and one corrected resource table (the alkali cliff sequence). The cross-anchor guarding against its recurrence is `tests/test_paper34_autopsy_baselines.py`.

Framework-native subtotal: $\nu_{\mathrm{HFS}}^{\mathrm{native}}(\text{He-3 } 2^3S_1) = 6653.41$ MHz at Slater-screened contact density. **Layer-2 net:** −12 804 ppm of ν_{exp} at this prescription. Decomposition:

- §3.20 multi-electron screening of 2s by 1s (±50 000 ppm spread across hydrogenic/Slater/full-screen): dominant; the framework's own multi-electron Layer-2 prescription is the load-bearing systematic.

- Multi-electron QED Pachucki–Drachman corrections (LS-8a multi- particle generalization, few hundred ppm): structural framework scope ends at one-loop on single-particle Dirac-S^3.
- He-3 nuclear polarizability (W3 inner-factor, few ppm): smaller than deuteron polarizability because alpha-particle-like binding is stronger than n+p; QCD-internal nuclear structure.
- Multi-loop QED $\alpha^2(Z\alpha)$ (LS-8a, ∼tens of ppm): the one-loop closure architecture wall.

Structural reading. The He-3 2^3S_1 HFS autopsy *restructures the catalogue's understanding of where the structural skeleton's residuals sit for HFS observables.* Single-electron HFS (H 21cm, D, T, μH, Mu) all have LS-8a multi-loop QED as the dominant Layer-2 wall, with framework residuals ranging from +2 ppm (μH) to −211 ppm (D). Multi-electron HFS (this entry; candidate future entries: Li-7, Cs-133) has §3.20 multi-electron screening as the dominant wall, with framework residuals 50–500× deeper (−1.28% here). The framework's one-loop QED scope is preserved at the same operator-level precision; what changes is which Layer-2 input dominates. **This is the first precision-catalogue entry where §3.20 is the load-bearing dictionary entry, validating its Sprint 1 dictionary-completion inclusion (May 15 2026).**

5.2.10 Helium $2^1P \to 1^1S$ oscillator strength

Reference. $f_{\text{length}} = 0.27616$ (Drake handbook; Theodosiou 1984), oscillator strength for the He $2^1P \to 1^1S$ electric-dipole transition; $\omega_{\text{exact}} = 0.7799$ Ha.

The first multi-electron extension of Sprint Calc-L (hydrogen Lyman α, +0.055% at depth-3) and Sprint Calc-P (hydrogen polarizability, machine-exact at $N_{\text{basis}} = 2$ in the Sturmian basis). The structural question: does the Sturmian continuum-closing property extend to multi-electron transition matrix elements?

The framework-native length-form oscillator strength is computed via

$$f = 2\omega\,|\langle 1^1S|z|2^1P_{m=0}\rangle|^2$$

in the graph-native CI sector at fixed M_L, with the 1^1S state in the ($l_1 = l_2 = 0$) ss singlet sub-block, the 2^1P state in the ($l_1 = 0, l_2 = 1, M_L = 0$) sp singlet sub-block, and the transition dipole assembled by Slater–Condon expansion of the multi-electron CI vectors.

Both ω and $|\langle z\rangle|^2$ *plateau* by $n_{\max} = 6$ at *wrong limits*: ω converges −12% low and $|\langle z\rangle|^2$ converges +82% high. The 2^1P state is variationally violated by ~ 0.08 Ha ($E(2^1P)_{\text{GeoVac}} = -2.204$ vs Drake −2.124 Ha) — the same κ-induced over-binding mechanism documented for the He $2^1S - 2^3S$ splitting (small-Z graph-validity-boundary artifact, $Z_c \approx 1.84$; He at $Z = 2$ is just above; cf. §5.1 He singlet-triplet row). The angular (Wigner $3j$) and vector-photon (§3.11) projections are exact; the failure is in the CI sector.

Structural reading. Sturmian's continuum-closing is a *single-focal-length* property. For multi-electron transitions where initial and final states have structurally different effective focal lengths (here, doubly-screened $1s$ and singly-screened $2p$), the single-exponent Sturmian basis cannot represent both, and the framework's CI eigenvectors overlap incorrectly even at large $n_{\max}$. The natural extension is the *multi-focal* architecture (`geovac/cross_register_vne.py` for chemistry; internal split-Z for atomic excited states is a named follow-on track per Sprint Calc-He recommendation).

The +61% residual at $n_{\max} = 7$ does not improve at higher $n_{\max}$ within the single-exponent architecture and is therefore tagged error code `B` (basis architecture) in Table 4. This entry is a *structural* negative result that complements the He $2^1S - 2^3S$ off-precision row: both expose the small-Z graph-validity-boundary at He as the dominant limitation for excited-state precision, separable from the framework's clean closures on hydrogenic and ground-state observables.

Internal multi-focal Phase D production result (May 9 follow-on). The extended-angular-CI architecture in `geovac/internal_multifocal.py` (1521 lines, 28 fast tests + 2 slow, 191/191 zero regression) lands at $f = 0.286$ (+3.4% vs Drake 0.276) on physically-motivated multi-exponent panels at well-conditioned $\mathrm{cond}(S) = 400$. The $E(2^1P)$ state is variationally violated by ~ 0.08 Ha (3.8%) due to κ-induced inter-shell over-binding, and the variational floor sits at +14%. An exponent sweet spot ($E1$ panel) gives -0.88% but is NOT robust (f varies 0.06–0.44 across plausible exponent panels). Sub-1% accuracy requires explicit r_{12} correlation (Hylleraas) on a P-state basis.

Hylleraas $\mathbf{r}_{12}$ closure status (Track 1 follow-on, May 9). The Hylleraas r_{12} explicit-correlation production module `geovac/hylleraas_r12.py` (Track 1 sprint, May 2026) reproduces the He 1^1S ground state to -0.016 mHa vs Drake exact NR at $\omega_{\max} = 4$ (22 basis functions) and the Hylleraas 1929 published 3p value to +0.79 mHa. Cusp condition variationally captured (cusp ratio at $u = 0$ converges $0.29 \to 0.49$ toward Kato 0.5). The single-exponent Hylleraas trial does not close the asymmetric-$\mathrm{Z}_{\mathrm{eff}}$ excited-state issue at modest basis size; the named structural closure was **Hylleraas–Eckart double-zeta** (Hylleraas–Eckart 1933).

Hylleraas–Eckart full Schwartz two-channel closure (May 2026, the structural endpoint of the autopsy). The Hylleraas–Eckart double-α implementation arc (CLAUDE.md §2 Track 5 backlog) closes with the full Schwartz 1961 two-channel 2^1P trial

$$\Psi^{M=0}_{2^1P} = \sum_q c_q^+ (z_1 + z_2)\, e^{-\alpha s} \cosh(\beta t)\, s^{l_q} t^{2m_q} u^{n_q} + \sum_q c_q^- (z_1 - z_2)\, e^{-\alpha s} \sinh(\beta t)\, s^{l_q} t^{2m_q} u^{n_q}$$

delivering Drake-class oscillator strength. At $\omega_s = 3$, $\omega_p = 2$ (14 P-state basis functions total: 7 sym + 7 antisym), the framework gives $E(1^1S) = -2.903659$ Ha (vs Drake -2.903724, +0.065 mHa), $E(2^1P) = -2.123744$ Ha (vs Drake -2.123843, +0.099 mHa), $\Delta E = 0.7799$ Ha (sub-mHa match), and $f = 0.2705$ **vs Drake** 0.2761**, residual** $\boxed{-2.02\%}$. The dipole decomposes as $D_{\mathrm{sym}} = +0.269$ and $D_{\mathrm{antisym}} = +0.148$ (35% antisym contribution; constructive addition); the antisym channel is genuinely required for Drake-class accuracy, not basis-size-limited. Sub-mHa energy convergence at the modest basis (14 functions) confirms the variational architecture is saturated; the residual -2% reflects the $\omega_s = 3$, $\omega_p = 2$ truncation. Production code: `geovac/hylleraas_eckart_pstate.py` (universal SO(3)-averaged 3D quadrature kinetic for the 3 non-trivial channel pairs $(-,-)$, $(+,-)$, $(-,+)$, plus the algebraic Hartree kinetic for $(+,+)$ from Track 3.5). Sprint memo: `debug/hylleraas_eckart_track5_closure_memo.md`. Driver: `debug/he_2p_oscillator_full_channel.py`. Data: `debug/data/he_2p_oscillator_full_omega3_2.json`. Tests: `tests/test_hylleraas_eckart_pstate.py::TestUniversalQuadratureKinetic` (6 new), `TestFullChannelOscillatorStrength` (2 new).

The closure validates two structural claims: (i) the **internal multi-focal architecture** (two electrons at distinct effective Z's) is verified operationally at the 2^1P transition level, not just at the angular-content level (§3.22); (ii) the **2-electron contact-density cliff** identified in the 2026-05-09 multi-track sprint (this autopsy at +61% with graph-native CI; the He 2^1S–2^3S splitting at +11.9%) is closed at Drake-class accuracy by Hylleraas–Eckart full Schwartz on the 2-electron problem.

Wigner–Eckart f-formula normalization. For $L = 0 \to L' = 1$ absorption, summing over final M_L states gives $f = (2/3)\Delta E\, |\langle L' \| r \| L\rangle|^2 = 2\Delta E\, |D_z|^2$ where $D_z = \langle L' = 1, M = 0 | r_z | L = 0\rangle$. The factor of 2 (rather than 2/3 on $|D_z|^2$) absorbs the M_L sum. Verified against hydrogen 1S$\to$2P at $f = 0.4162$.

Honest scope (closure does NOT extend to): (i) the Li $2^2S_{1/2}$ HFS cliff (§5.2.12, multi-electron 3-body system requires Hylleraas–CI hybrid); (ii) the Cs $Z > 20$ cliff (§5.2.11, heavy-atom screening cliff requires BBB93/KTT + Bohr–Weisskopf, structurally distinct mechanism per §5.2.12 closure-path discussion). The "three cliffs, one mechanism" framing that surfaced in the multi-track sprint was tighter than the math supports: Hylleraas–Eckart cleanly closes the 2-electron subset (He 1^1S, He 2^1S–2^3S splitting, He 2^1P→1^1S oscillator strength, and prospective He-3 2^3S$_1$ HFS), but the Li and Cs cliffs are separate downstream sprints.

5.2.11 Cesium 6S$_{1/2}$ hyperfine (atomic clock)

Reference. $\Delta\nu = 9\,192\,631\,770$ Hz (BIPM 1967, SI-second-defining transition), equivalently $A(6S_{1/2}) = \Delta\nu/4 = 2298.158$ MHz for the Cs-133 $6S_{1/2}$, $F = 4 \leftrightarrow F = 3$ hyperfine transition.

The framework's projection-chain machinery (`atomic_classifier`, [Xe] `FrozenCore`, Bohr–Fermi formula, generic $A \cdot \vec{I} \cdot \vec{J}$ Pauli wrapper) extends to $Z = 55$ without structural failure. The -47% residual is cleanly attributed to two named kernel gaps, not to architectural limitations.

Z-scaling assessment. The heavy-atom regime does not expose structural limitations beyond those already identified at $Z = 20$–56 in Sprint 7b screened SO splittings (Clementi–Raimondi single-zeta screening qualitative; W1c-residual core-polarization correction needed for sub-percent precision).

Z-cliff diagnostic refinement (Track 2, 2026-05-09 follow-on). A diagnostic-only sprint (post-multi-track Roothaan autopsy + screening kernel upgrade closeout) ran five hypothesis probes to localize the dominant cause of the -47% Cs HFS framework-native residual. All four candidate alternative causes were RULED OUT cleanly: (b) the FD solver `_solve_screened_radial_log` is faithful to $< 1.2\%$ in $|\psi(0)|^2$ across hydrogenic Z=1..80 with monotone n_grid convergence (probe b); (c) no factor-of-X convention bug in the $Z_{\rm eff}(r) \to \psi \to A_{\rm HF}$ chain (probe c); (d) the multi-zeta machinery in `geovac/multi_zeta_orbitals.py` itself is bug-free (STO normalization, MultiZetaOrbital reduction, density integration to $N_{\rm core}$ all correct, probe d). The **dominant cause is (a) sharpened by a structural mechanism (d): the missing radial nodes in the CR67 single-zeta hydrogenic basis.** Probe (a) extended on Cs Z=55 across all 11 shells of the [Xe] core shows the cliff is shell-resolved: inner shells (1s, 2s, 2p) overshoot by mean 1.01× (well-fit), outer shells (4d, 5s, 5p) overshoot by mean 2.68× (4d alone overshoots 4.14×). Probe (d) reveals the structural mechanism: the all-positive-coefficient single-zeta hydrogenic basis (and the heuristic two-zeta extension) produces s-orbitals that violate radial orthogonality at the level of $|\langle R_{2s}|R_{3s}\rangle| = 0.86$ — without negative coefficients, no radial nodes can form, and the orbital amplitude piles up near the inner-zeta peak. Real RHF outer-shell orbitals have $n - \ell - 1$ radial nodes whose physical role is to push amplitude OUT to the correct radial extent by orthogonalizing against inner shells. The two-zeta heuristic FAILED in the wrong direction ($-47\% \to -90\%$) because it inherited the no-nodes problem and made the screening transition steeper without fixing the position. Probe (e) v2 localizes the cliff onset cleanly at K (Z=19): H Z=1 ratio $|\psi_{\rm FW}(0)|^2/|\psi_{\rm HF\ lit}(0)|^2 = 0.999$ (perfect), Na Z=11 = 1.243 (mild over), K Z=19 = 0.309 (3.2× under, cliff onset), Rb Z=37 = 0.193 (deepest), Cs Z=55 = 0.625 (partial recovery). Source memo: `debug/z_cliff_diagnostic_track2_memo.md`; probe scripts in `debug/z_cliff_probe_{a,a_extended,b,c,d,e,e_v2}.py`.

BBB93/KTT GO with eyes open: The Track 2 diagnostic confirms the BBB93/Koga-Tatewaki-Thakkar full multi-zeta tabulation IS the correct address for the dominant cause (negative coefficients in BBB93 produce the radial nodes that all-positive heuristic two-zeta cannot). Expected outcome: BBB93 closes $\sim 20\%$ of the residual ($-47\% \to\sim -25$ to -30% for Cs), the full sub-percent closure requiring *both* the screening kernel upgrade AND the full Bohr–Weisskopf relativistic enhancement via spinor lift (§3.7). Self-consistent Hartree–Fock iteration in `geovac/neon_core.py` is the principled long-term alternative that closes the cliff for ALL Z (not just [Xe] core).

Diagnostic-instrument angle. The framework's projection-chain decomposition would isolate three known atomic-structure dependencies of the percent-level Cs PNC uncertainty:

- radial wavefunction at the origin (Bohr–Weisskopf via §3.18),
- core polarization (relativistic enhancement via §3.7),
- multi-loop QED in the heavy-atom regime (two-body Dirac/Breit Layer-2 input via §3.16),

each tied to a different Layer-2 input. This is the prospective catalogue's first heavy-atom test of the framework's projection-chain machinery in a regime where simple hydrogenic formulas break down, and connects to atomic-clock-grade precision (10^{-15} relative, the SI-second definition). Source memo: `debug/cs_hfs_v1_memo.md`.

5.2.12 Lithium-7 $2^2S_{1/2}$ hyperfine autopsy (first multi-electron HFS at $I = 3/2$)

Reference. $\nu_{\text{HFS}}(^7\text{Li}, 2^2S_{1/2}, F = 2 \leftrightarrow F = 1) = 803{,}504{,}086.0(34)$ Hz (Beckmann, Boklen, Elke 1974; confirmed Walls 2003). Nuclear: $I = 3/2$, $g_{^7\text{Li}} = 2.170951(4)$ (CODATA $\mu/I/\mu_N$ convention), $m_{^7\text{Li}} \approx 12{,}786.4\, m_e$.

This is the *first multi-electron HFS* in the catalogue at $I = 3/2$, exercising *three* projection-dictionary entries *simultaneously*: §3.18 (Zemach with $r_Z(^7\text{Li}) = 3.71$ fm), §3.20 (PK / core-valence orthogonality between the unpaired 2s valence and the $1s^2$ closed-shell core), and §3.22 (multi-electron Fermi contact via 1s–2s–2p CI mixing). All previous HFS catalogue entries are single-electron (H 21cm, μH, D HFS, Mu HFS, Ps HFS); Li-7 is the first three-electron HFS.

Operator-level $\hat{I} \cdot \hat{S}$ at $I = 3/2$. Eigenvalues $\{+3/4, -5/4\}$ on the 8-dim joint nuclear-electronic space $(2I + 1)(2J + 1) = 4 \cdot 2 = 8$. $F = 2$ to $F = 1$ splitting is $+3/4 - (-5/4) = 2$, the expected CG output at $I = 3/2$, $J = 1/2$ (from $E_F = A/2[F(F + 1) - I(I + 1) - J(J + 1)]$: $E_2 = +3A/4$, $E_1 = -5A/4$, $\Delta E = 2A$). Pauli encoding extends to $Q_{\text{tot}} = 3$ qubits (2 nuclear binary for $I = 3/2$ in 4-dim space + 1 electronic), 8 non-identity terms — *operator-level verdict: CG multiplicity correct to machine precision.*

Operator-level §3.18 at $I = 3/2$. Component 4 applies the same operator architecture as in the H 21cm and D HFS autopsies (§5.2.2 and §5.2.7): `geovac.magnetization_density` reproduces the Eides leading-order Zemach scalar at hydrogenic $Z = 1$ to machine precision, profile (Gaussian vs exponential) independence preserved, Pauli count = 4 (same as H and D, confirming §3.18 is I-independent at the operator level). The Li 2s contact density scales the operator output by $Z_{\text{eff}} = 1.279$ for the leading-order Zemach shift.

Multi-electron contact-density cliff (headline finding). The framework-native chain sits at -64.0% residual. This is *not* the $\sim 1.10\%$ Li atomic accuracy floor (Paper FCI-A graph-native CI at $n_{\max} = 4$); the cliff is an order of magnitude deeper. Z_{eff} sensitivity scan:

- $Z_{\text{eff}} = 1.279$ (CR67): $\nu = 288.91$ MHz (-64.0%)
- $Z_{\text{eff}} = 1.350$: $\nu = 339.75$ MHz (-57.7%)
- $Z_{\text{eff}} = 3.000$ (no screening): $\nu = 3728.37$ MHz ($+364.0\%$)
- Experimental $\nu = 803.5$ MHz $\Leftrightarrow Z_{\text{eff}}^{\text{effective}} \approx 1.80$

The effective Z_{eff} for Li 2s *contact density* sits near $Z = 1.8$, not $Z = 1.3$, despite the orbital's spatial extent being screened to $Z_{\text{eff}} \sim 1.3$. This is the well-known *Fermi contact enhancement* in alkali atoms: the §3.20 PK constraint $\langle 2s | 1s_{\text{core}} \rangle = 0$ pulls the 2s radial node inward and dramatically enhances $|\psi_{2s}(0)|^2$ over the hydrogenic Z_{eff} approximation. Clementi–Roetti 1974

SCF $|\psi_{2s}(0)|^2$ exceeds the hydrogenic-$Z_{\rm eff} = 1.279$ value; experiment caps the net enhancement at $\sim 2.8\times$ [corrected 2026-08-29: the retired $\sim 8\times$ figure came with the retired baseline], and §3.22 bipolar-harmonic core polarization (Lindgren 1985; Yan & Drake 2003) contributes a further $\sim$ 1–2% on top.

Why this is not the H/D HFS regime. Single-electron HFS observables (H 21cm at +18 ppm, μH 1S HFS at +2 ppm, D 1S HFS at −211 ppm) hit *nuclear-structure* Layer-2 walls (W3 inner-factor deuteron polarizability, multi-loop QED). Multi-electron HFS hits a prior *electronic-structure* wall: the framework's single-zeta hydrogenic basis cannot represent §3.20 core-valence orthogonality at the amplitude-at-origin level. Until this is engineered, neither r_Z nor multi-loop QED is independently testable in multi-electron HFS.

Two cliffs, one mechanism. The Li-7 multi-electron contact-density cliff (this autopsy) and the Cs HFS Z>20 cliff (§5.2.11 Track-2 diagnostic) are *two manifestations of the same structural limitation*: the all-positive-coefficient single-zeta hydrogenic basis cannot represent the PK-orthogonalized SCF radial wavefunction. The Z>20 cliff manifests as a $\sim$ 50% undershoot on Cs HFS (single-zeta gives −47%, heuristic two-zeta worse at −90%, BBB93 multi-zeta would close $\sim$ 20%). The Z= 3 cliff manifests as a −64.0% undershoot on Li HFS because the 2s valence is the *first* shell beyond the closed core, where PK orthogonality matters acutely. **Corrected 2026-08-28:** this passage previously read "$\sim$ 90%" (the retired baseline) and concluded "much larger magnitude at Z= 3 than at Z= 55". Both are withdrawn — see §5.3.8. The −47% quoted for Cs is a `FrozenCore` number and the −64.0% for Li a bare-hydrogenic one; compared like for like the cliff *deepens* monotonically with Z (−64.0 to −98.4%). The shared-mechanism reading stands; the inverse-Z magnitude reading does not.

New §V.D-class exposure (class iii). This autopsy surfaces a new exposure class distinct from the four existing §V.D entries (all literature convention mismatches): the *multi-electron contact-density cliff* is a framework-ARCHITECTURE limitation, not a literature convention nor a framework precision ceiling. Magnitude: $\sim 2.8\times$ on $A_{\rm hf}$ for Li-7, growing monotonically with Z (−64.0% at Li to −98.4% at Cs) — not a common factor across the alkali valence HFS series. See §5.3.8.

Named structural follow-on. *Hylleraas-Eckart double-zeta for Li 2^2S HFS:* extend `geovac/hylleraas_r12.` from the He singlet ground state (currently +0.0006% at $\omega = 4$; see §5.2.10) to 3-electron Hylleraas with r_{12}, r_{13}, r_{23}. Expected closure: framework Li atomic accuracy 1.10% → sub-0.1% *simultaneously* with closure of the $|\psi_{2s}(0)|^2$ cliff (Hylleraas explicitly satisfies PK orthogonality at the amplitude level, no $Z_{\rm eff}$ fitting required). Downstream impact: unlocks separable testing of §3.18 (Zemach via r_Z), §3.20 (PK orthogonality), §3.22 (bipolar core polarization), and multi-loop QED in multi-electron HFS. The alkali HFS frontier (Li, Na, K, Rb, Cs) then becomes a testbed for the framework's projection-chain dictionary at $Z \in [3, 55]$.

5.2.13 Positronium 1S–2S two-photon transition (first equal-mass two-body autopsy)

Reference. $\nu(1^3S_1 \to 2^3S_1)$ = 1,233,607,216.4(3.2) MHz (Fee, Chu, Mills, Mader, Mills, Chichester, *Phys. Rev. A* **48**, 192, 1993; 2.6 ppb).

The first § V.C autopsy of an *equal-mass two-body* system, and the first operator-level test of the just-added 16th projection (§ 3.16 Breit retardation, added 2026-05-08) since its introduction. Continues the multi-observable focal-length decomposition program (§ 1.8 directive) at the cleanest known instance of the multi-focal-composition wall.

Operator-level rest-mass projection (Component 1, FN). The ratio $\nu_{\rm Ps}/\nu_{\rm H} = 0.5002723085$ matches $m_{\rm red}(ee)/m_{\rm red}(ep) = 0.5002723085$ to relative residual 0.00×10^0 (bit-identical). This is the operator-level verification of § 3.14 ring-preservation in the equal-mass limit at the same observable type as Mu 1S-2S (May-8 Track 1 sprint, −0.11 ppm). Together the two cover the rest-mass projection across the lepton-mass-ratio axis (Mu, $\lambda_l/\lambda_n \sim 1/207$; Ps, $\lambda_l/\lambda_n = 1/1$).

Operator-level Breit retardation (Component 2, partial FN). The § 3.16 kernel is implemented as production module `geovac.breit_integrals` (26 tests, exact Fraction/sympy arithmetic). Closed-form matrix elements at $Z = 1$, $k = 0$, in the module's convention $R^k(a, b; c, d) = \iint P_{ab}(r_1)\, P_{cd}(r_2)\, K(r_1, r_2)$ — the first pair of labels is electron 1's density, the second pair electron 2's. **Corrected 2026-08-28:** the middle two rows of this table previously carried their labels swapped relative to that convention, so the label $(1s, 2s; 1s, 2s)$ appeared here with the log-bearing value while §3.16 gives it as 4/81. Both *values* were and remain correct; only the label assignment was inconsistent between the two locations, and the convention itself was never stated. Pinned by `tests/test_paper34_projection_spot_checks_batch1.py`:
The kernel lives in $\mathbb{Q}[\log 2, \log 3]$ (Paper 18 embedding-log content; the Mellin-regularized inner integral produces $\log p$ for small primes p from the negative-power singularity treatment).

Three operator-level subtests:

- **Test 1 (kernel implementation): PASS.** Closed-form expressions for all four tested orbital pairs; production-grade module.
- **Test 2 (Roothaan multipole termination at equal mass): PASS.** For $(1s, 1s)$ and $(2s, 2s)$ products $l_a = l_b = 0$, only $k = 0$ contributes by Wigner-3j triangle inequality (§ 3.8). The termination is angular content, *not* a small-parameter expansion — it holds independent of mass ratio, re-confirming the Ps HFS sprint finding for the 1S-2S observable.
- **Test 3 (state-specific bound-state energy at α^4): PARTIAL.** The framework implements (a) the radial kernel, (b) the angular projection via § 3.8, (c) the spinor sector via § 3.7. The full bound-state matrix element $\langle \mathrm{Ps}, 1S|H_B|\mathrm{Ps}, 1S\rangle - \langle \mathrm{Ps}, 2S|H_B|\mathrm{Ps}, 2S\rangle$ in the equal-mass two-body Dirac normalization requires the Bethe–Salpeter quasi-energy expansion machinery, which is structurally outside the framework's single-particle Eides § 3.2 SE bracket. This is the named structural extension of § 3.16.

§ 3.16 verdict: OPERATOR-LEVEL PARTIALLY VERIFIED. Kernel-level + Roothaan termination at equal mass = OK at operator level; full bound-state energy evaluation at α^4 requires Bethe–Salpeter machinery as the named extension target.

Layer-2 magnitude consistency check. The framework-extracted α^4 Breit residual (calibrated against the Pachucki–Karshenboim 1998 complete theoretical value) gives fraction $-79{,}861.9/(m_e c^2 \alpha^4/h) =$ -0.2279, vs canonical Karshenboim 2005 § 9 closed-form $-(11/48) = -0.2292$; agreement 0.57%. Match consistent with the calibration-by-construction caveat documented in `debug/precision_catalogue_positr`

Crystallization of the multi-focal-composition wall at α^4. This autopsy identifies the operator-level signature of the wall. The framework handles the α^2 Bohr level (Component 1) and the α^3 self-energy (Component 3) as native, with full operator-level realization. At α^4 (Component 2), the wall surfaces: the Breit retardation kernel is implemented and tested at operator level; the *bound-state matrix-element evaluation* in two-body Dirac normalization is the named structural extension.

This is the cleanest known instance of the multi-focal-composition wall because: (i) the wall surfaces at α^4 — one full order before the LS-8a renormalization wall at α^5 or higher; (ii) the wall is *isolated* from the renormalization wall (Components 4, 5 at the −8 to −9 MHz scale, summing to −17.8, while Component 2 is at the −80,000 MHz scale — over three orders of magnitude separation); (iii) the wall is at the operator level without being at the renormalization level (kernel implemented in closed form, angular structure preserved, multipole termination at equal mass preserved — only the bound-state evaluation machinery is the named extension).

The multi-focal-composition wall (CLAUDE.md § 1.7, Sprint HF May 2026) is therefore made visible at *the operator level* on the cleanest possible observable, transitioning from a structural-skeleton-scope *statement* to an operator-level *structural identification* with a named extension target.

Structural reading. Same architecture that closes the H 1S Lamb shift at § 5.2.1 and μH $2S$–$2P$ at § 5.2.3 closes Ps 1S-2S here, with the equal-mass two-body Dirac kinematic content (Component 2) being the structural extension specific to $\lambda_a = \lambda_b$ that has no analog in the unequal-mass cases. Together with the existing Ps 1S HFS sprint (framework-native Fermi-contact at 4/7 of LO, annihilation channel as 3/7 of LO Layer-2), the equal-mass leptonic axis is now spanned at both observable types (HFS spin-coupling + 1S-2S level energy) and the dominant Layer-2 mechanism shifts cleanly between observable types (annihilation for HFS, α^4 Breit for 1S-2S).

5.2.14 HD molecule $J = 1$ rotational hyperfine (first molecular precision-catalogue entry)

Reference. $\chi_d^{\mathrm{HD}}(J=1) = 224.540(20)$ kHz (Code-Ramsey 1971), the deuteron electric quadrupole coupling in HD ground-state vibrational $v = 0$ at rotational $J = 1$. Komasa–Pachucki 2020 ab initio reference at CCSDT-F12 + nonadiabatic ro-vibrational averaging: 224.51(5) kHz. Precision target $\sim 0.02\%$ (~ 50 Hz absolute).

First operator-level test of §3.19 (§III.19, rank-2 tensor multipole). Pre-track, §III.19 was the only Layer-2 projection with no operator-level test. Track 4 of the multi-track Roothaan autopsy sprint (2026-05-18) produced an order-of-magnitude Roothaan autopsy, the first molecular precision-catalogue entry, and the first concrete operator-level test of § 3.19.

The framework currently exposes the following ingredients of the χ_d chain analytically:

- Bare nuclear multipole of the partner H proton at the deuteron position via § 3.21: $q_{\mathrm{nuc}} = 2Z_H/R^3$ at $R = R_{\mathrm{eq}} = 1.4011$ bohr.
- $J = 1, m_J = 0$ rigid-rotor Wigner-3j angular reduction via § 3.8: $\langle V_{zz}\rangle = -(2/5)\, V_{zz}^{\mathrm{mol}}$ (Townes–Schawlow standard).
- $Q_d = 0.285699(15)(18)$ fm^2 as scalar Layer-2 input (Komasa–Pachucki 2020), in the same projection class as the $\langle r^2\rangle_E$ and r_Z scalars of § 3.17 and § 3.18.
- Born–Oppenheimer separation via § 3.24.

What the framework does NOT currently expose: the molecular electronic density $\rho_e(\vec{r})$ at the deuteron position in position space. Level 4 (`geovac/level4_multichannel.py`) returns hyperspherical (R, α, l, m) channel amplitudes, not $\psi(\vec{r}_1, \vec{r}_2)$, and consequently the EFG operator $V_{zz}^{\mathrm{elec}} = \partial^2 V_e/\partial z^2$ cannot yet be integrated against the molecular density.

What the OoM result establishes. The framework's existing multipole machinery (§3.21) captures the dominant LO contribution to within 13% of experiment in magnitude, with the right size and the right angular structure (the overall EFG sign convention is unresolved; see the table caption) (J=1 rigid-rotor Wigner-3j reduction is exact and framework-native). The 40% residual after adding the free-atom electronic screening is *specific and well-localized*: it attributes to the absence of position-space molecular electron density at the deuteron. This is not a framework gap at the projection-class level; it is one missing operator (electronic density at a point), structurally analogous to how the § 3.18 operator-level extension (May 2026) bridged r_Z as a scalar Layer-2 input to a full operator-level computation.

Named follow-on sprint: `HD-EFG-position-space` (beyond sprint scale; analog of the W1a-D Roothaan recoil and §3.18 operator-level extensions). Five components, each precedent-bounded:

(1) Expose Level 4 wavefunction at a nuclear coordinate (coordinate transform of existing (R, α) amplitudes); (2) Add EFG operator $V_{zz}(\vec{R}_D) = \int (3z_e^2 - r_e^2)/r_e^5 \, \rho_e(\vec{r}_e)\, d^3r_e$; (3) $|J=1, m_J=0\rangle$ rotational averaging via existing Wigner-3j infrastructure; (4) $v=0$ vibrational averaging over Level-4 hyperradial wavefunction; (5) $T_{ij}^{(2)}$ rank-2 spin tensor on $|I=1\rangle$ (extension of `hyperfine_a_pauli_for_atomic_hfs` from rank-1 to rank-2). Reference target: Komasa–Pachucki 2020 (J. Chem. Phys. 152, 154301). Realistic accuracy: $\sim 5\%$ at operator-level (sub-percent requires CCSDT-F12-class correlation that Level 4 does not provide; the residual would then attribute to non-adiabatic / electron-correlation Layer-2 input class, in the same way Komasa–Pachucki itemize their ab initio chain).

Structural reading. This is the catalogue's first molecular precision-catalogue entry (all prior § 5.2 autopsies are atomic) and the first operator-level test of § 3.19. The OoM result confirms the §III.19 entry's architectural claims: Q_d as Layer-2 scalar × Wigner-3j angular ($J = 1$ rigid-rotor) × multipole expansion × Born–Oppenheimer separation is consistent with the experimental observable at order-of-magnitude precision. §III.19 ceases to be *zero-operator-level-tests*; it now has one OoM-level autopsy with a named follow-on track for quantitative closure. The pattern matches the § 3.18 (Zemach) operator-level extension trajectory: scoping autopsy first, full operator-level sprint as named follow-on, sub-percent closure with explicit attribution of the remaining Layer-2 budget.

5.2.15 Muonic helium-4 ion $2S_{1/2}$–$2P_{1/2}$ Lamb shift autopsy

Reference. $\Delta E = 1378.521(48)$ meV (CREMA 2021, Krauth et al., *Nature* **589**, 527; muonic convention $E(2P_{1/2}) - E(2S_{1/2})$).

The CREMA 2021 muonic helium-4 ion Lamb shift completes the muonic-Z=2 cell of the precision catalogue. It is the most extreme mass-hierarchy × Z combination tested so far: muon on a spinless ($I = 0$) high-Z nucleus. Decomposes under the same five-component skeleton as the muonic hydrogen autopsy (§5.2.3) and muonic deuterium autopsy (§5.2.4), with three architectural substitutions: (a) nuclear charge $Z = 1 \to 2$ (Sommerfeld $\beta = 2/(Z m_{\text{red}}\alpha)$ scales as $1/Z$; $\beta = 1.475 \to 0.682$); (b) reduced mass $m_{\text{red},\mu p} \to m_{\text{red},\mu^4\text{He}} = 201.069\, m_e$ (factor 1.082); (c) nuclear charge radius $r_p \to r_\alpha = 1.6755(28)$ fm (factor 1.993, FNS scaling $\sim 80\times$ via $(Z\alpha)^4 m_{\text{red}}^3 r_E^2/12$); and one major structural simplification: (d) *nuclear spin* $I = 0$, so the magnetization-density operator (§3.18) is identically zero and no Zemach term is computed. This is the *cleanest pure-Lamb test in the muonic catalogue*: no HFS contamination, no $\hat{I}\cdot\hat{S}$ mixing, no nuclear-spin convention questions.

Headline framework computation. The full Uehling integration at $\beta = 0.682$ (vs $\beta = 1.475$ at μH, $\beta = 1.400$ at μd) matches the Diepold 2018 / Pachucki 1996 reference at -0.032% — the cleanest single-component muonic VP match in the catalogue at heavy-Z, transporting bit-identical kernel infrastructure from the μH regime to the $\mu^4\text{He}^+$ regime. This is the architectural transport theorem of the muonic Lamb catalogue: the same Uehling-kernel integral closes verbatim under $Z = 1 \to 2$ and $m_{\text{red}} = 185.84 \to 201.07$.

Contact-form Uehling failure escalates at heavy Z. At $\beta = 0.682$, the contact-form Uehling overshoots full kernel by a factor of **8.76**×, vs 3.55× at μH ($\beta = 1.475$). The contact approximation is fully out-of-regime when the bound-state Bohr radius becomes comparable to or smaller than the e^+e^- Compton wavelength. Full numerical kernel integration is *mandatory* for the $\mu^4\text{He}^+$ Lamb shift. This sharpens the architectural claim of §5.2.3: contact-form Uehling is correct only at $\beta \gtrsim 100$ (electronic regime); transitional at $\beta \sim 1$ (μH, μd); fails by an order of magnitude at $\beta < 1$ (heavy-Z muonic).

Same SE bracket recoil-mixing gap, shrunken at higher Z. The framework's one-loop SE Eides bracket gives -11.86 meV vs Diepold's -10.94 meV, a $+8.4\%$ overshoot. This is the same recoil-mixing gap identified in Sprint MH Track A (μH $+24\%$) and the multi-track Roothaan sprint Track 1 (μd +41%). The fractional gap shrinks monotonically with Z because the leading-log $(4/3)\ln(1/(Z\alpha)^2)$ term in the Eides bracket grows with Z while the missing sub-leading $\alpha(Z\alpha)^4 \times (m_{\mathrm{red}}/m_n)$ recoil-mixing pieces (Eides §4.2; Pachucki 1996) become a smaller percentage of the bracket. The absolute SE shift grows as $Z^4 \cdot m_{\mathrm{red}} \cdot$ (bracket ratio) (verified bit-exact in the Z-scaling cross-check below).

Foldy FNS overshoot reveals higher-Friar absorption. The framework's bare Eides Eq. 2.35 leading-order Foldy correction overshoots Diepold's tabulated "FNS leading" by +2.9% at $r_\alpha = 1.6755$ fm, and Diepold's *total* FNS (leading + HO + recoil) by +4.0%. Diepold's "FNS leading" line implicitly internalizes third-Zemach-moment and Darwin relativistic corrections (Friar 1979 Ann. Phys. 122:151) of order $(Z\alpha)^2$ at $Z = 2$, totaling ~ 11.4 meV (the dominant component of the -19.86 meV total residual). The framework's bare leading is structurally complete to its operator order ($\propto \langle r^2\rangle$); the higher-Friar absorption is a Layer-2 input that the bare-Eides formula doesn't natively generate. *Flagged as a new V.D-prediction candidate*: class-(ii) closed-form-vs-operator-level (Eides bare leading vs Diepold-tabulated FNS-leading-with-absorbed-Friar-HO), sibling to §5.3.4 (which addresses the magnetization-side profile-convention issue). At Z=2 the higher-Friar piece is 11.4 meV on a 295 meV leading; at μH Z=1 the same higher-Friar piece is ~ 0.13 meV on a 3.68 meV leading — a $\sim 3.5\%$ fraction at both Z, i.e. the higher-Friar fraction is essentially Z-independent between these two systems.

Z-scaling verification. The component-by-component Z-scaling of framework values from μH to $\mu^4\mathrm{He}^+$ reproduces the predicted forms exactly:

$$\frac{\Delta E_{\mathrm{SE}}(\mu^4\mathrm{He}^+)}{\Delta E_{\mathrm{SE}}(\mu p)} = Z^4 \cdot \frac{m_{\mathrm{red},\mu^4\mathrm{He}}}{m_{\mathrm{red},\mu p}} \cdot \frac{\mathrm{bracket}(Z=2)}{\mathrm{bracket}(Z=1)} = 17.31 \cdot 0.826 = 14.30 \quad (\text{observed: } 14.30) \tag{3}$$

$$\frac{\Delta E_{\mathrm{FNS}}(\mu^4\mathrm{He}^+)}{\Delta E_{\mathrm{FNS}}(\mu p)} = Z^4 \cdot \left(\frac{m_{\mathrm{red},\mu^4\mathrm{He}}}{m_{\mathrm{red},\mu p}}\right)^3 \cdot \left(\frac{r_\alpha}{r_p}\right)^2 = 16 \cdot 1.267 \cdot 3.969 = 80.45 \quad (\text{observed: } 80.45) \tag{4}$$

Both ratios are reproduced to seven-digit precision under the rest-mass projection §3.14, confirming the §III.14 ring- preservation theorem transports the entire Eides-bracket and Foldy-FNS machinery verbatim across $Z = 1 \to 2$ and isotope substitution.

Dominant Layer-2 wall shifts again. In μH the dominant Layer-2 line was Källén–Sabry two-loop VP (+1.51 meV, LS-8a renormalization wall). In μd it shifted to the deuteron polarizability (+1.69 meV, W3 inner-factor; §5.2.4). In $\mu^4\mathrm{He}^+$, the dominant Layer-2 lines are the *recoil corrections* (-5.94 meV, W1a sub-leading) and the α-particle polarizability (+3.10 meV alone, +4.06 meV with hadronic VP combined; W3 inner- factor). The α-particle polarizability per nucleon is much smaller than the deuteron polarizability per nucleon (the α-particle is tightly bound at 28.3 MeV, with no easily accessible photonuclear excitations compared to the deuteron's near-threshold ~ 2.2 MeV); this is the structural reason the W3 wall shrinks from μd to $\mu^4\mathrm{He}^+$ despite the heavier nucleus.

$\mathbf{r}_\alpha$ extraction sensitivity. The framework's FNS coefficient at $Z = 2$ in the muonic convention gives $d\Delta E/dr_\alpha = -352.94$ meV/fm. The CREMA 0.048 meV measurement uncertainty propagates to $\sigma(r_\alpha) = 0.14$ amf — the statistical precision of muonic atoms as nuclear-radius probes. Taking the framework's -19.86 meV residual at face value would shift the extracted radius by -0.056 fm to $r_\alpha = 1.619$ fm vs CREMA's 1.67824(13)(82) fm; this is *not* a framework extraction (which would use Diepold's Friar-HO-absorbed leading line, eliminating the +11.4 meV convention gap), but the size of the convention-mismatch correction at Z=2 muonic when applied to the bare Eides Eq. 2.35 leading directly.

Structural reading. The autopsy validates the multi-focal architecture under the most extreme mass-hierarchy / Z combination in the precision catalogue. Three framework-native components reproduce +1358.24 meV; four Layer-2 inputs (one less than μH since no Zemach is needed) close the remaining gap, with component-level convention attributions (Friar HO absorption +11.4 meV; SE recoil- mixing +0.92 meV; r_α extraction signal $\sim$ 7 meV) summing to the -19.86 meV residual within 1 meV. None of these is a framework architectural failure; all are documented convention surfaces with sibling entries in μH and μd. The rest-mass projection (§3.14) transports the entire five-projection-chain skeleton bit-identically; only the nuclear charge ($Z = 1 \to 2$), nuclear mass ($m_p \to m_{^4\mathrm{He}}$), and nuclear charge radius ($r_p \to r_\alpha$) inputs change between the muonic catalogue entries. The spinless-nucleus simplification eliminates one projection (§3.18) entirely; the remaining four projections give the same architecture at all three muonic systems. The same architecture that closes the H 1S Lamb shift at §5.2.1 closes the muonic Lamb shifts at §5.2.3, §5.2.4, and $\mu^4\mathrm{He}^+$ here, with full Uehling kernel kinematics (contact-form $\to$ full kernel) and Foldy r^2 scaling ($r_p \to r_d \to r_\alpha$) being the only two ingredients that change between the muonic regimes.

5.2.16 Hydrogen Lyman-α $1\,^2S_{1/2} \to 2\,^2P$ oscillator strength autopsy

Reference. NIST tabulates Lyman-α in two fine-structure components: $f(1\,S \to 2\,P_{1/2}) = 0.13881$ and $f(1\,S \to 2\,P_{3/2}) = 0.27764$, summing to 0.41645. The non-relativistic spin-averaged textbook value is 0.4162. The framework computes the non-relativistic spin-averaged value directly, so the relevant reference is the NIST sum.

The Lyman-α oscillator strength is the cleanest possible dipole-transition test in atomic physics: one electron, no nuclear structure, no multi-electron correlation, no continuum integration. The framework-native value comes out as an exact rational in $\mathbb{Q}$ after the squaring step ($|\langle d\rangle|^2$ removes the intermediate $\sqrt{2}, \sqrt{3}, \sqrt{6}$).

Walking the algebra. The dipole matrix element factorizes into radial and angular pieces. Hydrogenic wavefunctions $R_{1s}(r) = 2e^{-r}$ and $R_{2p}(r) = (1/(2\sqrt{6}))re^{-r/2}$ (atomic units, $Z = 1$) give the radial integral $\langle R_{2p}|r|R_{1s}\rangle = 128\sqrt{6}/243$ (sympy verified by direct integration). The angular factor $\langle Y_{10}|\cos\theta|Y_{00}\rangle = 1/\sqrt{3}$ closes to give $\langle 2p_0|z|1s\rangle = 128\sqrt{2}/243$ — exact in $\mathbb{Q}[\sqrt{2}]$. Squaring removes the irrational: $|\langle 2p_0|z|1s\rangle|^2 = 32768/59049 \in \mathbb{Q}$. The sum over $m_f \in \{-1, 0, +1\}$ multiplies by 3 (spherical symmetry), giving $\sum_{m_f} |\langle 2p_m|r|1s\rangle|^2 = 3 \times 32768/59049 = 32768/19683$. Energy gap $\Delta E = 3/8\,E_h$ exact. Combining via the oscillator-strength formula:

$$f^{\text{framework}}_{1s\to 2p} \;=\; \frac{2}{3}\cdot\frac{3}{8}\cdot\frac{32768}{19683} \;=\; \frac{8192}{19683} \;=\; \frac{2^{13}}{3^9} \;\approx\; 0.41620.$$

This is the framework-native value in exact rational form. Every intermediate transcendental ($\sqrt{6}, \sqrt{3}, \sqrt{2}$) cancels out at the squaring step.

Layer-2 residual attribution. Reduced-mass scaling $f \propto m_e \cdot \omega \cdot |\langle r\rangle|^2 \propto m_e \cdot \mu \cdot (m_e/\mu)^2 = m_e \cdot (m_e/\mu)$ multiplies the framework value by $m_e/\mu_{\text{red}} = 1 + m_e/m_p \approx 1.000545$. Framework + reduced mass: 0.41642. NIST 0.41645 gives a residual of +64 ppm, attributable to relativistic Dirac corrections at order $\alpha^2 Z^2 \approx 5 \times 10^{-5}$ (about 50 ppm) plus sub-Lamb radiative content.

Structural reading. This is the cleanest substrate-side autopsy in the catalogue. Layer 1 produces $f = 2^{13}/3^9$ exact in $\mathbb{Q}$, with all transcendentals cancelling at the squaring step. Layer 2 contributes $\sim$ 600 ppm split between reduced mass (§3.14, dominant) and LS-8a-class radiative. The autopsy demonstrates the generic pattern of dipole oscillator strengths in the framework: amplitudes can be irrational algebraic, but $|\text{amplitude}|^2$ is in $\mathbb{Q}$.

5.2.17 Hydrogen $2P_{1/2}$–$2P_{3/2}$ fine-structure splitting autopsy

Reference. NIST $E(2P_{3/2}) - E(2P_{1/2}) = 10{,}969.044$ MHz (includes all QED). This is arguably the cleanest precision QED test in atomic physics: no nuclear-structure budget, no multi-electron complication, no continuum integration — just spinor Dirac in a Coulomb field.

Walking the algebra. Sprint Dirac-on-S^3 Tier 3 Track T8 (April 2026) verified that Darwin + mass-velocity + spin-orbit on hydrogen sum exactly to the Sommerfeld/Dirac formula:

$$E_{n,j} \;=\; -\frac{m_e c^2 (Z\alpha)^2}{2n^2}\left[1 + \frac{(Z\alpha)^2}{n^2}\left(\frac{n}{j+1/2} - \frac{3}{4}\right) + O(\alpha^4)\right].$$

For $n=2$, $Z=1$: bracket factor is 5/4 at $j=1/2$ and 1/4 at $j=3/2$. Subtracting and simplifying (sympy):

$$\Delta E(2P_{3/2} - 2P_{1/2}) \;=\; \frac{m_e c^2 \alpha^4}{32} \;=\; \frac{\alpha^2}{32}\,E_h.$$

The coefficient 1/32 is intrinsic (in $\mathbb{Q}$, from the j-dependent angular structure); the α^2 factor is Class-1 calibration. With CODATA α: $\Delta E = 10{,}949.3$ MHz (infinite mass). Reduced-mass scaling $\Delta E \propto \mu \cdot \alpha^4$ multiplies by $\mu_{\text{red}}/m_e = 1 - m_e/m_p \approx 0.99946$, giving 10,943.3 MHz.

Layer-2 residual attribution. NIST 10,969.0 MHz minus framework 10,943.3 MHz = +25.7 MHz residual (+2349 ppm). This is the differential QED radiative correction between $2P_{3/2}$ and $2P_{1/2}$: one-loop self-energy and vacuum polarization act differently on the two j components (the famous Lamb shift of $2P_{1/2}$ relative to Dirac is part of this). Order-of-magnitude check: $\alpha \cdot \alpha^2 \cdot E_h/32 \sim 80$ MHz naively; observed +26 MHz reflects the partial cancellation between $2P_{1/2}$ and $2P_{3/2}$ Lamb shifts (similar magnitude, opposite-sign differential). The framework structurally cannot reach this — multi-loop QED counterterms (Z_2, δm, vertex renormalization) are required per the LS-8a closure pattern. Same wall as Lamb shift, Born rule (§3.28 "Born rule paragraph"), Yukawas.

Structural reading. Simplest non-trivial QED test in the framework. Layer 1 (substrate) produces exact $\Delta E = (1/32)\alpha^2 E_h$ in $\mathbb{Q}\times$ Class-1 calibration. Layer 2 contributes reduced mass (−545 ppm) plus differential QED (+2349 ppm, dominant). The 1/32 coefficient generalises across any $\Delta E(nj' - nj)$ derived from Sprint T8: Sommerfeld is exact in the framework, only the rational changes per (n, j, j'). Canonical "framework gets Dirac exactly, multi-loop is the wall" autopsy entry.

5.2.18 Helium $2\,^1S - 2\,^3S$ exchange splitting Roothaan walk (first three-architecture comparison)

Reference. NIST $E(2\,^1S) - E(2\,^3S) = 6{,}421.46\ \text{cm}^{-1} = 0.029257\,E_h = 192.51$ THz. The singlet sits above the triplet (standard Hund's-rule pattern). This observable isolates the exchange integral cleanly: $\Delta E(2\,^1S - 2\,^3S) = 2\,K(1s, 2s)$ at leading order (direct Coulomb $J(1s, 2s)$ cancels between singlet and triplet).

Texture difference from preceding autopsies. Lyman-α (§5.2.16) and H $2P$ FS (§5.2.17) have ONE clean framework value at the substrate level. This observable has THREE different framework values from three different basis architectures, spanning a factor of ~ 3 between best and worst. The autopsy structure is therefore "three-architecture reduction" rather than "single substrate value + Layer-2 correction."

Walking the architectures. Architecture A (graph-native CI with $\kappa = -1/16$ inter-shell, exact Slater integrals from `hypergeometric_slater`, single exponent $k = Z = 2$): κ inter-shell coupling adds constant amplitude between 1s and 2s regardless of orbital extent. For excited He $(1s, 2s)$, κ overbinds the diffuse 2s, breaking the $K(1s, 2s)$ computation. The 2^3S state is variationally violated at this architecture (CI eigenvalue below exact). Same mechanism as the small-Z graph-validity-boundary artifact ($Z_c \approx 1.84$, He at $Z = 2$ just above), specifically for excited states where the radial extent matters. Net effect on the splitting: +11.86% overestimate.

Architecture B (Hylleraas r_{12} single-α, $\zeta = Z = 2$ for both 1s and 2s, no κ, explicit r_{12}): the 2s orbital with $\zeta = 2$ is wildly too compact — physically 2s should have $\zeta_{\text{eff}} \approx 1$ because the 1s electron screens the nucleus. With $\zeta = 2$ on both, $K(1s, 2s)$ is roughly 3× too large; the splitting overshoots by +209%. Framework's Sprint Calc Track 1 first-pass identified this as the structural failure mode for excited states.

Architecture C (Hylleraas–Eckart double-α, Eckart 1933): each orbital gets its own variational exponent. $\zeta_{\text{inner}} = 2$ on 1s (full nuclear charge); $\zeta_{\text{outer}} \approx 1$ on 2s (screened nuclear charge). Variational bound preserved. Result −1.4% (slight undershoot at $\omega = 5$ basis truncation). The residual is dominated by *basis truncation*, not Layer-2 calibration — extending to $\omega = 8$ would close further (Drake's exhaustive calculation reaches sub-cm^{-1}).

Structural lesson. $K(1s, 2s)$ is fundamentally a *two-exponent problem* because the two electrons see different effective nuclear charges. Single-exponent architectures (A and B) fail by factors of 1.1 (with κ adjacency) or 3.1 (with explicit r_{12} correlation). Two-exponent architecture (C) succeeds at 1.4%. The framework's `hylleraas_r12.py` Sprint Calc Track 1 was DEFERRED 2026-05-18 in favor of Track 5 (oscillator strength); the scoping memo confirmed Gate 2 at −1.4% for double-α.

Layer-1 vs Layer-2 split (Architecture C). Layer 1 produces $\Delta E \approx 6{,}331$ cm^{-1} at $\omega = 5$ basis truncation. The 1.4% residual is *algorithmic* (basis truncation), not calibration — closeable by increasing ω. This is qualitatively different from Lyman-α and H $2P$ FS, where the framework substrate-side value was exact and Layer 2 residual was unavoidable LS-8a class. Here, LS-8a-class Layer 2 content (relativistic Breit, radiative QED) is sub-100 ppm — basis truncation is the dominant residual.

Connection to W1e wall pattern. Texture matches the W1e chemistry wall in atomic form. W1e (NaH 2026-05-23) discovered that multi-determinant correlation on a two-electron NaH valence active space cannot close binding even with κ adjacency, screened valence, multi-zeta, cross-block h_1, and Phillips–Kleinman — because multi-determinant correlation lives on the N-representable density-matrix manifold that the substrate's spectral-triple machinery does not natively engage. The He $2\,^1S - 2\,^3S$ autopsy is the same lesson applied to atomic precision physics: the framework's substrate-level CI fails by +12% because κ-adjacency cannot capture the differential exponent structure; closing to 1.4% requires the architectural extension (Hylleraas–Eckart double-α). The framework can close it, but only by extending the basis architecture, not by improving substrate-level CI.

Structural reading. Canonical "framework architecture choice matters for excited multi-electron states" autopsy entry. Three architectures produce three answers spanning factor ~ 30. The structural lesson is that excited-state He requires a basis that can host two different effective nuclear charges simultaneously — single-exponent bases fail by factors of 3 or 1.1 depending on whether κ adjacency or explicit r_{12} correlation is included. This is NOT an LS-8a-class wall: the production-architecture (−1.4%) residual is basis-truncation and would close to sub-cm^{-1} at $\omega = 8$. The bridge to the chemistry-side W1e wall is the deepest cross-domain structural pattern in the multi-focal-composition arc.

5.2.19 Muonic hydrogen 1S hyperfine structure autopsy (first muonic HFS; $r_Z(p)$ cross-link)

Reference. $E_{\text{HFS}}(\mu\text{H}, 1S) = 182.725$ meV $= 44\,183$ GHz (Karshenboim–Ivanov 2002, *Phys. Lett. B* **524**, 259; no direct experimental measurement — muon lifetime $\tau_\mu = 2.2\,\mu$s prevents conventional precision HFS spectroscopy). **First muonic HFS entry in the catalogue.** Creates the cross-link for $r_Z(p)$ identified in the cross-observable consistency matrix (Finding 5, May 31 2026).

The projection chain is *identical* to §5.2.2 (H 21cm): Fock ∘ spinor ∘ $3j$ ∘ rest-mass ∘ magnetization-density, with the single architectural substitution $m_e \to m_\mu$ via §3.14.

Zemach sensitivity enhancement: 186×. The Zemach correction at muonic mass is −7340 ppm (at $r_Z = 1.045$ fm) versus −39.5 ppm at electronic mass — an enhancement factor $m_{\text{red},\mu p}/m_{\text{red},ep} = 185.84/0.999 = 186$. The Eides ($r_Z = 1.045$ fm) vs Karshenboim ($r_Z = 1.054$ fm) convention split produces a 64 ppm = 2,792 MHz difference in the muonic HFS, versus ~ 0.3 ppm ≈ 0.5 kHz at electronic H 21cm [corrected 2026-08-29: an earlier version printed 56 kHz, the full Zemach line rather than the drift]. **The convention split is resolvable at muonic sensitivity**: 64 ppm is well above any plausible framework residual budget.

Cross-link with H 21cm (§5.2.2). The two autopsies share the same Layer-2 input ($r_Z(p)$) consumed through the same projection (§3.18), at mass ratios differing by 186×. A simultaneous fit constraining $r_Z(p)$ from both channels would:

- resolve the Eides/Karshenboim convention split (currently 0.34 ppm on H 21cm, far below the +18 ppm framework residual; ±64 ppm on μH HFS, resolvable);
- test whether the LS-8a multi-loop QED budget scales correctly from electronic to muonic mass (same physics, different kinematics);
- provide the first muonic-electronic HFS consistency check in the precision catalogue.

Alkali contact-density cliff: quantitative Z-sequence. The cross-observable consistency analysis (Finding 4, May 31 2026) computed the hydrogenic-Z_{eff} BF A-constant across all five stable alkali ground-state HFS systems, testing the cliff severity prediction from §5.2.12 and §5.2.11.

The rightmost column is the quantitative measure of the PK contact-enhancement cliff: 2.8× at Li (matching the §5.2.12 finding), growing to 63× at Cs.

The cliff in closed form (2026-08-29). With the corrected table it becomes possible to ask what the sequence actually measures, and the answer is not a power law in Z. Two observations fix it. First, the *needed* contact density is smooth and monotone ($0.233, 0.743, 1.106, 2.335, 3.896$ in units of a_0^{-3}) while the CR67 *estimate* is not (K's 0.050 falls below Na's 0.186, because CR67's $Z_{\text{eff}} = 2.162$ for the K 4s is smaller than Na's 2.507). The jump in the ratio at K is therefore an artifact of the denominator, not a feature of the physics — which is why a power-law fit through the ratio column is ill-founded. Second, the needed density is the textbook Fermi–Segré value $|\psi_{ns}(0)|^2 = Z/(\pi a_0^3 \nu^3)$, with $\nu = \sqrt{\text{Ry}/E_I}$ the effective quantum number read off the measured ionization energy and $\mathrm{d}\nu/\mathrm{d}n$ taken as unity. Including the standard relativistic contact enhancement $F_{\text{rel}} = [\gamma(2\gamma - 1)]^{-1}$, $\gamma = \sqrt{1 - (Z\alpha)^2}$, it reproduces the needed density to $\leq 10\%$ across the whole series. The cliff is then the ratio of that value to the framework's hydrogenic estimate:

$$\frac{|\psi(0)|^2_{\text{needed}}}{|\psi(0)|^2_{\text{CR67}}} = \underbrace{\frac{Z}{Z_{\text{eff}}^3}}_{\text{charge}} \cdot \underbrace{\left(\frac{n}{\nu}\right)^3}_{\text{quantum defect}} \cdot F_{\text{rel}}(Z\alpha), \tag{5}$$

giving $2.9, 4.4, 22.3, 41.5, 56.9$ against the tabulated $2.8, 4.0, 22.0, 43.1, 63.0$ — errors $2, 10, 1, 4, 10\%$, versus 126% for the retired power law.

What drives it. Decomposing the growth from Li to Cs: the charge factor Z/Z_{eff}^3 spans only 2.7× and is non-monotone; F_{rel} spans 1.3×; the quantum-defect factor $(n/\nu)^3$ spans 17× and is monotone. The cliff is therefore dominated by a single structural fact: the alkali valence electron has $\nu = 1.59, 1.63, 1.77, 1.80, 1.87$ along the series — *essentially constant near* 1.7 — while the framework labels it with the integer principal quantum number $n = 2, \ldots, 6$. The quantum defect $\delta = n - \nu$ grows by almost exactly one per added core s shell ($0.41, 1.37, 2.23, 3.20, 4.13$), which is the correct form of the "inner shells" reading: the shells do not pile amplitude at the origin directly, they shift ν — and it is n/ν, cubed, that the contact density sees. The existing Cs autopsy (§5.2.11) achieved -47% using the full `FrozenCore` screened wavefunction (which captures *some* core penetration); the -98.4% here uses the bare hydrogenic-$Z_{\rm eff}$ contact density (which captures *none*). The gap between -47% and -98.4% at Cs quantifies the partial recovery from the `FrozenCore` infrastructure.

5.3 Literature convention exposures (running catalogue)

The framework's projection-chain decomposition acts as a unique diagnostic instrument for precision atomic and molecular physics: multi-observable global fits using a single structural dictionary (the § 3 projection family) expose itemization choices in standard-physics compilations that single-observable analyses cannot see. This is the directive's class (a) finding type — "literature convention mismatches" between independent compilations (Eides–Grotch–Shelyuto 2007 [48], Karshenboim 2005, Krauth 2017, Pachucki–Yerokhin 2010, Antognini 2013, Friar–Payne 2005) on Layer-2 itemization at the sub-percent precision the framework operates at. Frameworks that consume single-observable Layer-2 inputs in their native conventions cannot expose these mismatches structurally. This subsection consolidates the nine exposures catalogued to date (V.D.1–V.D.4 from the 2026-05-09 multi-track Roothaan autopsy sprint; V.D.5–V.D.6 from the 2026-05-10 V.D-prediction sprint; V.D.7–V.D.9 subsequently); each future sprint that surfaces a class (a) finding adds a new § V.D.N subsection with the five-element template *Observable* | *Source compilations* | *Convention difference* | *Magnitude* | *Sprint reference* + *Closure verdict.*

5.3.1 D 1S HFS Layer-2 itemization (Eides vs Pachucki–Yerokhin)

Observable. Deuterium 1S hyperfine splitting $\nu_{\rm HFS}(D) = 327\,384\,352.522$ Hz (Wineland–Ramsey 1972), used in the Sprint Calc-rZG global fit (§5.1, row "Self-consistent r_Z global extraction") to extract $r_Z(D)$ via the framework's §3.18 operator.

Source compilations. Eides–Grotch–Shelyuto 2007 [48] Tab. 7.3 (used as the canonical Layer-2 budget for H 21cm in this paper) and Pachucki–Yerokhin 2010 (the standard atomic-physics itemization for deuterium HFS).

Convention difference. The two compilations use different groupings for the recoil-NLO + multi-loop QED + finite-size contributions to ν_F^D. Eides Tab. 7.3 itemizes the leading Zemach line at -150 ppm; Pachucki–Yerokhin 2010, when fully expanded with the Friar–Payne 2005 closure analysis, places the itemization at -286 ppm. Both compilations close to the same total $\nu_{\rm HFS}(D)$ at the precision the deuterium HFS measurement reports; the difference is in how sub-leading recoil and multi-loop contributions are aggregated.

Magnitude. $\sim 0.01\%$ of ν_F, propagating to ~ 25 mfm in extracted $r_Z(p)$ via the global fit.

Sprint reference and closure. The original Sprint Calc-rZG fit (2026-05-09 morning) used the Eides -150 ppm itemization for deuterium; this gave $r_Z(D) = 6.18(132)$ fm, the well-known ~ 5 fm artifact. The diagnostic memo `debug/rzg_bug_diagnosis_memo.md` verified that the production cross-register V_{eN} kernel (`geovac.cross_register_vne`) reproduces leading-order Bethe–Salpeter recoil at 2.86% for hydrogen, 2.03% for deuterium, and 8.18% for muonium, species-agnostically and correctly. The artifact came from the Layer-2 budget specification for D HFS, not from the framework kernel. Re-anchoring to the Pachucki–Yerokhin 2010 budget at -286 ppm closed the extraction to $r_Z(D) = 2.583(1588)$ fm (within 0.01σ of Friar–Payne 2.593(16) fm). **Closure verdict:** resolved by Layer-2 budget re-anchoring. The original "recoil double-counting" diagnosis was a misdiagnosis; the issue was Layer-2 itemization specification, not a kernel bug.

Operator-level autopsy refinement (May 9 2026). The Track 5 operator-level autopsy of D 1S HFS (§5.2.7) operates exclusively at the Pachucki–Yerokhin 2010 / Friar–Payne 2005 -286 ppm Layer-2 budget convention. Component-by-component the framework reproduces (a) the $\hat{I}\cdot\hat{S}$ Hamiltonian at I=1 on the 6-dim joint nuclear-electronic Hilbert space at machine precision (multiplicity ratio D/H = 3/2 to 12 digits); (b) the §3.18 operator-level Zemach at $r_Z(D) = 2.593$ fm to $\sim 1.45 \times 10^{-14}$% of the LO shift; and (c) the cumulative chain residual, corrected on 2026-08-28 to -210.9 ppm. **The previously reported match between a $+285.6$ ppm residual and the -286 ppm PY 2010 Layer-2 budget magnitude is withdrawn**: it rested on a Bohr–Fermi baseline that was high by 496.5 ppm (§5.2.7). With the corrected baseline the residual changes sign and the budget itemization does not close; whether it closes under either compilation's convention is now an open item. No new convention mismatch surfaced at the operator level. This sharpens the $\sim 0.01\%$ / ~ 25 mfm magnitude estimate to: framework-side architecture is identical for H and D (same operators, same chains, same scope boundaries); the Layer-2 budget magnitudes differ by $\sim 16\times$ because the deuteron's nuclear-structure complexity (NN binding dynamics inside a weakly bound n+p system) makes its polarizability, multi-loop QED, and recoil NLO contributions much larger than the proton's at the same precision. The convention exposure is intrinsically a Layer-2 itemization split, NOT a framework architecture issue.

Cross-references. Originating documentation: §5.1 Sprint Calc-rZG row. Operator-level elaboration: §5.2.7. Source memos: `debug/calc_track_rZG_global_zemach_memo.md`, `debug/rzg_bug_diagnosis_memo` `debug/d_hfs_autopsy_track5_memo.md`.

5.3.2 μH $2S_{1/2}$–$2P_{1/2}$ VP one-loop split (Antognini vs Krauth)

Observable. Muonic hydrogen Lamb shift VP one-loop component, the dominant ($\sim +205$ meV out of $+202.37$ meV total) contribution to the CREMA 2010 measurement. Documented in the §5.2.3 Roothaan autopsy.

Source compilations. Antognini 2013 (the canonical muonic-hydrogen reference for CREMA Lamb-shift theory; used as the Layer-2 anchor in §5.2.3) and Krauth 2017 (the canonical muonic-hydrogen reference for the parallel HFS measurement and the most thorough recent itemization).

Convention difference. Antognini reports VP one-loop at $+205.0074$ meV with "VP-VP mixed" itemized separately at $+0.150$ meV. Krauth absorbs ~ 0.021 meV of higher-order VP into the one-loop line and reports $+205.0282$ meV. Both compilations agree on the bottom-line μH Lamb total to printed precision; the difference is in per-component itemization.

Magnitude. ~ 100 ppm in per-component split (totals agree at much tighter precision).

Sprint reference and closure. Sprint MH Track A (§5.1, row "μH $2S$–$2P$ Lamb shift") and Sprint Calc-muH Lamb autopsy v1 (§5.2.3) both operationally surfaced the convention difference: the framework's full-Uehling kernel matches Antognini 2013 at -0.10 ppm, which is $1000\times$ tighter than the Antognini-vs-Krauth ~ 100 ppm itemization gap. This means the framework's match precision is sensitive to the choice of reference compilation for Layer-2 inputs. **Closure verdict:** anchored to Antognini 2013 throughout the μH Lamb autopsy. No reconciliation against Krauth attempted at this time; flagged for future sprint if Krauth-anchored cross-checks become relevant.

Cross-references. Originating documentation: §5.2.3 Convention-mismatch surface paragraph. Source memo: `debug/muH_Lamb_autopsy_v1_memo.md` (LCM tag).

5.3.3 μ Lamb shift SE-recoil aggregation (framework vs Karshenboim)

Observable. Muonium $2S_{1/2}$-$2P_{1/2}$ Lamb shift self-energy bracket. Documented in the §5.1 row "Mu $2S_{1/2}$-$2P_{1/2}$ Lamb 1047.63 MHz vs Mariam 1982 experiment 1054(22) MHz."

Source compilations. Framework Eides §3.2 SE bracket (used in this paper's μ Lamb-shift framework calculation) and Karshenboim 2005 (the canonical muonium-Lamb reference).

Convention difference. The framework's m_{red} rescaling at the Eides bracket level absorbs the leading recoil-correction beyond reduced-mass automatically, giving SE bracket = +1074.13 MHz. Karshenboim itemizes "SE one-loop" at +1085.84 MHz with a separately-itemized "Recoil NLO" line at −11.30 MHz. The two reach exact agreement at the bracket level: $+1085.84 - 11.30 = +1074.54$ MHz vs framework +1074.13 MHz (within bracket-evaluation precision).

Magnitude. ~ 11 MHz on the μ Lamb-shift SE bracket ($\sim 1\%$ of the SE component, ~ 0.01 of the total Lamb shift); the difference is exactly the recoil-NLO itemization line.

Sprint reference and closure. Sprint precision-catalogue 2026-05-08 (Mu Lamb shift row, §5.1) operationally surfaced this: the framework total at 1047.63 MHz matches Karshenboim 2005 theory at 1047.49(5) MHz to +0.013%. Naively adding Karshenboim's separately-itemized "Recoil NLO −11.30 MHz" to the framework total double-counts the same recoil correction that the framework's m_{red} rescaling already absorbs. **Closure verdict:** resolved by recognizing the framework's m_{red}-bracket convention is structurally equivalent to Karshenboim's SE one-loop + Recoil NLO sum, and not adding the recoil NLO twice.

Cross-references. Originating documentation: §5.1 Mu $2S_{1/2}$-$2P_{1/2}$ Lamb row.

5.3.4 Friar moment profile convention (Eides closed-form vs operator-level)

Observable. Muonic hydrogen Friar moment, the $O((Z\alpha)^2)$ finite-nuclear-size correction in the §5.2.3 Lamb-shift autopsy (Component 3).

Source compilations. Eides Eq. 2.35 closed-form derivation (used as the Antognini 2013 reference value −3.84 meV at $r_p = 0.84087$ fm) and the framework's operator-level realization via the §3.18 module `geovac.magnetization_density`.

Convention difference. Eides Eq. 2.35 is derived assuming the input radius is $\langle r^2\rangle^{1/2}$ (RMS charge radius); the operator-level §3.18 module is calibrated to the first moment $\langle r\rangle = r_p$ (consistent with its existing HFS Zemach use case). For a three-dimensional Gaussian profile, the two calibrations differ by exactly the factor $\langle r^2\rangle/\langle r\rangle^2 = 3\pi/8 \approx 1.178$ [corrected 2026-08-29: an earlier version quoted the two-dimensional constant $4/\pi \approx 1.27$].

Magnitude. +17.7% overshoot on the Friar moment vs the Eides closed form (operator-level −4.33 meV vs −3.68 meV at fixed r_p; +12.7% vs the Antognini itemization value −3.84 meV). This is $(3\pi/8)\times$ the closed-form value to 0.1% relative precision.

Sprint reference and closure. Sprint Calc-muH Lamb autopsy v1 (2026-05-08, §5.2.3 Operator-level Friar moment paragraph) surfaced the discrepancy by computing the Friar moment both ways: closed-form Eides reproduction matches Antognini 2013 at −4.3%, while operator-level realization overshoots Antognini by +12.7%, with the ratio between the two paths equal to the three-dimensional Gaussian profile factor $3\pi/8$ to 0.13% [corrected 2026-08-29: an earlier version said $4/\pi$ to sub-percent precision; $4/\pi$ is the two-dimensional constant and fails by 8%]. **Closure verdict:** flagged for a sibling `MagnetizationDensitySpec` calibrated to RMS radius for the Lamb-shift FNS channel, distinct from the existing first-moment calibration for the HFS Zemach channel. The HFS Zemach channel is unaffected (cf. §5.2.2, where the operator-level Zemach result −39.495 ppm matches Eides −39.500 ppm at 0.012%). Resolution is mechanical (sprint-scale).

Cross-references. Originating documentation: §5.2.3 Operator-level Friar moment paragraph. Source memo: `debug/muH_Lamb_autopsy_v1_memo.md` (profile-convention finding).

5.3.5 D 1S HFS polarizability sub-component (Friar–Payne vs Pachucki–Yerokhin)

Observable. Deuterium 1S hyperfine splitting $\nu_{\rm HFS}(D) = 327\,384\,352.522$ Hz, with focus on the *deuteron polarizability* sub-component of the Layer-2 budget entering the framework's -210.9 ppm chain residual (§5.2.7 Track 5 autopsy). This refines the broader §5.3.1 entry to the dominant polarizability piece.

Source compilations. Friar–Payne 2005 (PRC **72**, 014002; building on Friar 1979 PRC **20**, 325) versus Pachucki–Yerokhin 2010 (the modern itemization framework adopted in Kalinowski–Pachucki–Yerokhin 2018 PRA **98**, 062513 for muonic D and applied analogously for electronic D), and the recent Bonilla et al. 2025 chiral-EFT deuteron nuclear-structure update (arXiv:2508.18776).

Convention difference. Friar–Payne 2005 aggregates the "low-energy nuclear-structure + polarizability + dispersion + Friar-moment recoil" contributions into a single low-energy aggregate term (e.g., $\delta E_{\rm Low}(eD) = 87.3$ kHz in Friar–Payne's reporting style). Pachucki–Yerokhin 2010+ separates polarizability ($\sim +240 \times 10^{-6}\, E_F \approx +78.6$ kHz) and Zemach ($\sim -100 \times 10^{-6}\, E_F \approx -32.7$ kHz) explicitly as distinct itemized lines, using chiral-EFT-derivable nuclear-structure operators rather than Friar–Payne effective Friar moments. Both compilations close to similar bottom-line totals at the experimental precision (Wineland–Ramsey 1972 ∼few ppb), but per-component itemization differs structurally. The disagreement reaches operational consequences: Bonilla et al. 2024/2025 documents a 5σ disagreement with previous calculations on muonic D HFS, while the underlying convention difference applies to electronic D HFS as well.

Magnitude. ~ 240 ppm of E_F for the polarizability term in the PY itemization [**note 2026-08-29: the chiral-EFT evaluation of Bonilla *et al.* gives the polarizability channel as $+110.16(38)$ kHz = $+336.5$ ppm and the *total* TPE as $44.5(1.1)$ kHz = $+135.9$ ppm; the spread between these itemizations is exactly the convention exposure this entry documents, and the *total* is the convention-independent quantity**]; the convention split (bundled-vs-separated) is ∼tens-to-hundreds of ppm of the observable; ~ 80 kHz absolute on the 327 MHz observable. Comparable to V.D.1's ~ 25 mfm in $r_Z(p)$ precedent; both at the 10^{-4} to 10^{-3} relative scale.

Sprint reference and closure. V.D-prediction sprint, 2026-05-10 (`debug/v_d_prediction_sprint_memo.md` Candidate A); refines §5.3.1 to the polarizability-specific sub-component, addressing the dominant Layer-2 budget item. **Closure verdict** (corrected 2026-08-29)**:** the framework's projection chain does *not* consume any polarizability input — the only Layer-2 scalar entering the D chain is $r_Z(D)$ (§5.2.7), which is why the residual is -210.9 ppm and why this row is an open case for Prediction 2. The convention choice documented here therefore does not yet propagate into the residual; it becomes live the moment a polarizability channel is added, and at that point the itemization spread (+336.5 ppm inelastic vs the +135.9 ppm total) is the leading systematic on the input. Currently anchored to Pachucki–Yerokhin 2010+ itemization (matching the framework's Track 5 autopsy assumption); the Friar–Payne 2005 convention would shift the budget bucketing without changing bottom-line totals.

Cross-references. Originating documentation: this sprint memo. Refines §5.3.1 (broader D HFS itemization) to the polarizability-specific axis. Cross-references §5.2.7 "Why D Layer-2 is deeper than H Layer-2" paragraph (deuteron polarizability is the dominant component there).

Honest caveat: overlap with V.D.1. V.D.1 documents the broader Eides Tab. 7.3 vs Pachucki–Yerokhin 2010 D HFS itemization mismatch. V.D.5 refines V.D.1 to the polarizability-specific sub-component on a distinct convention axis (Friar–Payne 2005 vs Pachucki–Yerokhin 2010+). The two entries differ in which compilation pair they compare; kept separate per V.D protocol since the convention axes differ. Future audit may merge V.D.5 into V.D.1 as a sub-paragraph if the two axes prove operationally equivalent.

5.3.6 He 2^3P relativistic Bethe-log reevaluation (Drake vs Pachucki–Yerokhin)

Observable. Helium 2^3P fine-structure intervals $\nu_{01} = 29\,616\,946.2(1.6)$ kHz and $\nu_{12} = 2\,291\,177.3(1.6)$ kHz (current theoretical precision), used historically as a determination of the fine-structure constant α to ~ 30 ppb. Documented in the §5.2.6 Roothaan autopsy.

Source compilations. Drake 1990 / Drake–Yan 1995 (Drake–Yan PRA **46**, 2378, 1992; Drake–Yan PRL **74**, 4791, 1995; Drake in *Long-Range Casimir Forces*, ed. F. S. Levin and D. A. Micha, Plenum, NY, 1993): original itemization of relativistic Bethe-log term and combining coefficients $(3/50, -2/5, 3/2, -1)$ that GeoVac uses sympy-exactly (Sprint 4 DD). Pachucki–Yerokhin 2009 (PRA **79**, 062516, 2009 "Reexamination of helium fine structure"; cf. Pachucki–Yerokhin 2011 Can. J. Phys. **89**, 95, arXiv:1011.1370): reevaluated the relativistic Bethe-log term and resolved a 3σ disagreement on ν_{01} that Drake's earlier itemization had with experiment.

Convention difference. Pachucki and Yerokhin (PRA **79**, 062516, 2009) explicitly state they "eliminated this [3σ] disagreement [vs Drake on ν_{01}] by reevaluating the relativistic Bethe logarithm term." Pachucki and Sapirstein "pointed out several computational mistakes and inconsistencies in earlier calculations, and eventually the results turned out to be consistent after Drake revised the sign" (per literature retrieval). This is a textbook example of inter-compilation itemization correction at the $\alpha^3(Z\alpha)^2$ multi-loop level, sitting inside the LS-8a renormalization-wall budget that the framework does not capture.

The relativistic Bethe-log term is part of the $\alpha^3(Z\alpha)^2$ multi-loop sector that the GeoVac framework attributes to its LS-8a structural wall. The framework's residual on $P_0 - P_1$ is -0.014%, on $P_0 - P_2$ is -0.20%, on $P_1 - P_2$ is -2.62% from partial-cancellation amplification (see §5.2.6). The Drake-vs-Pachucki–Yerokhin convention difference lives inside this same budget.

Magnitude. $\sim$kHz on the 29 GHz ν_{01} interval (Pachucki–Yerokhin's revised value 29 616 946.2 kHz vs Drake's older value disagreeing at the 3σ $\sim$few kHz level). In ppm: ~ 0.1 ppm on ν_{01}. In α-determination units: the helium 2^3P fine structure determines α to $\sim$ 27-31 ppb; the Bethe-log reevaluation was the load-bearing fix for the discrepancy that earlier α-determination work had against alternative determinations.

Sprint reference and closure. V.D-prediction sprint, 2026-05-10 (`debug/v_d_prediction_sprint_memo.md` Candidate C); refines the §5.2.6 Three-class diagnosis paragraph's "Class A: NEGATIVE" statement, which was correct at LO Breit-Pauli only, not at the $\alpha^3(Z\alpha)^2$ multi-loop level where the relativistic Bethe-log reevaluation lives. **Closure verdict:** this entry is the strongest of the V.D-prediction sprint candidates because the literature explicitly documents both the disagreement (Drake–Pachucki–Yerokhin original 3σ split) and its resolution (Pachucki–Yerokhin 2009 Bethe-log reevaluation, with Drake's subsequent agreement after sign revision per Pachucki–Sapirstein). It is not an inferred convention split but an explicitly noted correction in the published record.

Cross-references. Originating documentation: this sprint memo. Refines §5.2.6 "Three-class diagnosis" paragraph ("Class A: NEGATIVE" statement now read as "Class A: NEGATIVE at LO Breit-Pauli, POSITIVE at $\alpha^3(Z\alpha)^2$ multi-loop level"). Cross-references §5.1 He 2^3P catalogue rows.

5.3.7 μd $2S$–$2P$ Lamb deuteron polarizability (Carlson–Gorchtein–Vanderhaeghen vs Carlson–Vanderhaeghen)

Observable. Muonic deuterium $2S_{1/2}$–$2P_{1/2}$ Lamb shift $\Delta E = 202.8785(34)$ meV (Pohl et al., CREMA 2016). The dominant Layer-2 contribution to the theoretical prediction is the deuteron nuclear polarizability, a QCD-internal evaluation of virtual photonuclear excitations between the bound $n + p$ system.

Source compilations. Carlson–Gorchtein–Vanderhaeghen 2014 (*Phys. Rev. A* **89**, 022504): +1.690(20) meV. Carlson–Vanderhaeghen 2011 (earlier evaluation, distinct sub-channel decomposition): +1.94 meV. Two distinct QCD-internal evaluations, both anchored to deuteron photoabsorption data, with the 2014 update including refined sub-leading contributions (elastic-inelastic interference, subtraction-function refinement, finite-nuclear-size corrections to the polarizability kernel).

Convention difference. The 2014 evaluation explicitly splits the nuclear-structure correction into three sub-contributions (elastic intermediate states, inelastic photoabsorption, subtraction function); the 2011 evaluation used a broader-aggregate treatment that produced a ~ 0.25 meV larger value. Both evaluations converge to the same physical mechanism (virtual photoabsorption between $n + p$ via the muonic Lamb shift kinetic energy); they differ in how the QCD content is itemized.

Magnitude. $\sim 13\%$ spread on the *dominant* Layer-2 contribution at μd Lamb shift (+1.69 meV vs +1.94 meV; ~ 0.25 meV absolute). At the framework's -0.12% Lamb-shift residual, the convention choice between the two QCD evaluations sets whether the framework reads as "sub-percent closure" (with CGV 2014 anchoring, -0.12%) or "over-shoots experiment by $+0.12\%$" (with CV 2011 anchoring, $+0.12\%$). The framework's own residual is at the same precision as the spread between the two QCD evaluations.

Three-class tag. Class (i) inter-compilation itemization, distinct from §5.3.5 (deuteron HFS polarizability sub-component, Friar–Payne vs Pachucki–Yerokhin): this entry is between two QCD nuclear-structure evaluations of the deuteron polarizability rather than between atomic-physics itemization frameworks. Cross-references the §5.2.4 "Convention-mismatch surface" paragraph and W3 inner-factor wall (CLAUDE.md §1.7).

Verdict (multi-track Roothaan sprint Track 1, 2026-05-18): CONFIRMED. The two QCD evaluations explicitly disagree at $\sim 13\%$ on the dominant Layer-2 line of a precision-frontier observable. This is the first explicit Class (i) convention exposure originating in a QCD-internal sub-component evaluation rather than an atomic-physics itemization compilation. Refines the broader W3 inner-factor calibration question to a specific operational instance: until the deuteron polarizability convention is reconciled at the framework-residual scale, multi-observable extractions involving the deuteron (μd Lamb, D 1S HFS, isotope-shift extractions of r_d) cannot reach beyond the $\sim 13\%$ Layer-2-determined precision.

Cross-references. Originating documentation: `debug/muD_Lamb_autopsy_track1_memo.md`. Refines §5.2.4 "Convention-mismatch surface" paragraph. Sibling to §5.3.5 (D HFS polarizability sub-component; different observable, same W3 inner-factor wall). Cross-references CLAUDE.md §1.7 multi-focal-composition wall taxonomy W3.

5.3.8 Alkali ns valence HFS multi-electron contact-density cliff (class (iii): framework architecture)

Observable. Alkali-atom ground-state hyperfine splittings on single-zeta hydrogenic basis; concretely Li-7 $2^2S_{1/2}$ HFS = 803.504086(34) MHz (Beckmann–Boklen–Elke 1974), with generalization to Na, K, Rb, Cs (all alkali ns valence HFS share the same mechanism). Originating sprint: multi-track Roothaan sprint Track 5, 2026-05-18 (§5.2.12).

Source of mismatch. Hydrogenic-Z_{eff} approximation (Clementi–Raimondi 1963 for alkalis: $Z_{\text{eff}} = 1.279$ for Li 2s) vs SCF + multi-electron CI (Clementi–Roetti 1974; Yan & Drake 2003) at the level of the contact density $|\psi_{ns}(0)|^2$ *specifically.* Spatial extent ($\langle r \rangle$, polarizability) of the orbital is well-approximated by Z_{eff}; contact density is not.

Convention difference. The hydrogenic-Z_{eff} contact density $|\psi_{2s}(0)|^2 = Z_{\text{eff}}^3/(8\pi)$ at $Z_{\text{eff}} =$

1.279 gives $|\psi_{2s}(0)|^2 \approx 0.083$ in atomic units. Clementi–Roetti 1974 SCF Li 2s gives $|\psi_{2s}(0)|^2$ enhanced by a factor of $\sim 2.8\times$ [**corrected 2026-08-28: the retired** 0.665 / $\sim 8\times$ **pair came with the suppressed baseline and is excluded by experiment, which caps the enhancement at** $2.78\times$]. The mechanism is the §3.20 core-valence orthogonality constraint $\langle 2s|1s_{\rm core}\rangle = 0$ *exactly*, which pulls the 2s radial node inward and dramatically enhances amplitude at the origin. This is the well-known Fermi contact enhancement in alkali atoms.

Magnitude. $\sim 2.8\times$ on $A_{\rm hf}$ for Li-7 ($A_{\rm hf}^{\text{framework, hydro-}} = 144.6$ MHz vs $A_{\rm hf}^{\rm exp} = 401.75$ MHz), growing with Z to $\sim 63\times$ at Cs-133 (Table 24). **Corrected 2026-08-28:** this paragraph previously gave 41.5 MHz and $\sim 10\times$, and asserted "the same factor-$\sim 10\times$ undershoot *for all* alkali valence HFS observables". Those came from the suppressed driver (the value $41.5 = 144.6/3.4818$), and the uniformity claim is refuted by the corrected table: the undershoot is *not* a common factor but grows monotonically with Z. What survives is the structural prediction that the single-zeta hydrogenic architecture undershoots every alkali valence HFS, with the deficit increasing along the series.

Class. This is a class (iii) exposure — *framework architecture limitation*, distinct from class (i) inter-compilation itemization (entries V.D.1, V.D.2, V.D.3, V.D.5, V.D.6, V.D.7) and class (ii) closed-form vs operator-level convention (entry V.D.4 Friar profile RMS-vs-first-moment). Class (iii) exposes a structural property of the GeoVac graph-native basis choice (single-zeta hydrogenic with empirical $Z_{\rm eff}$) that the projection-chain dictionary alone cannot fix; closing it requires extending the graph-native basis to multi-zeta with explicit PK orthogonality at the orbital level.

Two cliffs, one mechanism. This exposure shares a structural mechanism with the §5.2.11 Track-2 Z>20 cliff (single-zeta basis with no radial nodes cannot represent the PK-orthogonalized SCF radial wavefunction). The Cs cliff manifests as a -47% undershoot on $A_{\rm hf}$ at $Z = 55$ *using the `FrozenCore` wavefunction*; the Li cliff manifests as a -64.0% undershoot at $Z = 3$ on the bare hydrogenic basis. The two cliffs are not independent walls but two manifestations of the same structural limitation. **Corrected 2026-08-28:** this passage previously read that the cliff scales "with Z inversely to expectation," the $Z = 3$ case being more affected than $Z = 55$. That comparison set a bare-hydrogenic Li against a `FrozenCore` Cs. On a like-for-like bare-hydrogenic basis the corrected sequence is *monotonically increasing* in Z (-64.0, -75.2, -95.5, -97.7, -98.4% for Li through Cs), i.e. in the expected direction: more inner shells to orthogonalize against, deeper cliff. The inverse-scaling reading is **withdrawn**; the shared-mechanism reading stands.

Named structural closure. Hylleraas–Eckart double-zeta extension for $n = 2$ multi-electron systems (`geovac/hylleraas_r12.py`, currently He singlet at $\omega = 4$ giving $+0.0006\%$ on $E_{\rm tot}$ — §5.2.10). The Hylleraas basis explicitly satisfies PK orthogonality at the amplitude level (no $Z_{\rm eff}$ fitting required), and closure of the multi-electron contact-density cliff and the Li atomic accuracy floor (1.10%, Paper FCI-A) is expected *simultaneously.* Sibling closure to the BBB93/KTT multi-zeta tabulation closure for the §5.2.11 Z>20 cliff (also named in §5.2.11).

Cross-references. Originating documentation: `debug/Li7_HFS_autopsy_track5_memo.md`. Cross-references §5.2.12 (Li-7 HFS autopsy) and §5.2.11 (Cs HFS scoping + Z>20 cliff diagnostic). Generalizes to Na ($Z = 11$), K ($Z = 19$), Rb ($Z = 37$), Cs ($Z = 55$) valence HFS as a precision-frontier challenge for the framework's atomic-spectroscopy reach.

5.3.9 Multi-focal-composition wall: bimodule cross-shifts vs. module endomorphisms (Sub-sprint M-Z, 2026-05-23)

Observable. The multi-focal-composition wall named in CLAUDE.md § 1.7: five independent observables (Sprint H1 Yukawa selection, LS-8a multi-loop QED renormalization, HF-3 recoil, HF-4 Zemach, HF-5 multi-loop a_e) hitting one structural wall on the framework's ability to compose multiple Fock-style projections.

Source compilations / framing perspectives. The single-observation reading (each observable a separate failure mode) versus the Hilbert C^*-bimodule reading of the modular propinquity literature (Latrémolière 2019, 2021; arXiv:1608.04881, arXiv:1811.04534).

Convention difference. Single-observation reading treats each instance as an independent kernel-approximation gap; the bimodule reading splits the wall cleanly into two structural pieces: *bimodule cross-shifts* (the framework's production code does compose them: HF-3 recoil via `geovac/cross_register` HF-4 Zemach via `geovac/magnetization_density.py`, HF-5 Parker–Toms first-order curvature on Dirac-S^3, FNS via Foldy–Friar, cross-register V_{eN}) versus *module endomorphisms* (the framework does NOT compose them: LS-8a renormalization counterterms Z_2-1 and δm, Sprint H1 Yukawa Y, inner-factor calibration data). The cross-shift piece is empirically handled (e.g., HF-3 at sub-ppm match with Bethe–Salpeter recoil); the endomorphism piece is the LS-8a wall precisely. This is an architecture-class (iv) exposure (framework-level structural taxonomy refinement); not a literature-convention mismatch in the strict sense.

Magnitude. Qualitative structural finding at this level. Quantitative tests live in the constituent sprints: LS-8a yields the two-loop QED renormalization gap (Paper 35 § VII.3 Refined Prediction 1); the W1c Layer 3 finding (Paper 19 (W1c-residual), 2026-05-23 α-PES test) is the most recent concrete demonstration that the module-endomorphism piece can be invisible at finite basis size when the FCI variational state does not occupy the affected subspace.

Sprint reference and closure. Sub-sprint M-Z (`debug/sprint_modular_propinquity_mZ_bethe_log_memo.m` 2026-05-23) surfaced the bimodule reading; the modular propinquity synthesis memo (`debug/sprint_modular_prop` consolidates it as the load-bearing structural finding of the modular sprint's chemistry-side thread. The Track β synthesis (`debug/sprint_modular_alpha_arc_synthesis_memo.md`) extends the consolidation with the W1c Layer 3 finding from the follow-on α arc.

5.3.10 Three-bucket M-Z partition hypothesis (basis-closable cross-shift as intermediate class) — FALSIFIED via NaH max_n=3 test

Observable. The W1c $\times$ M-Z partition bridge sprint (`debug/sprint_w1c_mz_partition_analysis_memo.md`, 2026-05-23) proposed a three-bucket refinement of the M-Z partition (§5.3.9): cross-shift (handled) / *basis-closable cross-shift* (handled at sufficient basis size) / endomorphism (external input). The middle bucket was introduced to classify W1c sub-layer 3 (orbital-pair flexibility at NaH max_n=2), on the reading that the bonding combination [H 1s $\pm$ Na 3s]$_\pm$ is structurally a bimodule cross-shift (left+right linear combination in the bimodule basis) that the FCI cannot realize at max_n=2 due to dimensional poverty alone.

Source compilations / framing perspectives. Bridge sprint's three-bucket hypothesis (the refinement to test) vs. the original two-bucket M-Z partition (the null hypothesis). Three falsifiable predictions for NaH max_n=3 were prespecified and frozen into `debug/data/sprint_w1c_mz_partition_predictio` before the test ran (curve-fit-audit discipline at inter-sprint scope): **P1** (internal PES minimum at $R_{\text{eq}} \in [3.0, 4.5]$ bohr), **P2** (binding $D_e \in [0.0375, 0.150]$ Ha within 2$\times$ experimental $D_e \approx 0.075$ Ha), **P3** (multi-zeta differential $\in [0.02, 0.30]$ Ha at R_{eq}).

Convention difference and verdict. The F1 max_n=3 sprint (`debug/sprint_f1_maxn3_predictions_test_` 2026-05-23) returned 1 of 3 predictions passing: P1 FAIL ($R_{\min} = 2.0$ bohr, the smallest tested R, on the exact falsification criterion); P2 FAIL (spurious-binding descent depth $D_e^{\text{PES}} = +0.7097$ Ha, $\sim 10\times$ experimental); P3 PASS (multi-zeta differential 0.2066 Ha at $R = 2.0$, consistent with max_n=2's 0.21 Ha at the same R). **CLEAN NEGATIVE per the prespecified gate.** The proposed three-bucket refinement is falsified; the two-bucket M-Z partition stands.

Magnitude (structural finding). The substantive new content the prediction test did not anticipate is the structural success / energetic failure split. At max_n=3 under combined W1c+mz, the dominant natural orbital IS a true bonding combination: $\phi_{\text{dom}} = -0.698 \cdot \phi_{\text{Na, 3s}} - 0.687 \cdot \phi_{\text{H, 1s}}$ with 50/50 amplitude^2 split (the 2nd NO is the antibonding combination). At max_n=2 the dominant NO was 100% H-localized at every architecture. *The basis enlargement DID provide the orbital-mixing flexibility predicted by the bridge sprint.* But the constructed bonding orbital has higher energy than the separated configuration at every R; natural occupations remain $[1.0, 1.0]$ (open-shell singlet), NOT $[2.0, 0.0]$ (closed-shell bond). The framework can BUILD the right orbital but cannot energetically PREFER it. Sub-layer 3 reclassifies from the bridge sprint's "basis-closable cross-shift" to "module endomorphism (basis-irreducible at tested scales)" and joins sub-layer 2 (multi-zeta basis-shape) in the endomorphism bucket.

Class. This is an architecture class (iv) exposure (framework-level structural taxonomy refinement), sibling to §5.3.9 (the parent M-Z partition entry). The exposure is FALSIFICATION of a proposed sub-classification, not discovery of a new convention mismatch. It is retained in the running catalogue because the methodological lesson — the structural success / energetic failure split is a robust new diagnostic, with the two-bucket partition surviving while the three-bucket refinement evaporates — sharpens where chemistry endomorphisms localize in the M-Z framework.

Same-day refinement: F2 + F3 sharpen sub-layer 3 further. F1 max_n=3's verdict (sub-layer 3 as "module endomorphism, basis-irreducible at tested scales") is structurally correct but was further sharpened by two same-day same-arc follow-on sprints: F2 (cross-V_{ne} kernel-shape diagnostic, KERNEL-NOT-IT verdict) ruled out kernel-shape substitution as a closure mechanism (multipole kernel bit-faithful to converged 3D quadrature within $\sim 10^{-5}$ Ha, six orders below wall depth) and discovered the framework's h1 in `composed_qubit` is strictly block-diagonal — the cross-block off-diagonal h1 slot is *architecturally absent*, not just kernel-error-flawed. F3 (cross-block h1 architectural extension) CLOSED W1d (the architectural absence named by F2) at the FCI level: naturals transform $[1.0, 1.0] \to [1.9991, 0.0007]$ at NaH max_n=2 (four-orders-of-magnitude advance in dominant occupation signature). But F3 surfaced a new sub-layer W1e (inner-region overattraction): the bonding pair lacks Pauli repulsion against the [Ne] frozen core; PES still monotonically descending with $D_e^{\text{F3}} = 4.37$ Ha $= 58\times$ experimental. **Net refined reading:** chemistry sub-walls split into THREE structurally distinct sub-categories within the two-bucket M-Z partition (cross-shift / endomorphism / **architectural-absence**), with architectural-absence sub-walls conditionally closable via architectural extension (W1d via `geovac/cross_block_h1.py`). Chemistry endomorphisms split further into basis-shape data (sub-layer 2, mitigable via FrozenCore + atomic-physics fits), kernel-energetics (F1's reading of sub-layer 3, now sharpened to architectural-absence + Pauli-repulsion candidates), and Pauli-class constraints (W1e, F4 target via bonding-orbital PK extension). See §5.3.11 for the F2+F3 follow-on entry.

Sprint reference and closure. W1c × M-Z bridge sprint (`debug/sprint_w1c_mz_partition_analysis_memo.` F1 max_n=3 prediction test (`debug/sprint_f1_maxn3_predictions_test_memo.md`); Sprint β-neg synthesis (F1-maturity, superseded for framing by the full-arc memo) (`debug/sprint_beta_neg_synthesis_memo.m` F2 architectural-absence diagnostic (`debug/sprint_f2_cross_vne_kernel_memo.md`); F3 cross-block h1 closure (`debug/sprint_f3_cross_block_h1_memo.md`); comprehensive full-arc synthesis (`debug/sprint_w1c_ful` Cross-references Paper 19 (W1c-residual wall + F1-F2-F3 arc extension), CLAUDE.md §1.7 (multi-focal-composition wall taxonomy), CLAUDE.md §3 (dead-ends rows for the three-bucket hypothesis and kernel-shape substitution as W1c-residual closure mechanism), and §5.3.9 (parent M-Z partition entry whose three-bucket refinement is here falsified).

5.3.11 Architectural-absence sub-category for chemistry sub-walls (F2 diagnosis + F3 closure of W1d; W1e newly named) — W1d CLOSED at FCI level, W1e OPEN as F4 target

Observable. Same-day same-arc follow-on to the three-bucket falsification entry above (§5.3.10): two sprints (F2 + F3) tightened the W1c-residual sub-layer hierarchy at NaH max_n=2 beyond F1 max_n=3's verdict. F2 (cross-V_{ne} kernel-shape substitution diagnostic) executed a Step 1 algebraic gate before any production extension: five matrix elements computed via converged 3D quadrature vs the framework's default multipole expansion at $L_{\max} = 4$. Largest converged differential 2.0×10^{-5} Ha on the H 1s diagonal cross-V_{ne} contribution ($2.7 \times 10^{-4} = 0.027\%$ of bond-energy scale 0.075 Ha). F3 (cross-block h1 architectural extension) implemented the missing matrix slot: new module `geovac/cross_block_h1.py` ($\sim$ 437 lines, 18 new tests) computes off-diagonal h1 elements $\langle \psi_a^A \mid T + \sum_C(-Z_C/|r - R_C|) \mid \psi_b^B \rangle$ between orbitals on different centers $A \neq B$ via 2D axial Gauss-Legendre quadrature. `build_composed_hamiltonian` and `build_balanced_hamiltonian` gain `cross_block_h1` kwargs; bit-exact backward compat at default `cross_block_h1=False`.

Source compilations / framing perspectives. F2's diagnostic-first gate logic ruled out the obvious kernel-error hypothesis at six orders of magnitude below wall depth, ruling out kernel-shape substitution as a W1c-residual closure mechanism. F3's algebraic Step 1 diagnostic predicted strong bonding splitting (h1 lower-eigenvalue 3.46 Ha for hydrogenic and 4.20 Ha for multi-zeta, both $\gtrsim 70\times$ the 50 mHa decision-gate threshold). F3 Steps 3+4 ran the production-wired FCI + mini-PES tests.

Convention difference and verdict. F2 verdict: **KERNEL-NOT-IT**. The multipole expansion is bit-faithful where it lives; the W1c-residual wall is NOT a kernel-truncation artifact within the existing block-diagonal architecture. F2's substantive new content (**architectural-absence finding, W1d named**): the framework's h1 in `composed_qubit` is strictly block-diagonal — cross-block h1 matrix elements for orbitals on different centers are *architecturally absent*, not just kernel-error-flawed. F3 verdict: **PARTIAL-CLOSURE** — bonding-orbital construction CLOSED at the FCI level; inner-region overattraction OPEN (**new sub-layer W1e**). Adding cross-block h1 transforms FCI naturals $[1.0000, 1.0000] \to [\mathbf{1.9991}, \mathbf{0.0007}]$ at NaH max_n=2 R=3.5 — four-orders-of-magnitude advance in the dominant natural-orbital occupation signature, with the bonding combination as a doubly-occupied ground state. Concurrently the h1 lowest eigenvector character flips from H-localized at -0.802 Ha to bonding combination Na/H = 0.51/0.49 at -3.161 Ha ($\sim$ 2.4 Ha lower).

Magnitude (substantive structural finding). F3 mini-PES across $R \in [2, 10]$ bohr is monotonically descending with $R_{\min} = 2.0$ (smallest tested) and well depth $D_e^{\mathrm{F3}} = +4.37$ Ha vs experimental NaH $D_e \approx 0.075$ Ha ($58\times$ too deep). Dominant NO is bonding (50/50 Na/H) at every R — the cross-block h1 construction is robust across the full PES, not a localized feature. The new well-depth is WORSE than W1c+mz alone (0.69 Ha): cross-block h1 enables bonding without supplying opposing Pauli repulsion against the [Ne] frozen core. This is the W1e mechanism: 2-electron FCI on NaH max_n=2 has NO explicit frozen-core electrons (the [Ne] core is a screening potential via W1c, not occupied orbitals); no Pauli-exclusion mechanism prevents the bonding pair from collapsing toward small R. Structural analog of the missing-Pauli-repulsion mechanism that standard Phillips-Kleinman pseudopotentials supply, but the framework's existing PK cross-center operates on partner-side valence orbital diagonal — not on the bonding-combination orbital that cross-block h1 newly constructs.

Class. This is an architecture class (iv) exposure (framework-level structural taxonomy refinement) sibling to §5.3.10 (the parent F1-arc entry) and §5.3.9 (the grandparent M-Z partition entry). The exposure CONFIRMS the F2 architectural-absence diagnosis at the FCI level (F3) and surfaces W1e (inner-region overattraction) as a new structural sub-layer to be addressed by F4 (bonding-orbital PK extension, sprint-scale). Refines the methodological lesson of §5.3.10: chemistry sub-walls have a richer sub-structure than the two-bucket M-Z partition alone captures — architectural-absence sub-walls are CONDITIONALLY CLOSABLE via architectural extension (W1d demonstrated), and the architectural-extension ladder (W1c → W1c-multi-zeta → W1d → W1e?) is the chemistry-side analog of refinements that the QED side does not natively support (QED endomorphisms strictly external-input via UV-completion physics).

Sprint reference and closure. F2 cross-V_{ne} kernel-shape diagnostic (`debug/sprint_f2_cross_vne_kernel_me` F3 cross-block h1 architectural extension (`debug/sprint_f3_cross_block_h1_memo.md`); comprehensive full-arc synthesis (`debug/sprint_w1c_full_arc_synthesis_memo.md`). Cross-references Paper 19 (W1c-residual wall + F1-F2-F3 arc extension with revised five-sub-layer hierarchy), Paper 17 §6.10 (composed-architecture cross-reference, updated to F3 maturity), Paper 32 §VIII Sprint M-Z addendum (updated with W1d/W1e architectural-extension reading), CLAUDE.md §2 (F1-F2-F3 comprehensive sprint bullet), CLAUDE.md §3 (dead-ends rows for three-bucket hypothesis and kernel-shape substitution), and §5.3.10 (parent entry).

5.3.12 W1e three-sub-sub-mechanism refinement after F4–F6 (β-update, 2026-05-23)

Observable. Same-day same-arc follow-on to the F3 W1d-CLOSED + W1e-newly-named verdict above (§5.3.11): three sprints (F4, F5, F6) tested all three candidate W1e closure mechanisms F3 §6 had named, and refined the W1e mechanism characterization into a structurally distinct multi-sub-mechanism reading. **F4** (bonding-orbital Phillips–Kleinman extension on the F3-constructed bonding orbital): RULED OUT at Step 1 algebraic diagnostic + FCI rank-1 PK sensitivity injection. **F5** (explicit-core Hartree treatment of the [Ne] core electrons via $\sum_c (J-K)$): RULED OUT at Step 1 algebraic diagnostic. **F6** (valence basis enlargement to NaH max_n=4 with full F3 stack): PARTIAL_IMPROVEMENT only.

Source compilations / framing perspectives. The three sprints each ran a Step 1 diagnostic before any production implementation, per the diagnostic-before-engineering discipline (CLAUDE.md §1.7, `feedback_diagnostic_before_engineering.md`). F4 Step 1 computed bonding-core overlap from the full F3 h1 diagonalization and ran an FCI sensitivity test injecting a rank-1 PK shift at varying magnitudes. F5 Step 1 computed the cross-block 2-body Coulomb correction $\sum_c (J-K)_{\mathrm{bonding},c}$ between the F3 bonding orbital and the five Na [Ne] core orbitals via Hartree mean-field plus Slater–Condon exchange. F6 executed full production wiring at max_n=4 (Q=120, M=60, FCI dim 3600; ∼5 min/single-point) across a 5-point PES.

Convention difference and verdict.

F4 verdict: RULED OUT. Bonding orbital has substantial Na 2s core overlap ($|S| = 17.5\%$, well above the 5% strong-gate threshold) but the Phillips–Kleinman barrier at strict absolute reference $E_v = 0$ is only +0.194 Ha (22× smaller than 4.37 Ha wall depth). FCI sensitivity test injecting rank-1 PK shift $\Delta H = c \cdot |\mathrm{bond}\rangle\langle\mathrm{bond}|$ shows 0.05% closure at the Step 1-predicted $c = +0.19$ Ha, and saturates at 43% closure even at $c = 2.0$ Ha (10× predicted). **W1e is NOT a single-particle Pauli-orthogonality wall.**

F5 verdict: RULED OUT. Hartree mean-field $J = +1.13$ Ha; Slater–Condon exchange $K = +0.008$ Ha; predicted correction $J - K = +1.12$ Ha. Right SIGN (repulsive, would lift inner-region collapse) but only 25.7% of wall depth. Linear extrapolation predicts residual $D_e^{\mathrm{F5}} \approx +3.25$ Ha = 43× experimental NaH $D_e \approx 0.075$ Ha. **W1e is NOT addressable by Hartree-level core-bonding $J - K$ interaction alone.**

F6 verdict: PARTIAL_IMPROVEMENT. At max_n=4 with full F3 stack, PES well depth $D_e^{\mathrm{PES,\ F6}} = +3.929$ Ha at $R_{\min} = 2.0$ bohr (smallest tested) vs F3 max_n=2 baseline $D_e^{\mathrm{F3}} =$

+4.374 Ha — 10.2% **closure**. The 2-point gate at R=3.5/R=10 (the standard F1/F2/F3 quick-gate) gives 26.1% apparent closure but **this is ARTIFACTUAL**: the inner-region collapse continues from R=3.5 down to R=2.0 with another 0.696 Ha descent the 2-point gate misses. Bonding signature preserved at every R (dominant NO $\in [1.9971, 1.9992]$; Na/H amplitude $\approx 50/50$). **W1e is NOT addressable by basis enlargement at max_n=4 alone.**

Magnitude (substantive structural finding). W1e decomposes into three structurally INDEPENDENT sub-sub-mechanisms with closure ceilings:

- **W1e-basis-truncation:** ∼10.2% closure ceiling at max_n=4 PES level (F6 measured).
- **W1e-Hartree-pressure:** ∼25.7% closure ceiling at max_n=2 2-point level (F5 predicted; PES-level may be lower per F6 2-pt-vs-PES caveat).
- **W1e-deep-correlation-residual:** ∼65–90% of the wall remains beyond mean-field + single-determinant + basis-enlargement architecture. Remaining closure candidates (Schmidt orthogonalization, fully-correlated [Ne] cores) are beyond-sprint-scale architectural investments.

Methodological side finding: 2-point-vs-PES discrepancy. F6 reveals that the 2-point gate at $R = 3.5$ bohr vs $R = 10$ bohr (the standard F1/F2/F3 quick-gate) **overestimates PES-derived closure by 2.5× on monotonically-descending PES in this regime**. The inner-region collapse extends below $R = 3.5$ down to $R_{\min} = 2.0$ with significant additional descent (∼0.7 Ha at max_n=4). Future quick-gate tests in this regime should use a 3-point or 5-point mini-PES with $R_{\min}$ unbounded below 3.5 bohr, not a 2-point differential at fixed reference geometry. The F3-reported $D_e^{\mathrm{F3}} = +4.37$ Ha is itself a 2-pt value; the corresponding F3 PES well depth at $R_{\min} = 2.0$ is likely larger (untested but expected by analogy to F6).

Class. This is an architecture class (iv) exposure (framework-level structural taxonomy refinement) sibling to §5.3.11 (the parent F3 entry) and §5.3.10 (the grandparent F1-arc entry). The exposure refines the W1e mechanism characterization from "missing Pauli repulsion" (F3 working hypothesis) to "deep multi-determinant correlation requiring architectural extension beyond mean-field + single-determinant + basis-enlargement." The three candidate closure mechanisms F3 §6 ranked by priority are now all empirically tested: priority 1 (F5 explicit-core Hartree) closes at most 25.7%; priority 2 (F4 bonding-orbital PK) closes at most 43% in the FCI sensitivity test; priority 3 (F6 basis enlargement to max_n=4) closes only 10.2% at PES level. The remaining candidates not yet tested are beyond-sprint-scale architectural lifts (Schmidt orthogonalization at basis-construction level; fully correlated [Ne] cores requiring occupancy-restriction FCI infrastructure not present in framework).

Sprint reference and closure. F4 bonding-orbital PK STOP-at-Step-1 (`debug/sprint_f4_bonding_pk_memo` F5 explicit-core Hartree STOP-at-Step-1 (`debug/sprint_f5_explicit_core_memo.md`); F6 max_n=4 NaH PARTIAL_IMPROVEMENT (`debug/sprint_f6_maxn4_nah_memo.md`); Sprint β-update F4-F6-APPENDED-TO-F3-BASELINE synthesis (`debug/sprint_beta_update_f4f6_memo.md`). Cross-references Paper 19 (W1e refinement, F4–F6) (refined six-sub-layer hierarchy with the three W1e sub-sub-mechanisms), Paper 17 §6.10 (composed-architecture cross-reference, updated to β-update F4–F6 maturity), Paper 32 §VIII Sprint M-Z addendum (extended with F4–F6 refinement), CLAUDE.md §2 (F4–F6 β-update bullet appending to the F3-maturity baseline), CLAUDE.md §3 (three new dead-ends rows for the three ruled-out W1e closure mechanisms).

5.3.13 Cross-domain instance of the multi-focal-composition wall (W1e chemistry, Schmidt + core correlation closeout, 2026-05-23)

Observable. Closeout of the W1e (inner-region overattraction) chemistry-arc sprint-scale candidate space, with cross-domain pattern recognition. Two Day-1 diagnostics tested the two beyond-sprint-scale candidate closure mechanisms named in §5.3.12 (Schmidt orthogonalization at basis-construction level; fully correlated [Ne] cores), executing the diagnostic-before-engineering rule at sprint-cycle scale. **Both ruled out.** Schmidt produces a +8.6 mHa diagonal-element differential at NaH $R = 3.5$ bohr (only 0.2% of wall depth, 12× below the 100 mHa GO threshold); literature order-of-magnitude estimates for [Ne] core-correlation differential at NaH R_{eq} converge across three independent anchors to 0.001–0.003 Ha (14–1000× below the 0.1 Ha DEFER threshold). The W1e wall is now characterized as deep multi-determinant correlation beyond all sprint-scale closure mechanisms.

Source compilations / framing perspectives. The W1e wall joins five previously-identified structural walls, all in physics observables: (1) Sprint H1 Yukawa (the framework's AC extension admits a Higgs structurally but does not autonomously select Y_F; Marcolli–van Suijlekom Yang-Mills-without-Higgs reading at the structural level); (2) Sprint LS-8a $Z_2/\delta m$ (iterated Connes–Chamseddine spectral action reproduces the UV-divergent integrand of two-loop QED self-energy with correct prefactor, sign, and divergence structure, but does not autonomously generate the renormalization counterterms required for the finite $C_{2S} = +3.63$ extraction); (3) HF-3 recoil (no cross-register two-body coordinate operator $V_{ne}(\hat{r}_e, \hat{R}_n)$ in the framework; the Track-NI nuclear–electronic composed architecture couples spin labels across registers but not spatial coordinates); (4) HF-4 Zemach (no magnetization-density operator at the proton spatial coordinate; r_Z must enter as Layer-2 calibration input); (5) HF-5 multi-loop a_e (same renormalization wall as LS-8a, vertex sector instead of self-energy). **The W1e wall is the first chemistry-domain instance of this pattern.**

Convention difference and verdict. The unified structural reading across the six instances: the framework determines the *structural skeleton* of each observable (selection rules, transcendental classes, scaling laws, the build of the right basis states, the correct projection-chain decomposition), but does *not* autonomously generate the calibration data that connects the structural skeleton to specific numerical observable values (Yukawa values, renormalization counterterms, cross-register spatial-coordinate operators, magnetization radii, multi-loop QED finite parts, second-row binding energetics). *Calibration data in both physics and chemistry domains is genuinely external input within the framework's current scope.*

Magnitude (chemistry-side closeout). W1e closure-mechanism candidate space at sprint scale (F4 + F5 + F6 + Schmidt + core correlation): joint upper-bound closure $\sim$ 10–25% of the 4.37 Ha wall depth at NaH $n_{\max} = 2$; experimental $D_e = 0.075$ Ha requires $\sim$ 98% closure. Scales mismatch by $\sim 50\times$. Remaining candidates are major architectural commitments beyond sprint scale (self-consistent Z_{eff} depending on cross-center electronic density; fully-correlated all-electron treatment beyond the active-space architecture) or acceptance of the structural-skeleton scope as the framework's autonomous boundary.

Class. This is an architecture class (iv) exposure (framework-level structural taxonomy refinement) and *the first explicit cross-domain instance documented in the catalogue.* The pattern's cross-domain transferability — physics (five observables) plus chemistry (W1e), structurally distinct calibration-data classes hitting the same structural-skeleton-scope boundary — is the substantive new content this entry contributes: not a new convention exposure on a single observable, but a refinement of the framework's structural-skeleton scope statement (CLAUDE.md §1.7) from a five-physics-observable pattern to a six-observable cross-domain pattern.

Sprint reference and closure. Schmidt orthogonalization Day-1 diagnostic STOP (`debug/sprint_w1e_schmi` [Ne] core-correlation literature order-of-magnitude estimate DEFER (`debug/sprint_w1e_core_correlation_estin` Cross-references Paper 19's W1e Schmidt / core-correlation closeout section (W1e Schmidt + core correlation closeout), Paper 32 §VIII Sprint M-Z addendum (cross-domain wall extension), Paper 17 §6.10 (composed-architecture cross-reference, updated to Schmidt + core correlation closeout), and

CLAUDE.md §1.7 multi-focal-composition wall taxonomy.

Cross-pattern observation. The exposures partition into *four* structural classes. *Class (i) inter-compilation itemization* (entries §5.3.1, §5.3.2, §5.3.3, §5.3.5, §5.3.6, §5.3.7): six of the nine exposures are between Friar–Payne 2005, Eides–Grotch–Shelyuto 2007 [48], Krauth 2017, Karshenboim 2005, Drake 1990 / Drake–Yan 1995, Pachucki–Yerokhin 2009/2010, Antognini 2013, Carlson–Gorchtein–Vanderhaeghen 2014 / Carlson–Vanderhaeghen 2011 compilations on items at sub-percent precision. The pattern is consistent: most precision-physics compilations agree on bottom-line totals but differ in itemization at the 10^{-4} to 10^{-3} level. This is exactly the precision range the framework's projection-chain decomposition operates at, hence the surface contact between the framework's diagnostic instrument and the literature compilations. *Class (ii) closed-form vs operator-level convention* (entry §5.3.4): one exposure is between Eides closed-form derivation (which assumes RMS-radius input) and the framework's operator-level realization (which calibrates to first-moment input), with the two paths differing by a profile-dependent factor that the closed-form derivation hides. Class (ii) is internal to the framework's operator-level realization choices, not a literature convention mismatch in the strict sense, but it is the same kind of finding: sub-percent agreement on totals masks per-channel calibration choices that surface only when the two paths are compared at operator level. *Class (iii) framework-architecture limitation* (entry §5.3.8): a new class introduced 2026-05-18 by the multi-track Roothaan Track-5 Li-7 HFS autopsy. The single-zeta hydrogenic basis with empirical Z_{eff} cannot represent the PK-orthogonalized contact density $|\psi_{ns}(0)|^2$ at the alkali valence; the cliff is $\sim 2.8\times$ for Li 2s (growing to $\sim 63\times$ at Cs) and same mechanism as the §5.2.11 Z>20 cliff. Class (iii) is not a literature convention mismatch nor an operator-level calibration choice but a property of the framework's graph-native basis architecture itself. Closing it requires extending the basis (Hylleraas–Eckart for low-Z, BBB93/KTT multi-zeta or self-consistent HF for high-Z), not the projection-chain dictionary. Eq. (5) makes the target precise: what the extended basis has to reproduce is the *effective* quantum number $\nu \approx 1.7$ of the valence orbital, not a larger Z_{eff} and not more shells. A single-zeta hydrogenic function carrying an integer label n and a screened charge cannot produce $\nu \approx 1.7$ at $n = 6$ by any choice of Z_{eff} — the deficit is in the radial scale, which is what makes this an architecture-class exposure rather than a parameter-fitting one. *Honest scope:* Eq. (5) is a diagnosis, not a framework prediction — it consumes ν from the measured ionization energy, so it is a Layer-2-fed account of the cliff's size, and its value is that it identifies *which* structural quantity the basis is missing. All four classes vindicate the directive's finding types (class (iv), the architectural-absence family, is tagged row-by-row in the table): multi-observable global fits using a single dictionary expose itemization choices that single-observable analyses cannot see.

Pattern broadening (V.D-prediction sprint, 2026-05-10). V.D.5 (D HFS polarizability sub-component) is HFS-class, consistent with the original four entries. V.D.6 (He 2^3P relativistic Bethe-log) is *Lamb-class precision spectroscopy*, NOT strictly HFS. The running pattern is therefore best read as "precision-spectroscopy observables with multi-component Layer-2 decomposition" rather than "HFS-only." All six entries remain in the same general category: observables where the framework operates at sub-percent precision and where multiple independent compilations exist with potentially different itemizations. The next falsifier-test for this pattern is to attempt § V.D entries on (a) heavy-atom HFS at coarse experimental precision (Cs, Rb), (b) molecular vibrational/rotational spectroscopy, or (c) non-precision atomic spectra. If 0 of those land class-(i), Pattern C tightens to "exclusive to high-precision atomic spectroscopy with multi-loop QED + nuclear-structure budgets."

Protocol for adding new exposures. Each future sprint that surfaces a class (a) finding adds a new § V.D.N subsection following the five-element template: *Observable* | *Source compilations* | *Convention difference* | *Magnitude* | *Sprint reference + Closure verdict.* The exposure is also added as a row to Table 25. Each entry should cross-reference the § 3 projection chain whose operator-level decomposition surfaced the convention difference, and the originating documentation in the catalogue (§5, §5.1, or §5.2). The catalogue is a running register; expect it to grow as the multi-observable focal-length decomposition program (CLAUDE.md § 1.8) advances.

Predictive refinement: where to look for the next exposures. The four entries above all involve hyperfine-class or muonic-Lamb observables. The mechanism is structural: HFS observables decompose into many sub-leading Layer-2 corrections at sub-percent precision (leading Zemach, NLO recoil-mixing, finite-size, multi-loop QED, polarizability, hadronic VP), each compiled differently across Eides–Grotch–Shelyuto 2007 [48], Karshenboim 2005, Krauth 2017, Pachucki–Yerokhin 2010, and Friar–Payne 2005. Sub-percent itemization differences compound at exactly the precision the framework operates at. This suggests three pre-named candidate observables for the next class (a) findings, in decreasing prior probability:

1. **Deuteron polarizability convention.** Pachucki–Yerokhin 2010 versus Friar–Payne 2005 itemization of the deuteron polarizability contribution to D HFS. Expected magnitude: tens of ppm. **Verdict (V.D-prediction sprint, 2026-05-10): CONFIRMED → V.D.5 (§5.3.5).**
2. **Positronium HFS annihilation channel itemization.** Czarnecki–Melnikov–Yelkhovsky 2000 versus Karshenboim 2005 itemization of the α^5 corrections to the Ps 1S annihilation contribution. Expected magnitude: $\sim$ 100 ppm of LO HFS. **Verdict (V.D-prediction sprint, 2026-05-10): INCONCLUSIVE** — the three relevant compilations (CMY 2000, Karshenboim 2005, Adkins–Czarnecki 2014) operate at three different orders of the expansion (mα^6, review-aggregate, mα^7) and are not in convention conflict at the sprint scope; flagged for follow-up PDF-level diagnostic (sprint-scale) on Karshenboim 2005 § 6.2-6.5 [section anchor under re-verification 2026-08-29: the review's TOC places positronium in § 9 and § 6 covers light hydrogen-like HFS; cf. the § 4→§ 9 correction] vs CMY 2000 vs Adkins–Czarnecki 2014.
3. **Helium fine-structure $\alpha^3(Z\alpha)^2$ itemization.** Drake 1971 versus Pachucki–Yerokhin 2010 itemization of higher-order Breit–Pauli plus multi-electron spin-spin / spin-other-orbit content. Expected magnitude: tens of MHz on GHz-scale intervals. This is the only Lamb-class candidate in the predicted set; the other two are HFS-class. **Verdict (V.D-prediction sprint, 2026-05-10): CONFIRMED → V.D.6 (§5.3.6)** — Pachucki–Yerokhin 2009 (PRA **79**, 062516) explicitly resolved a 3σ disagreement with Drake on ν_{01} by reevaluating the relativistic Bethe-log term, with magnitude $\sim$kHz on the 29 GHz observable ($\sim$ 30 ppb on α determination). This is the strongest of the three V.D-prediction sprint candidates because the literature explicitly documents both the disagreement and its resolution.

The protocol classifies entries by predicted observable class: class-(i) literature-itemization (V.D.1, V.D.2, V.D.3, V.D.5, V.D.6 of the existing catalogue) versus class-(ii) closed-form-vs-operator-level (V.D.4 of the existing catalogue). Future sprints adding entries should tag which class the new entry falls under.

Pattern C status (2026-05-10). After the V.D-prediction sprint: 6/6 of the *then-catalogued* § V.D entries (V.D.1–V.D.6; the 2026-08 catalogue has nine, and the later architecture/wall-class additions are not spectroscopy observables, so Pattern C is a statement about the original six) are precision-spectroscopy class observables (HFS, muonic-Lamb, or Lamb-class fine structure).

Pattern C is **CONFIRMED at n=6 and STRENGTHENED**; the pattern broadens slightly from “HFS-only” to “precision-spectroscopy with multi-component Layer-2 decomposition” (V.D.6 is Lamb-class He 2^3P fine structure, not strictly HFS).

What would falsify Pattern C. If the next sprints produce § V.D entries from heavy-atom HFS at coarse experimental precision (Cs, Rb), molecular vibrational/rotational spectroscopy, or non-precision atomic spectra, Pattern C is precision-spectroscopy-specific rather than HFS-specific (a refinement, not a falsification). A genuine falsifier requires *0 of those land class-(i)*, in which case Pattern C tightens to “exclusive to high-precision atomic spectroscopy with multi-loop QED + nuclear-structure budgets” rather than broadening further. Conversely, a class-(i) finding on a *non*-precision-spectroscopy observable would falsify the current pattern reading entirely.

5.4 Falsified candidate: W3 spectral-zeta identification of SM calibration data (sprint 2026-05-08, falsified same evening)

This subsection preserves the record of a single-session candidate that was *falsified* the same evening by three independent follow-up tracks. It is retained in the catalogue because the methodological lesson – how a 6–10σ signal under a hand-curated prespecified basis can vanish under a mechanically generated basis – is a sharper instance of the curve-fit-audit discipline (see `docs/curve_fit_audit_memo.md`) than any abstract description.

The original candidate. On 2026-05-08 evening, a conversational sprint produced candidate spectral-zeta identifications of all four CKM Wolfenstein parameters with master-Mellin-engine M1 and M2 ring elements: $\lambda^2 \approx 1/\mathrm{Vol}(S^3) = 1/(2\pi^2)$ and $\bar{\rho} \approx 1/(2\pi)$ in the M1 Hopf-base family; $\bar{\eta} \approx (\ln 2)/2 = -\zeta'(0, 1/2)$ and $A^2 \approx \ln 2$ in the M2 chirality family. The four-form fit gave 7 hits at $< 1\%$ in a hand-curated 30-form prespecified basis (random expectation 0.7 ± 1.0); a derived structural relation $A^2 = 2\bar{\eta}$ survived PDG at 0.52σ; a closed form $\tan(\delta_{CP}) = \pi \cdot \ln(2)$ at 0.02σ provided a striking single-number target. The aggregate signal was reported as approximately 6–10σ above chance for that basis size. The candidate motivated a draft addition to CLAUDE.md §1.7 (WH7) and prompted a forward plan in `debug/w3_forward_plan_memo.md` listing six tests.

Three-track falsification (same evening). The first three tests of the forward plan ran in parallel as independent sub-agent dispatches. All three returned negatives.

- **PMNS recheck.** The same 30-form prespecified basis applied to the PMNS angles $(\sin^2\theta_{12}, \sin^2\theta_{23}, \sin^2\theta_{13},$ at PDG 2024 NH best-fit values returned *zero* matches at $< 2\%$ across all four parameters. Aggregate signal: -0.68σ below chance. PMNS sits closer to tri-bimaximal pure rationals $(1/3, 1/2, 0)$ than to any M1/M2 form. The cleanest single PMNS hit is $\theta_{12} \approx \arcsin(1/\sqrt{3})$ at $0.11\,\sigma$ – the TBM solar-angle prediction, not in the master Mellin rings. W3 universality across mixing matrices is falsified. Files: `debug/w3_pmns_recheck.{py,_memo.md}`.
- **Lepton mass spectrum.** The same methodology applied to $\sim$30 charged-lepton mass observables (ratios, log spacings, sqrt-mass ratios, Koide K, cone position) crossed with the 30-form basis and 10 small-rational factor multipliers (9000 tests total) returned only 2 within-1σ matches, both encoding the same Koide 45° cone fact via algebraically equivalent identities $(2/3 \times (\varphi^2 - \varphi) = 2/3$ and $4 \times \pi/16 = \pi/4)$. Statistical signal: $+0.85\sigma$ above chance, indistinguishable from random. Hit-density gap CKM vs lepton: 12% vs 0.22% (50×). W3 identification of mass spectra is falsified. Files: `debug/w3_lepton_mass_recheck.{py,_memo.md}`.

- **Mechanically-generated basis sweep.** The hand-curated 30-form basis was replaced by a mechanically generated 21,448-form basis: combinations of M1/M2/M3 seeds (Hopf-base measures, Hurwitz shifts, Catalan G, Dirichlet β, ζ_R at integer points) with small-integer rational prefactors and small algebraic factors ($\sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi$), plus a pure-rational control class. The full basis was frozen and saved to JSON before any test run. Re-running the four-Wolfenstein search returned aggregate $z = -0.52$ (slightly *below* random chance); per-parameter z-scores all within $\pm 1\sigma$ of chance (λ +0.74, A −0.02, $\bar{\rho}$ −1.01, $\bar{\eta}$ +0.82). The original M1↔real / M2↔CP-imaginary structural reading does not survive: $\bar{\eta}$'s top mechanical match is in the M1+algebraic class ($\pi^{3/2}/16$), not M2; A's original $\sqrt{\ln 2}$ does not appear in the top-20 sigma-ordered slice; and λ's top match is the pure rational $9/40 = 0.225$ exactly, beating every transcendental form. Selection-bias verdict: fully attributable. Files: `debug/w3_mechanical_basis.{py,_memo.md}`.

Three independent tests, three negatives. The W3 spectral-zeta candidate is falsified. The original signal was basis-selection bias. WH7 is not promoted to CLAUDE.md §1.7.

Methodological lesson. The original "5–10σ above chance" calculation was statistically correct *for that basis size.* The error was the implicit assumption that the 30 hand-curated forms were drawn without knowledge of the data; in fact, the basis was disproportionately near the empirical parameter values. Mechanical generation removes the implicit selection, and when removed, the signal vanishes. This is the cleanest illustration the project has produced of why the curve-fit-audit discipline (independent re-run with strictly mechanically-generated basis) is necessary, not optional, for empirical pattern claims that touch transcendental rings of any size.

Surviving observations independent of W3. The charged-lepton Koide 45° cone (1 arcsec) is independent of the W3 sprint and stands. One isolated match is preserved as a record (no structural reading attached): the CKM Wolfenstein parameter $\lambda = 0.225$ is exactly 9/40 to 5-digit PDG precision, which in GeoVac vocabulary reads as $\lambda = 9\Delta$ with $\Delta^{-1} = g_3^{\mathrm{Dirac}} = 40$ (Paper 2 Dirac-mode boundary count); whether this is structural or Diophantine coincidence at PDG precision is not addressable from a single data point and the framework does not commit to either reading.

Standing position on calibration data. GeoVac's structural-skeleton scope (Bertrand truncation, inner-factor Mellin trivialization, H1/G3 verdicts, LS-8a renormalization gap) is restored as the standing position on calibration data: parameter values that are not autonomously generated by the framework are genuinely external input. The second-packing-axiom question for calibration data remains open at the same level it was on 2026-05-07.

5.5 Lorentzian transfer audit (Sprint L0, May 2026)

This subsection is a structural partition table, not an empirical match. It is placed in § 5 because it accumulates sprint-by-sprint refinements parallel to § 5.1 (off-precision matches), § 5.2 (Roothaan autopsies), and § 5.3 (literature convention exposures). The Sprint L0 audit (`debug/lorentzian_l0_audit_memo.md`, May 2026) classifies this section's twenty-eight projections (§ 3) by their transfer behavior under the Riemannian $(3,0) \to$ Lorentzian $(3,1)$ extension in the Bizi–Brouder–Besnard [41] signature classification. The full structural framing of the signature partition is in Paper 31 § 8 [17]. The principal falsifier of the audit's structural-correspondence verdict on the Wick-rotation projection (§ 3.27) is the operator-system-level computation flagged in § 9 open question 6.

The audit's trigger was the Nieuviarts 2025 twisted-spectral-triple morphism [45], which provides an explicit (Riemannian twisted spectral triple) $\to$ (pseudo-Riemannian spectral triple) morphism via a Krein-product construction. Whether the construction applies cleanly to GeoVac $S^3 = \mathrm{SU}(2)$ (odd-dimension caveat) is the parallel L2-Nieuviarts-scoping pass and is *not* assumed in the audit.

Classification buckets. Four buckets are used (see Paper 31 § 8.1 for the structural framing in $\mathcal{A}/D$ partition language):

transfers freely (TF). Signature-blind: lives on the V or D axis of § 4 only, no Riemannian volume form, no Mellin transform of $\mathrm{Tr}\, e^{-tD^2}$, no Latrémolière propinquity. Holds at $(3, 0)$ and $(3, 1)$ identically.

wick-map free (WMF). Reduces to the M1 sub-mechanism of the master Mellin engine via the four-witness Wick-rotation theorem (Hawking + Sewell + Bisognano-Wichmann + Unruh codified in Sprint Unruh-pendant, May 2026; see Paper 32 § VIII Remark on the Bisognano-Wichmann reading). Lorentzian reading inherited at no structural cost at the metric-functional level.

euclidean specific (ES). Requires Krein-space machinery, Lorentzian-native heat-trace formalism, or Lorentzian analog of Latrémolière propinquity. Named blockers B1 (Lorentzian propinquity, deep original NCG-mathematics work, possibly shortened by Nieuviarts 2025 morphism), B2 (Krein spectral triple at $(3, 1)$ per BBB 2018), B3 (Lorentzian-native heat-trace formalism); see Paper 31 § 8.2.

mixed (M). Parts transfer freely, parts require extension. Sub-mechanism-specific decomposition required.

The 28-projection table. Table 26 gives the per-projection classification with one-line mechanism. The machine-readable form with confidence ratings is at `debug/data/lorentzian_l0_audit.json`.

Headline distribution. Seventeen projections classify as TRANSFERS FREELY (61%); four as WICK-MAP FREE (14%); five as EUCLIDEAN SPECIFIC (18%); two as MIXED (7%). Three-quarters of the dictionary inherits a Lorentzian reading at no further structural cost at the metric-functional level (TRANSFERS FREELY + WICK-MAP FREE = 75%). The remaining 25% (EUCLIDEAN SPECIFIC + MIXED) sits exactly at the load-bearing structural extensions that define future Sprints L1 (operator-system literal-identification of Wick rotation, multi-sprint), L2 (BBB Krein lift, deep), and L3 (Lorentzian propinquity, deep, possibly skippable; see Paper 31 § 8.5).

Two predicted trivializations at $(3, 1)$. The audit produces one structural prediction worth flagging: under the BBB sign table at $(m, n) = (3, 1)$, the M3 sub-mechanism of the master Mellin engine (vertex-parity Hurwitz / Dirichlet-L content in Camporesi-Higuchi vertex sums, Paper 28 § QED-vertex) becomes structurally empty on chirality-symmetric spectrum. The $\{J, \gamma_5\}$ anticommutation flip at $(3, 1)$ removes the asymmetry that drives M3's half-integer-Hurwitz content. This would simplify Paper 32 § VIII's case-exhaustion theorem from three mechanisms (M1, M2, M3) to two (M1, M2) on the Lorentzian side — not a contradiction with the Riemannian theorem, but a structural sharpening that becomes a Sprint L2 consistency check (cf. Paper 31 § 8.4 and Paper 32 § VIII.E).

Sprint L2-E: Krein-space period closure of the Wick-rotation projection — signature-blind (2026-05-17; the "literal identification" reading was withdrawn 2026-06, below). The Wick-rotation projection (§ 3.27) was named at Sprint L0 as the BRIDGE projection from Riemannian to Lorentzian content; the operator-system-level literal identification on the Lorentzian Krein side was named there as the principal Sprint L1 target (Riemannian side) and Sprint L2-E target (Lorentzian side). Sprint L2-E (`geovac/modular_hamiltonian_lorentzian.py`, Paper 32 § VIII.E.E [18]) closes the Lorentzian-side identification at finite cutoff: on $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ at signature $(3,1)$, with the Lorentzian wedge $W_L = P_W^{\mathrm{spatial}} \otimes P_{t\geq 0}$ and the BW choice $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$, the Krein-space period-closure identities close bit-exactly: BW-α period closure, BW-γ Tomita period closure, flow conjugacy $\sigma_t^{L,\mathrm{TT}} = \sigma_{-t}^{L,\alpha}$, and six-witness collapse all at residual $\leq 4 \times 10^{-16}$ across $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. At $N_t = 1$ the construction reduces bit-identically to Paper 42's Riemannian closure [64] (the $N_t = 1$ slice), on the same compact KMS $\beta = 2\pi$ circle. The structural finding that survives: Paper 42 § 7.2 "$H_{\mathrm{local}} \neq D_W$" open question O3 is **signature-INDEPENDENT at the Riemannian limit** ($\|H_{\mathrm{local}} - D_L^W\|_F|_{N_t=1}$ equals the Riemannian-side residual bit-exact at every tested $n_{\max}$); the spectral-action–vs–modular-Hamiltonian generator distinction is a deeper structural feature of the framework, not a peculiarity of the round-S^3 Riemannian truncation. *Status (2026-06):* the "literal identification at the Krein operator-system level" reading is **withdrawn** — the Paper 45 K^+-annihilation theorem (2026-06-09) and the compact-boost closure (2026-06-19) show the truncated BW boost $K = \mathrm{diag}(2m_j)$ has integer spectrum ($\mathrm{e}^{2\pi \mathrm{i} K} = I$ bit-exact), so the period closure is the compact KMS $\beta = 2\pi$ circle and the Lorentzian signature is metrically invisible at finite cutoff (Euclidean/convention; see the status update in § 3.27). What survives is the period-closure computation as a compact-side KMS statement. No finite-cutoff Lorentzian propinquity construction exists.

Sprint L2-D refinement of the M3 trivialization prediction (2026-05-16). The L0 prediction above was tested directly by Sprint L2-D (Paper 32 § VIII.D, computational driver `debug/data/l2_d_connes_a` and found to be *convention-dependent*. Under the n_{Fock}-parity convention — the Paper 28 § QED-vertex convention that accesses Catalan G and Dirichlet $\beta(4)$ via quarter-integer Hurwitz shifts at $a = 3/4$ and $a = 5/4$ — the vertex-parity sum $D_{\mathrm{even}}(4) - D_{\mathrm{odd}}(4)$ on the chirality-symmetric $(3,1)$ truncation is bit-identical to the Riemannian value at every $n_{\max} \in \{1,\ldots,5\}$: the L0 prediction is *falsified* under this reading. Under the *chirality-pairing* convention (pair "even" with $\chi = +1$ Weyl and "odd" with $\chi = -1$ anti-Weyl), the chirality-symmetric truncation gives the difference $= 0$ bit-exact, but tautologically by symmetry alone; the L0 prediction is *trivially confirmed* under this reading but does not access Paper 28's M3 content. **Structural refinement:** the BBB sign-flip $\{J, \gamma_5\} = 0$ at $(m,n) = (4,6)$ is a spectral-triple-axiom statement, not a vertex-parity-sum statement. M3's mechanism is sectional in n_{Fock}-parity (the chirality-independent label used in Paper 28's vertex-parity sum), not in spacetime signature. This is the corresponding informational annotation on the M3 entries under § III.6 / § III.7 / § 3.27; it is not a status flip on a Table 26 row. Paper 18 § III.7 records the corresponding sharpening of the master Mellin engine domain partition.

Scope and limits. This is a first-pass classification by an analyst, not a theorem. Confidence ratings flag medium-confidence rows that may shift to either side under Sprint L1 / L2 actual computations. In particular, two MIXED rows (Wilson plaquette #10, Breit retardation #16) and one TRANSFERS_FREELY row at medium confidence (coupled-channel #25, which inherits an algebraic-implicit π from Fock) may move under finer audit. The audit explicitly does not assume that Nieuviarts 2025's morphism applies cleanly to GeoVac $S^3 = \mathrm{SU}(2)$ (odd-dim caveat); that question is the L2-Nieuviarts-scoping pass. The Bekenstein-Hawking entropy sprint (`debug/bekenstein_hawking_sprint_scoping_memo.md`) is correctly DEFERRED and outside the L0/L1/L2/L3 sprint scope: BH entropy is Euclidean spectral-action work that does not require Lorentzian extension.

Cross-references. The structural framing of the Sig/Op partition is Paper 31 § 8 [17]. The bridge projection (§ 3.27) already names the operator-system-level deepening as the principal Sprint L1 falsifier; § 9 open question 6 carries the forward-looking task list. Paper 32 § VIII.E sharpens the master Mellin engine's M1 sub-mechanism as the free-transfer engine via the four-witness theorem.

6 Dimensional Axiomatization

The projection dictionary of § 4 treats each projection as a graded map along three axes: variables added, dimensions added, and transcendental class introduced. Sprint TX-A (May 2026) treated this graded structure as an axiomatic system and tested three theorems analytically. This section consolidates the verdicts; full data are in `debug/data/tx_a_projection_table.json` and `debug/data/tx_a_composition_table.json`, with verification driver `debug/tx_a_signature_arithmetic.py`.

Theorem 1 (Dimensional Consistency)**.** *For any projection chain $S = P_n \circ \cdots \circ P_1$ applied to the bare graph (Layer 1, dimensionless), the variable, dimension, and transcendental signatures of S equal the union of the signatures of the constituent P_i:*

$$\mathrm{sig}(P_n \circ \cdots \circ P_1) = \bigcup_{i=1}^{n} \mathrm{sig}(P_i).$$

The catalogue rows of § 5 satisfy this consistency check across all eight fully-tagged entries with non-trivial chains (Lamb shift, $K = \pi(B+F-\Delta)$, Stefan-Boltzmann, He CI, hydrogen E_n, scalar/Dirac spatial Casimir, Uehling VP).

The theorem is true by construction at the level of the dictionary metadata: each projection's signature is fixed in § 3, so set/list union is a tautology. Its non-trivial content is the empirical check that the catalogue's expected target signature matches the chain signature; this passes for all eight tested rows.

Theorem 2 (Weak Minimality)**.** *Two chains containing the same multiset of projections produce the same signature, hence are observationally indistinguishable at the level of dimensional types and transcendental classes. They may differ at the operator-system level when bundle-non-trivial projections (Hopf, spectral action, Wilson) are composed with the spinor lift; these inequivalences are not visible to signature arithmetic and require explicit operator-system analysis (cf. the WH1 spectral truncation rounds 1–3 documented in `CLAUDE.md` § 2).*

The strong form of minimality (uniqueness up to commutation as operators on the Hilbert space) fails for the Paper 2 K-chain: `hopf` composed with `spinor_lift` acts differently in the two orderings (Hopf-then-spinor produces Hopf-S^2 with Dirac fiber; spinor-then-Hopf gives a $\mathbb{Z}_2$-quotient of Dirac S^3 with half-integer charges). After the addition of `breit_retardation` (§ 3.16), a second non-commuting pair appears: `rest_mass` composed with `breit_retardation` is ordered, with rest-mass first. The non-commutation is semantic: Breit retardation's variable m_l/m_n is a refinement of the rest-mass variable m (it is the ratio between two species' masses, both of which are introduced by the rest-mass projection), so it is not well-defined until rest-mass has acted. The metadata-level test correctly flags both (`hopf`, `spinor_lift`) and (`rest_mass`, `breit_retardation`) as the non-commuting pairs in the worked examples; the underlying mechanisms differ (bundle-non-trivial composition for the first, variable-refinement dependence for the second).

Theorem 3 (Composition Algebra)**.** *The 16×16 composition table classifies each ordered pair* (P_1, P_2) *as one of:* commute *(admissible in both orders, disjoint variables);* one-way *(admissible in only one order, constrained by Layer transitions or by semantic dependence between variables);* not composable*;* idempotent *(Layer-1 self-composition);* self-compose *(non-trivial diagonal);* variable conflict *(same variable name added by both);* inverse pair *(one is the inverse of the other). Of* $16^2 = 256$ *cells:*

- ~ 196 *commute (*$\sim 76.6\%$*; updated from 15×15 count of 182 by adding the 14 new commuting cells in* P_{16}*'s row and column with projections that compose layer-cleanly under disjoint variable axes);*
- ~ 26 *one-way pairs (one-way cells flagged either by Layer transitions or by semantic dependence between variables; updated by +4 from the 15×15 count of 22 to account for the four new one-way cells flagged below in* P_{16}*'s row and column);*
- 6 *variable conflicts (energy-scale-conflating compositions of Fock, Bargmann–Segal, and spectral action; the new* P_{16} *introduces no further variable conflicts since its variable* m_l/m_n *is distinct from the existing variable set);*
- 4 *idempotent projections on the diagonal (Sturmian, Wigner* $3j$*, Drake–Swainson, and now two-body Dirac / Breit retardation: applying the Breit Hamiltonian twice gives the same two-body output);*
- 3 *inverse pairs (stereographic, Wigner* $3j$*, Wigner* D*; unchanged by the new projection).*

The previous version of this theorem flagged `drake_swainson` *as the unique projection with input_layer = output_layer = 2. After the addition of* `breit_retardation` *(§ 3.16),* two *projections now have input_layer = output_layer = 2: Drake–Swainson and Breit retardation. The structurally distinguishing fact is that they operate on disjoint Layer-2 content (Drake–Swainson regularizes divergent Bethe-log spectral sums; Breit retardation adds two-body kinematic content to the framework's single-particle Lamb-shift bracket), so they commute as a* (P_1, P_2) *pair; their non-uniqueness on the layer index is harmless for signature arithmetic.*

New structural finding (the load-bearing T3 update from this addition). *The pair* $(P_{14}, P_{16}) = (\text{rest-mass projection}, \text{Breit retardation})$ *does* not *commute, but the non-commutation is* not *a Layer transition — both projections output to Layer 2 from input layers 1 and 2 respectively, and the Layer indices alone would mark this as a one-way pair under the existing classification. The* semantic *reason for one-way ordering is sharper: the Breit projection's variable* m_l/m_n *is the rest-mass ratio between two species of particle in the bound state, which is well-defined only after rest-mass projection has assigned masses to each species. In the reverse order* $P_{14} \circ P_{16}$ *the variable* m_l/m_n *would have to be introduced before the masses themselves exist, which is semantically inadmissible. This is the first instance in the Paper 34 dictionary of a one-way pair driven by a* variable refinement *dependence rather than by a Layer transition. The signature-arithmetic test correctly flags both the* (P_{14}, P_{16}) *and the* (P_{16}, P_{14}) *cells, and the test agrees with the structural reading: Breit retardation must follow rest-mass projection in any admissible chain.*

Erratum (Track TS-C, 2026-05-04): an earlier draft of this section claimed that `spectral_action` *and* `observation_window` *were mutually one-way, encoding the field-theoretic UV/IR ordering anomaly. That claim was retracted on direct inspection of the composition data: both projections* *target output_layer=2 from input_layer=1 with disjoint variable sets, and signature arithmetic correctly returns* `commute` *in both directions. The UV/IR anomaly ($\beta \to \infty$ does not commute with $\Lambda \to \infty$) is real physics that the current coarse layer-grading cannot see; refining the output_layer index to distinguish UV-cutoff, thermal, and flow Layer-2 outputs is the natural next step.*

Note on cell counts. The exact 16×16 cell counts are stated above as approximate (∼) totals; the load-bearing update is the 15×15 → 16×16 expansion plus the qualitative shifts (new idempotent, new one-way pair driven by variable refinement, no new variable conflicts). Definitive recomputation against the metadata in `debug/data/tx_a_composition_table.json` *is deferred until the next TX-A-style audit.*

Pending T3 expansion (2026-05-09). With the addition of the three intra-nuclear projections (§ 3.17, § 3.18, § 3.19), the T3 composition table needs expansion from 16×16 to 19×19. The qualitative expectation: the three new projections all introduce intra-nuclear $[L]$ (or $[L]^2$) variables disjoint from the existing length axes, so they should commute layer-cleanly with most existing projections; expected new entries are ~ 50 commute cells in each new row and column, with possible variable-refinement one-way pairs against rest-mass (the intra-nuclear distribution radii are defined per nuclear species, so they require rest-mass projection to assign mass to each species first — structurally analogous to the (P_{14}, P_{16}) one-way pair documented above). No new variable conflicts are expected since intra-nuclear $[L]$ scales are categorically disjoint from atomic / internuclear $[L]$ scales. Definitive cell-by-cell verification deferred to the next TX-A audit.

Observation 5 (axes are not independent)**.** *Of the three axes (variable, dimension, transcendental), only the transcendental axis is genuinely independent. Cross-tabulating the twenty-eight projections by axis-flag pattern, no projection adds a dimension without also adding the variable that carries it (perfect V/D correlation, 12/12); zero projections fall into the* (*D-only*) *bucket. The transcendental axis admits an independence witness: vector-photon promotion (§ 3, projection 11) introduces* $1/(4\pi)$ *without adding any variable or dimension. Two projections (*`spectral_action`*,* `observation_window`*) move all three axes; these are the "heavy" projections, carrying the most physical content per step. The newly added* `breit_retardation` *(§ 3.16) preserves the V/D correlation (adds the variable m_l/m_n together with the dimension energy) and lives in the ring-preserving $\alpha^4 \cdot \mathbb{Q}$ tier on the transcendental axis; it is therefore axis-pattern-equivalent to the rest-mass projection at the V/D level, with the transcendental class differing only by the leading α^4 factor inherited from its calibration sibling, the spectral action.*

This observation has structural force. The variable axis and dimension axis collapse to a single *physical-content* axis without information loss; the transcendental axis remains independent. The three-axis dictionary is therefore mildly overcomplete on the V/D side but irreducible on the transcendental side. See Remark 3 in § 4 for the parallel statement at the projection-row level.

7 Falsifiable Prediction: Layer-2-Presence Bound on Catalogue Residuals

The catalogue (§ 5, § 5.1) does *not* arrange itself by projection depth. Re-tagging each catalogue row by the number of Layer-2 inputs consumed in its chain (Layer-2 inputs in the sense of § 3.29: external scalars such as r_Z, g_p, deuteron polarizability, multi-loop QED coefficients) reveals a much sharper structural ordering:

- **L2-count** $= 0$ **(framework-native chains).** Residuals run from machine-exact (Stefan–Boltzmann $\pi^2/90$ at depth 3, hydrogen $1S$ static polarizability $\frac{9}{2}a_0^3$ at depth 3, spatial S^3

Casimir 1/240 at depth 2) to converged-basis ceilings (-0.534% for the H Lamb shift one-loop closure, $\approx 0.024\%$ for He NR variational). The residual is bounded by basis quality alone — error code B in the § 5.1 system — when the chain consumes no external Layer-2 input.

- **L2-count $= 1$ (one external scalar).** Residuals tighten to tens of ppm or sub-ppm — with the known exception of the corrected D 1S HFS row at -210.9 ppm (§5.2.7), an open case for this class, and one now diagnosed: the D chain is missing a nuclear-structure channel rather than mis-consuming its input, and a chain consuming the sourced two-photon-exchange total would land at $+23.0$ ppm, inside this class — (H 21cm HF-4 with Zemach r_Z at $+18$ ppm; muonium 1S–2S under rest-mass projection at -0.11 ppm; μH 1S Bohr–Fermi at $+2$ ppm). The single Layer-2 input is always a scalar parameter and its uncertainty propagates linearly.
- **L2-count $= 2+$ (multi-focal-wall regime).** Residuals are dominated by the largest single Layer-2 contribution. Mu HFS at $+199$ ppm is the cleanest LS-8a-class isolation (no QCD nuclear budget); positronium 1S–2S at $+64.75$ ppm framework-native is closed by the α^4 Breit Layer-2 input as the § 3.16 sixteenth projection; μH Lamb at -0.10% closes by Antognini 2013 multi-loop QED plus deuteron polarizability plus recoil NLO with the Källén– Sabry two-loop VP dominating.

Prediction 2 (Layer-2-presence bound on catalogue residuals)**.** *The residual error of a catalogue row is bounded above by the larger of (i) the framework's basis-quality residual at zero Layer-2 inputs in the chain, and (ii) the largest single Layer-2 input consumed by the chain:*

$$|\epsilon(observable)| \;\leq\; \max\!\big(\epsilon_{basis}(framework),\; \max_i |L_2^{(i)}|\big).$$

The genuine ordering axis is observable-class (zero-, one-, multi-Layer-2-input) crossed with the framework's basis-quality limit at zero Layer-2 inputs. Chained projection depth is a downstream proxy: it correlates with opportunity for compounding when the chain consumes Layer-2 inputs, but it fails in both directions when the chain does not (deep zero-L2 chains can be exact, shallow multi-L2 chains can have large residuals).

This is a falsifiable structural claim about the framework, not a phenomenological fit. The catalogue rows above are consistent with it, with one known exception introduced by the 2026-08-28 correction: the D 1S HFS row (§5.2.7), whose corrected -210.9 ppm residual against a single $r_Z(D)$ Layer-2 input is not bounded on the ppm-class reading, and is therefore an open case for this Prediction rather than support for it. *Status (2026-08-29):* the exception is now diagnosed rather than merely recorded. It is a *missing-channel* failure, not a counterexample to the bound's logic: the D chain omits the polarizability channel entirely (§5.2.7), and a chain consuming the sourced total two-photon-exchange correction (44.5(1.1) kHz = $+135.9$ ppm; Bonilla *et al.*, arXiv:2508.18776) would sit at $+23.0$ ppm — the same sign and class as the architecturally identical H 21cm row at $+18.4$ ppm. That is a *prediction* of what building the channel should produce, and it is the sharpest available test of this Prediction: if the channel is built and the row does not land in the tens-of-ppm class, the Prediction is falsified on a row that was chosen for it in advance. The bound, the depth-linear falsification below, and the three exact L2-count $= 0$ anchors (the spatial S^3 Casimir $\frac{1}{240} = \frac{1}{2}\zeta_R(-3)$, the Stefan–Boltzmann $\pi^2/90$, and the H $1S$ polarizability $\frac{9}{2}a_0^3$ via a Dalgarno–Lewis symbolic solve, each landing at residual 0) are pinned in `tests/test_paper34_l2_presence_bound.py` (the polarizability by a genuine symbolic solve there; the two zeta anchors as identities whose from-scratch derivations live in `tests/test_paper35_kg_panel.py`). Tests for the prediction:

- Any new framework-native chain (L2-count = 0) producing a rational-analytic answer should land at residual 0 at converged basis (Sprint Calc-P generalization).
- Any new chain with a single Layer-2 input should land at residual bounded by the uncertainty on that input.
- Any new chain with two or more Layer-2 inputs should have its residual dominated by the largest single Layer-2 input, not by the sum or quadrature.

A failure of the prediction would take one of three forms: (a) a catalogue row with L2-count = 0 and residual greatly exceeding the framework's basis-quality limit (would indicate a structural artefact not yet identified); (b) a catalogue row with L2-count = 2+ whose residual is smaller than the largest single Layer-2 input by more than an order of magnitude (would indicate an unexplained cancellation between Layer-2 inputs); (c) a statistical fit of the catalogue showing that depth alone provides a tighter bound than Layer-2-presence.

Remark 6 (Naive depth-vs-residual prediction, falsified). *A naive earlier form of this prediction asserted that residual scales roughly linearly with chain depth, $\epsilon_k \sim k \cdot \epsilon_1$ with $\epsilon_1 \sim 1\%$. The May 2026 synthesis pass falsified this on the catalogue itself. At depth 3 alone the catalogue contains the H 1S polarizability ($\frac{9}{2}\, a_0^3$ exact, residual 0, Sprint Calc-P), the Stefan–Boltzmann $\pi^2/90$ (Sprint TD Track 1, residual 0), positronium 1S–2S framework-native (+64.75 ppm), D HFS Bohr–Fermi strict (−456 ppm; the +40 ppm quoted here until 2026-08-28 was an artifact of the retired baseline), and the He oscillator strength (+3.4%). At depth 4 the catalogue contains μH Bohr–Fermi (+2 ppm) and the H Lamb shift one-loop closure (−0.534%). Residuals at every depth ≥ 2 are scattered across more than four orders of magnitude. Depth correlates with* opportunity *for compounding when Layer-2 inputs are consumed, but the residual is controlled by the Layer-2-presence count and the framework's basis-quality limit, not by depth itself.*

Remark 7 (Sprint Calc-P, the cleanest L2-count = 0 exact example). *The H 1S static dipole polarizability calculation (Sprint Calc-P, § 5 three-projection chain row) returned $\alpha_p(1S) = \frac{9}{2}\, a_0^3$* exactly *via a depth-3 chain (Fock ∘ Wigner 3j ∘ Sturmian), residual = 0. Under the Layer-2-presence bound of Prediction 2 this is the structurally clean "L2-count = 0, rational analytic answer in finite-basis span" class: there are no external Layer-2 inputs in the chain, the analytic answer lies exactly within the framework's truncated basis, so the residual sits at the basis-quality limit which is 0 for this observable.*

The mechanism is sharp. The Dalgarno–Lewis solution $F|1s\rangle = -\frac{1}{2}\, z(r+2)|1s\rangle$ has p-wave radial part $(r^2+2r)e^{-r}$ that lies exactly *in the span of the first two Sturmian basis functions $\{S_2, S_3\}$ at $\lambda = Z$. Combined with the framework's symbolic-exact matrix-element machinery (rational arithmetic via `geovac/hypergeometric_slater.py` and `geovac/dirac_matrix_elements.py`), the projection chain produces the analytic answer with no truncation residual.*

This refines the L2-count = 0 class of Prediction 2 along the transcendental axis: zero-Layer-2-input observables whose analytic answer is a pure rational with a closed-form representation in the framework's truncated basis land at residual 0 at finite n_{max}; zero-Layer-2-input observables whose analytic answer requires irrational content (continuous integration over a temporal/spectral parameter per Paper 35 Prediction 1, or limits inaccessible at finite n_{max}) land at the framework's basis-quality residual. Polarizability is the first identified example of the rational class; static linear-response observables that admit Dalgarno–Lewis-type closed forms are candidates for the same exact-at-finite-n_{max} behaviour. The dividing line is empirically diagnosable from the transcendental signature: rational-class observables have pure $\mathbb{Q}$ transcendental signature, irrational-class observables have at least one transcendental factor.

8 Connection to Paper 18 (Transcendentals) and Paper 32 (Spectral Triple)

This paper does not supersede Paper 18 or Paper 32; it extends them with a complementary axis.

Versus Paper 18. Paper 18 classifies transcendental content of GeoVac observables (intrinsic, calibration, embedding, flow, composition). Paper 34 classifies the projections that *produce* that content. The two classifications are duals: Observation 2 maps each Paper 18 tier to specific Paper 34 projections. Reading direction A (transcendental $\rightarrow$ projection) lets you predict which projection must be active given an observed transcendental. Reading direction B (projection $\rightarrow$ transcendental) lets you predict the transcendental signature of a calculation before running it. The audit memo of 2026-05-02 documents what happens when this dictionary is unavailable: PSLQ searches against an arbitrary basis surface near-misses that look like identifications, because the right basis (determined by the projection) was never specified.

Versus Paper 32. Paper 32 constructs the GeoVac spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ and audits it against the Connes axioms. In that language, Layer 1 of this paper is the algebra $\mathcal{A}_{\mathrm{GV}}$ and the Hilbert space $\mathcal{H}_{\mathrm{GV}}$, while Layer 2 is the family of spectral functions, gauge sectors, and embedding maps that take $(\mathcal{A}, \mathcal{H}, D)$ to physical observables. Paper 34 is therefore the projection-side reading of the spectral triple synthesis.

Versus Paper 31. Paper 31's universal/Coulomb partition is the same observation phrased on the algebra/Dirac side: features that depend only on $\mathcal{A}$ (e.g. angular sparsity) are universal, features that depend on D_{GV} (e.g. calibration π, the Hopf bundle's projection-π) are potential-specific. Paper 34's claim is the same in projection language: zero-projection matches and Wigner-$3j$-only matches transfer between potentials, while spectral-action and Fock-conformal matches do not.

9 Open Questions

1. **Closure of the projection list.** Twenty-eight projections are now documented. The thirteenth (Drake–Swainson asymptotic subtraction) was added in sprint LS-4 to close the 2P Bethe-log structural floor of LS-3. The fourteenth (rest-mass projection) and fifteenth (observation / temporal-window projection) were added in sprint KG / Paper 35 (May 2026) after verifying that the bare relativistic Klein-Gordon spectrum on $S^3 \times \mathbb{R}$ is π-free in $\mathbb{Q}[\sqrt{d_1}, \sqrt{d_2}, \ldots]$ and that π enters the spectrum at exactly one step — temporal compactification. These two projections were previously bundled implicitly with other parameter-projection rows; the split was forced by the Casimir computation $E_{\mathrm{Cas}}^{S^3} = 1/240$ (spatial only, exact rational, no π) versus the Stefan-Boltzmann $\pi^2/90$ (high-T limit on $S^3 \times S^1_\beta$, π from Matsubara). The sixteenth (two-body Dirac / Breit retardation, § 3.16) was added in the precision-catalogue sprint of 2026-05-08, motivated by the positronium 1S–2S framework-native residual at +64.75 ppm (the sixth and structurally cleanest instance of the multi-focal-composition wall). The seventeenth, eighteenth, and nineteenth (Foldy/Friar charge density, Zemach magnetization-density, tensor multipole; §§ 3.17, 3.18, 3.19) were added in the multi-track Roothaan autopsy sprint of 2026-05-09, formalizing the intra-nuclear Layer-2 input mechanisms that surfaced as operator-level constructions during the H 21cm and D 1S HFS autopsies. The twentieth through twenty-fifth projections (§§ 3.20–3.25: Phillips–Kleinman / core–valence orthogonality; multipole expansion / Gaunt termination; bipolar

harmonic / Drake combining; symmetry / Young tableau; adiabatic / Born–Oppenheimer; coupled-channel / adiabatic curve) were added in the Sprint 1 dictionary-completion sprint of 2026-05-15. These six projections are transcription of pre-existing framework structures rather than new mechanisms: the framework had been using each of them implicitly for years, but they were not previously named in the §III dictionary. The six together close the textbook-completeness gap identified in the dictionary's structural audit and bring the catalogue from nineteen to twenty-five.

The twenty-sixth through twenty-eighth projections (§§ 3.26–3.28: gauge choice / Coulomb-Lorenz-Feynman; Wick rotation / signature change; apparatus identity / state-side reduction) were added in the Sprint 2 axis-completeness sprint of 2026-05-15. These three close the V/D/transcendental axis-grid completeness identified by the TX-A audit: gauge choice provides a second independence witness next to vector-photon promotion in the transcendental-only corner; Wick rotation promotes the metric-functional-level content of the Bisognano–Wichmann reading (Sprint TD Track 4 + Sprint Unruh-pendant, May 2026) from a §VIII open-question candidate to a named §III projection slot; and apparatus identity opens the state-side complement of the dictionary, which Sprint TD Track 5's PSLQ negative proved is categorically distinct from the spectral-side master Mellin engine.

Sprint 3 (2026-05-15) closed three additional scoping candidates as ABSORBED into existing entries rather than promoted to §III slots: (s1) Tomita–Takesaki modular flow is absorbed into the § 3.15 BW-reading footnote and § 3.27, with the framework-internal realization named as the operator-system-level Lorentzian extension below (Sprint 3 Track 10); (s2) loop expansion / α-power-sorting is absorbed into § 3.6 (where it is now named explicitly in the Structural reading paragraph), with the renormalization-counterterm gap as the natural future-direction *separate* projection candidate (Sprint 3 Track 11); (s3) heat-kernel regularization / Schwinger proper-time is absorbed into § 3.6 (where it is now named explicitly with the master Mellin engine framing) since proper-time integration and the Connes–Chamseddine spectral action are the same operation under two names; the Mellin transform itself is the meta-mechanism organizing the spectral-side projection family, not a §III peer (Sprint 3 Track 12). The Sprint 3 verdicts collectively confirm that the dictionary's twenty-eight projections are structurally complete: the master Mellin engine of Paper 32 § VIII / Paper 18 § III.7 is fully accounted for through M1+M2+M3 at the existing §III entries, and the operator-system-level Lorentzian extension and state-side enumeration are correctly classified as open extensions rather than missing projections.

Are there projections in the framework that are still not yet in this list? Two pre-Sprint-3 candidates remain on review: (a) the Mellin / heat-kernel evaluation at fractional s (currently treated as part of spectral-action; Sprint 3 Track 12 confirmed this candidate is structurally separate from the heat-kernel absorption verdict and remains open); (b) the direction-resolved Hodge decomposition of vector-photon edges (Paper 28 § vq_sprint). Plus two open extensions surfaced by Sprint 2: (c) the full operator-system-level Lorentzian extension that would deepen Wick rotation (§ 3.27) from structural-correspondence to literal-identification status (see "Operator-system-level Lorentzian extension" below; this replaces the previously-listed candidate-Lorentz-boost-projection entry, which has been subsumed by the §III.27 promotion); (d) state-side dictionary enumeration — mutual information, conditional entropy, fidelity, trace distance, relative entropy $S(\rho\|\sigma)$, Wasserstein–Kantorovich distance on $P(\mathcal{H})$ — candidates as future §III entries on the state-side complement opened by § 3.28. Decision pending on (a), (b); structural progress on (c), (d) named in their respective entries below.

2. **Quantitative form of Prediction 2.** The Layer-2-presence bound replaces the falsified naive depth-vs-residual prediction (see Remark 6). Outstanding work: (i) quantify the framework's basis-quality residual ϵ_{basis}(framework) for each major chain class (one-loop QED Lamb-shift class, NR variational class, graph-native CI class) so that the L2-count = 0 side of the prediction has a per-class quantitative bound; (ii) test whether multi-Layer-2 chains are dominated by the largest single Layer-2 input (the prediction's claim) versus quadrature-sum or worst-case sum; (iii) classify whether "rational analytic answer with finite-basis representation" (Sprint Calc-P, Remark 7) is a sub-class of L2-count = 0 that always lands at residual 0, distinct from "irrational analytic content" that lands at the basis-quality limit.

   The LS-4 Lamb shift datapoint adds historical context. The LS-4 chained projection (Fock + spectral action + Sturmian + Drake–Swainson) achieves -0.39% residual at the sweet-spot $N = 40$. At converged N, the LS-4 result reverts to the LS-1 ceiling -3.10% (LS-4, pre-LS-6a convention); the LS-6a fix supersedes this with -0.534%. Under Prediction 2 the -0.534% is the ϵ_{basis}(H one-loop QED Lamb) value and is the framework's converged-basis quality limit at zero Layer-2 inputs; the -0.39% at $N = 40$ is a transient finite-basis cancellation, not a genuine reduction of the basis-quality limit. Prediction 2 applies to converged-basis residuals.

3. $K = \pi(B + F - \Delta) \approx 1/\alpha$. The Paper 2 Observation is the most extreme multi-projection coincidence in the catalogue: three structurally independent projections (finite Casimir trace at $m = 3$, infinite scalar Dirichlet at $s = d_{\max}$, Dirac mode degeneracy at the spinor cutoff), all evaluated on the same S^3, summing to $1/\alpha$ at 4.8×10^{-7} relative residual (the tighter 8.8×10^{-8} figure is the residual of Paper 2's self-consistency root $1/\alpha + \alpha^2 = K$ against CODATA, not of the plain sum). Under the (now falsified) naive depth-vs-residual prediction this was the central anomaly: a three-projection match carrying 10^{-7} residual versus $\sim 3\%$ expected. Under the Layer-2-presence bound of Prediction 2 the chain has L2-count = 0 (no external Layer-2 inputs are consumed), so the residual should sit at the framework's basis-quality limit for the chain class. The framework's basis-quality limit for a zero-Layer-2-input chain involving the K combination is not characterized: K is a single number, not a converging sequence in $n_{\max}$. Either something structural we have not identified is forcing the combination, or the match is a coincidence below the framework's intrinsic resolution. Working hypothesis WH5 (Paper 2 in Observations status as of 2026-05-02) reads it as a coincidence; this paper records the structural anomaly without resolving it.

   Track TS-C update (2026-05-04). The "depth model under-counts" reading was tested by examining whether the three chains share a fourth implicit projection. They do not: F is derived from a Dirichlet zeta on the (n, l) lattice with no Hopf invocation (Phase 4F α-J); Δ^{-1} is a Camporesi–Higuchi Dirac mode degeneracy with no Hopf invocation (Phase 4H SM-D); and the Hopf-quotient graph at $n_{\max} = 3$ does not encode K through any independent spectral invariant beyond a Casimir relabeling (Phase 4B α-D). The outer π prefactor of K *is* identified with the Hopf-base measure (Sprint A α-PI), but this is already counted as the Hopf projection in the depth-3 chain. Under-counting is therefore not the mechanism. Combined with the K-CC clean negative on a unified Connes–Chamseddine spectral-action source, twelve mechanisms total have been eliminated. The remaining live readings are (ii) and (iii) above; the standing interpretation is (iii), per WH5.

4. **Unit-introduction asymmetry between Fock and Bargmann.** Both Fock and Bargmann–Segal projections introduce energy as the dimension; the former adds Z and produces π via $\mathrm{Vol}(S^3)$, the latter adds $\hbar\omega$ and produces no π at all. The asymmetry is captured by the HO/Coulomb rigidity theorems (Paper 24, Paper 31) at the operator-order level (first-order

complex versus second-order real). Whether the projection-dictionary axis can independently predict the asymmetry without invoking operator order is open.

5. **Pure-axis projections.** The current dictionary contains no projection that moves only the dimension axis (e.g., a unit-rescaling that introduces a length scale without a coordinate variable). Whether such projections are structurally absent from the GeoVac framework, or merely absent from the current implementation, is open. Two candidate additions surfaced by the TX-A axiomatization sprint: (a) a gauge-choice projection (Coulomb vs Lorenz) that would sit in the pure-transcendental bucket alongside vector-photon, (b) a spin-statistics projection that splits boson/fermion Matsubara content out of observation/temporal-window. Neither is forced by current data.

6. **Operator-system-level Lorentzian extension (deepening of § 3.27).**

   Sprint L0 update (May 2026). The May 2026 NO-GO scoping on full Lorentzian extension was reopened after a named trigger (`debug/lorentzian_literature_update_2026_05_16.md`) fired: the Nieuviarts 2025 twisted-spectral-triple morphism (arXiv:2502.18105) provides a Riemannian → pseudo-Riemannian construction via Krein-product twist that may significantly shorten the blocker-B1 budget. Sprint L0 (`debug/lorentzian_l0_audit_memo.md`, May 2026) executed a documentation-only pass: a 28-projection transfer audit classifying this section's projections by their behavior under the Riemannian $(3,0) \to$ Lorentzian $(3,1)$ extension. Headline distribution: 17 TRANSFERS_FREELY, 4 WICK_MAP_FREE, 5 EUCLIDEAN_SPECIFIC, 2 MIXED. Full per-projection table at § 5.5; structural framing of the Sig/Op partition at Paper 31 § 8 [17].

   Sprint L1 closure (2026-05-16): Riemannian operator-system identification verified. The principal Sprint L1 falsifier — "compute the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state, verify $\sigma_{2\pi}(O) = O$ at finite $n_{\max}$" — is now closed at the strongest level on the Riemannian side (`debug/l1_modular_hamiltonian_results_memo.md`, Paper 32 § VIII.F). Sprint L1 constructs K via the hemispheric-wedge BW-α geometric realization ($K = J_{\text{polar}}$ with integer spectrum on the full-Dirac basis), verifies the period closure $\sigma_{2\pi}(O) = O$ bit-exactly at $n_{\max} \in \{2, 3, 4, 5\}$ for all four physical witnesses (BW, HH, Sew, Unruh), with cross-witness collapse bit-exact (residuals match across witnesses at machine precision; period closure is independent of κ_g). This lifts the four-witness Wick-rotation theorem from "structural correspondence" to "literal identification at the operator-system level (Riemannian)" at every finite cutoff. (This Riemannian period-closure survives the 2026-06 descope of the *Lorentzian* "literal identification at the Krein level" reading — see the status update in § 3.27 — because $\sigma_{2\pi}$ = identity is precisely the compact KMS $\beta = 2\pi$ circle; only the Lorentzian-signature promotion was withdrawn.) § 3.27 honest-scope paragraph updated; § 5.1 new "Modular periodicity" row added as machinery-witness for the closure.

   Residual open scope (full Lorentzian extension). The remaining open question now is the Lorentzian extension to signature $(3,1)$ via the Bizi–Brouder–Besnard 2018 [41] Krein-space spectral triple — this is the named Sprint L2 follow-up (long-range scale per the Sprint L0 audit blocker-B1 estimate; the Nieuviarts 2025 twisted-spectral-triple morphism is the named trigger that may shorten this budget if it applies cleanly to $S^3 = \mathrm{SU}(2)$). Expected outcome (per Sprint L0 prediction): M3 trivializes at $(3,1)$ but M1 stays, so the four-witness theorem extends to Lorentzian. Sprint L1's Riemannian closure is the prerequisite that confirms the operator-system identification is sound before lifting signature.

A natural candidate projection is a Lorentz boost relating two observers at relative rapidity v/c on the $T_{S^3} \otimes T_{S^1_\beta}$ tensor-product spectral triple of Sprint TD Track 1: under the boost, $\lambda_x \to \lambda_x/\gamma$ and $\beta \to \beta\gamma$ with $\gamma = 1/\sqrt{1 - v^2/c^2}$. The tagged signature would be: *variable* v/c, *dimension* dimensionless, *transcendental* algebraic over $\mathbb{Q}(v/c)$ via the relation $\gamma^2(1 - (v/c)^2) = 1$.

Verdict: REQUIRES-EXTENSION at the framework level, but the M1 mechanism already does Lorentz-boost-class physics (Bisognano–Wichmann reading). The GeoVac spectral triple is Wick-rotated — the Camporesi–Higuchi Dirac spectrum is on Euclidean S^3, not on Minkowski space. Lorentz boosts are Minkowski-signature transformations and have no natural action on a Riemannian spectral triple at the metric / curvature level: under Wick rotation $t \to i\tau$ a Lorentz boost in the (t, x) plane becomes a rotation in the Euclidean (τ, x) plane that mixes the spatial S^3 factor with the temporal S^1_β factor; in Track 1's tensor-product construction the two factors are independent with disjoint operators, and a connection between them would be a structural extension, not a projection refinement.

However, the temperature mechanism IS already Lorentz-boost physics. The Bisognano–Wichmann theorem [35] identifies the modular flow of a vacuum state restricted to a Rindler wedge with the unitary representation of Lorentz boosts preserving the wedge. Sewell 1980 [37] extends this to Schwarzschild geometry: the KMS state at $\beta = 8\pi M$ on the Euclidean cigar IS the modular flow of the Lorentz boost (more precisely, the Killing flow generated by the timelike Killing vector outside the horizon). Sprint TD Track 4 reproduced $T_H = 1/(8\pi M)$ on the Euclidean Schwarzschild cigar via the master Mellin engine M_1 Hopf-base measure mechanism, with $\beta = 8\pi M$ forced by regularity at $r = 2M$. Under the Bisognano–Wichmann reading, **the 2π in $\beta = 2\pi/\kappa_g$ (or in the Unruh formula $\beta = 2\pi/a$ for a Rindler observer at acceleration a) IS the period of the boost orbit, and the framework's M_1 Hopf-base measure signature is the same 2π.** Sprint TD Track 4's reproduction of the Hawking temperature is therefore not just a Wick-rotation observation; it is the framework's instantiation of Lorentz-boost / modular-flow physics at the level the master Mellin engine controls.

What the framework does NOT do is the metric / curvature side at finite v: length contraction, Wigner rotation of angular momentum under boost, the spinor-content side of how the Camporesi–Higuchi spinors transform under boost, and the continuous-spectrum side (Lorentzian Dirac has continuous spectrum, breaking compact resolvent without temporal compactification). A clean Lorentz-boost projection acting on *all* particles simultaneously and relating two $T_{S^3} \otimes T_{S^1_\beta}$ constructions at different rapidities requires a Lorentzian spectral triple at signature $(m, n) \in (\mathbb{Z}/8)^2 = (3, 1)$ in the Bizi–Brouder–Besnard 2018 [41] framework (currently $(3, 0)$), with Hilbert space replaced by Krein space carrying fundamental symmetry $\mathcal{J}^2 = I$, and the real structure J extended via the $(\mathbb{Z}/8) \times (\mathbb{Z}/8)$-valued sign table.

The Lorentzian-NCG literature provides several published prescriptions (Strohmaier 2006 [40]; Franco 2014 [42] temporal Lorentzian spectral triples; Bizi–Brouder–Besnard 2018 (m, n) classification; van den Dungen 2016 [43] Krein fermionic action; Devastato–Lizzi–Martinetti 2018 [44] twisted-emergence) for what a Lorentzian extension would look like at the spectral-triple level. *None* provides a Lorentzian analog of the Latrémolière propinquity used in Paper 38 [63] for the WH1 PROVEN result; the propinquity literature is purely Riemannian as of May 2026 (Latrémolière, Toyota, Hekkelman–McDonald, Leimbach–van Suijlekom). Constructing a Lorentzian propinquity is original NCG-mathematics work at long-range scale.

Distinct from existing γ-extensions in the catalogue. The Dirac-sector $\gamma = \sqrt{1 - (Z\alpha)^2}$

in § 3.7 is a *bound-state radial Lorentz factor* for a single-particle hydrogenic wavefunction in a central potential; it is not a frame transformation between two observers. Similarly, the rest-mass projection (§ 3.14) rescales a single particle's mass under $m_e \to m_{\text{red}}$, and the Breit retardation projection (§ 3.16) introduces the rest-mass ratio m_l/m_n as a two-body kinematic control parameter. None of these is a frame-level boost. A Lorentz-boost projection would act on *all* particles in the system simultaneously and would relate two $T_{S^3} \otimes T_{S^1_\beta}$ constructions at different observer rapidities.

Non-Lorentzian uniform-rescaling alternative. A non-Lorentzian "rapidity-like" projection that uniformly rescales all spatial focal lengths λ_x by $1/\gamma$ and the temporal compactification scale β by γ is admissible in the current Riemannian framework: the rescaling preserves the master Mellin engine ring assignments (M1 Hopf-base measure, M2 Seeley–DeWitt heat-kernel coefficients, M3 vertex-restricted Hurwitz Dirichlet content) and the Roothaan multipole termination $L_{\max} = 2\,l_{\max}$ which is pure-angular. However, this uniform-rescaling alternative does not surface a new structural feature beyond what the existing § 3.14 and § 3.16 variable-refinement projections already encode. In the V/D/P tagging it would be V—, with no new dimension or transcendental witness, structurally similar to the existing variable-refinement projections.

Recommendation: not adding now. The framework cannot host a clean Lorentz-boost projection at the metric / curvature level without a Minkowski-signature spectral triple (a substantial NCG program, dominated by original Lorentzian-propinquity construction). The non-Lorentzian uniform-rescaling alternative is admissible but reducible to existing variable-refinement projections. Sprint TD Track 4's $T_H = 1/(8\pi M)$ result already captures the Lorentz-boost / Bisognano–Wichmann content of the M_1 mechanism, so the framework loses nothing strategically important by deferring the full Lorentzian extension. This open question is flagged for future PI decision: *if a Minkowski spectral triple is built (a major structural extension), the literal operator-system-level identification of the Wick-rotation projection (§ 3.27) becomes possible at the metric level; pre-Minkowski, the metric-functional-level Bisognano–Wichmann reading of β-circle physics is what § 3.27 already encodes, and the open extension here is the literal-identification verifier (compute the modular Hamiltonian K on $\mathcal{T}_{n_{max}}$, verify $\sigma_{i\cdot 2\pi}$ = identity at finite cutoff, take the GH limit using Paper 38's lemmas).* No near-term track; the operator-system-level falsifier is named in § 3.27's Honest scope. Equivalently, this extension is the framework-internal realization of Tomita–Takesaki modular flow on the truncated wedge / static-exterior algebra of $\mathcal{T}_{n_{\max}}$, with the modular Hamiltonian K and modular conjugation J as the principal operator-system invariants; Sprint 3 Track 10's absorption verdict (2026-05-15) classifies Tomita–Takesaki as the algebraic vocabulary of this extension rather than as an independent projection candidate.

7. **Candidate state-side dictionary enumeration.** § 3.28 (apparatus identity / state-side reduction) is the first state-side entry in the dictionary; all prior twenty-seven projections operate on the spectral data $(\mathcal{A}, \mathcal{H}, D)$ while this one operates on density matrices ρ on $\mathcal{H}$. Sprint TD Track 5's PSLQ negative established that state-side correlation entropies are categorically distinct from the spectral-side master Mellin engine, opening a structurally distinct dictionary class. Natural candidate state-side projections for future enumeration: *mutual information* $I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$, *conditional entropy* $S(A|B) = S(\rho_{AB}) - S(\rho_B)$, *relative entropy* $S(\rho\|\sigma) = \mathrm{Tr}\rho(\log\rho - \log\sigma)$, *trace distance* $\frac{1}{2}\|\rho-\sigma\|_1$, *fidelity* $F(\rho,\sigma) = \mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}$, *Wasserstein–Kantorovich distance* on the state space $P(\mathcal{H})$ identified by Paper 38 as the natural metric container for the propinquity limit. These candidates have appeared in the framework only as scaffolding inside specific calculations (Paper 26 bipartite entanglement; Paper 38

Wasserstein–Kantorovich identification of the propinquity limit) and have not been promoted to dictionary entries. Whether the state-side admits its own internal sub-mechanism partition analogous to the spectral-side M1 / M2 / M3 split (and if so, whether the four-mechanism structure — M1, M2, M3 spectral, plus a state-side fourth — is the right extension of the case-exhaustion theorem of Paper 32 § VIII) is the deeper structural question. No near-term track; this open question is the natural Sprint 3 (or future-sprint) target if the dictionary-completion arc continues.

Falsifier. This open-question entry can be retired in either direction. (a) If a Lorentzian propinquity is published (monitor Latrémolière / Toyota / Hekkelman–McDonald arXiv listings for "Lorentzian / Krein-space propinquity"), or the twisted-spectral-triple emergence approach (Nieuviarts 2024–2025 [45, 46]) stabilizes as a standard prescription, the dominant cost shrinks dramatically and the structural extension becomes tractable. (b) If a Sprint TD-class GeoVac construction reproduces a quantitative Lorentz-boost observable that is NOT already an M_1 2π-circle output (e.g., a Wigner-rotation matrix element on the Camporesi–Higuchi spinor sector, or a length-contraction matrix element relating two truncations at different β), the framework already does more boost-class physics than this entry recognizes and the entry needs sharpening. Neither falsifier is currently engaged.

Track D documentation closure (May 2026). The Track B scoping pass (`debug/lorentz_boost_scoping` May 2026) recommended a sprint-scale documentation deliverable (Candidate A) on the Bisognano–Wichmann reading of Sprint TD Track 4. Track D (`debug/bisognano_wichmann_track_d_memo.m` May 2026) delivered Candidate A: five-section structural analysis documenting that the framework's M1 mechanism on the cigar's S^1_τ factor reproduces $\beta = 8\pi M$, and that under the verified Wick-rotation chain (Hartle–Hawking 1976 [38]; Sewell 1982 [36]; Bisognano–Wichmann 1976 [35]) this 2π IS the modular-flow / Lorentz-boost orbit period of a stationary observer outside the Schwarzschild horizon. Verdict: *structural correspondence*, not literal identification (the operator-level statement $\sigma_{i\cdot 2\pi}$ = identity on the truncated metric spectral triple $\mathcal{T}_{n_{\max}}$ is the principal Track D falsifier, follow-up at MEDIUM-HIGH priority). Paper 32 § VIII gained a Bisognano–Wichmann-reading remark (`rem:bisognano_wichmann_reading`); Paper 35 § VIII gained a Bisognano–Wichmann subsection (`subsec:bisognano_wichmann_reading`); § 3.15 (temporal-window projection) gained a footnote naming the Bisognano–Wichmann reading; the Hawking-temperature off-precision row in § 5.1 catalogues the Wick-rotated thermal mechanism as a machinery-witness (error class C, M external input). Two literature corrections inherited from Track D's verification pass: Sewell's publication year is 1982 (not 1980 as the Track B scoping data had), and the Bisognano–Wichmann 1976 paper title is "On the duality condition for quantum fields" ("On the duality condition for a Hermitian scalar field" is the 1975 paper title). Track D *does not* change the REQUIRES-EXTENSION verdict on the operator-system-level Lorentzian extension itself; it closes the documentation loop on the Bisognano–Wichmann reading without lifting projection admissibility.

The Unruh pendant (`debug/unruh_pendant_memo.md`, May 2026, pendant follow-up to Track D) added the flat-space face $T_U = a/(2\pi)$ from the same M1 mechanism with $\beta_U = 2\pi/a$, completing the Hawking–Sewell–BW–Unruh four-witness chain at the framework's M1 sub-mechanism level (cf. Paper 32 §VIII Unruh extension to `rem:bisognano_wichmann_reading` and Paper 35 § VIII `subsec:unruh_pendant`). The REQUIRES-EXTENSION verdict on a literal Lorentz-boost projection is unchanged, since the same operator-level falsifier ($\sigma_{i\cdot 2\pi}$ = identity on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted state) tests all four Wick-rotation faces simultaneously — the bridge runs through the same KMS-Tomita-Takesaki algebra in every case.

8. **Born rule as calibration data — CLOSED structural position (added 2026-05-26).** The question "does GeoVac derive the Born rule from substrate primitives?" was attacked in sprint `ahha_born_rule_attempt` (2026-05-26, generative pass + audit + literature check + explicit construction) and *closed in the negative as a structural position*, not left as an open question. The §III.28 apparatus identity / state-side reduction projection produces Born-rule probabilities via the standard GNS construction; the squared-modulus structure is inherited from the complex inner product on $\mathcal{H}_{\mathrm{GV}}$ (taken as input in the spectral-triple framework of Paper 32 § III), and the Born rule follows from Hilbert-space + GNS via Gleason's theorem (1957). No GeoVac-specific operation reduces Born rule to more primitive structures. Three attempted derivation routes (§III.28 + GNS; Pythagorean orthogonality + apparatus preferred basis; master Mellin engine M_1 modular trace) all return the same Born expression by the same standard argument, with the Pythagorean route adding decoherence-flavored content (preferred basis = centralizer of H_{local} in $\mathcal{A}_W$) but not modifying the Born weighting. The framework's structural position is therefore: the Born rule sits at the calibration-data tier (Paper 18 § IV.6 inner-factor input data; CLAUDE.md § 1.7 WH8; the external-input three-class partition documented in memory file `external_input_three_class_partition.md`), alongside the Yukawa couplings, α, the gauge-group lower-bound calibration, and the multi-loop QED counterterms. Probabilistic weighting of measurement is external input to the structural skeleton, not derivable from the packing axiom + Fock projection + spectral-triple data. This places GeoVac in the Wigner / calibration-tier camp on the Born rule rather than the Deutsch–Wallace / Zurek / Bub–Pitowsky derivation camps; in particular, GeoVac uses Gleason 1957 via standard Hilbert inheritance, does not improve on Gleason, and does not claim to. What the framework *does* contribute to the measurement-problem literature is the operator-algebraic realization with Pythagorean orthogonality $\langle H_{\mathrm{local}}, D_W^L \rangle_{\mathrm{HS}} = 0$ (Paper 43 § 10.2, formal theorem with $1/\pi^2$ master Mellin engine M_1 signature): a concrete realization of relational-QM-style measurement structure where the modular (observer-defined) generator and the spectral-action (substrate) generator decouple at the Hilbert–Schmidt inner-product level. This is novel as a theorem; it is not novel as a Born rule derivation. The honest contribution of the framework on this question is therefore a positional clarification: Born rule is calibration data in the same sense that Yukawas are calibration data, structurally consistent with the framework's broader scope statement. The corresponding §III.28 catalogue paragraph documents this in the dictionary; this open- questions entry records the closure. No further derivation work on Born rule is planned at the substrate level.

   Quantitative converse tested (2026-07-01, WH8 Step-1 probe). The closure above is derivation-direction (no route from substrate primitives to the Born expression). The generation direction — could the skeleton's own native measure *substitute* for Born? — is now also tested negative and pinned. The graph's only native measures are counting-type (rational, built from discrete invariants); the probe shows (i) the counting measure coincides with Born exactly on, and only on, the *flat locus* (states of uniform amplitude over their support; for amplitudes $\propto (1, 2, 3, 4)$ on a 4-fold multiplet the deviation is the exact rational 1/3), (ii) it diverges at total-variation distance 0.86 from the He graph-native CI ground state ($n_{\mathrm{max}} = 3$, zero-parameter, exact Slater integrals; 90.8% of the Born weight sits on the $1s^2$ configuration, against $1/N_{\mathrm{supp}}$ uniform), and (iii) the flat locus is dynamically unstable under the graph's own Hamiltonian (TV grows as t^2; measured small-t exponent 2.00), so no state-independent skeleton rule can track Born through dynamics. Frozen falsifier `tests/test_wh8_born_probe.py`: a skeleton-side rule reproducing Born statistics on non-flat physical states would break these pins. Registered internally as WH8 (CLAUDE.md § 1.7, PI direction): the Born measure is the *exchange constant* of the observation projection (§ III.28) — the projection's irreducible imported content, unique given the projection lattice (Gleason) — exactly as π is the import of the Coulomb projection and $2\pi/\beta$ of the temporal one.

10 Conclusion

GeoVac is a two-layer framework. Layer 1 is a bare combinatorial graph: dimensionless integers, rational matrix elements, no physics, no π. Layer 2 is a finite family of named projections; each adds specific physical variables, specific physical dimensions, and specific transcendental content. Every physical quantity in any GeoVac calculation enters through exactly one projection in Layer 2 and is fully traceable to it.

The dictionary in this paper makes that statement operational. Twenty-eight projections are listed, each tagged by variables, dimension, and transcendental signature. An empirical matches catalogue uses these tags to organise the framework's known machine-precision agreements with standard physics values. Sprint TX-A's dimensional axiomatization (§ 6) verified three structural theorems on this dictionary: dimensional consistency holds across all worked examples, weak minimality holds at the signature level (with the strong form failing on two non-commuting pairs — the Paper 2 `hopf`–`spinor_lift` pair under bundle-non-trivial composition, and the `rest_mass`–`breit_retardation` pair under variable-refinement dependence). The 16×16 composition table expanded from the previous 15×15 reflects the new sixteenth projection (two-body Dirac / Breit retardation, added 2026-05-08); under this expansion Drake–Swainson is no longer the unique projection with input_layer = output_layer = 2, with Breit retardation joining as the second. Both projections continue to account for all asymmetric (one-way) cells in the 16×16 table. The 2026-05-09 addition of three intra-nuclear projections (charge-density / Foldy–Friar, magnetization-density / Zemach, tensor multipole) extends the dictionary further to nineteen, and sharpens the L–E asymmetry of § 4.1: $[L]$ admits six scalar instances plus one rank-2 tensor instance, while $[E]$ admits four. Sprint 1 (2026-05-15) added six more projections that the framework had been using implicitly (§§ 3.20–3.25: Phillips–Kleinman / core–valence orthogonality, multipole expansion / Gaunt termination, bipolar harmonic / Drake combining, symmetry / Young tableau, adiabatic / Born–Oppenheimer, coupled-channel / adiabatic curve), closing the textbook-completeness gap. Sprint 2 (2026-05-15) added three more (§§ 3.26–3.28: gauge choice / Coulomb-Lorenz-Feynman; Wick rotation / signature change; apparatus identity / state-side reduction), closing the axis-grid completeness gap: gauge choice provides a second independence witness in the transcendental-only corner of the V/D/transcendental grid, Wick rotation promotes the Bisognano–Wichmann reading of the M1 mechanism from a § VIII open-question candidate to a named §III slot, and apparatus identity opens the state-side complement of the dictionary (which Sprint TD Track 5's PSLQ negative proved is categorically distinct from the master Mellin engine). These nine additions bring the dictionary count from nineteen to twenty-eight. Sprint 3 (2026-05-15) tested three further candidates — Tomita–Takesaki modular flow, loop expansion / α-power-sorting, heat-kernel regularization / Schwinger proper-time — and returned ABSORBED verdicts on all three: TT is absorbed into §§ 3.15+3.27; loop expansion is absorbed into §§ 3.6+3.11+3.16+3.26; heat-kernel regularization is absorbed into § 3.6 where the master Mellin engine framing is now made explicit in its Structural reading paragraph. The Sprint 3 scoping verdicts confirm structural completeness at twenty-eight: the dictionary does not require additional §III slots to cover the master Mellin engine, the loop-order sorting of QED, or the Tomita–Takesaki algebraic vocabulary of the Wick-rotation chain. The new projections add no new base units and preserve the V/D correlation property; T3 expansion to 28×28 is flagged for the next TX-A audit. A boundary subsection (§ 3.29) names the structural distinction between intrinsic graph-derived variables (which the framework projects natively) and Layer-2 inputs (external classical fields, multi-loop QED coefficients, lattice-QCD-derived radii); the same day's external-field hunting verified that uniform Zeeman / Stark fields are Layer-2, not new projections. T3 expansion to 19×19 is flagged for the next TX-A audit. The field-theoretic UV/IR ordering anomaly between `spectral_action` and `observation_window` is real physics but is *not* encoded by the current coarse layer-grading (Track TS-C, 2026-05-04); refining the output_layer index to distinguish UV-cutoff, thermal, and flow Layer-2 outputs is the natural next step. Of the three axes, only the transcendental axis is genuinely independent — variable and dimension are perfectly correlated. A falsifiable structural prediction (§ 7) emerges from the depth-grouped catalogue: the corrected reading is the Layer-2-presence bound articulated in §7 (the earlier "error compounds with projection depth" conjecture was falsified by the §5.2 autopsy panel and replaced). The corrected prediction is consistent with current data — with the one known exception recorded in §7, the corrected D 1S HFS row — including the 208/208-confirmation panel of Paper 35 Prediction 1 (Sprint TX-B, May 2026), and refinable by future calculations.

Apart from Remarks 1 and 2, added after certification, this paper introduces no new computational result. Its contribution is a vocabulary that the framework has been needing. With the two-layer view in place, the question "where does π come from in this observable?" has a definite answer for every observable GeoVac produces. The same is true for α, Z, $\hbar\omega$, Λ, $\zeta(3)$, Catalan's G, and any other quantity that enters from outside the bare graph.

Citation status of the Layer-2 inputs

The Layer-2 values this catalogue consumes come from the external precision literature, and until 2026-08-29 most were carried as inline "Author Year" attributions with no bibliography entry. That layer is where every external-citation defect found by the 2026-08 review arc actually lived: a phantom edition ("Eides 2024" — no such edition exists; the work is Eides–Grotch–Shelyuto 2007), a section anchor wrong at eight loci (positronium is § 9 of Karshenboim 2005, not § 4), ab-initio Zemach radii credited to a paper that does not contain them (they are King *et al.* 2026), and a coefficient characterised as a source's "closed form" that does not appear in that source's text.

Resolved. Thirteen works are now in the bibliography with volume/page/year verified against the primary text: Eides–Grotch–Shelyuto 2007 [48], Karshenboim 2005 [49], Antognini 2013 [50], Friar–Payne 2005 [51], Clementi–Roetti 1974 [52], Wineland–Ramsey 1972 [53], Lundeen–Pipkin 1981 [54], Sick 2014 [55], Nevo Dinur *et al.* 2019 [56], King *et al.* 2026 [57], Bonilla *et al.* 2025 [58], Pachucki–Karshenboim 1998 [59] and Carlson–Nazaryan–Griffioen 2008 [60]. These carry the experimental anchors and the nuclear-structure evaluations that the autopsies of §5.2 actually consume.

Not yet resolved, and tracked. The remaining inline attributions still lack bibliography entries. Two deserve naming because they are load-bearing and ambiguous rather than merely unlisted. (i) The "Pachucki–Yerokhin 2010" itemization invoked for the deuterium hyperfine budget: that author pair's 2009/2010 papers are helium fine-structure calculations, and the modern deuterium-HFS itemization is Kalinowski–Pachucki–Yerokhin 2018 (and now Bonilla *et al.* 2025 [58]), so the attribution needs re-sourcing rather than a bibitem transcribed from it. (ii) The $-(11/48)$ positronium Breit coefficient: the section anchor is now correct, but the literal fraction does not appear in that section's text and is the framework's own reduction, verified numerically at 0.57% (§5.1).

The unresolved set is not left to memory: the deterministic check `debug/qa/check_inline_attributions.py` enumerates every inline attribution, marks those with no resolvable `\bibitem`, and holds the current set as a ratchet baseline — a *new* unresolved attribution fails the check by name, while the existing backlog is recorded rather than silently tolerated.

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Projection	Variables added	Dimension	Transcendental signature
Fock conformal	Z, E	energy	$\kappa = -1/16$; π via $\mathrm{Vol}(S^3)$
Hopf bundle	α	dimensionless	$\pi = \mathrm{Vol}(S^2)/4$
Bargmann–Segal	$\hbar\omega$	energy	*none* (graph π-free)
Stereographic	r (length)	length	conformal factor (Coulomb $1/r$)
Sturmian reparam	Z, n (re-use)	preserves	rational preserved
Spectral action	Λ, α	energy via Λ	$\sqrt{\pi} \times \mathbb{Q}$ SD coefficients; $\pi^{2k} \times \mathbb{Q}$ observables
Camporesi–Higuchi spinor	α via $(Z\alpha)^2$	dimensionless	half-integer Hurwitz $\rightarrow$ odd ζ; Catalan G, $\beta(4)$ at vertex
Wigner $3j$	none	dimensionless	$\mathbb{Q}[\sqrt{2k{+}1}]_k$
Wigner D rotation	$\vec{R}_{AB}$	length	$\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$
Wilson plaquette	$\beta = 1/g^2$	dimensionless	SU(2) Haar
Vector-photon promotion	none	dimensionless	$1/(4\pi)$ per loop
Molecule-frame hypersph.	R	length	piecewise-smooth in R
Drake–Swainson asymptotic subtraction	K (cancels)	dimensionless	flow tier; $D_{\mathrm{drake}} = 2(2\ell + 1)Z^4/n^3$ rational
Rest-mass projection	m	mass	*trivial* (ring-preserving for $m^2 \in \mathbb{Q}$)
Observation / temporal-window	β (or $T = 1/\beta$)	time	$2\pi \cdot \mathbb{Q}$ per Matsubara mode; $\pi^{2k} \cdot \mathbb{Q}$ in integrated quantities; Stefan-Boltzmann $\pi^2/90$ in high-T limit
Two-body Dirac / Breit retardation	m_l/m_n	energy	$\alpha^4{\cdot}\mathbb{Q}$ (ring-preserving over $\mathbb{Q}(\alpha)$)
Charge-density (Foldy/Friar)	r_E per nucleus	length	$\mathbb{Q}(\alpha)$ ring-preserving
Magnetization-density (Zemach)	r_Z per nucleus	length	$\mathbb{Q}(\alpha)$ ring-preserving (leading); profile-shape sub-features at sub-leading
Tensor multipole (quadrupole, etc.)	Q_N per nucleus	length-squared	$\mathbb{Q}$ via Wigner $3j$ at angular level
Phillips–Kleinman / core–valence orthogonality	$\{\phi_c, E_c\}$ core orbitals (Layer-2 input)	dimensionless	ring-preserving over $\mathbb{Q}(\alpha)$; no transcendental injected
Multipole expansion / Gaunt termination	L (integer; runs $0, \ldots, L_{\max} = 2l_{\max}$)	preserves	ring-preserving over $\mathbb{Q}[\sqrt{2k{+}1}]_k$; exact termination, not asymptotic
Bipolar harmonic / Drake combining	(k_1, k_2, K) bipolar triple	preserves	pure rational over $\mathbb{Q}[\sqrt{2k{+}1}]_k$ (sibling of Wigner $3j$)
Symmetry / Young tableau	irrep tableau λ	dimensionless	*none* (integer characters; projector entries in $\mathbb{Z}/N!$)
Adiabatic / Born–Oppenheimer	$\vec{R}$ (slow); m_e/M_n small parameter	length	*none* at projection step (transcendentals appear downstream in § 3.25)
Coupled-channel / adiabatic curve	R (radial), ν (channel index)	length	algebraic-implicit at Level 3 (over $\mathbb{Q}(\pi, \sqrt{2})$); piecewise-smooth at Level 4
Gauge choice (Coulomb / Lorenz /	none (selection of representative)	preserves	gauge-dependent at intermediate-propagator

Projection	Variable	Dim	Base-unit injection	Role
Fock conformal	Z, E	$[E]$	foundational anchor for atomic units; sets a_0 ($[L]$) and Ry ($[E]$) jointly via $p_0 = Z/n$; $[Q]$ as integer label Z	foundational
Hopf bundle	α (Paper 2)	dim'less	none	pure
Bargmann–Segal	$\hbar\omega$	$[E]$	foundational anchor (HO sibling of Fock); sets oscillator length ($[L]$) and $\hbar\omega$ ($[E]$) jointly	foundational (sibling)
Stereographic	r	$[L]$	$[L]$ as continuous coordinate; no new scale (coordinatizes the foundational $[L]$)	coordinate-explicit
Sturmian reparam	Z, n (re-use)	—	none	reparam
Spectral action	Λ, α	$[E]$	new $[E]$ scale (UV cutoff Λ); α dimensionless	scale-extension
Camporesi–Higuchi spinor	α via $(Z\alpha)^2$	dim'less	none	pure
Wigner $3j$	none	dim'less	none	pure
Wigner D rotation	$\vec{R}_{AB}$	$[L]$	new $[L]$ scale (internuclear position vector)	scale-extension
Wilson plaquette	$\beta = 1/g^2$	dim'less	none	pure
Vector-photon promotion	none	dim'less	none	pure
Mol-frame hyperspherical	R	$[L]$	new $[L]$ scale (internuclear distance; same physical length as $\lvert\vec{R}_{AB}\rvert$)	scale-extension
Drake–Swainson	K (cancels)	$[E]$	transient $[E]$ scale (cancels in $\beta_{\text{low}}(K) + \beta_{\text{high}}(K)$)	scale-extension (transient)
Rest-mass	m	$[M]$	first $[M]$ scale beyond foundational $m_e = 1$; one projection per particle species	scale-extension (new base)
Observation / temporal-window	β (or T)	$[T]$	first $[T]$ scale beyond foundational $\tau_0 = \hbar/\text{Ry}$; one projection per measurement window	scale-extension (new base)
Two-body Dirac / Breit	m_l/m_n	dim'less	ratio of two prior $[M]$ scales (lepton, nucleon); no new base content	composition
Charge-density (Foldy/Friar)	r_E per nucleus	$[L]$	new $[L]$ scale (intra-nuclear, charge distribution); modifies V_{ne} via convolution	scale-extension
Magnetization-density (Zemach)	r_Z per nucleus	$[L]$	new $[L]$ scale (intra-nuclear, magnetization distribution); modifies $A_{\text{hf}}^{\text{contact}}$	scale-extension
Tensor multipole (rank-≥ 2)	Q_N per nucleus	$[L]^2$	rank-L multipole moment, dim $[L]^L$; couples to electronic gradients ($l \geq 1$ on electronic side)	scale-extension (rank-2)

Table 2: Base-unit decomposition of the twenty-eight projections (the six Sprint 1 additions of 2026-05-15 add no new base units; they preserve or re-use existing dimensions; the three Sprint 2 additions of 2026-05-15 also add no new base units — gauge choice, Wick rotation, and apparatus identity all preserve dimension). "Variable" is the new physical content; "Dim" repeats the compound dimension from Table 1; "Base-unit injection" shows what enters in $[L]$, $[M]$, $[T]$, $[Q]$, with $[E]$

Match	Projection(s)	Residual	Source	Notes
He ground state (graph-native CI, $Z=2$)	Fock + Sturmian	0.20% at $n_{\max}=9$	S	Small-Z graph-validity-boundary artifact (Sec. 2 of CLAUDE.md, v2.9.2 CUSP-2).
He ground state (2D variational + cusp)	Fock + spectral cusp	0.019% at $\ell_{\max}=7$	T	Track DI Sprint 1; cusp-corrected 0.004% at $\ell_{\max}=4$.
He ground state (Hylleraas r_{12}, $\omega_{\max}=4$)	Fock $\circ$ explicit r_{12} correlation	-0.016 mHa = 0.0006% vs Drake NR at 22 basis functions	T	Track 1 sprint 2026-05-09 (`geovac/hylleraas_r12.py`); cusp ratio 0.49 vs Kato 0.5 at $u=0$ (variationally captured); 3p reproduces Hylleraas 1929 -2.90324 Ha to $+0.79$ mHa. Ground-state architecture *categorically distinct* from graph-native CI (no $\kappa=-1/16$ inter-shell coupling artifact, no small-Z graph-validity-boundary limitation). Excited states (2^1S, 2^3S, 2^1P) deferred — single-α trial inadequate, named structural extension is Hylleraas-Eckart double-zeta.
LiH $R_{\rm eq}$ (composed, PK)	Fock $\circ$ molecule-frame	5.3% at $\ell_{\max}=2$	B	PK $\ell_{\max}=2$ optimal operating point; higher ℓ degrades structurally (Track BQ, Paper 17).
BeH_2 $R_{\rm eq}$	composed + exchange	11.7%	B	Same PK structural ceiling as LiH; full 1-RDM exchange.
H_2O $R_{\rm eq}$	composed + lone pairs	19.4%	B	6:1 charge asymmetry overwhelms angular basis at $\ell_{\max}=2$.
H_2 D_e (Level 4)	molecule-frame hyperspherical	4.0%	T	96.0% of experimental D_e at $\ell_{\max}=6$; CBS $\sim 97\%$.
LiH 4-electron FCI energy	balanced coupled $\circ$ Fock	0.20% at $n_{\max}=3$	T	Improvement 1.8% $\to$ 0.20% from $n_{\max}=2\to 3$; structural drift in $R_{\rm eq}$ remains.
Schwinger $\alpha/(2\pi)$ from graph-native vertex	vector-photon $\circ$ graph-native	33% level overshoot	C	Wrong projection constant; $C_{\rm VP}\times F_2$ diverges (sprint VP-1; CLAUDE.md graph-native QED bullet).
Anomalous moment $C\times F_2$ asymptotic	graph-native	12.7× overshoot at $n_{\max}=3$	C	Vector promotion gets 3.6× closer to scalar baseline but does not close projection mismatch (sprint VP-1).
Bethe logarithm $\ln k_0(2S)=2.812$	Sturmian (acc.) $\circ$ Fock	-0.92% at $N=40$	T	Acceleration form 3.3× tighter than LS-2 velocity at $N=50$; sprint LS-3.
Bethe logarithm $\ln k_0(1S)=2.984$	Sturmian (acc.) $\circ$ Fock	$+0.60\%$ at $N=20$	T	Acceleration form crosses Drake at smaller N than velocity; sprint LS-3.
Bethe logarithm $\ln k_0(2P)=-0.0307$	Drake–Swainson $\circ$ Sturmian $\circ$ Fock	$+2.4\%$ at $N=40$	T	**LS-4 closes the LS-3 structural floor.** New Drake–Swainson projection (§ 3.13) produces K-independent value via $\ln k_0 = J_v/D_{\rm drake}$ with structural denominator $D_{\rm drake}(n,\ell)$ =

Component	MHz	Projection chain	Status
SE $2S_{1/2}$ (Bethe + Eides 10/9)	+1066.44	§3.1 ∘ §3.5 ∘ §3.6 ∘ §3.13	FN
VP $2S_{1/2}$ (Uehling)	−27.13	§3.1 ∘ §3.6	FN
SE $2P_{1/2}$	+12.88	§3.1 ∘ §3.5 ∘ §3.6 ∘ §3.7 ∘ §3.13	FN
VP $2P_{1/2}$	0	§3.1 (parity zero)	FN
α^5 multi-loop QED	+1.20	§3.1 ∘ §3.5 ∘ (§3.6)^2 ∘ §3.7 + Layer-2 C_{2S}	L2
FNS r_p contribution	+0.138	§3.1 ∘ §3.7 ∘ **§3.17** + Layer-2 r_p	L2
Recoil NLO (beyond reduced-mass)	−2.40	§3.1 ∘ §3.14 ∘ §3.16 + Layer-2 R_{nl}	L2
Hyperfine averaging (F-state mixing)	∼ +5.00	§3.1 ∘ §3.7 ∘ §3.8 ∘ §3.14	FN
Sum	+1056.13		
Experimental	+1057.845(9)		
Residual	+1.72	(was +0.68 at the retired FNS value; the "within quoting precision" reading no longer follows)	

Table 5: Hydrogen $2S_{1/2} - 2P_{1/2}$ Lamb shift Roothaan autopsy. **Corrected 2026-08-28:** the FNS row previously read +1.18 MHz, which is the *1S* finite-size shift (at $r_p \approx 0.868$ fm); the $2S$ value is +0.138 MHz at $r_p = 0.8409$ fm, an $n^3 = 8$ factor smaller. The error was visible inside this paper: the He$^+$ autopsy (§5.2.5) uses a Lamb$_{\text{FNS}}$ ratio of 63.599, which implies an H reference of 8.7913/63.599 = 0.13823 MHz. Each component is tagged to its §3 projection chain. All entries are contributions to the $2S_{1/2} - 2P_{1/2}$ splitting: the $2P_{1/2}$ self-energy is a −12.88 MHz level shift, entering the splitting as +12.88 MHz. Status: FN = framework-native; L2 = Layer-2 input. Source memo: `debug/calc_track_LAR_lamb_autopsy_memo.md`.

Component	MHz	Projection chain	Status
Bohr–Fermi Dirac (point nucleus, $g_e = 2$, no recoil)	+1421.1595	§3.1 ∘ §3.7 ∘ §3.8	FN
+ Schwinger a_e (Parker–Toms-verified at +0.5%)	$\times(1 + \alpha/2\pi)$	§3.1 ∘ §3.7 ∘ §3.6	FN (with calibration)
+ Reduced-mass $(1 + m_e/m_p)^{-3}$	×0.998366	§3.1 ∘ §3.14	FN
+ Zemach $r_Z = 1.045$ fm via §3.18 operator-level	×(1 − 39.495 ppm)	§3.1 ∘ §3.7 ∘ **§3.18**	FN at op-level + L2 (r_Z)
Final $A_{\mathbf{HF}}$	+1420.4318		
Experimental	+1420.4058	(12-digit precision)	
Residual	+0.026	= +18.4 ppm; marginally above the Eides Tab. 7.3 itemized +12–+18 ppm band	

Table 6: Hydrogen 21 cm hyperfine Roothaan autopsy. Each component is tagged to its §3 projection chain. Status: FN = framework-native; L2 = Layer-2 input. Source memo: `debug/h21_autopsy_v1_memo.md` (Sprint Calc-H21-Autopsy v1, May 2026).

Table 7: Five-component Roothaan decomposition of the muonic hydrogen Lamb shift (muonic convention $E(2P_{1/2}) - E(2S_{1/2})$). The framework-native subtotal +200.50 meV comes from full Uehling integration via §3.6, the self-energy bracket under rest-mass projection (§3.14 + §3.5), and the operator-level Friar moment via the magnetization-density module (§3.18; cf. §3.17 for the parent charge-density projection). The +1.67 meV gap to the +202.17 meV framework-plus-literature total is closed by Antognini 2013 literature inputs covering multi-loop QED (Källén–Sabry, mixed VP-VP/VP-SE, α^7) and QCD-internal proton polarizability (W3 inner-factor). Source memo: `debug/muH_Lamb_autopsy_v1_memo.md`.

Component	Value (meV)	Projection chain	Status
1. Full Uehling VP	+205.0074	§3.1 ∘ §3.6	FN (−0.10 ppm vs Antognini)
2. Self-energy (Eides bracket)	−0.8295	§3.1 ∘ §3.5 ∘ §3.14	FN (+24% vs Antognini, recoil-mixing ga
3. Friar moment (closed-form)	−3.6752	§3.1 ∘ §3.17 / §3.18	FN (−4.3% vs Antognini)
4. Källén–Sabry 2-loop VP	+1.5081	Layer-2 input	LS-8a renormalization wall
5. Proton polarizability	+0.0129	Layer-2 input	W3 inner-factor (QCD)
Companions (recoil + multi-loop)	+0.144	Layer-2 input	Antognini 2013 catalogue
Total	+202.17	Residual −0.10% vs CREMA +202.3706(23) meV	

Table 8: Five-component Roothaan decomposition of the muonic deuterium Lamb shift (muonic convention $E(2P_{1/2}) - E(2S_{1/2})$). The framework-native subtotal +198.91 meV comes from full Uehling integration via §3.6, the self-energy bracket under rest-mass projection (§3.14 + §3.5), and the closed-form Friar moment at the deuteron's measured RMS charge radius via §3.17. The +3.71 meV gap to the +202.63 meV framework-plus-literature total is closed by Krauth 2016 literature inputs covering multi-loop QED (Källén–Sabry, iterated Uehling, α^7), higher-Friar moments and recoil-mixing, NLO recoil corrections, and—dominant contribution—deuteron polarizability (Carlson–Gorchtein– Vanderhaeghen 2014). Source memo: `debug/muD_Lamb_autopsy_track1_memo.md`.

Component	Value (meV)	Projection chain	Status
1. Full Uehling VP	+227.6347	§3.1 ∘ §3.6	FN (−0.50% vs Krauth)
2. Self-energy (Eides bracket)	−0.8737	§3.1 ∘ §3.5 ∘ §3.14	FN (+41% vs Krauth, same recoil-mixin
3. Friar moment leading (closed-form Eides)	−27.8466	§3.1 ∘ §3.17	FN (+0.024% vs Krauth at r_d, bit-tight
4. Multi-loop QED (KS + iterated + mixed + 3-loop)	+1.8578	Layer-2 input	LS-8a renormalization wall
5. Friar HO + recoil-mixing	+0.1030	Layer-2 input	higher Friar moments
6. Recoil corrections (NLO + HO)	+0.0637	Layer-2 input	W1a sub-leading recoil
7. **Deuteron polarizability**	**+1.6900**	Layer-2 input	**W3 inner-factor (QCD, dominant)**
Total	+202.63	Residual −0.12% vs CREMA +202.8785(34) meV	

Table 9: Five-component Roothaan decomposition of the He^+ $2S_{1/2}$–$2P_{1/2}$ Lamb shift (normal-H convention $E(2S_{1/2}) - E(2P_{1/2})$). The framework-native subtotal +13 841.11 MHz comes from the same projection-chain machinery that closed the H Lamb shift (§5.2.1) with $Z = 2$ substituted throughout: Bohr-Fock Dirac (zero by $2S_{1/2} = 2P_{1/2}$ degeneracy), one-loop self-energy via Eides Sec. 3.2 bracket, one-loop vacuum polarization via full Uehling kernel, and Foldy-Friar finite nuclear size via §3.17. The −200 MHz deficit against experiment is the Z-amplified bracket-completion gap (α^3 truncation amplified by $Z^4 \times \ln$ from the −0.55% residual at Z=1), closed at Layer-2 by Drake-Yan 1992 multi-loop QED ($\alpha^2(Z\alpha)^4 + \alpha(Z\alpha)^5 + \alpha(Z\alpha)^6 \ln(Z\alpha)^{-2}$). Source memo: `debug/he_plus_lamb_autopsy_track_a_memo.md`.

Component	Value (MHz)	Projection chain	Status
1. Bohr-Fock Dirac baseline	+0.0000	§3.1	FN (zero by Dirac $2S_{1/}$
2. One-loop SE (Eides Sec. 3.2 bracket)	+14 258.5252	§3.1 ∘ §3.5 ∘ §3.6 ∘ §3.13	FN ($\alpha^3 Z^4$ truncation)
3. One-loop VP (full Uehling kernel)	−426.2096	§3.1 ∘ §3.6	FN ($\beta = 137$, contact f
4. Foldy/Friar FNS ($r_E(^4\mathrm{He})^2$)	+8.7913	§3.1 ∘ §3.17 + Layer-2 r_E	FN
5a. $\alpha^2(Z\alpha)^4$ two-loop QED	+6.71	Layer-2 input	LS-8a renormalization
5b. $\alpha(Z\alpha)^5$ higher-order	+0.18	Layer-2 input	LS-8a
5c. $\alpha(Z\alpha)^6$ ln relativistic Bethe-log (Drake-Yan)	+14.7	Layer-2 input	bracket-completion gap
6. Recoil NLO m_e^2/m_α	−3.18	Layer-2 input	W1a sub-leading
7. ^{4}He polarizability	+0.06	Layer-2 input	W3 inner-factor (small
Framework-native subtotal	**+13 841.11**	(1+2+3+4)	
Framework + literature total	+13 859.58	(1+2+3+4+5a+5b+5c+6+7)	
Experimental	**+14 041.13(17)**	van Wijngaarden 2000	
Theory	**+14 041.18(13)**	Pachucki 2001	
Residual (framework+lit)	**−181.55** MHz / −1.29%	Z-amplified bracket-completion gap	

Interval	E_{SO} (MHz)	E_{SS} (MHz)	E_{SOO} (MHz)	Total (MHz)	Projection chain
$P_0 - P_1$	−29 198.090	+23 747.770	+35 063.226	+29 612.906	§3.1 ∘ §3.7 ∘ §3.5 (SO); + (SS, SOO chains via §3.19 ∘ §3.8)
$P_1 - P_2$	−58 396.180	−9 499.108	+70 126.452	+2 231.165	(same as P_0-P_1)
$P_0 - P_2$	−87 594.270	+14 248.662	+105 189.679	+31 844.070	(same as P_0-P_1)
NIST					
$P_0 - P_1$				+29 616.951(6)	
$P_1 - P_2$				+2 291.176(15)	
$P_0 - P_2$				+31 908.131(6)	
Residual					
$P_0 - P_1$				−4.05 MHz / −0.014%	
$P_1 - P_2$				−60.01 MHz / −2.62%	partial cancellation 61.9×
$P_0 - P_2$				−64.06 MHz / −0.20%	

Table 10: He 2^3P fine-structure operator-level Roothaan autopsy. The angular content (Drake J-pattern, spin reduced m.e., bipolar Gaunt selection) is sympy-exact at machine precision; residuals live entirely in the radial sector (multi-electron correlation, higher-order Breit-Pauli, α^3/π QED, recoil). Source memo: `debug/he_2_3P_autopsy_v1_memo.md`.

Component	MHz	Projection chain	Status
Bohr–Fermi Dirac (point nucleus, $g_e = 2$, no recoil), I=1 multiplicity 3/2	+327.234993	§3.1 ∘ §3.7 ∘ §3.8 ($\hat{I} \cdot \hat{S}$, 3/2 from CG)	FN
+ Schwinger a_e (Parker–Toms-verified at +0.5%)	$\times(1 + \alpha/2\pi)$	§3.1 ∘ §3.7 ∘ §3.6	FN (with calibration)
+ Reduced-mass $(1 + m_e/m_d)^{-3}$	×0.999183	§3.1 ∘ §3.14 (variable m_n: $m_p \to m_d$)	FN
+ Zemach $r_Z(D) = 2.593$ fm via §3.18 operator-level	$\times(1 - 98.001$ ppm$)$	§3.1 ∘ §3.7 ∘ **§3.18** (at I=1)	FN at op-level + L2 (r_Z sc
Final $\nu_{\mathbf{HFS}}(D)$	+327.315305		
Experimental	+327.384353	(Wineland–Ramsey 12-digit precision)	
Residual	−0.069	= −210.9 ppm; itemization does not close (see below)	

Table 11: Deuterium 1S hyperfine Roothaan autopsy. Each component tagged to its §3 projection chain. Status: FN = framework-native; L2 = Layer-2 input. Source memo: `debug/d_hfs_autopsy_track5_memo.md` (multi-track sprint Track 5, May 9 2026).

Component	MHz	Projection chain	Status
Bohr–Fermi Dirac (point nucleus, $g_e = 2$, no recoil), I=1/2 multiplicity 1	+1515.865482	§3.1 ∘ §3.7 ∘ §3.8 ($\hat{I} \cdot \hat{S}$, multiplicity 1 for I=1/2)	FN
+ Schwinger a_e (Parker–Toms-verified at +0.5%)	$\times(1 + \alpha/2\pi)$	§3.1 ∘ §3.7 ∘ §3.6	FN (with calibration
+ Reduced-mass $(1 + m_e/m_t)^{-3}$	×0.999454	§3.1 ∘ §3.14 (variable m_n: $m_p \to m_t$)	FN
+ Zemach $r_Z(t) = 1.762$ fm via §3.18 operator-level	$\times(1 - 66.594$ ppm$)$	§3.1 ∘ §3.7 ∘ **§3.18** (at I=1/2, $r_Z(t)$)	FN at op-level + L2 ($r_Z(t)$
Final $\nu_{\mathbf{HFS}}(T)$	+1516.696899		
Experimental	+1516.701471	(Greene 2017, 12-digit precision)	
Residual	−0.005	= −3.0 ppm‡; inside Karshenboim 2005 Layer-2 budget $+12 \pm 5$ ppm	

Table 12: (‡ provisional: rests on $r_Z(t) = 1.762$ fm, under review — see the status note below.) Tritium 1S hyperfine Roothaan autopsy. Each component tagged to its §3 projection chain. Status: FN = framework-native; L2 = Layer-2 input. Source memo: `debug/T_HFS_autopsy_track2_memo.md` (multi-track sprint Track 2, May 18 2026).

Component	MHz	Projection chain	Status
Bohr–Fermi 2^3S_1 (multi-electron, Slater-screened $Z_{\mathrm{eff,2s}} = 1.15$, $\hat{I}\cdot\hat{J}$ multiplicity 3/2 at I=1/2, J=1)	+6650.31	§3.1 ∘ §3.7 ∘ §3.8 ∘ §3.22 (multi-electron triplet) ∘ §3.20 (Slater screen)	FN at op-level (PK conventi
+ Schwinger a_e	$\times(1+\alpha/2\pi)$	§3.6	FN (with calibrat
+ Reduced-mass $(1+m_e/m_{\mathrm{He3}})^{-3}$	$\times 0.999454$	§3.14 ($m_p \to m_{\mathrm{He3}}$)	FN
+ Zemach r_Z(He-3) = 1.965 fm **[under review; see §5.2.9]** via §3.18 (Z=2 scaling)	$\times(1-148.5\ \mathrm{ppm})$	§3.1 ∘ §3.7 ∘ **§3.18** (at I=1/2)	FN at op-level + L2 (r
Final $\nu_{\mathbf{HFS}}$(He-3 2^3S_1)	+6653.41		
Experimental	+6739.70	(Schuessler/Prior, 14-digit precision)	
Residual	−86.29	= −1.28% (−12 804 ppm); dominant wall is §3.20 multi-electron screening, NOT LS-8a multi-loop QED	

Table 13: Helium-3 2^3S_1 hyperfine structure Roothaan autopsy. **First multi-electron HFS in the catalogue.** The Slater-screened contact density is a sprint-scope placeholder; the actual ab initio §3.20 calculation would give the framework-preferred unique value within the 10.5% spread documented below. Source memo: `debug/He3_HFS_autopsy_track3_memo.md` (multi-track sprint Track 3, 2026-05-18).

$n_{\max}$	configs	ω (Ha)	$\lvert\langle z\rangle\rvert^2$	f_{length}	err vs Drake
2	5	0.9044	1.107	2.002	+625%
3	12	0.7045	0.420	0.592	+114%
4	22	0.6930	0.397	0.550	+99%
5	35	0.6891	0.323	0.445	+61%
6	51	0.6893	0.334	0.460	+67%
7	70	0.6893	0.322	0.444	+61%
exact	—	0.7799	0.177	0.27616	0

Table 14: Coulomb Sturmian convergence for He $2^1P \to 1^1S$ oscillator strength at single exponent $k_{\mathrm{orb}} = Z = 2$. Both ω and $\lvert\langle z\rangle\rvert^2$ plateau by $n_{\max} = 6$ at *wrong limits* (ω low by −12%, $\lvert\langle z\rangle\rvert^2$ high by +82%). Source memo: `debug/he_oscillator_v1_memo.md`.

Component	A (MHz)	Residual	Status
C1: Bohr–Fermi strict ($g_e = 2$, point nucleus)	783.0	−65.9%	FN
C2: Screened $\lvert\psi_{6s}(0)\rvert^2$ via [Xe] `FrozenCore` $Z_{\text{eff}}(r)$ (§3.1 + §3.20)	1.328 bohr^{-3} (Richardson extrap.)		FN
C3: Schwinger a_e (§3.11)	783.9	−65.9%	FN
C4: Casimir relativistic enhancement $F_R = 1.555$ (§3.7)	**1219.0**	**−47.0%**	FN
C5: Zemach $r_Z = 6.7$ fm (§3.18)	−0.1		L2
C5: Bohr–Weisskopf +1.2% (Roberts–Ginges 2015)	+14.6		L2
L: Multi-loop QED $\sim +100$ ppm (§3.16)	+0.1		L2
Framework-native total	**1219.0**	**−47.0%**	
With Layer-2 corrections	**1233.8**	**−46.3%**	

Table 15: Cs-133 $6S_{1/2}$ hyperfine five-component Roothaan autopsy. The −47% residual is dominated by two identified gaps: (a) Clementi–Raimondi single-zeta screening produces $\lvert\psi(0)\rvert^2 \approx 1.33$ bohr^{-3} (outer-shell overshoot 2.68× from missing radial nodes); (b) Casimir $F_R = 1.555$ is leading-order only (full Bohr–Weisskopf ≈ 2.6 at $Z = 55$). Source: `debug/data/cs_hfs_v2.json`.

Component	MHz	Projection chain	Status
Bohr–Fermi Dirac at $I = 3/2$, $J = 1/2$, hydrogenic $2s$ with $Z_{\text{eff}} = 1.279$ (CR67), multiplicity $\Delta\nu = 2A_{\text{hf}}$	+288.913	§3.1 ∘ §3.7 ∘ §3.8 ($\hat{I} \cdot \hat{S}$ at $I = 3/2$; eigenvalues $\{+3/4, -5/4\}$, mult 2)	FN (with Z_{eff} L2)
+ Schwinger a_e	$\times(1 + \alpha/2\pi)$	§3.1 ∘ §3.6	FN (with calib)
+ Reduced-mass $(1 + m_e/m_{^7\text{Li}})^{-3}$	×0.999765	§3.14 at $m_n = m_{^7\text{Li}}$	FN
+ Zemach $r_Z(^7\text{Li}) = 3.71$ fm via §3.18 (Z_{eff}-scaled)	×(1 − 179 ppm)	**§3.18** at $I = 3/2$	FN at op-level + L2 (r_Z scalar)
Final framework-native ν_{HFS}	+289.13		
Experimental	+803.504	(Beckmann–Boklen–Elke 12-digit precision)	
Residual	−514.4	= −64.0% ($\sim 2.78\times$ cliff factor)	

Table 16: Lithium-7 $2^2S_{1/2}$ hyperfine Roothaan autopsy. **Corrected 2026-08-28:** the Bohr–Fermi baseline previously read +82.977 MHz. That value carried the same Track-5 convention error as the deuterium autopsy (§5.2.7): the $g^{\text{atomic}} = 2\mu_I/\mu_N$ convention combined with a nuclear-mass magneton gives correct $\times\, 2(m_p/m_N)$, and $82.977/288.913 = 0.287204 = 2(m_p/m_{^7\text{Li}})$ to six digits — a pure mass-and-convention ratio with no Z_{eff} content, so screening cannot account for it. The same formula reproduces deuterium's +496.5 ppm excess and the factor-3/2 discrepancy noted in §5.2.9. Each component tagged to its §3 projection chain. The framework-native chain at hydrogenic-Z_{eff} sits $\sim 2.78\times$ below experiment — a large cliff, NOT the $\sim 1.10\%$ framework atomic accuracy floor (Paper FCI-A). Source memo: `debug/Li7_HFS_autopsy_track5_memo.md`.

Table 17: Five-component Roothaan decomposition of the positronium 1S–2S two-photon transition. Framework-native subtotal (1 + 3) = 1,233,687,096.0775 MHz reproduces the May-8 Ps 1S-2S sprint baseline at +64.75 ppm vs Fee 1993; the +80 GHz overshoot is the missing α^4 Breit retardation content (Component 2 = first operator-level test of § 3.16). Cumulative $(1+2+3+4+5)$ closes to +0.0075 MHz, within Fee 1993 ±3.2 MHz uncertainty. Source memo: `debug/ps_1s2s_autopsy_track4_memo.md`.

Component	Value (MHz)	Projection chain	Status
1. Bohr at $m_{\text{red}}(ee) = 0.5$	+1,233,690,735.0941	§3.1 ∘ §3.5 ∘ §3.14	FN
2. α^4 Breit / two-body Dirac	−79,861.9	§3.1 ∘ §3.5 ∘ §3.14 ∘ **§3.16** + Layer-2 BS expansion	L2 (op-level kernel: FN)
3. Eides § 3.2 SE Lamb-shift differential	−3,639.0166	§3.1 ∘ §3.5 ∘ §3.14 ∘ §3.6 ∘ §3.13	FN
4. Annihilation channel ($e^+e^- \to \gamma$)	−9.3100	Layer-2 input (LS-8a vertex sector)	L2
5. Multi-loop $\alpha^6 + \alpha^7$	−8.4600	Layer-2 input (LS-8a renormalization)	L2
Framework-native subtotal (1+3)	+1,233,687,096.08	+64.75 ppm vs Fee 1993	
Cumulative (1+2+3+4+5)	+1,233,607,216.41	$+5 \times 10^{-6}$ ppm; within Fee 1993 ±3.2 MHz	

Orbital pair	$R^0_{\text{BP}}(Z=1)$
$(1s, 1s; 1s, 1s)$	$-5 + 8\log 2 \approx 0.5452$
$(1s, 2s; 1s, 2s)$	$4/81 \approx 0.0494$
$(1s, 1s; 2s, 2s)$	$-4\log 2 - 19/9 + 9\log(3)/2 \approx 0.0601$
$(2s, 2s; 2s, 2s)$	$-175/256 + \log 2 \approx 0.0096$

Component	χ_d^{mol} (kHz)	$\chi_d^{\text{J=1}}$ (kHz)
Bare proton at R_{eq} (§3.21)	+488.13	−195.25
Free H-atom 1s electron (§3.1)	−150.52	+60.21
Free-atom OoM total (with $\lvert J = 1, m_J = 0\rangle$ via §3.8)	+337.61	**−135.04**
Experimental	–	+224.54
Komasa–Pachucki 2020 ab initio	–	+224.51(5)
Framework OoM residual	–	**−89.5** kHz, **−39.9**%

Table 18: HD $J = 1$ deuteron quadrupole coupling order-of-magnitude autopsy. $Q_d = 0.285699(15)(18)$ fm^2 as Layer-2 scalar input. The bare-nuclear-only contribution alone reaches 87% of experiment in magnitude (|−195| vs 224 kHz; the −2/5 rotational sign carried by the J=1 column is not resolved against the experimental sign convention by this OoM walk); the 40% residual is the absence of molecular bond-charge redistribution from the free-atom electronic picture. Source: `debug/HD_rotational_HFS_autopsy_track4.py`.

Table 19: Five-component Roothaan decomposition of the muonic helium-4 ion Lamb shift (muonic convention $E(2P_{1/2}) - E(2S_{1/2})$). The framework-native subtotal +1358.24 meV comes from full Uehling integration via §3.6, the self-energy bracket under rest-mass projection (§3.14 + §3.5), and the closed-form Foldy charge-density at the α-particle's measured RMS charge radius via §3.17. No magnetization-density (§3.18) is activated, since ^{4}He has $I = 0$. Layer-2 inputs from Diepold 2018 / Krauth 2021 (multi-loop QED, recoil NLO, α-particle polarizability) close the remaining gap to the CREMA 2021 measurement modulo two convention surfaces (SE bracket recoil-mixing, Friar HO absorption) documented below. Source memo: `debug/mu_he4_lamb_autopsy_track_b_memo.md`.

Component	Value (meV)	Projection chain	Status
1. Full Uehling VP	+1665.7731	§3.1 ∘ §3.6	FN (−0.032% vs Diepold 2018)
2. Self-energy (Eides bracket)	−11.8583	§3.1 ∘ §3.5 ∘ §3.14	FN (+8.4% vs Diepold, recoil-mixing ga
3. Foldy FNS (closed-form Eides)	−295.6714	§3.1 ∘ §3.17 ∘ §3.14	FN (+2.9% vs Diepold leading, +4.0% v
4. Multi-loop QED (KS + 3L VP + VPVP/VPSE mixed + 2L SE + α^7)	+0.787	Layer-2 input	LS-8a renormalization wall
5. Recoil corrections (relativistic + binding)	−5.941	Layer-2 input	W1a sub-leading recoil
6. **α-particle polarizability** (+ hadronic VP)	**+4.061**	Layer-2 input	**W3 inner-factor (QCD; small)**
7. FS-NS cross + residual higher-Friar	+1.510	Layer-2 input	FNS cross-channel sub-leading
Total	+1358.66	Residual −1.44% vs CREMA +1378.521(48) meV (Krauth 2021)	

Table 20: Six-component Roothaan decomposition of the Lyman-α oscillator strength. Components 1–4 produce the framework-native $f = 2^{13}/3^9 = 0.41620$ exactly. Component 5 (reduced mass) multiplies by $m_e/\mu_{\text{red}} \approx 1.000545$, giving $f^{\text{framework}+\mu} = 0.41642$. Component 6 (LS-8a-class) closes the remaining +64 ppm to NIST.

#	Component	Value	Layer	Projection
1	Bohr energy gap $E_2 - E_1$	$3/8\,E_h$ exact	1	§3.1
2	Radial dipole $\langle R_{2p}\|r\|R_{1s}\rangle$	$128\sqrt{6}/243\,a_0$	1	Algebraic Hydrogenic integration
3	Angular factor $\langle Y_{10}\|\cos\theta\|Y_{00}\rangle$, $\sum_{m_f}$	$1/\sqrt{3}\times 3$	1	§3.8
4	Oscillator-strength prefactor $(2/3)\times(1/g_i)$	2/3	2	§3.11
5	Reduced mass μ/m_e in $m_e \cdot \omega \cdot \|\langle r\rangle\|^2$ scaling	+545 ppm	2	§3.14
6	Relativistic Dirac + radiative QED	$\sim$ +64 ppm	2	LS-8a class

Table 21: Six-component Roothaan decomposition of the H $2P$ fine-structure splitting. Layer 1 produces the closed-form $\Delta E = \alpha^2/32\, E_h$ exactly. CODATA α gives the infinite-mass value 10,949.3 MHz; reduced-mass correction gives 10,943.3 MHz. Differential QED radiative closes the remaining +25.7 MHz to NIST.

#	Component	Value	Layer	Projection
1	Camporesi–Higuchi spinor Dirac on S^3, $\lvert\lambda_n^D\rvert = n + 3/2$	Spectrum exact	1	§3.7
2	Fock projection $p_0 = Z/n$, maps to H bound states	Exact	1	§3.1
3	Sommerfeld/Dirac expansion (T8 sympy-verified)	Closed form	1	Algebraic Tier-3
4	Coefficient $1/32 \in \mathbb{Q}$ from j-dependent bracket	1/32	1	j-angular structure
5	α^2 factor (CODATA)	5.32514×10^{-5}	2	External Class-1 calibration
6	Reduced mass μ/m_e in Dirac equation	≈ 0.99946	2	§3.14
7	Differential QED radiative on 2P states	$\approx +25.7$ MHz	2	LS-8a class

Table 22: Three-architecture framework reduction on the He $2\,^1S - 2\,^3S$ splitting. Single-α architectures (A, B) fail by factors 1.1 and 3.1 respectively; double-α (C) closes to 1.4% at sprint-scale basis truncation.

#	Architecture	Basis	ΔE (cm^{-1})	Error vs NIST
A	Graph-native CI, single-α ($k = Z = 2$, $\kappa = -1/16$) at $n_{\max} = 8$	1,218 configs	7,183	+11.86%
B	Hylleraas r_{12} single-α ($\zeta = Z = 2$ for both 1s and 2s)	$\omega = 5$	$\sim 19{,}900$	+209%
C	Hylleraas–Eckart double-α ($\zeta_1 = 2$, $\zeta_2 \approx 1$; Eckart 1933)	Variational	$\sim 6{,}331$	−1.4%

Component	GHz	Projection chain	Status
BF strict ($g_\mu = 2$, point nucleus)	+44 115	§3.1 ∘ §3.7 ∘ §3.8 ∘ §3.14	FN
+ Schwinger a_μ ($g_\mu = 2.0023318$)	×1.001166	§3.6	FN (with calibration)
+ Reduced mass $(m_p/(m_\mu + m_p))^3 = 0.7261$	(folded into BF)	§3.14	FN
+ Zemach $r_Z = 1.045$ fm (Eides–Grotch–Shelyuto 2007)	×(1 − 7340 ppm)	§3.1 ∘ §3.7 ∘ **§3.18**	FN at op-level + L2 (r_Z)
Final (Eides r_Z)	+43 842		
Final (Karshenboim r_Z = 1.054)	+43 839		
Literature (Karshenboim–Ivanov)	+44 183		
Residual	−341	= −0.77%; LS-8a + recoil-mixing budget	

Table 23: Muonic hydrogen 1S HFS Roothaan autopsy. First muonic HFS entry in the catalogue; creates the $r_Z(p)$ cross-link. The −0.77% residual is the same LS-8a + recoil-mixing budget as electronic H 21cm (§5.2.2), scaled to muonic mass. Source: `debug/muh_hfs_and_na_cliff.py`.

Atom	Z	ns	$Z_{\rm eff}$(CR67)	$A_{\rm fw}$ (MHz)	$A_{\rm exp}$ (MHz)	Cliff	$\lvert\psi(0)\rvert^2_{\rm needed}/\lvert\psi(0)\rvert^2_{\rm CR67}$
Li-7	3	2s	1.279	144.6	401.8	−64.0%	2.8×
Na-23	11	3s	2.507	219.8	885.8	−75.2%	4.0×
K-39	19	4s	2.162	10.5	230.9	−95.5%	22.0×
Rb-85	37	5s	2.771	23.5	1011.9	−97.7%	43.1×
Cs-133	55	6s	3.475	36.5	2298.2	−98.4%	63.0×

Table 24: Alkali ground-state HFS cliff sequence using hydrogenic-$Z_{\rm eff}$ contact density (CR67 single-zeta). The rightmost column shows the ratio of true contact density to the $Z_{\rm eff}^3/(\pi n^3)$ estimate — the quantitative measure of core-penetration contact enhancement from PK orthogonality (§3.20). [**Diagnosed 2026-08-29: the enhancement is *not* a power law in Z — a fitted $\sim Z^{1.2}$ misses Na by 126%, because the CR67 estimate itself is non-monotone in Z (K dips below Na). The closed form is given below the table.**] **Corrected 2026-08-28:** the source driver evaluated the Fermi-contact mass factor as m_e/m_N; the nuclear magneton $\mu_N = e\hbar/(2m_p)$ is defined with the *proton* mass for every nucleus, so each entry was suppressed by m_N/m_p (7.0× at Li-7 rising to 131.9× at Cs-133). The tell was the three identical 0.3 MHz entries for K, Rb and Cs. This is the mass-slot half of the same convention defect corrected in §5.2.7 and §5.2.12; the corrected Li-7 row now agrees with that autopsy's independently corrected residual, which it did not before. Source: `debug/muh_hfs_and_na_cliff.py`.

Table 25: Literature convention exposures catalogued to date (V.D.1–V.D.6 through 2026-05-10; V.D.7–V.D.9 subsequently). Each row is documented in detail in the indicated subsection (originating sprint memo) and cross-references the § 3 projection chain whose operator-level decomposition exposed it. V.D.5 and V.D.6 added 2026-05-10 by the V.D-prediction sprint testing Pattern C from the 2026-05-09 synthesis review.

ID	Observable	Source compilations	Convention difference	Magnitude	Sprint ref.
§5.3.1	D 1S HFS Layer-2 budget	Eides–Grotch–Shelyuto 2007 [48] Tab. 7.3 vs Pachucki–Yerokhin 2010	itemization of recoil NLO + multi-loop QED + finite-size correction	$\sim$ 0.01% of $\nu_F \to \sim$ 25 mfm in $r_Z(p)$	Calc-rZG (§5.1)
§5.3.2	μH $2S$-$2P$ VP one-loop	Antognini 2013 vs Krauth 2017	absorption of higher-order VP into one-loop line vs separate "VP-VP mixed" itemization	$\sim$ 100 ppm in per-component split (totals agree)	Sprint MH Track A / Calc-muH Lamb autopsy v1
§5.3.3	μ Lamb shift SE bracket	Framework Eides §3.2 vs Karshenboim 2005	$m_{\rm red}$ rescaling absorbs recoil NLO at bracket level vs separately-itemized "Recoil NLO" line	$\sim$ 11 MHz at μ Lamb (match within bracket-evaluation precision, 0.4 MHz, between framework +1074.13 and Karshenboim +1085.84−11.30)	Sprint precision-catalogue Mu Lamb 2026-05-08
§5.3.4	μH Friar moment	Eides Eq. 2.35 closed-form vs operator-level §3.18	$\langle r^2\rangle^{1/2} = r_p$ (RMS) vs $\langle r\rangle = r_p$ (first moment) for the Gaussian profile	$\langle r^2\rangle/\langle r\rangle^2 = 3\pi/8 \approx 1.178 \to$ +17.7% on Friar moment (vs Eides; +12.7% vs Antognini)	Sprint MH / Calc-muH Lamb autopsy v1
§5.3.5	D 1S HFS polarizability subcomponent	Friar–Payne 2005 vs Pachucki–Yerokhin 2010+	low-energy aggregate (FP) vs explicit polarizability + Zemach split (PY)	$\sim$ 240 ppm of E_F; $\sim$ 80 kHz absolute	V.D-prediction sprint, 2026-05-10
§5.3.6	He 2^3P fine structure (29 GHz; α-determination)	Drake 1990 / Drake–Yan 1995 vs Pachucki–Yerokhin 2009 (PRA **79**, 062516)	Drake's original relativistic Bethe-log itemization was numerically incorrect; PY	$\sim$kHz on 29 GHz; $\sim$ 30 ppb on α	V.D-prediction sprint, 2026-05-10

Table 26: Lorentzian transfer audit, Sprint L0. Per-projection classification of the twenty-eight entries of § 3. T.F. = TRANSFERS FREELY; W.M.F. = WICK-MAP FREE; E.S. = EUCLIDEAN SPECIFIC; M. = MIXED. Confidence: H high, MH medium-high, M medium. Mechanism column gives one-line rationale; for full discussion see `debug/lorentzian_l0_audit_memo.md`.

#	Projection (§ III)	Class	Conf	Mechanism / Blocker
1	Fock conformal	E.S.	H	Riemannian sphere quotient; B2.
2	Hopf bundle	E.S.	H	Riemannian fibration; $\mathrm{Vol}(S^2)$ is M1 generator.
3	Bargmann–Segal	T.F.	M	1st-order complex Euler op; π-free certificate signature-blind.
4	Stereographic / conformal	E.S.	H	Riemannian conformal map; B2.
5	Sturmian reparameterization	T.F.	H	Algebraic re-labeling on $\mathcal{A}$.
6	Spectral action (CC)	E.S.	H	Tr $f(D^2/\Lambda^2)$ Euclidean by construction; B3.
7	Camporesi–Higuchi spinor lift	E.S.	MH	Riemannian spin manifold; B2 Krein spinor at $(3,1)$.
8	Wigner $3j$ angular coupling	T.F.	H	SU(2) rep theory; canonical universal sector.
9	Wigner D rotation	T.F.	H	Spatial rotations; SU(2).
10	Wilson plaquette	M.	M	Combinatorial vocab T.F.; Marcolli-vS = YM E.S.
11	Vector-photon promotion	W.M.F.	M	$1/(4\pi)$ per loop = M1 Hopf-base.
12	Mol-frame hyperspherical	T.F.	MH	Angular separation on $\mathcal{A}$.
13	Drake–Swainson asymptotic subtraction	T.F.	H	Central tier (Paper 31 TS-C); signature-blind.
14	Rest-mass	T.F.	H	$D^2 \to D^2 + m^2 \not{1}$ central; signature-blind.
15	Observation / temporal-window	W.M.F.	H	The Wick-engine: Matsubara $2\pi k/\beta$.
16	Two-body Dirac / Breit retardation	M.	M	Angular T.F.; kinematic needs Lorentzian re-derivation.
17	Charge density (Foldy/Friar)	T.F.	H	Layer-2 scalar input; signature-blind.
18	Magnetization (Zemach)	T.F.	H	Layer-2 scalar input.
19	Tensor multipole Q_N	T.F.	H	Wigner-Eckart; pure SU(2).
20	Phillips–Kleinman	T.F.	H	Pseudopotential projection on $\mathcal{A}$.
21	Multipole / Gaunt termination	T.F.	H	3j triangle inequality; pure SU(2).
22	Bipolar / Drake combining	T.F.	H	Multi-electron angular; pure SU(2).
23	Symmetry / Young tableau	T.F.	H	S_N rep theory; signature-blind.
24	Adiabatic / Born-Oppenheimer	T.F.	MH	Scale separation; relativistic BO well-defined.
25	Coupled-channel / adiabatic curve	T.F.	M	ODE machinery; transcendental inherits from Fock.
26	Gauge choice (Coulomb/Lorenz/Feynman)	W.M.F.	MH	Coulomb-gauge $1/(4\pi)$ = M1; Ward identity holds.
27	Wick rotation / signature change	W.M.F.	H	THE BRIDGE; Krein-space period closure bit-exact but signature-blind — Lorentzian "literal identification" withdrawn 2026-06 (compact-boost / K^+ descope; § 3.27).
28	Apparatus identity / state-side	T.F.	MH	State-side; PSLQ-disjoint from master Mellin engine.